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Anatomical Position Definition

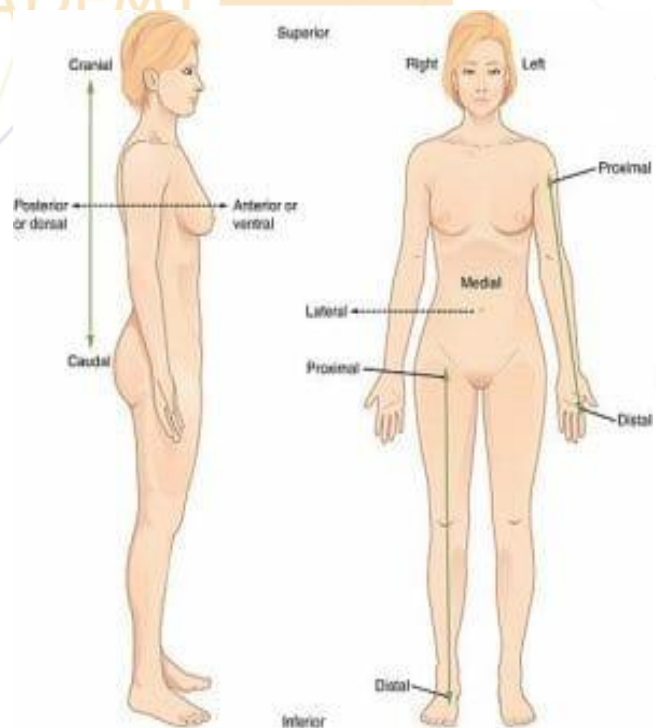
Standard anatomical position is a way of describing the anatomy of an organism so that it is easy to understand what part of the body is being talked about no matter what direction the organism is facing or where its appendages/limbs are. In humans, standard anatomical position is defined as standing up straight with the body at rest.

Function of Standard Anatomical Position

The purpose of standard anatomical position is to be able to clearly talk about different parts of moving organisms no matter how they are moving or what position they are in. This makes it easier to avoid confusion when discussing anatomy. In humans, standard anatomical position is with the body standing up straight and facing forward, with arms by the sides of the body and palms facing forward. The legs are straight, and the feet are slightly apart from one another and turned outward slightly. This is used as a starting point for describing the body. In animals that are quadrupedal, or walk on four legs, standard anatomical position is with all four feet on the ground so that the belly of the animal is roughly parallel with the ground.

Relative Locations of the Body

This diagram depicts the body in standard anatomical position and uses positional labels. Let's take a more detailed look at the labels used for different parts of the anatomy. They are regarded as axes because they are used to describe how near or far a body part is from an end of the organism.



Superior-Inferior

Superior and inferior come from the Latin words meaning “above” and below”. In standard anatomical position, the head is the most superior part of the body, and the feet are the most inferior.

Medial-Lateral

Lateral describes the parts of the body that are toward the sides, while medial describes the middle of the body. If a part of the body is lateral, it can mean left lateral or right lateral. The left and right sides of an organism refer to the left and right sides as viewed by that organism. In standard anatomical position, a person’s right hand is on the viewer’s left, since they are facing opposite the viewer.

Dorsal-Ventral

The dorsal side is the back side of an organism; think of the dorsal fin of a fish. The ventral side is the belly side of an organism. When a human is in standard anatomical position, the ventral side is facing the viewer. When a four-legged animal such as a dog is in standard anatomical position, the ventral side is their belly, which is parallel to the ground.

Cranial-Caudal

Cranial refers to being closer to the head, while caudal refers to being closer to the tail. These terms are usually used to describe the anatomy of quadrupedal organisms with tails, and not humans.

The terms rostral and caudal are used when referring to the skull. The area toward the frontal lobes of the brain is rostral, while the area toward where the spine attaches is caudal.

Proximal-Distal

Proximal and distal are used to describe bodily appendages such as limbs. The proximal part of a limb is the part close to where it joins the body, while the distal part is the part furthest away. For example, the tips of our fingers are the most distal part of our arms.

All of these terms are used for relative positions as well as standard anatomical position; for example, the neck is superior to the chest, but inferior to the head. The hands are distal to the arms. The eyes are rostral to the back of the skull.

Planes of the Body

There are three main planes that transect, or divide, the human body into sections to make it easier to describe the locations of body parts and movements. They are the sagittal plane, the coronal plane, and the transverse plane. The sagittal plane runs down the center of the body and divides the body into left and right sides. The coronal plane divides the body into dorsal and ventral sides. The transverse plane is at the waist and runs parallel to the ground, dividing the body into cranial and caudal sections.

This diagram shows the three anatomical planes of the body.

Anatomical Planes

A plane is a 2D slice through 3D space, which can be thought of as a glass sheet. The anatomical planes are different lines used to divide the human body. You will commonly see them when looking at anatomical models and prosections. Using anatomical planes allows for accurate description of a location, and also allows the reader to understand what a diagram or picture is trying to show.

There are three planes commonly used; sagittal, coronal and transverse.

- **Sagittal plane** – a vertical line which divides the body into a left section and a right section.
- **Coronal plane** – a vertical line which divides the body into a front (anterior) section and back (posterior) section.
- **Transverse plane** – a horizontal line which divides the body into an upper (superior) section and a lower (inferior) section.

For example, a diagram may be labelled as a transverse section, viewed superiorly. This indicates that you are looking downwards onto a horizontal section of the body.

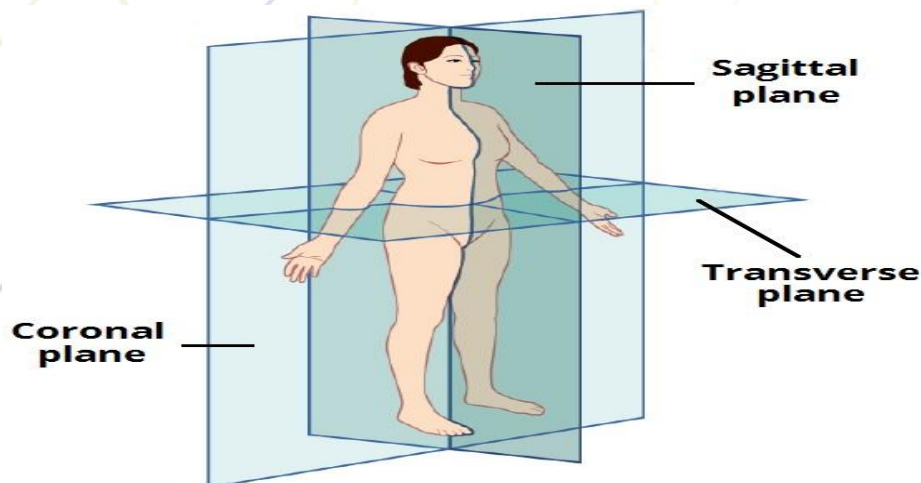


Fig 1 – The anatomical planes of the human body

Anatomical Terminology

Before we get into the following learning units, which will provide more detailed discussion of topics on different human body systems, it is necessary to learn some useful terms for describing body structure. Knowing these terms will make it much easier for us to understand the content of the following learning units. Three groups of terms are introduced here:

- Directional Terms
- Planes of the Body
- Body Cavities

Directional Terms

Directional terms describe the positions of structures relative to other structures or locations in the body.

Superior or cranial - toward the head end of the body; upper (example, the hand is part of the superior extremity).

Inferior or caudal - away from the head; lower (example, the foot is part of the inferior extremity).

Anterior or ventral - front (example, the kneecap is located on the anterior side of the leg).

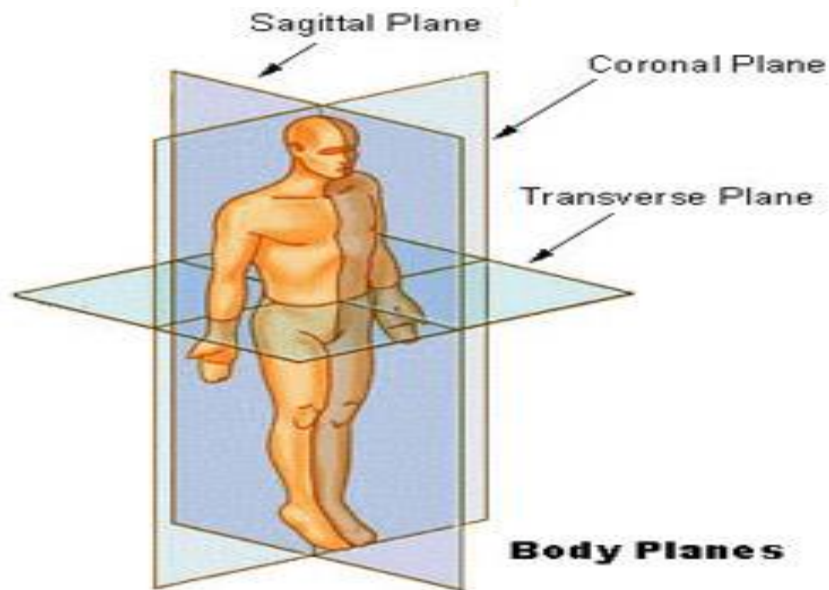
Posterior or dorsal - back (example, the shoulder blades are located on the posterior side of the body).

Medial - toward the midline of the body (example, the middle toe is located at the medial side of the foot).

Lateral - away from the midline of the body (example, the little toe is located at the lateral side of the foot).

Proximal - toward or nearest the trunk or the point of origin of a part (example, the proximal end of the femur joins with the pelvic bone).

Distal - away from or farthest from the trunk or the point or origin of a part (example, the hand is located at the distal end of the forearm).



Planes of the Body

Coronal Plane (Frontal Plane) - A vertical plane running from side to side; divides the body or any of its parts into anterior and posterior portions.

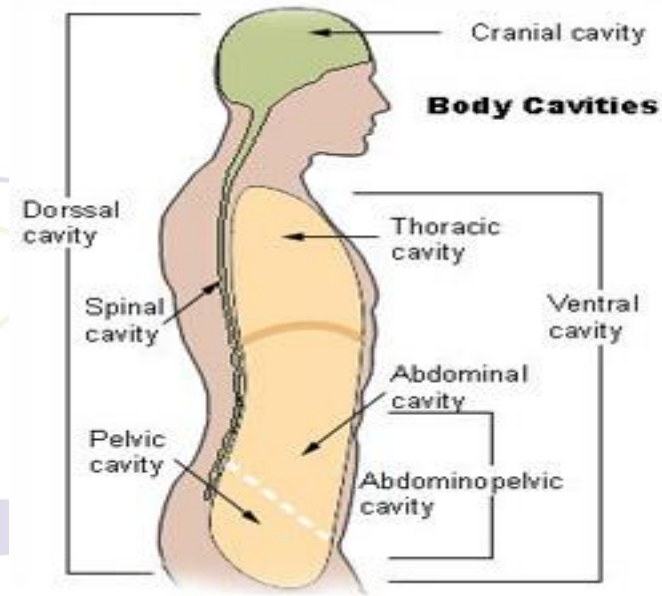
Sagittal Plane (Lateral Plane) - A vertical plane running from front to back; divides the body or any of its parts into right and left sides.

Axial Plane (Transverse Plane) - A horizontal plane; divides the body or any of its parts into upper and lower parts.

Median plane - Sagittal plane through the midline of the body; divides the body or any of its parts into right and left halves.

Body Cavities

The cavities, or spaces, of the body contain the internal organs, or viscera. The two main cavities are called the ventral and dorsal cavities. The ventral is the larger cavity and is subdivided into two parts (thoracic and abdominopelvic cavities) by the diaphragm, a dome-shaped respiratory muscle.



Thoracic cavity

The upper ventral, thoracic, or chest cavity contains the heart, lungs, trachea, esophagus, large blood vessels, and nerves. The thoracic cavity is bound laterally by the ribs (covered by costal pleura) and the diaphragm caudally (covered by diaphragmatic pleura).

Abdominal and pelvic cavity

The lower part of the ventral (abdominopelvic) cavity can be further divided into two portions: abdominal portion and pelvic portion. The abdominal cavity contains most of the gastrointestinal tract as well as the kidneys and adrenal glands. The abdominal cavity is bound cranially by the diaphragm, laterally by the body wall, and caudally by the pelvic cavity. The pelvic cavity contains most of the urogenital system as well as the rectum. The pelvic cavity is bounded cranially by the abdominal cavity, dorsally by the sacrum, and laterally by the pelvis.

Dorsal cavity

The smaller of the two main cavities is called the dorsal cavity. As its name implies, it contains organs lying more posterior in the body. The dorsal cavity, again, can be divided into two portions. The upper portion, or the cranial cavity, houses the brain, and the lower portion, or vertebral canal houses the spinal cord.

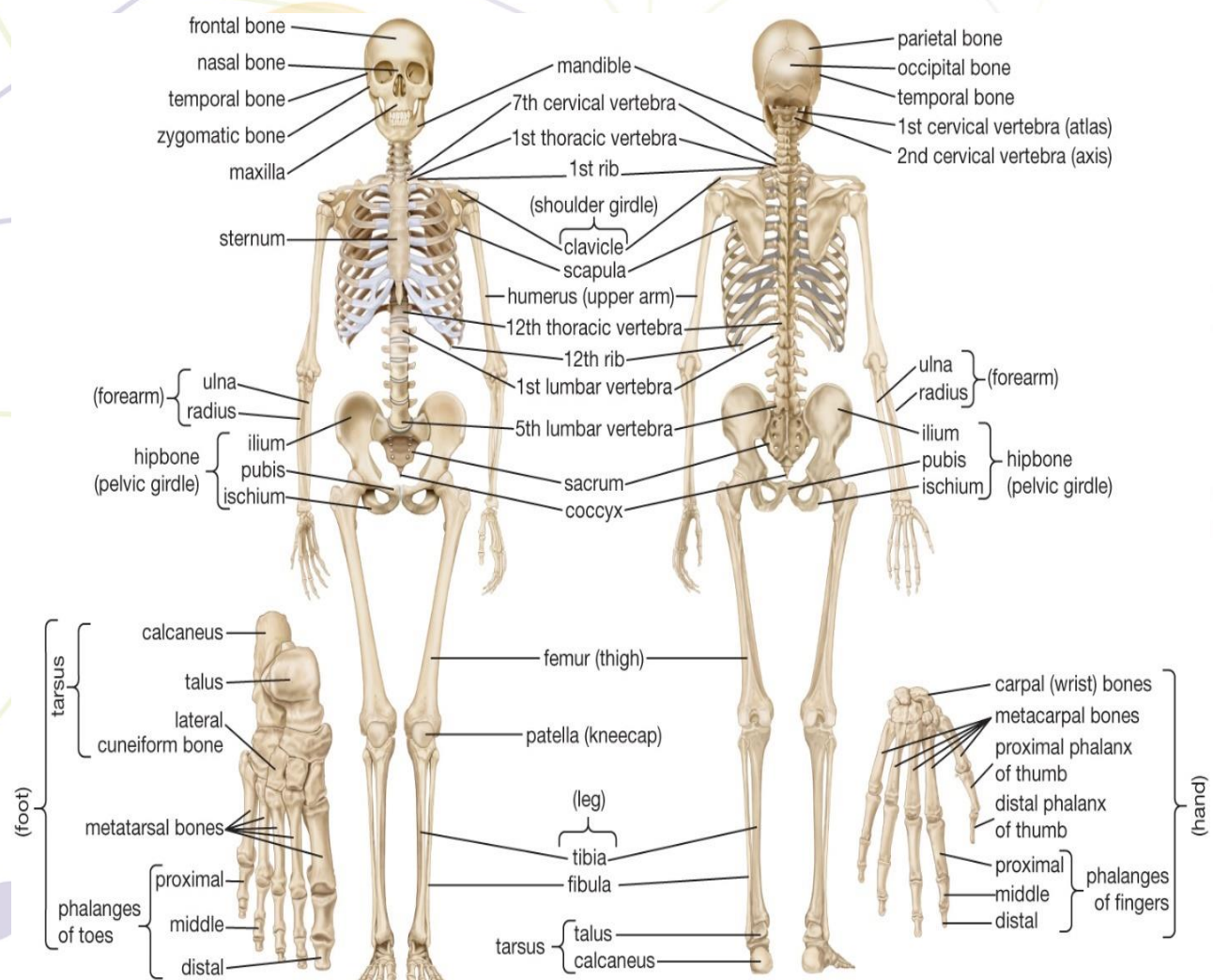
What is the skeletal system?

The human skeletal system consists of all of the bones, cartilage, tendons, and ligaments in the body. Altogether, the skeleton makes up about 20 percent of a person's body weight.

An adult's skeleton contains 206 bones. Children's skeletons actually contain more bones because some of them, including those of the skull, fuse together as they grow up.

There are also some differences in the male and female skeleton. The male skeleton is usually longer and has a high bone mass. The female skeleton, on the other hand, has a broader pelvis to accommodate for pregnancy and child birth.

Regardless of age or sex, the skeletal system can be broken down into two parts, known as the axial skeleton and the appendicular skeleton.



Axial skeleton anatomy

The adult axial skeleton consists of 80 bones. It's made up of the bones that form the vertical axis of the body, such as the bones of the head, neck, chest, and spine.

Skull bones: The adult skull comprises 22 bones. These bones can be further classified by location:

- **Cranial bones.** The eight cranial bones form the bulk of your skull. They help to protect your brain.
- **Facial bones.** There are 14 facial bones. They're found on the front of the skull and make up the face.

Auditory ossicles: The auditory ossicles are six small bones found within the inner ear canal in the skull. There are three auditory ossicles on each side of the head, known as the:

- malleus (hammer)
- incus (anvil)
- stapes (stirrup)

They work together to transmit sound waves from the surrounding environment to the structures of the inner ear.

Hyoid: The hyoid is a U-shaped bone found at the base of the jaw. It serves as a point of attachment for muscles and ligaments in the neck.

Vertebral column: The vertebral column is made up 26 bones. The first 24 are all vertebrae, followed by the sacrum and coccyx (tailbone).

The 24 vertebrae can be further divided into the:

- **Cervical vertebrae.** These seven bones are found in the head and neck.
- **Thoracic vertebrae.** These 12 bones are found in the upper back.
- **Lumbar vertebrae.** These five bones are found in the lower back.

The sacrum and coccyx are both made up of several fused vertebrae. They help support the weight of the body while sitting. They also serve as attachment points for various ligaments.

Thoracic cage: The thoracic cage is made up of the sternum (breastbone) and 12 pairs of ribs. These bones form a protective cage around the organs of the upper torso, including the heart and lungs.

Some of the ribs attach directly to the sternum, while others are linked to the sternum via cartilage. Some have no attachment point and are referred to as "floating ribs."

What is bone?

Bone is living tissue that makes up the body's skeleton. There are 3 types of bone tissue, including the following:

Anatomy of a bone, showing the subchondral tissue, marrow, cancellous tissue, and compact tissue.

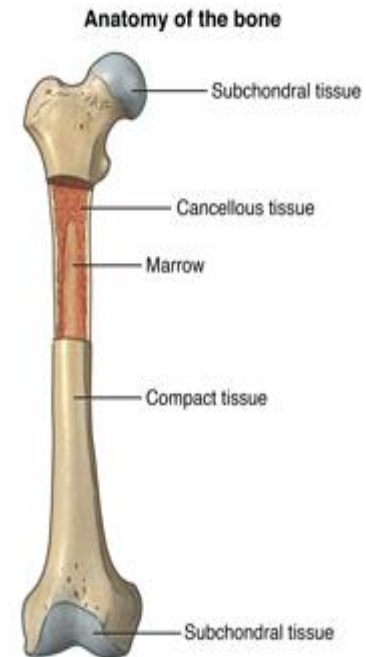
Compact tissue. The harder, outer tissue of bones.

Cancellous tissue. The sponge-like tissue inside bones.

Subchondral tissue. The smooth tissue at the ends of bones, which is covered with another type of tissue called cartilage. Cartilage is the specialized, gristly connective tissue that is present in adults. It is also the tissue from which most bones develop in children.

The tough, thin outer membrane covering the bones is called the periosteum. Beneath the hard outer shell of the periosteum are tunnels and canals through which blood and lymphatic vessels run to carry nourishment for the bone. Muscles, ligaments, and tendons may attach to the periosteum.

Bones are classified by their shape—as long, short, flat, and irregular. Primarily, they are referred to as long or short.



What are the functions of bone?

Bone provides shape and support for the body, as well as protection for some organs. Bone also serves as a storage site for minerals and provides the medium—marrow—for the development and storage of blood cells.

What are the different types of bone cells?

The different types of bone cells include the following:

Osteoblast. Found within the bone, its function is to form new bone tissue.

Osteoclast. A very large cell formed in bone marrow, its function is to absorb and remove unwanted tissue.

Osteocyte. Found within the bone, its function is to help maintain bone as living tissue.

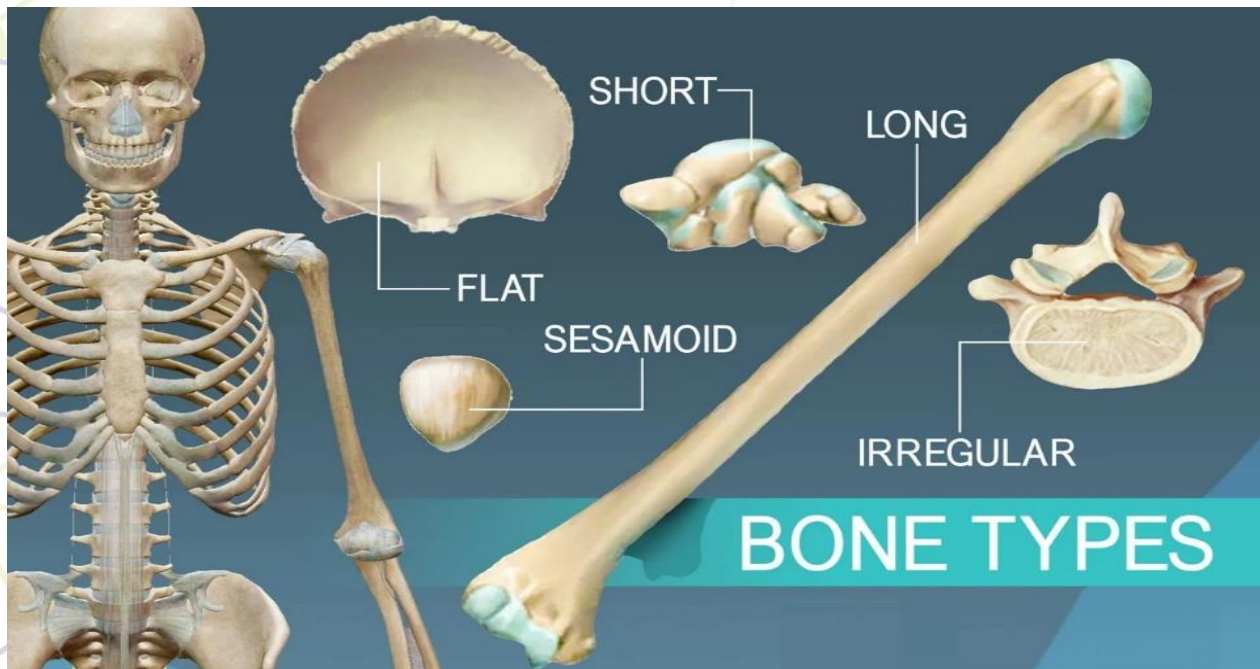
Hematopoietic. Found in bone marrow, its function is to produce red blood cells, white blood cells, and platelets.

Fat cells are also found within the bone marrow.

Because of the complexities of a bone's function, from providing strength and support for the body, to serving as a site for development and storage of blood cells.



bone types:



The human skeleton has a number of functions, such as protection and supporting weight. Different types of bones have differing shapes related to their particular function.

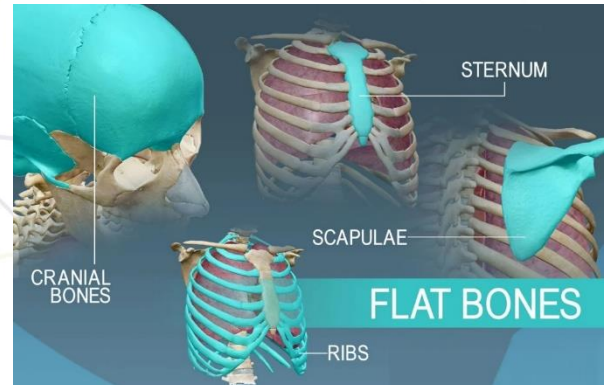
So, what are the different types of bones? How are they categorized?

There are five types of bones in the skeleton: flat, long, short, irregular, and sesamoid.

Let's go through each type and see examples.

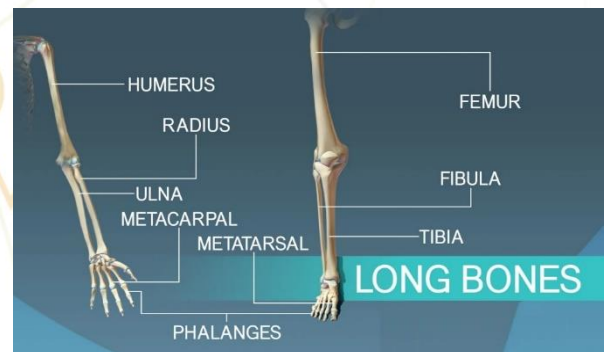
1. Flat Bones Protect Internal Organs:

There are **flat bones** in the skull (occipital, parietal, frontal, nasal, lacrimal, and vomer), the thoracic cage (sternum and ribs), and the pelvis (ilium, ischium, and pubis). The function of flat bones is to protect internal organs such as the brain, heart, and pelvic organs. Flat bones are somewhat flattened, and can provide protection, like a shield; flat bones can also provide large areas of attachment for muscles.



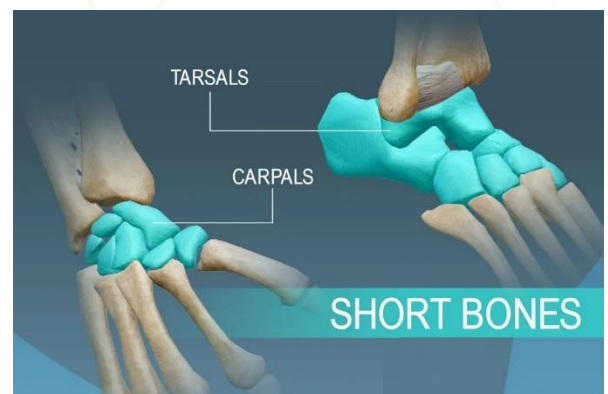
2. Long Bones Support Weight and Facilitate Movement

The **long bones**, longer than they are wide, include the femur (the longest bone in the body) as well as relatively small bones in the fingers. Long bones function to support the weight of the body and facilitate movement. Long bones are mostly located in the appendicular skeleton and include bones in the lower limbs (the tibia, fibula, femur, metatarsals, and phalanges) and bones in the upper limbs (the humerus, radius, ulna, metacarpals, and phalanges).



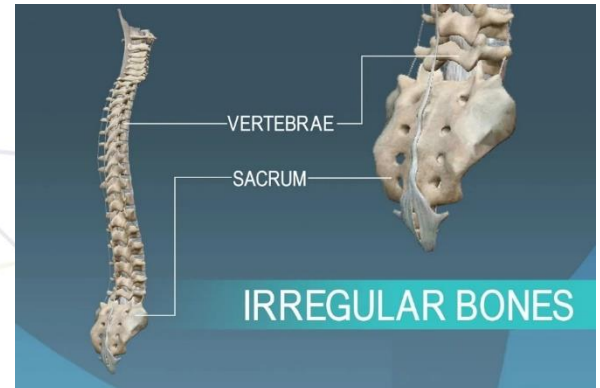
3. Short Bones Are Cube-shaped

Short bones are about as long as they are wide. Located in the wrist and ankle joints, short bones provide stability and some movement. The carpals in the wrist (scaphoid, lunate, triquetral, hamate, pisiform, capitate, trapezoid, and trapezium) and the tarsals in the ankles (calcaneus, talus, navicular, cuboid, lateral cuneiform, intermediate cuneiform, and medial cuneiform) are examples of short bones.



4. Irregular Bones Have Complex Shapes

Irregular bones vary in shape and structure and therefore do not fit into any other category (flat, short, long, or sesamoid). They often have a fairly complex shape, which helps protect internal organs. For example, the vertebrae, irregular bones of the vertebral column, protect the spinal cord. The irregular bones of the pelvis (pubis, ilium, and ischium) protect organs in the pelvic cavity.

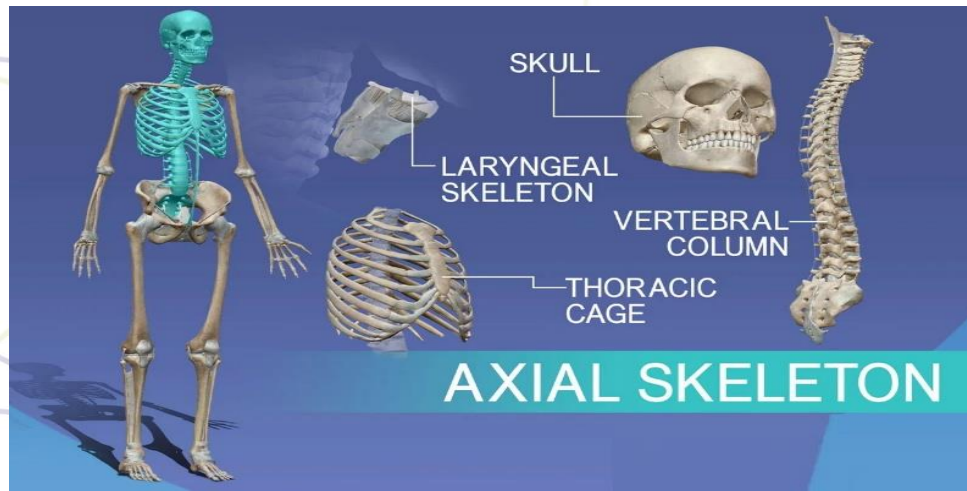


5. Sesamoid Bones Reinforce Tendons

Sesamoid bones are bones embedded in tendons. These small, round bones are commonly found in the tendons of the hands, knees, and feet. Sesamoid bones function to protect tendons from stress and wear. The patella, commonly referred to as the kneecap, is an example of a sesamoid bone.



Along the Axis of the Axial Skeleton: Bones That Form the Axial Skeleton

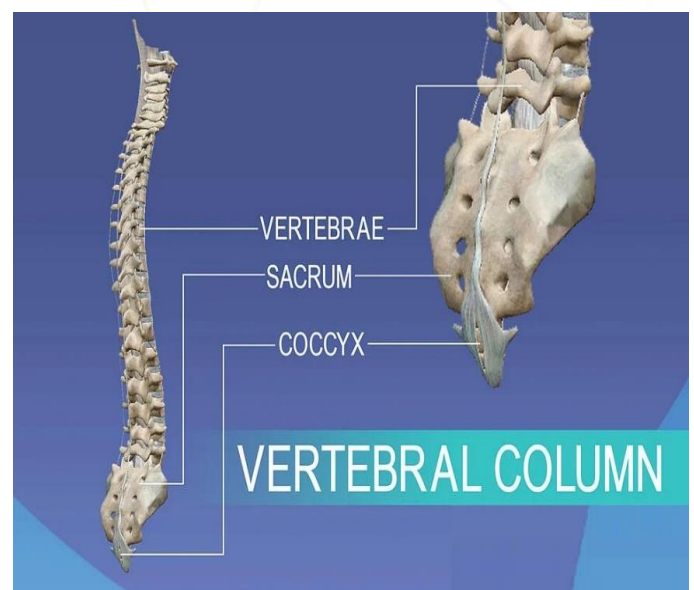


The bones of the human skeleton are divided into two groups. The **appendicular skeleton** includes all the bones that form the upper and lower limbs, and the shoulder and pelvic girdles. The **axial skeleton** includes all the bones along the body's long axis. Let's work our way down this axis to learn about these structures and the bones that form them.

The **axial skeleton** includes the bones that form the skull, laryngeal skeleton, vertebral column, and thoracic cage. The bones of the appendicular skeleton (the limbs and girdles) "append" to the axial skeleton.

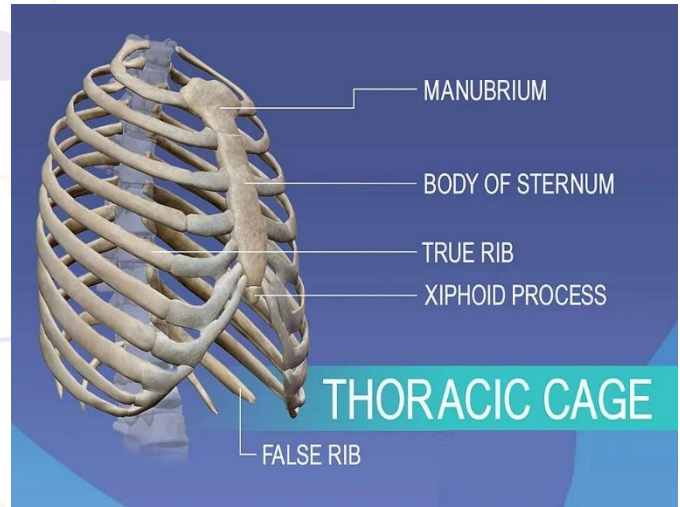
The Bones of the Vertebral Column: The Vertebrae, Sacrum, and Coccyx

The **vertebral column** is a flexible column formed by a series of 24 vertebrae, plus the **sacrum** and **coccyx**. Commonly referred to as the spine, the vertebral column extends from the base of the skull to the pelvis. The spinal cord passes from the foramen magnum of the skull through the vertebral canal within the vertebral column. The vertebral column is grouped into five regions: the cervical spine (C01-C07), the thoracic spine (T01-T-12), the lumbar spine (L01-L05), the sacral spine, and the coccygeal spine.



The Bones of the Thoracic Cage Protect Internal Organs

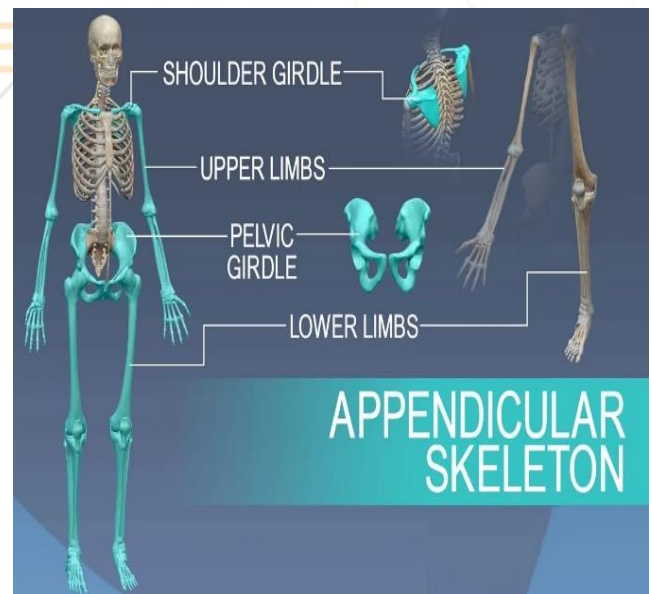
The **thoracic cage**, formed by the ribs and sternum, protects internal organs and gives attachment to muscles involved in respiration and upper limb movement. The **sternum** consists of the manubrium, body of the sternum, and xiphoid process. Ribs 1-7 are called true ribs because they articulate directly to the sternum, and ribs 8-12 are known as false ribs.



Bones of the Appendicular Skeleton

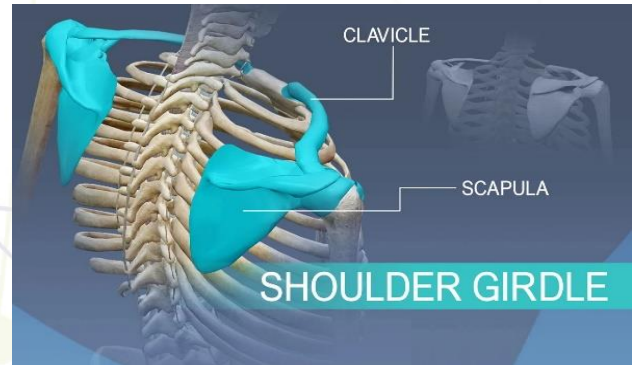
The bones of the human skeleton are divided into two groups. The axial skeleton includes all the bones (that form bony structures) along the body's long axis. The bones of the appendicular skeleton make up the rest of the skeleton, and are so called because they are appendages of the axial skeleton. The appendicular skeleton includes the bones of the shoulder girdle, the upper limbs, the pelvic girdle, and the lower limbs.

Let's take a look at the bones of the appendicular skeleton.



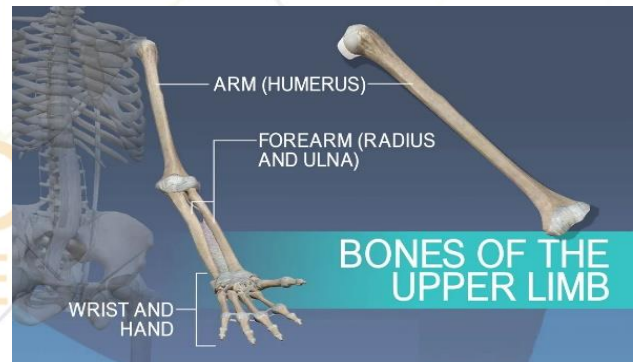
1. The Bones of the Shoulder Girdle

The **pectoral** or **shoulder girdle** consists of the scapulae and clavicles. The shoulder girdle connects the bones of the upper limbs to the axial skeleton. These bones also provide attachment for muscles that move the shoulders and upper limbs.



2. Bones of the Upper Limbs

The **upper limbs** include the bones of the arm (humerus), forearm (radius and ulna), wrist, and hand. The only bone of the arm is the humerus, which articulates with the forearm bones—the radius and ulna—at the elbow joint. The ulna is the larger of the two forearm bones.



Wrist Bones.

The wrist, or carpus, consists of eight carpal bones. One mnemonic to remember the carpal bones is the sentence: **Some Lovers Try Positions That They Can't Handle**. The eight carpal bones of the wrist are the Scaphoid, Lunate, Triquetral, Pisiform, Trapezoid, Trapezium, Capitate, Hamate.



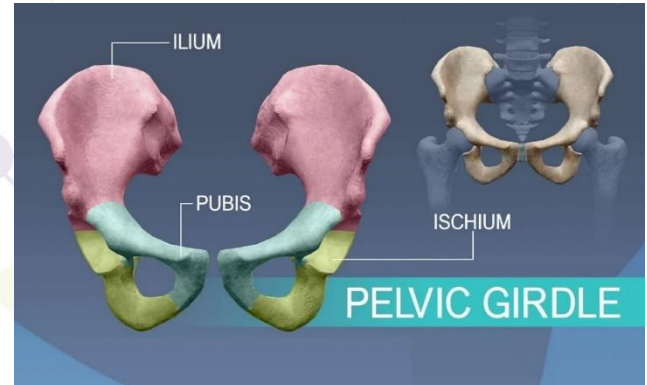
Hand Bones.

The hand includes 8 bones in the wrist, 5 bones that form the palm, and 14 bones that form the fingers and thumb. The wrist bones are called **carpals**. The bones that form the palm of the hand are called **metacarpals**. The **phalanges** are the bones of the fingers.



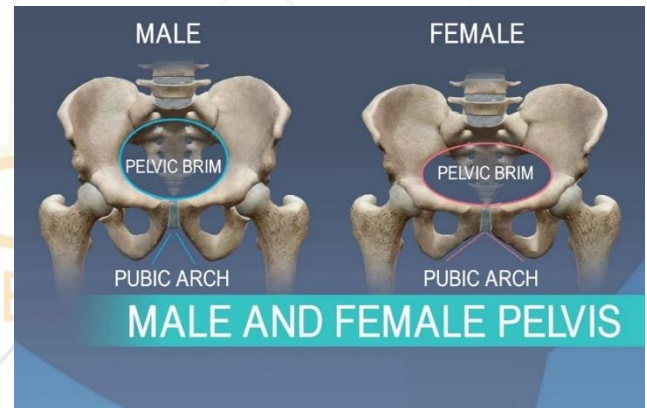
3. The Bones of the Pelvis

The **pelvic girdle** is a ring of bones attached to the vertebral column that connects the bones of the lower limbs to the axial skeleton. The pelvic girdle consists of the right and left hip bones. Each hip bone is a large, flattened, and irregularly shaped fusion of three bones: the ilium, ischium, and pubis.



Female and Male Pelvis. The female and male pelvises differ in several ways due to childbearing adaptations in the female.

- The female pelvic brim is larger and wider than the male's.
- The angle of the pubic arch is greater in the female pelvis (over 90 degrees) than in the male pelvis (less than 90 degrees).
- The male pelvis is deeper and has a narrower pelvic outlet than the female's.

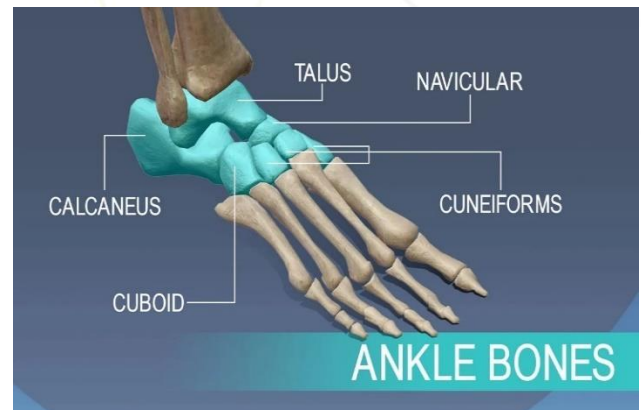
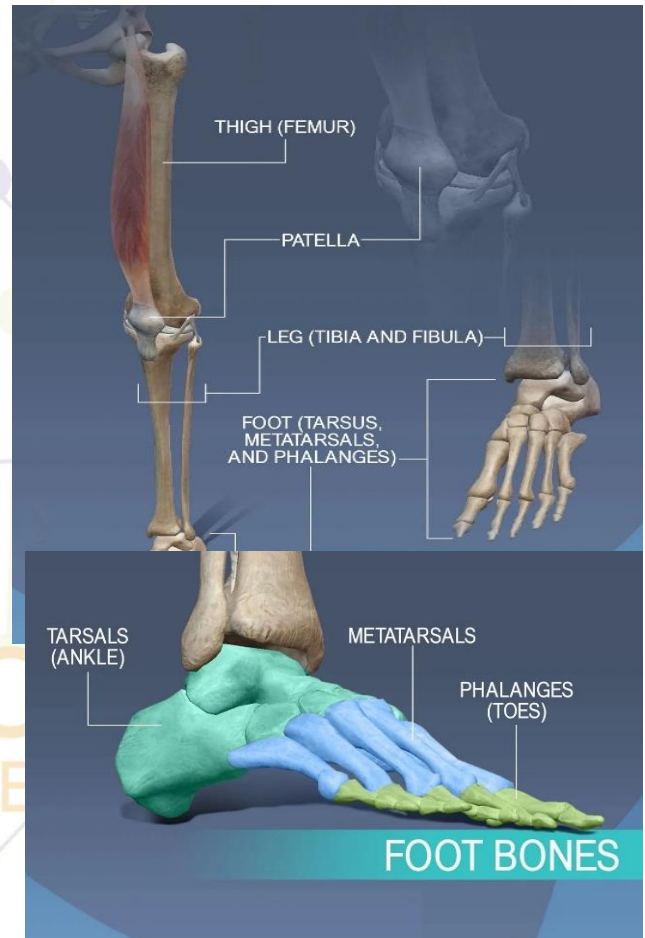


4. The Bones of the Lower Limbs

The **lower limbs** include the bones of the thigh, leg, and foot. The femur is the only bone of the thigh. It articulates with the two bones of the leg—the larger tibia (commonly known as the shin) and smaller fibula. The thigh and leg bones articulate at the knee joint that is protected and enhanced by the patella bone that supports the quadriceps tendon. The bones of the foot include the tarsus, metatarsus, and phalanges.

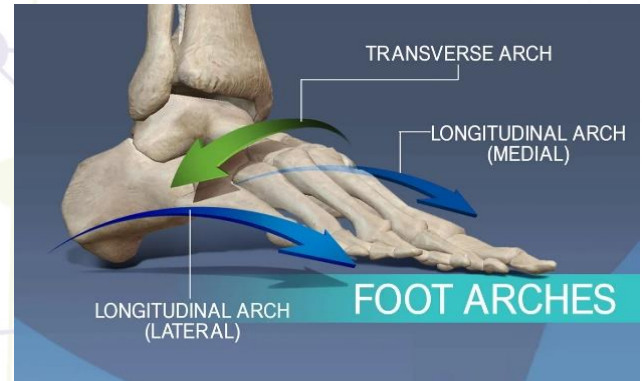
Foot Bones. The bones of the foot consist of the tarsal bones of the ankle, the phalanges that form the toes, and the metatarsals that give the foot its arch. As in the hand, the foot has five metatarsals, five proximal phalanges, five distal phalanges, but only four middle phalanges (as the foot's "big toe" has only two phalanges).

Ankle Bones. The ankle, or tarsus, consists of seven tarsal bones: the calcaneus, talus, cuboid, navicular, and three cuneiforms.

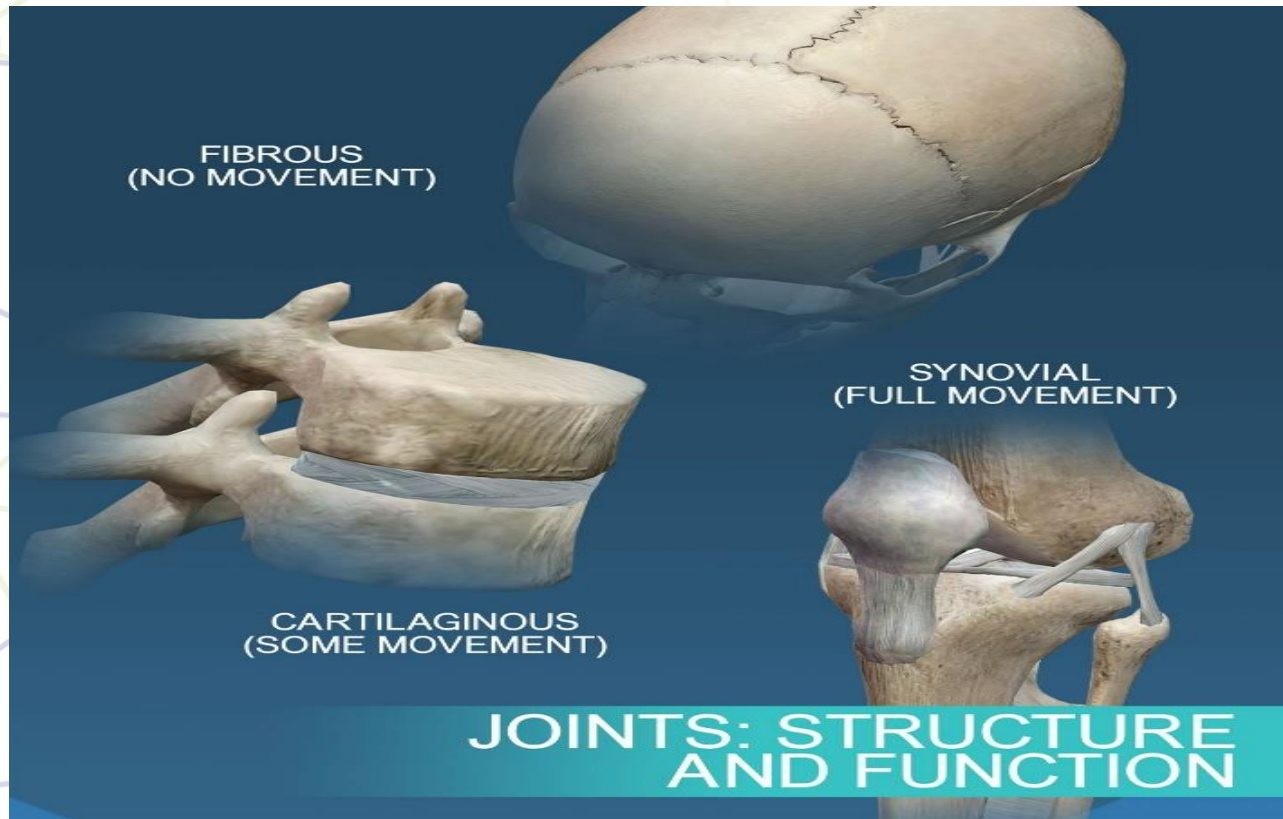


Foot Arches. The arches of the foot are formed by the interlocking bones and ligaments of the foot. They serve as shock-absorbing structures that support body weight and distribute stress evenly during walking.

- The longitudinal arch of the foot runs from the calcaneus to the heads of the metatarsals, and has medial and lateral parts.
- The transverse arch of the foot runs across the cuneiforms and the base of the metatarsal bones.



Joints and ligaments



Joints hold the skeleton together and support movement. There are two ways to categorize joints. The first is by **joint function**, also referred to as **range of motion**. The second way to categorize joints is by the material that holds the bones of the joints together; that is an organization of **joints by structure**.

Joints in the human skeleton can be grouped by function (range of motion) and by structure (material). Here are some joints and their categorizations.

Joint	Range of Motion and Material
Skull Sutures	Immovable fibrous joints
Knee	Full movement synovial capsule hinge joint
Vertebrae	Some movement cartilaginous joint

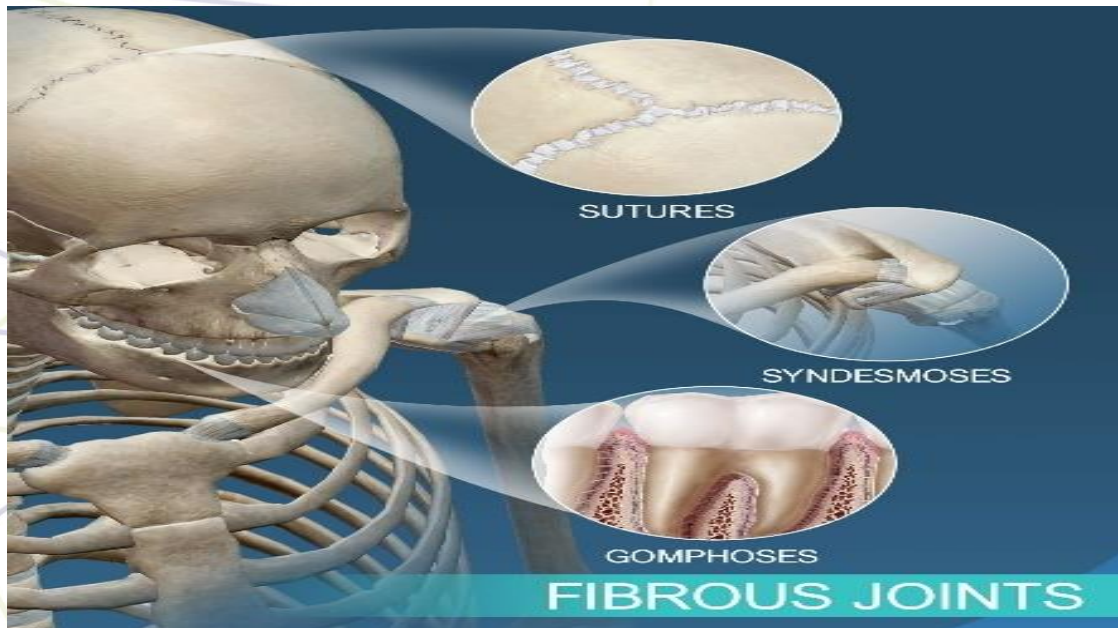
1. Joints Can Be Grouped By Their Function into Three Ranges of Motion



Immovable joints (called **synarthroses**) include skull sutures, the articulations between the teeth and the mandible, and the joint found between the first pair of ribs and the sternum. Examples of joints allowing slight movement (called **amphiarthroses**) include the distal joint between the tibia and the fibula and the pubic symphysis of the pelvic girdle. Joints allowing full movement (called **diarthroses**) include many bone articulations in the upper and lower limbs. Examples of these include the elbow, shoulder, and ankle.

Type of Joint Function	Examples
Synarthrosis (range of joint motion: no movement)	Skull Sutures, articulations of bony sockets and teeth in facial skeleton
Amphiarthrosis (range of joint motion: little Movement)	distal joint between the tibia and the fibula and the pubic symphysis
Diarthrosis (range of joint motion full movement)	Elbow, shoulder, ankle

2. Joints Can Be Grouped By Their Structure into Fibrous, Cartilaginous, and Synovial Joints



Fibrous Joints. Between the articulations of **fibrous joints** is thick connective tissue, which is why most (but not all) fibrous joints are immovable (synarthroses). There are three types of fibrous joints:

- (1) **Sutures** are nonmoving joints that connect bones of the skull. These joints have serrated edges that lock together with fibers of connective tissue.
- (2) The fibrous articulations between the teeth and the mandible or maxilla are called **gomphoses** and are also immovable.
- (3) A **syndesmosis** is a joint in which a ligament connects two bones, allowing for a little movement (amphiarthroses). The distal joint between the tibia and fibula is an example of a syndesmosis.

Cartiliginous Joints.

Joints that unite bones with cartilage are called **cartilaginous joints**. There are two types of cartilaginous joints:

- (1) A **symphondosis** is an immovable cartilaginous joint. One example is the joint between the first pair of ribs and the sternum.
- (2) A **symphysis** consists of a compressable fibrocartilaginous pad that connects two bones. This type of joint allows for some movement. The hip bones, connected by the pubic symphysis, and the vertebrae, connected by intervertebral discs, are two examples of symphyses.



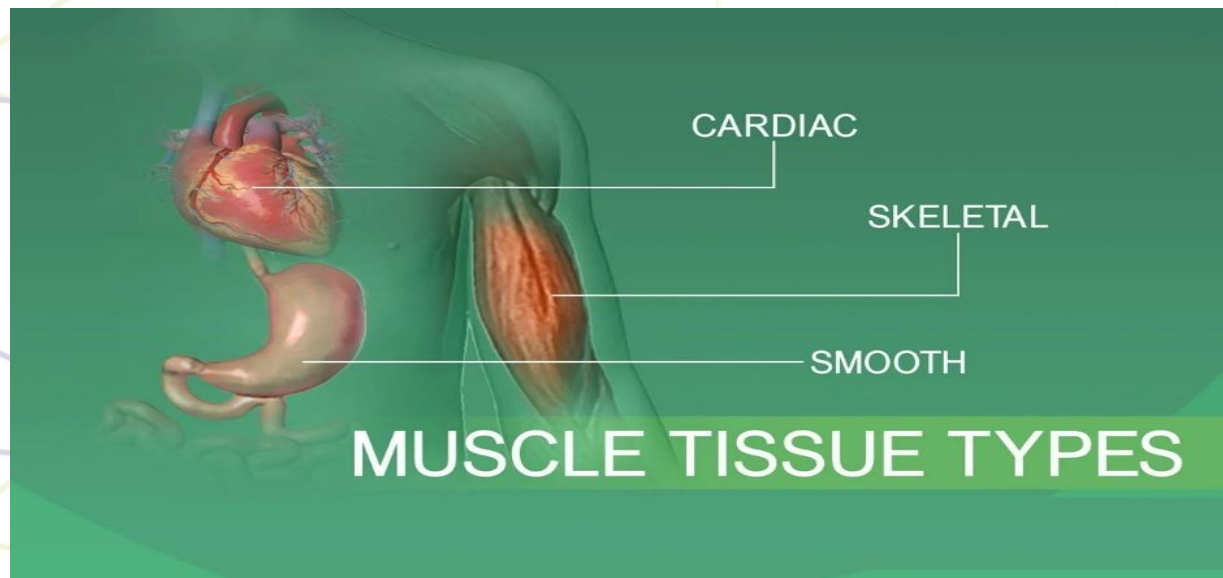
Synovial Joints. Synovial joints:

are characterized by the presence of an articular capsule between the two joined bones. Bone surfaces at synovial joints are protected by a coating of articular cartilage. Synovial joints are often supported and reinforced by surrounding ligaments, which limit movement to prevent injury. There are six types of synovial joints:

- (1) **Gliding** joints move against each other on a single plane. Major gliding joints include the intervertebral joints and the bones of the wrists and ankles.
- (2) **Hinge** joints move on just one axis. These joints allow for flexion and extension. Major hinge joints include the elbow and finger joints.
- (3) A **pivot** joint provides rotation. At the top of the spine, the atlas and axis form a pivot joint that allows for rotation of the head.
- (4) A **condyloid** joint allows for circular motion, flexion, and extension. The wrist joint between the radius and the carpal bones is an example of a condyloid joint.
- (5) A **saddle** joint allows for flexion, extension, and other movements, but no rotation. In the hand, the thumb's saddle joint (between the first metacarpal and the trapezium) lets the thumb cross over the palm, making it opposable.
- (6) The **ball-and-socket** joint is a freely moving joint that can rotate on any axis. The hip and shoulder joints are examples of ball and socket joints.



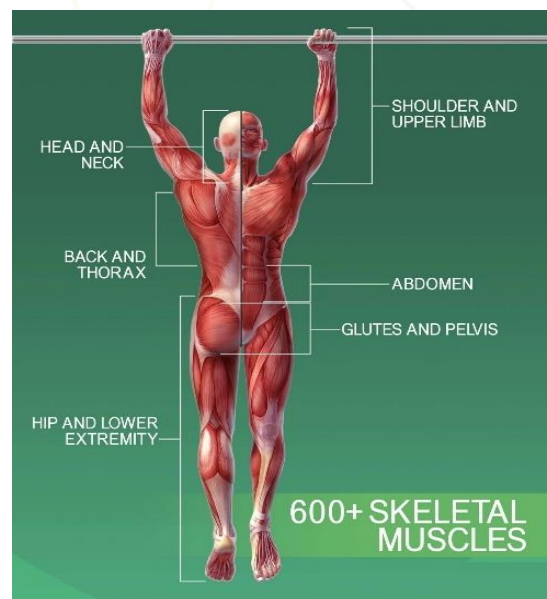
Muscular System and How Muscles Move



At the simplest level, muscles allow us to move. Smooth muscle and cardiac muscle move to facilitate body functions like heartbeats and digestion. The movement of these muscles is directed by the autonomic part of the nervous system—those are the nerves that control organs. Skeletal muscles move our bodies in space. They take direct instruction from the specific nerves that innervate each muscle. Want to learn more about the muscles in the human body? Here are five other facts to keep in mind about the muscular system.

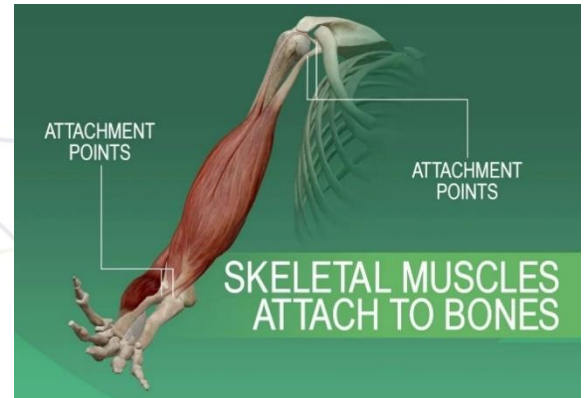
1. Over 600 Skeletal Muscles Account for about Half of Our Body Weight

The muscles that move the human skeleton vary greatly in shape and size and extend to every part of our bodies. The muscular system contains over 600 skeletal muscles alone, which make up about 40% of our mass. Blood vessels and nerves run to every muscle, helping control and regulate each muscle's function.



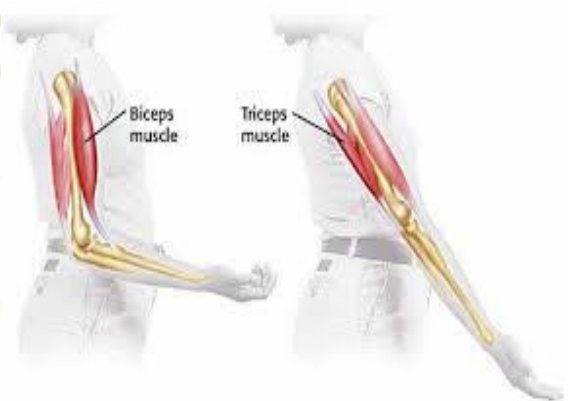
2. Skeletal Muscles Attach to Bones

In the muscular system, skeletal muscles are connected to the skeleton, either to bone or to connective tissues such as ligaments. Muscles are always attached at two or more places. When the muscle contracts, the attachment points are pulled closer together; when it relaxes, the attachment points move apart.



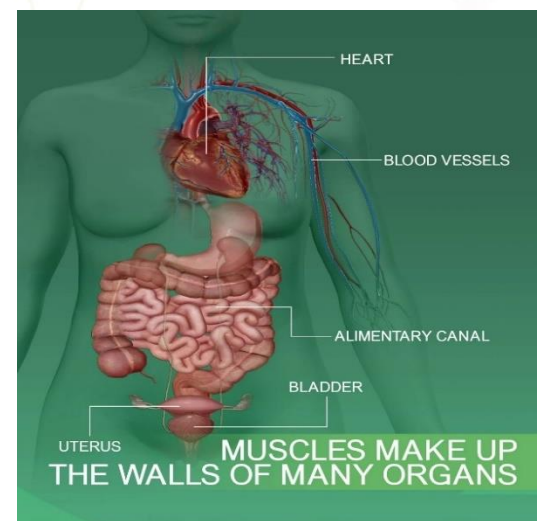
Muscles Pull on Bones to Move the Body

Muscles contract and relax to move bones. The elbow joint bends (flexes) when muscles pull on the radius and ulna of the arm. Muscles contract when messages travel from nerves to muscles and trigger chemical reactions. These reactions change the internal structures of muscle fiber cells, a process that shortens the muscle. The muscle fibers relax when the nervous system signal is no longer present, thus reversing the shortening.



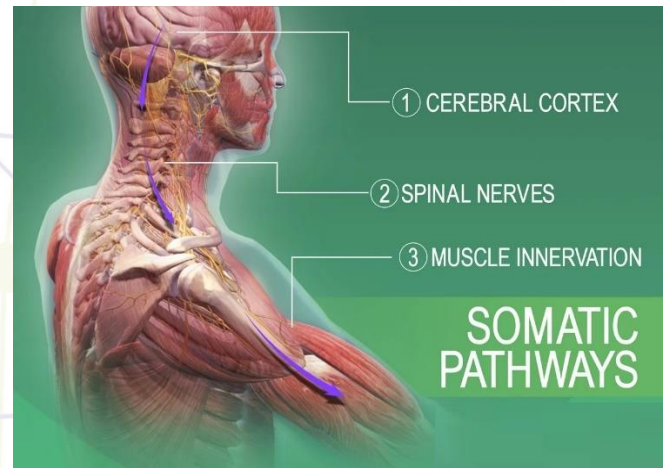
4. Muscles Make Up the Walls of Many Organs

Not all muscle tissue is skeletal muscle. Smooth muscle tissue is in the walls of many human body organs and helps those organs move to facilitate body functions. The alimentary canal (esophagus, stomach, and intestines) includes muscle tissue that contracts and relaxes to move nutrients through the digestion process. The urinary bladder also includes muscle tissue that contracts and relaxes to hold and release urine. Heartbeats are the result of the contraction and relaxation of muscle tissue in the heart wall. Smooth muscles in the walls of arteries help move blood through the body.



5. Somatic Motor Signals Move Skeletal Muscles

How do skeletal muscles move? It happens when the muscular system and the nervous system work together: Somatic signals are sent from the cerebral cortex to nerves associated with specific skeletal muscles. Most signals travel through spinal nerves that connect with nerves that innervate skeletal muscles throughout the body. Want to flex an elbow joint? Your cerebral cortex sends a signal through your spinal nerve to the nerves that innervate muscles around the elbow joint. When that signal reaches the muscle tissue its cells reorganize, causing a contraction that bends the elbow joint!

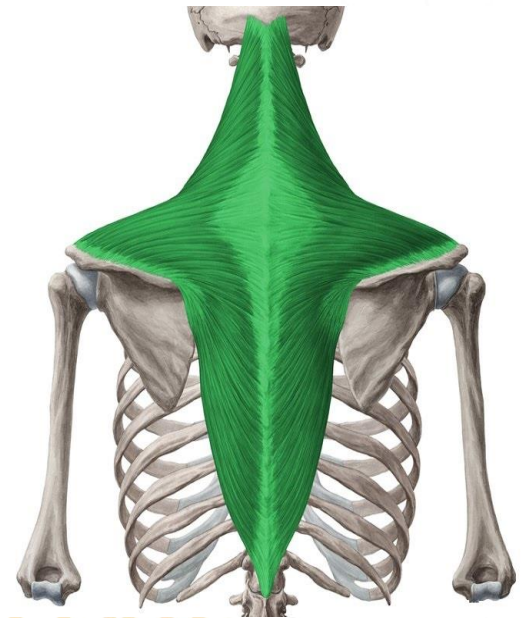


Muscles anatomy :

Trapezius:

The trapezius muscle is a large, triangular, paired muscle located on the posterior aspect of the neck and thorax. When viewed together, this pair forms a diamond or trapezoid shape, hence its name. The trapezius has many attachment points, extending from the skull and vertebral column to the shoulder girdle.

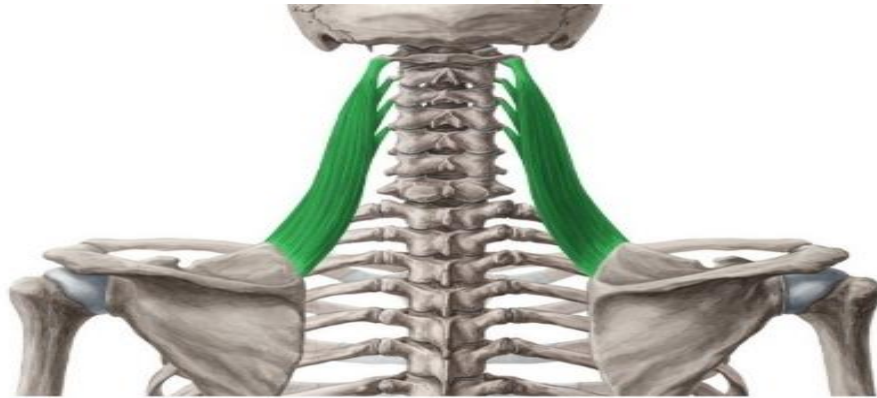
The trapezius belongs to the **superficial layer** of the **extrinsic muscles of the back**, along with latissimus dorsi, rhomboid major and minor, and levator scapulae muscles. The trapezius is largely involved in movements of the shoulder girdle, and is therefore functionally considered as a muscle of the upper limb rather than of the back.



In this article, we will discuss the detailed anatomy and function of the trapezius muscle.

Key facts about the trapezius muscle	
Origin	Descending part (superior fibers): medial third of the superior nuchal line, external occipital protuberance Transverse part (middle fibers): nuchal ligament attached to the spinous processes of C1-C6 vertebrae, spinous processes and supraspinous ligaments of vertebrae C7-T3 Ascending part (inferior fibers): spinous processes and supraspinous ligaments of vertebrae T4-T12
Insertion	Descending part (superior fibers): lateral third of clavicle Transverse part (middle fibers): medial acromial margin, superior crest of spine of scapula Ascending part (inferior fibers): lateral apex of the medial end of scapular spine
Innervation	Motor: accessory nerve (CN XI) Motor/Sensory: ventral rami of spinal nerves C3-C4 (via cervical plexus)
	Descending part (superior fibers) - Scapulothoracic joint: draws scapula superomedially - Atlantooccipital joint: extension of head and neck, lateral flexion of head and neck (ipsilateral) - Atlantoaxial joint: rotation of head (contralateral) Transverse part (middle fibers) - Scapulothoracic joint: draws scapula medially Ascending part (inferior fibers) - Scapulothoracic joint: Draws scapula inferomedially
Blood supply	Occipital artery (descending part), superficial or transverse cervical artery (transverse part), dorsal scapular artery (ascending part)

Levator Scapulae



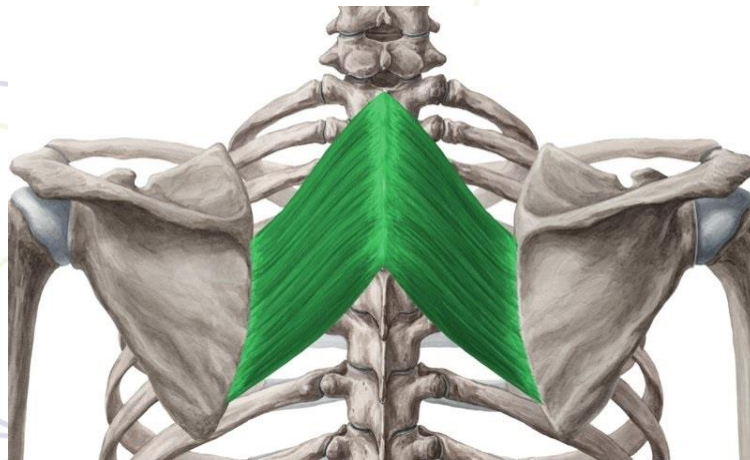
Levator scapulae is a long and slender muscle that anatomically belongs to the superficial layer of **extrinsic muscles of the back**. Functionally, however, it is considered to be a muscle of **scapular motion** along with the rhomboids, serratus anterior, serratus posterior superior and serratus posterior inferior muscles. The levator scapulae muscle extends from the transverse processes of vertebrae C1-C4 to the medial border of scapula.

As its name suggests, the main function of this muscle is to **elevate the scapula**. Additionally, levator scapulae acts with the trapezius, latissimus dorsi, rhomboids, pectoralis major and pectoralis minor muscles to inferiorly rotate the glenoid cavity, stabilize the spine, and extend and laterally flex the neck.

In this article, we will discuss the anatomy and function of the levator scapulae muscle.

Key facts about the levator scapulae muscle	
Origin	Transverse processes of vertebrae C1-C4
Insertion	Medial border of scapula (from superior angle to root of spine of scapula)
Action	Scapulothoracic joint: Draws scapula superomedially, rotates glenoid cavity inferiorly; Cervical joints: Lateral flexion of neck (ipsilateral), extension of the neck
Innervation	Anterior rami of the nerves C3 and C4, dorsal scapular nerve (branch of the C5)
Blood supply	Transverse cervical and ascending cervical arteries (branches of the thyrocervical trunk)

Rhomboids

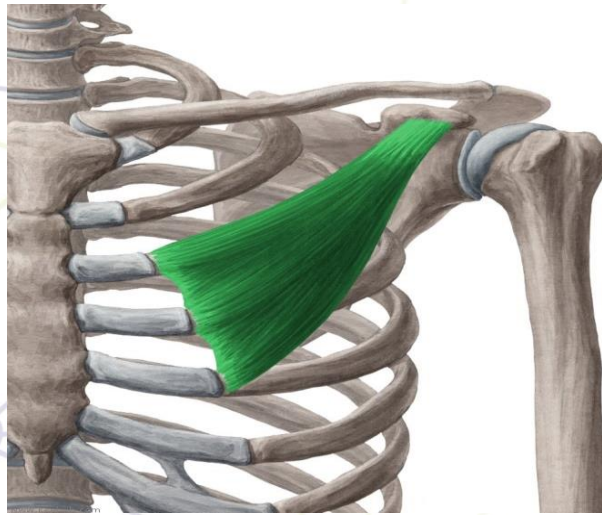


The rhomboids are two bilateral, superficial muscles located in the upper back. They consist of two functionally similar muscles called **rhomboid minor** and **rhomboid major**. Together with trapezius, levator scapulae and latissimus dorsi, they comprise the superficial layer of the extrinsic back muscles.

The rhomboid muscles extend between the nuchal ligament, spinous processes of thoracic vertebrae and scapula. Under innervation from the **dorsal scapular nerve**, the rhomboids primarily retract the scapula superomedially and rotate the **glenoid cavity**. Therefore, the rhomboids are important to stabilize the scapula into position and reinforce the shoulder.

Key facts about the rhomboid muscles	
Origin	Rhomboid minor: Nuchal ligament, Spinous processes of vertebrae C7-T1 Rhomboid major: Spinous process of vertebrae T2-T5
Insertion	Rhomboid minor: Root (medial end) of spine of scapula Rhomboid major: Medial border of scapula (from inferior angle to root of spine of scapula)
Action	Scapulothoracic joint: Draws scapula superomedially, Rotates glenoid cavity inferiorly; Supports position of scapula
Innervation	Dorsal scapular nerve (C4-C5)
Blood supply	Dorsal scapular artery, deep branch of transverse cervical artery, dorsal branch of upper five or six posterior intercostal arteries

Pectoralis Minor:



The pectoralis minor muscle (often called just pec minor) is one of the most **superficial** muscles on the anterior aspect of the chest or thoracic wall, located deep only to the **pectoralis major muscle**. It is one of the anterior axioappendicular (thoracoappendicular) muscles, together with the pectoralis major, subclavius and serratus anterior.

Together with other muscles of the region it produces various movements of the scapula and can be used as an accessory muscle of respiration.

This article will discuss the anatomy and function of the pectoralis minor muscle.

Key facts about the pectoralis minor muscle	
Origin	Anterior surface, costal cartilages of <u>ribs</u> 3-5
Insertion	Medial border and coracoid process of scapula
Innervation	Medial and lateral pectoral nerves (C5-T1)
Blood supply	Thoracoacromial a. (pectoral and deltoid branches), superior thoracic a., lateral thoracic a.
Function	Scapulothoracic joint: draws scapula anteroinferiorly, stabilizes scapula on thoracic wall

Serratus Anterior



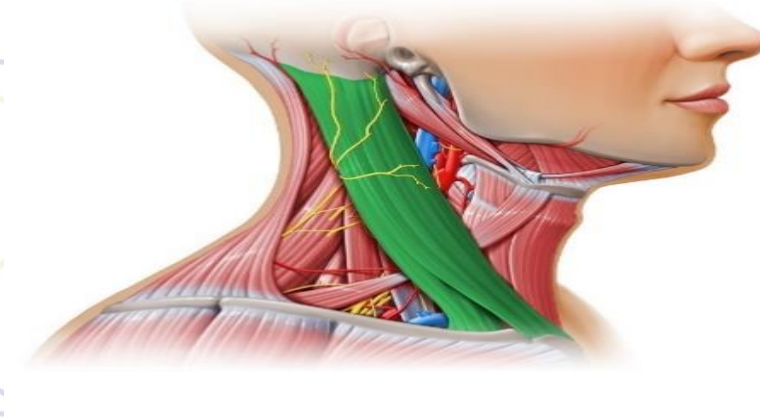
The **serratus anterior muscle** is a fan-shaped muscle at the lateral wall of the thorax. Its main part lies deep under the scapula and the pectoral muscles. It is easy to palpate between the pectoralis major and latissimus dorsi muscles. In athletic bodies the muscle may be even visible to the naked eye along the ribs underneath the axilla.

The serratus anterior muscle originates from the 1st to 10th rib and inserts at the anterior surface of the scapula. Due to its course it has a “serrated” or “saw-toothed” appearance.

This article will discuss the anatomy of the serratus anterior muscle.

Key facts about the serratus anterior muscle	
Origin	Superior part: Ribs 1-2, Intercostal fascia Middle part: Ribs-3-6 Inferior part: Ribs 7-8/9/10 (+ <u>external oblique muscle</u>)
Insertion	Superior part: Anterior and posterior surface of superior angle of scapula Middle part: Anterior surface of entire medial border of scapula Inferior part: Anterior and posterior surface of inferior angle of scapula
Innervation	Long thoracic nerve (C5- C7) <i>Mnemonics: 'SALT' (stands for serratus anterior - long thoracic) & 'C5, 6, 7 raise your arms to heaven!'</i>
Blood supply	Superior and lateral thoracic arteries, thoracodorsal artery branches
Function	<u>Scapulothoracic joint:</u> Draws scapula anterolaterally, Suspends scapula on thoracic wall, Rotates scapula (draws inferiorly angle laterally)

Sternocleidomastoid:

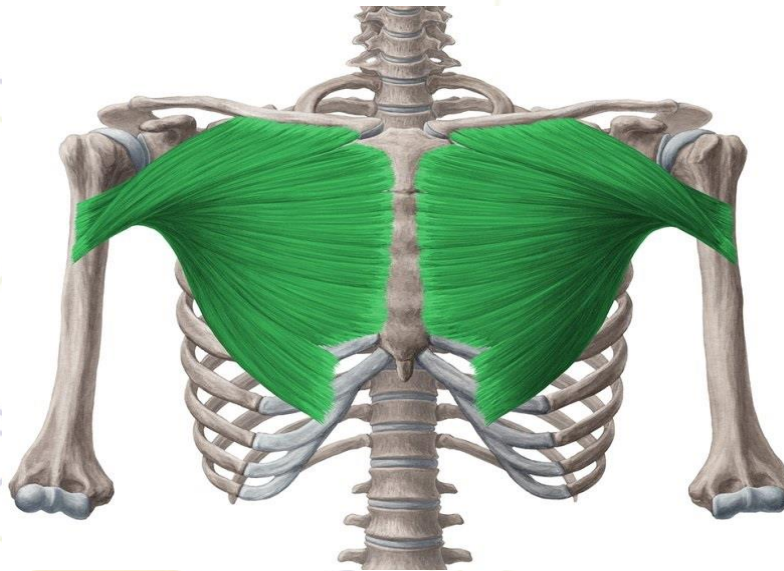


The **sternocleidomastoid muscle** is a two-headed neck muscle, which true to its name bears attachments to the manubrium of sternum (sterno-), the clavicle (-cleido-), and the mastoid process of the temporal bone (-mastoid).

It is a long, bilateral muscle of the neck, which functions to flex the neck both laterally and anteriorly, as well as rotate the head contralaterally to the side of contraction. The muscle is closely related to certain neurovascular structures that pass through the neck on their way either to the head or to the periphery of the body.

Key facts about the Sternocleidomastoid muscle	
Origins	Sternal head: superior part of anterior surface of manubrium sterni Clavicular head: superior surface of medial third of the clavicle
Insertions	Lateral surface of mastoid process of the temporal bone, Lateral half of superior nuchal line of the occipital bone
Innervation	Accessory nerve (CN XI), branches of cervical plexus (C2-C3)
Functions	Unilateral contraction: cervical spine: neck ipsilateral flexion, neck contralateral rotation Bilateral contraction: atlantooccipital joint/ superior cervical spine: head/neck extension; Inferior cervical vertebrae: neck flexion; <u>sternoclavicular joint</u> : elevation of clavicle and manubrium of sternum
Clinical relations	Wryneck, torticollis

Pectoralis major :

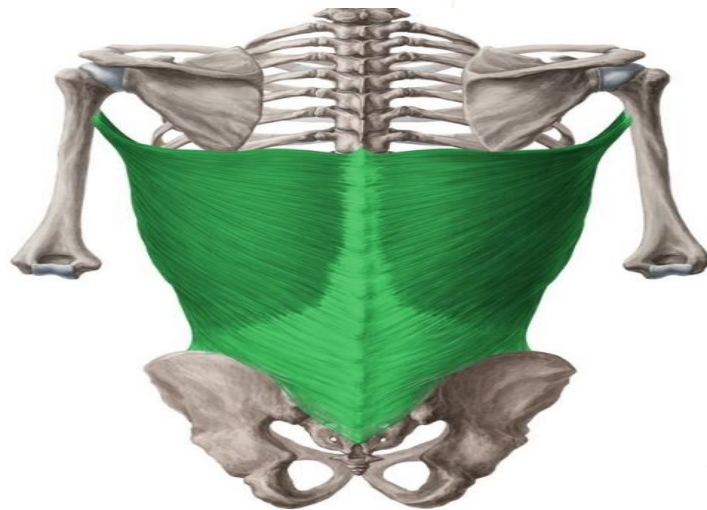


The **pectoralis major** is a paired, superficial muscle located on the anterior surface of the thoracic cage. If you're a gym lover, you'll hear these muscles also being referred to as the *pecs muscles*. The pectoralis major has a broad origin, based on which it is divided into three parts: clavicular part, sternocostal part and abdominal part. All three parts converge laterally and insert onto the greater tubercle of humerus.

The **main function** of this chest muscle as a whole is the adduction and internal rotation of the arm on the shoulder joint. Acting independently, the clavicular part helps to flex the extended arm up to 90°, while the sternocostal part facilitates the extension of the flexed arm by pulling it downwards.

Key facts about the pectoralis major muscle	
Origin	Clavicular part: anterior surface of medial half of clavicle Sternocostal part: anterior surface of sternum, Costal cartilages of ribs 1-6 Abdominal part: Anterior layer of rectus sheath
Insertion	Crest of greater tubercle of humerus
Action	Shoulder joint: Arm adduction, Arm internal rotation, Arm flexion (clavicular head), arm extension (sternocostal head); Scapulothoracic joint: Draws scapula anteroinferiorly
Innervation	Lateral and medial pectoral nerves (C5-T1)
Blood supply	Pectoral branches of thoracoacromial artery, perforating branches of internal thoracic artery

Latissimus dorsi:



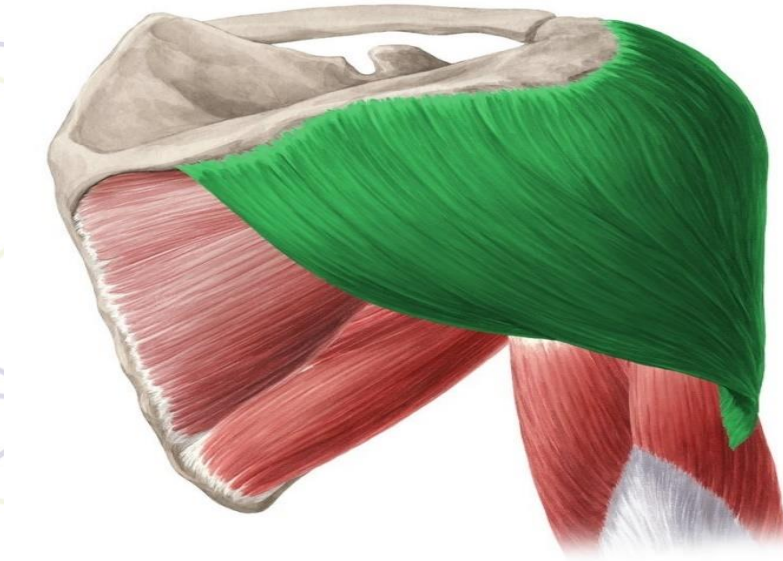
The **latissimus dorsi muscle** (AKA: 'the lats muscle' or 'the lats') is the widest muscle in the human body. It is relatively thin and covers almost all back muscles at the posterior trunk, except the trapezius.

Along with the levator scapulae, trapezius and rhomboid muscles, the latissimus dorsi belongs to the superficial layer of the extrinsic back muscles.

Functionally, the latissimus dorsi muscle belongs to the **muscles of the scapular motion**. This muscle is able to pull the inferior angle of the scapula in various directions, producing movements on the shoulder joint; internal rotation, adduction and extension of the arm. Moreover, it is an accessory respiratory muscle, as well as one of the main stabilizers of the spine during its various movements.

Key facts about the latissimus dorsi muscle	
Origin	Vertebral part: Spinous processes of vertebrae T7-T12, Thoracolumbar fascia Iliac part: Posterior third of crest of ilium Costal part: Ribs 9-12 Scapular part: Inferior angle of scapula
Insertion	Intertubercular sulcus of the humerus, between pectoralis major and teres major muscles <i>Mnemonic: Lady between two majors</i> (lady refers to latissimus dorsi)
Innervation	Thoracodorsal nerve (C6-C8)
Blood supply	Thoracodorsal artery, perforating arteries of the 9th-11th posterior intercostal arteries, and 1st-3rd lumbar arteries
Functions	Shoulder joint: Arm internal rotation, Arm adduction, Arm extension; Assists in respiration

Deltoid :

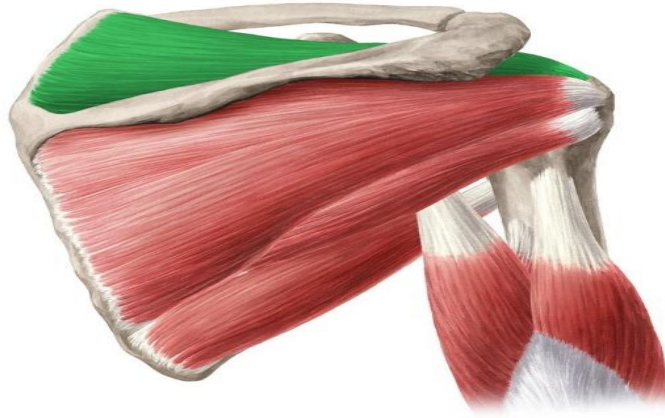


The deltoid is a thick, triangular **shoulder muscle**. It gets its name because of its similar shape to the Greek letter 'delta' (Δ). The muscle has a wide origin spanning the clavicle, acromion and spine of scapula. It passes inferiorly surrounding the glenohumeral joint on all sides and inserts onto the humerus.

The deltoid is formed of acromial, clavicular and scapular spinal parts. Acromial part (middle fibres) **abducts** the **arm**, while the clavicular and scapular spinal parts play a significant role in stabilization, ensuring a steady plane of abduction. Additionally, the clavicular part (anterior fibers) can act as a flexor and internal rotator of the arm, while the scapular spinal part (posterior fibers) can extend and externally rotate the arm.

Key facts about the deltoid muscle	
Origins	Lateral 1/3 of Clavicle (clavicular part), Acromion (acromial part), Spine of Scapula (spinal part) <i>Mnemonic: 'Deltoid helps you carry SACS'</i>
Insertion	Deltoid tuberosity of humerus
Innervation	Axillary nerve (C5, C6)
Blood supply	Deltoid and acromial branches of thoracoacromial artery, subscapular artery, anterior and posterior circumflex humeral arteries, deltoid branch of deep brachial artery
Function	<i>Clavicular part:</i> flexion and internal rotation of the arm, <i>Acromial part:</i> abduction of the arm beyond the initial 15° <i>Spinal part:</i> extension and external rotation of the arm.

Supraspinatus:



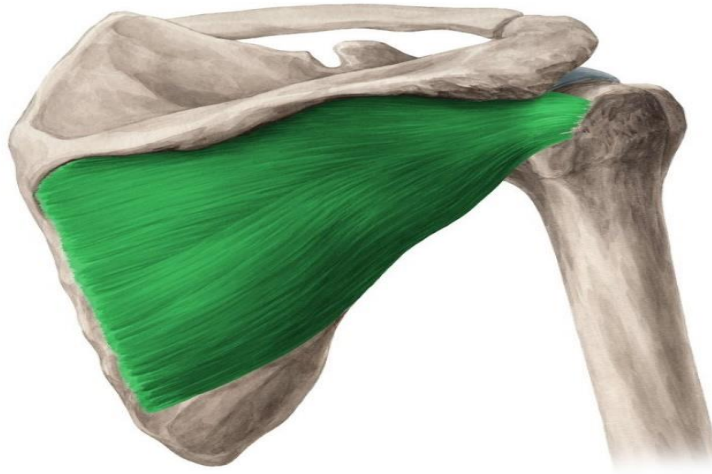
Supraspinatus is one of the rotator cuff muscles, along with infraspinatus, teres minor and subscapularis muscles. Supraspinatus is located deep to the trapezius muscle in the posterior scapular region, extending from the supraspinous fossa of scapula to the proximal humerus.

Together with the other rotator cuff muscles, supraspinatus stabilizes the glenohumeral (shoulder) joint during the movements of the upper limb. Additionally, it assists in the abduction of the arm.

This article will discuss the anatomy and function of supraspinatus muscle.

Key facts about supraspinatus muscle	
Origin	Supraspinous fossa of scapula
Insertion	Greater tuberosity of humerus
Action	Shoulder joint: abduction of arm, stabilization of the humeral head in the glenoid cavity
Innervation	Suprascapular nerve (C5, C6)
Blood supply	Suprascapular artery

Infraspinatus:



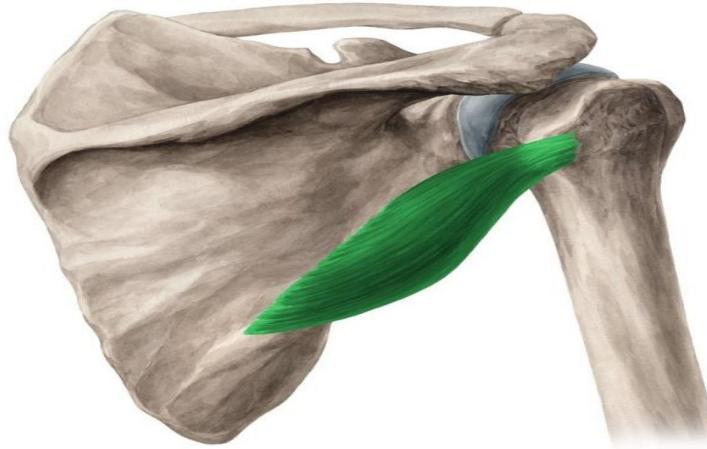
Infraspinatus is a thick triangular muscle that occupies the majority of the dorsal surface of the scapula. It arises mostly from the infraspinous fossa of scapula, and connects it to the proximal humerus.

Infraspinatus is one of the four **rotator cuff muscles**, alongside subscapularis, teres minor and supraspinatus muscles. As a part of the rotator cuff muscles, the main function of infraspinatus muscle is external rotation of the humerus, as well as stabilization of the glenohumeral, or shoulder joint.

This article will discuss the anatomy and functions of the infraspinatus muscle.

Key facts about the infraspinatus muscle	
Origin	Infraspinous fossa of scapula
Insertion	Greater tubercle of humerus
Action	Shoulder joint: Arm external rotation; Stabilizes humeral head in glenoid cavity
Innervation	Suprascapular nerve (C5, C6)
Blood supply	Suprascapular artery, circumflex scapular arteries

Teres minor:



Teres minor is a posterior muscle of the shoulder that extends between the scapula and the head of humerus. It is one of the four muscles of the **rotator cuff**, along with the supraspinatus, infraspinatus and subscapularis.

Rotator cuff muscles act together to control the movements of the humeral head and stabilize it within the glenoid cavity of the shoulder joint, as the arm moves in various directions. Every muscle within the group also has its own specific role, and that of teres minor is to **externally rotate** and **adduct** the **arm**.

This article will discuss the anatomy and function of teres minor muscle.

Key facts about the teres minor muscle	
Origin	Lateral border of scapula
Insertion	Greater tubercle of humerus
Action	Shoulder joint: Arm external rotation, arm adduction; Stabilizes humeral head in glenoid cavity
Innervation	Axillary nerve (C5, C6)
Blood supply	Suprascapular artery, dorsal scapular artery

Subscapularis:



Subscapularis is a triangular shoulder muscle located in the subscapular fossa of scapula. Attaching between the scapula and the proximal humerus, it is one of the four muscles of the **rotator cuff**, along with supraspinatus, infraspinatus and teres minor.

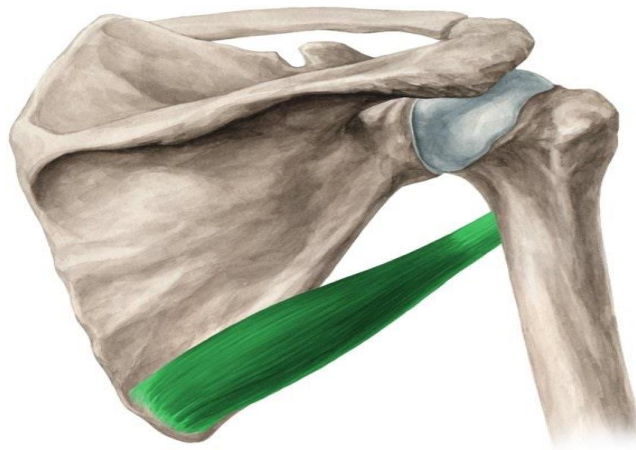
Rotator cuff muscles act together to stabilize and steer the humeral head within the glenoid cavity during various movements of the upper limb.

Each muscle of the rotator cuff also performs its own specific role, and the precise action of subscapularis is to **internally rotate the arm** on the shoulder joint.

This article will discuss the anatomy and function of subscapularis muscle.

Key facts about the subscapularis muscle	
Origin	Subscapular fossa of scapula
Insertion	Lesser tubercle of humerus
Action	Shoulder joint: Arm internal rotation Stabilizes humeral head in glenoid cavity
Innervation	Upper and lower subscapular nerves (C5 - C6)
Blood supply	Suprascapular artery, axillary artery, subscapular artery

Teres major:



The teres major is a thick muscle of the **shoulder joint**. It spans from the inferior aspect of the scapula to the proximal part of the humeral shaft. Unlike the teres minor, the teres major muscle does not attach to the capsule of the glenohumeral joint. Thus it is not regarded as part of the rotator cuff muscles.

The main **function** of this muscle is to produce movements of the arm on the shoulder joint. By contracting, it pulls the humerus posteriorly (extension) and rotates it medial towards the trunk (internal rotation). Additionally, it contributes to the stabilization of the shoulder joint.

In this article, we will discuss the anatomy and function of the teres major muscle.

Key facts about the teres major muscle	
Origins	Inferior angle and lower part of the lateral border of the scapula
Insertions	Intertubercular sulcus (medial lip) of the humerus
Innervation	Lower subscapular nerve (C5-C7)
Blood supply	Thoracodorsal branch of subscapular artery and posterior circumflex humeral artery
Function	Extension and internal rotation of the humerus (arm)

Biceps brachii :



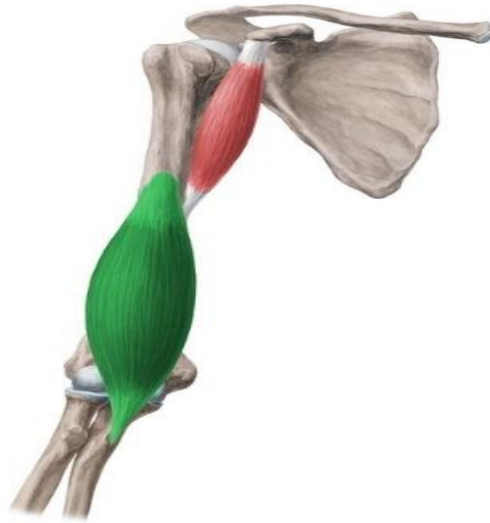
The **biceps brachii muscle** is one of the chief muscles of the arm.

The origin at the scapula and the insertion into the radius of the biceps brachii means it can act on both the shoulder joint and the elbow joint, which is why this muscle participates in a few movements of the arm.

It derives its name from its **two heads** which merge in one unique distal body, defining the unusual structure of the muscle.

Key facts about the biceps brachii muscle	
Origin	Short head - Apex of the Coracoid process of the scapula Long head - Supraglenoid tubercle of the scapula <i>Mnemonic: 'You walk Shorter to a street Corner. You ride Longer on a Superhighway'</i>
Insertion	Radial tuberosity of the radius Deep fascia of forearm (insertion of the bicipital aponeurosis)
Innervation	Musculocutaneous nerve (C5- C6)
Blood supply	Branches of brachial artery
Function	Flexion and supination of the forearm at the elbow joint, weak flexor of the arm at the glenohumeral joint

Brachialis:

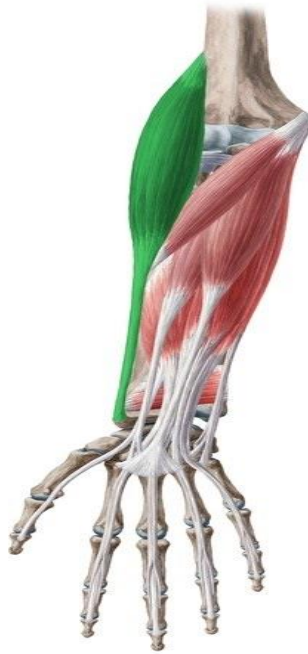


The **brachialis muscle** is a prime flexor of the forearm at the elbow joint. It is fusiform in shape and located in the anterior (flexor) compartment of the arm, deep to the biceps brachii. The brachialis is a broad muscle, with its broadest part located in the middle rather than at either of its extremities.

It is sometimes divided into two parts, and may fuse with the fibers of the biceps brachii, coracobrachialis, or pronator teres muscles. It also functions to form part of the floor of the cubital fossa.

Key facts about the brachialis muscle	
Origin	Distal half of anterior surface of humerus
Insertion	Coronoid process of the ulna; Tuberosity of ulna
Innervation	Musculocutaneous nerve (C5,C6); Radial nerve (C7)
Blood supply	Brachial artery, radial recurrent artery, (occasionally) branches from the superior and inferior ulnar collateral arteries
Functions	Strong flexion of forearm at the elbow joint

Brachioradialis:



Brachioradialis is a fusiform muscle located in the lateral part of the posterior forearm. Along with extensor carpi radialis brevis and extensor carpi radialis longus, it comprises the radial group of forearm muscles, which belong to the superficial layer of posterior forearm muscles.

Although anatomically part of the posterior forearm muscles, which are known to be forearm extensors, brachioradialis' fiber orientation enables it to rather flex the forearm, especially when the forearm is semi pronated. The function of this action is seen in various ordinary activities such as hammering or rowing.

Key facts about the brachioradialis muscle

Origin	Lateral supracondylar ridge of humerus, lateral intermuscular septum of arm
Insertion	(Proximal to) styloid process of radius
Action	Elbow joint: Forearm flexion (when semi pronated)
Innervation	Radial nerve (C5-C6)
Blood supply	Radial artery, radial recurrent arteries, radial collateral artery
Mnemonic	'BrachioRadialis follows the BR rule' Function: Its the Beer mug Raising muscle (i.e. flexes elbow, strongest when the wrist is held for holding a beer mug.) Innervation: Breaks Rule in that it's a flexor muscle, But Radial. (Radial nerve usually is for extensors). One relation: Behind it is the Radial nerve in the cubital fossa.

Triceps brachii :



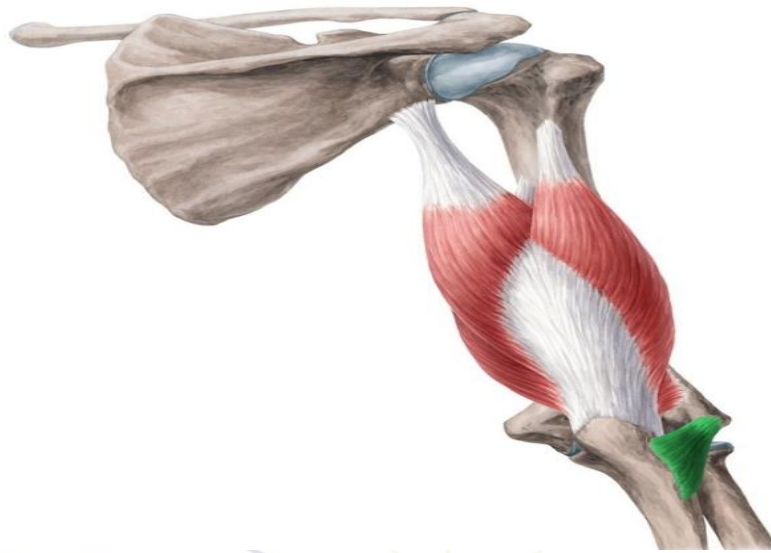
Triceps brachii is a three-headed (**tri** - three, **cep** - head) muscle of the arm. It represents the only constituent of the posterior muscle group of the arm, spanning almost the entire length of the humerus. The triceps brachii muscle consists of a long, medial and lateral head, that originate from their respective attachments on the humerus and scapula, and insert via a common tendon on the ulna.

The main function of triceps brachii is **extension of the forearm** at the elbow joint. In addition, its long head contributes to the **extension** and **adduction of the arm** at the shoulder joint. Besides movement, the triceps brachii also plays a role in creating anatomical spaces which are traversed by neurovascular structures. This makes the triceps brachii muscle an important surgical landmark.

This article will discuss the anatomy and function of the triceps brachii muscle.

Key facts about the triceps brachii muscle	
Origins	Long head - infraglenoid tubercle of the scapula Medial head - posterior surface of the humerus (inferior to radial groove) Lateral head - posterior surface of the humerus (superior to radial groove)
Insertion	Olecranon of ulna and fascia of forearm
Action	Elbow joint: extension of the forearm Shoulder joint: extension and adduction of the arm (long head)
Innervation	Radial nerve (C6-C8)
Blood supply	Deep brachial artery, superior ulnar collateral artery

Anconeus:



The **anconeus** is a small, triangular muscle of the arm. It is located at the posterior aspect of the elbow, extending from the distal humerus to the proximal ulna.

The anconeus muscle belongs to the **superficial extensor compartment**, along with the brachioradialis, extensor carpi radialis longus, extensor carpi radialis brevis, extensor digitorum, extensor digiti minimi and extensor carpi ulnaris muscles.

The anconeus muscle assists in the extension of the forearm and provides support for both the dorsal capsule of the humero-ulnar joint and the ulna itself.

In this article, we will discuss the anatomy and function of the anconeus muscle.

Key facts about the anconeus muscle	
Origin	Lateral epicondyle of humerus
Insertion	Lateral surface of olecranon
Action	Assists in forearm extension at the elbow joint; Stabilization of elbow joint
Innervation	Radial nerve (C7-C8)
Blood supply	Posterior interosseous recurrent artery

Supinator :



Supinator is a spiral muscle contained in the posterior compartment of the forearm, along with brachioradialis, and the superficial and deep forearm extensors.

Supinator muscle curls around the proximal part of radius, connecting it with the ulna. In doing so, the muscle crosses the sagittal axis of forearm. This interesting anatomy enables the supinator muscle to rotate the radius laterally, producing a movement known as forearm supination. Its famous antagonist is the pronator quadratus muscle which rotates the radius medially, producing the opposite movement of pronation.

Key facts about the supinator muscle	
Origin	Lateral epicondyle of humerus, radial collateral ligament, annular ligament, supinator crest of ulna
Insertion	Lateral, posterior, and anterior surfaces of proximal third of radius
Action	Proximal radioulnar joint: Forearm supination
Innervation	Posterior interosseous nerve (C7, C8)
Blood supply	Radial recurrent artery, posterior interosseous artery, posterior interosseous recurrent artery

Pronator teres:



Pronator teres is a fusiform muscle found in the anterior forearm. It belongs to the group of **superficial flexors of the forearm**, together with flexor carpi radialis, palmaris longus, flexor digitorum superficialis and flexor carpi ulnaris muscles. Pronator teres is the most lateral muscle of this group. It is composed of two heads named after the bones from which they originate;

- **Humeral head** that arises from the distal aspect of humerus
- **Ulnar head** that arises from coronoid process of ulna

The main action of pronator teres is the pronation of the forearm, while it also contributes to the flexion of the forearm at the elbow joint.

Key facts about pronator teres muscle	
Origin	Humeral head: medial supracondylar ridge of humerus Ulnar head: Coronoid process of ulna
Insertion	Lateral surface of radius (distal to supinator)
Action	Pronation of forearm at the proximal radioulnar joint, flexion of the forearm at the elbow joint
Innervation	Median nerve (C6, C7)
Blood supply	Branches of brachial, radial and ulnar arteries

Pronator quadratus:



Pronator quadratus is a quadrangular, thin, short and flat muscle lying within the **anterior compartment of forearm**. It is part of the **deep group** of forearm flexors, together with flexor digitorum profundus and flexor pollicis longus. These three muscles are overlaid by the superficial group of forearm flexors.

Pronator quadratus extends across the distal parts of the radius and ulna. As the name implies, the main function of this muscle is **forearm pronation**.

Thanks to your pronator quadratus, you can turn your forearm and palm in order to write or type on a computer. These actions are possible due to nervous transmission by the **median nerve** which supplies this muscle.

Key facts about the pronator quadratus muscle	
Origins	Distal anterior surface of ulna
Insertion	Distal anterior surface of radius
Actions	Proximal radioulnar joint: Forearm pronation
Innervation	Median nerve (anterior interosseous nerve, C7, C8)
Blood supply	Anterior interosseous artery

Flexor carpi radialis:



Flexor carpi radialis is a fusiform muscle located in the anterior forearm. It belongs to the **superficial layer** of the **anterior forearm compartment**, along with the pronator teres, flexor carpi ulnaris, palmaris longus and flexor digitorum superficialis muscles.

All these muscles share the function of **flexing the hand** on the wrist, which is why the entire compartment is also called the flexor compartment of the forearm. These muscles all have individual functions besides wrist flexion, and those of the flexor carpi radialis are wrist flexion and wrist abduction.

Key facts about the flexor carpi radialis muscle	
Origin	Medial epicondyle of humerus
Insertion	Bases of metacarpal bones 2-3
Action	Wrist joint: Wrist flexion, wrist abduction
Innervation	Median nerve (C6, C7)
Blood supply	Anterior/posterior recurrent ulnar artery, radial artery

Flexor carpi ulnaris:



Flexor carpi ulnaris is a fusiform muscle located in the anterior compartment of the forearm. It belongs to the **superficial flexors of the forearm**, along with pronator teres, palmaris longus, flexor digitorum superficialis and flexor carpi radialis. Flexor carpi ulnaris is the most medial of the superficial flexors.

Together with other muscles of the anterior forearm, flexor carpi ulnaris **flexes the hand** at the wrist. Besides flexing the hand, flexor carpi ulnaris is also involved in **adduction** of the hand at the wrist.

This article will discuss the anatomy and function of the flexor carpi ulnaris muscle.

Key facts about the flexor carpi ulnaris muscle	
Origin	Medial epicondyle of humerus, olecranon and posterior border of ulna
Insertion	Pisiform bone, hamate bone, base of metacarpal bone 5
Action	Wrist joint: Wrist flexion, wrist adduction
Innervation	Ulnar nerve (C7-T1)
Blood Supply	Posterior ulnar recurrent artery, ulnar artery

Extensor carpi radialis brevis:



Extensor carpi radialis brevis is a fusiform muscle found in the lateral part of the posterior forearm. Together with anconeus, brachioradialis, extensor carpi radialis longus, extensor digitorum, extensor digiti minimi and extensor carpi ulnaris, it belongs to the superficial forearm extensor group.

Crossing the elbow and wrist joints from their posterior aspects, these muscles produce **extension** at one or both of these joints. The specific action of extensor carpi radialis brevis is to extend and abduct the hand at the wrist joint.

Key facts about the extensor carpi radialis brevis muscle	
Origin	Lateral epicondyle of humerus (common extensor tendon)
Insertion	Posterior aspect of base of metacarpal bone 3
Action	Wrist joints: Hand extension, hand abduction (radial deviation)
Innervation	Radial nerve (C5- C6)
Blood supply	Radial recurrent artery, radial artery, deep brachial artery

Extensor carpi radialis longus:



Extensor carpi radialis longus is an extensor muscle occupying the **posterior compartment** of the forearm. Based on their location, all muscles off the forearm can be classified into layers (superficial to deep) and groups (radial and ulnar). Extensor carpi radialis is classified into the following:

- **Superficial layer**, together with brachioradialis, extensor carpi radialis brevis, extensor digitorum, extensor digiti minimi and extensor carpi ulnaris.
- **Radial group**, together with brachioradialis and extensor carpi radialis brevis.

This muscle extends between the humerus and the second metacarpal bone. It acts on the wrist joint causing extension and abduction (radial deviation) of the hand while powered by the radial nerve. These two actions are vital for effective **hand gripping**.

Key facts about the extensor carpi radialis longus muscle	
Origin	Lateral supracondylar ridge of humerus, lateral intermuscular septum of arm
Insertion	Posterior aspect of base of metacarpal bone 2
Actions	Wrist joints: Hand extension, hand abduction (radial deviation)
Innervation	Radial nerve (C5-C8)
Blood supply	Radial recurrent artery, radial collateral artery, radial artery

Extensor carpi ulnaris:



Extensor carpi ulnaris is a fusiform muscle in the posterior forearm. It spans between the **elbow** and base of the **little finger**.

This muscle belongs to the superficial forearm extensor group, along with anconeus, brachioradialis, extensor carpi radialis longus, extensor carpi radialis brevis, extensor digitorum and extensor digiti minimi muscles.

Like all the muscles of this compartment, extensor carpi ulnaris works as an **extensor** of the wrist. Moreover, due to its specific course, this muscle also acts to adduct the hand.

This article will discuss the anatomy and function of the extensor carpi ulnaris muscle.

Key facts about the extensor carpi ulnaris muscle	
Origin	Lateral epicondyle of humerus, posterior border of ulna
Insertion	Base of metacarpal bone 5
Action	Wrist joint: Hand extension and adduction
Innervation	Posterior interosseous nerve (C7, C8)
Blood supply	Radial recurrent artery, posterior interosseous artery

Extensor digitorum:



Extensor digitorum is a long muscle located in the posterior compartment of the forearm. Together with extensor carpi ulnaris and extensor digiti minimi, it comprises the group of **superficial extensors** of the **forearm**. These muscles can be easily palpated in the lateral aspect of the posterior forearm, especially during the extension of hand when they are contracted.

Extensor digitorum runs from the lateral epicondyle of humerus to the medial four phalanges of the hand. In this way, it generates the pull for the extension of the four medial fingers in their metacarpophalangeal and both interphalangeal joints. Extensor digitorum also participates in the extension of the wrist.

Key facts about the extensor digitorum muscle

Origin	Lateral epicondyle of humerus (common extensor tendon)
Insertion	Extensor expansions of digits 2-5
Action	Metacarpophalangeal / Interphalangeal joints 2-5: Finger extension
Innervation	Posterior interosseous nerve (C7, C8)
Blood supply	Posterior interosseous artery, radial recurrent artery, anterior interosseous artery

Flexor digitorum superficialis :



Flexor digitorum superficialis is the largest muscle of the **anterior compartment of the forearm**. It belongs to the **superficial flexors** of the forearm, together with pronator teres, flexor carpi radialis, flexor carpi ulnaris and palmaris longus. Some sources alternatively classify this muscle as an independent middle/intermediate layer of the anterior forearm, found between the superficial and deep groups.

Based on its origin sites, flexor digitorum superficialis is divided into two heads; a **humeroulnar head** and **radial head**. Its large muscular belly courses distally towards the wrist, where it splits into four tendons and attaches to the middle phalanges of the second through fifth digits of the hand. Due to their superficial location, these tendons can be easily palpated on the distal part of the forearm.

The main action of this muscle is **flexion of the digits 2-5** at both the metacarpophalangeal and proximal interphalangeal joints.

Key facts about the flexor digitorum muscle	
Origin	Humeroulnar head: Medial epicondyle of humerus, coronoid process of ulna Radial head: Proximal half of anterior border of radius
Insertion	Sides of middle phalanges of digits 2-5
Action	Metacarpophalangeal and proximal interphalangeal joints 2-5: Finger flexion
Innervation	Median nerve (C8, T1)
Blood supply	Ulnar artery, radial artery, median artery

Extensor pollicis longus:

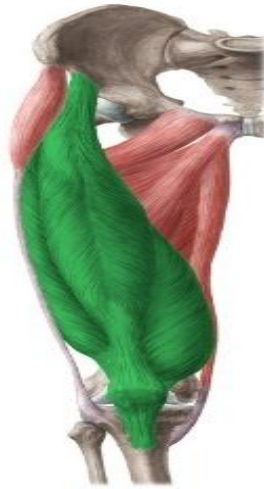


Extensor pollicis longus is part of the deep extensors of the forearm together with extensor pollicis brevis, abductor pollicis longus, extensor indicis and supinator muscles. It is located on the posterior aspect of forearm, extending from the middle third of the ulna, and adjacent interosseous membrane, to the distal phalanx of the thumb.

As the name of the muscle suggests, the main **function** of this muscle is extension of the thumb at the metacarpophalangeal and interphalangeal joints. Also, this muscle participates in extension and abduction of the wrist joint.

Key facts about extensor pollicis longus muscle	
Origin	Posterior surface of middle third of ulna and interosseus membrane
Insertion	Posterior aspect of base of distal phalanx of thumb
Action	Wrist joints: Weak hand extension Metacarpophalangeal and interphalangeal joint of thumb: Thumb extension
Innervation	Posterior interosseous nerve (C7, C8)
Blood supply	Posterior interosseous artery, anterior interosseous artery

Quadriceps femoris:



The **quadriceps femoris muscle**, commonly known as the quad muscle, is the strongest muscle of the human body. It is located in the anterior compartment of the thigh, together with the sartorius.

The quadriceps femoris muscle translates to “four-headed muscle” from Latin. It bears this name because it consists of **four individual muscles**; rectus femoris, vastus medialis, vastus lateralis, and vastus intermedius. Out of all four muscles, only the rectus femoris crosses both the hip and knee joints. The others cross only the knee joint. These muscles differ in their origin, but share a common quadriceps femoris tendon which inserts into the patella. The function of the quadriceps femoris muscle is to extend the leg at the knee joint and to flex the thigh at the hip joint.

This article will discuss the anatomy and function of the quadriceps femoris muscle.

Key facts about the quadriceps femoris muscle	
Parts	Rectus femoris, vastus lateralis, vastus medialis, vastus intermedius
Origins	Rectus femoris: Anterior inferior iliac spine, supraacetabular groove Vastus medialis: Intertrochanteric line, pectineal line of femur, linea aspera, medial supracondylar line of femur Vastus lateralis: Intertrochanteric line, greater trochanter, gluteal tuberosity, linea aspera of femur Vastus intermedius: Anterior surface of femoral shaft
Insertions	Rectus femoris and vastus intermedius: Tibial tuberosity (via patellar ligament), patella Vastus lateralis: Tibial tuberosity (via patellar ligament), patella, (lateral condyle of tibia) Vastus medialis: Tibial tuberosity (via patellar ligament), patella, (medial condyle of tibia)
Innervation	Femoral nerve (L2-L4)
Function	Hip joint: Thigh flexion (rectus femoris only); Knee joint: Leg extension

Popliteus:



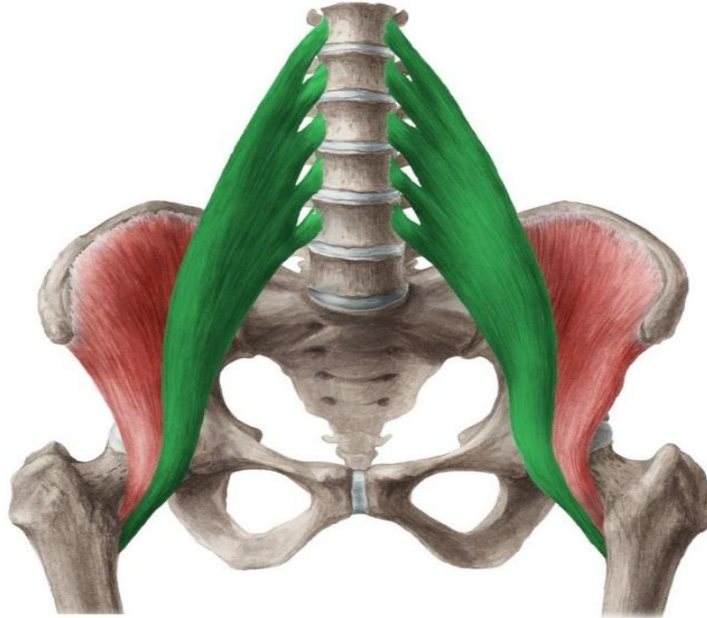
The popliteus muscle is a small muscle that forms the floor of the popliteal fossa. It belongs to the **deep posterior leg muscles**, along with tibialis posterior, flexor digitorum longus and flexor hallucis longus.

The popliteus muscle extends over the posterior aspect of the knee joint. It originates from the femur and the posterior horn of the lateral meniscus, and inserts on the proximal tibia.

The popliteus muscle plays an important role in the gait cycle by initiating the flexion of the fully extended (“locked”) knee. Thus, the popliteus is referred to as the **“key to unlock the knee”**. In addition, the popliteus muscle is the main stabilizer of the dorsal knee region.

Key facts about the popliteus muscle	
Origin	Lateral condyle of femur, posterior horn of lateral meniscus of knee joint
Insertion	Posterior surface of proximal tibia
Innervation	Tibial nerve (L4-S1)
Blood supply	Inferior medial and lateral genicular arteries (popliteal artery), posterior tibial recurrent artery, posterior tibial artery, nutrient artery of tibia
Function	Unlocks knee joint; Knee joint stabilization

Iliopsoas:



Iliopsoas is a large compound muscle of the **inner hip** composed of the iliacus and psaos major muscle. Aside from the iliopsoas, other muscles of the inner hip include the psaos minor, obturator externus, obturator internus, superior gemellus, inferior gemellus, piriformis and quadratus femoris.

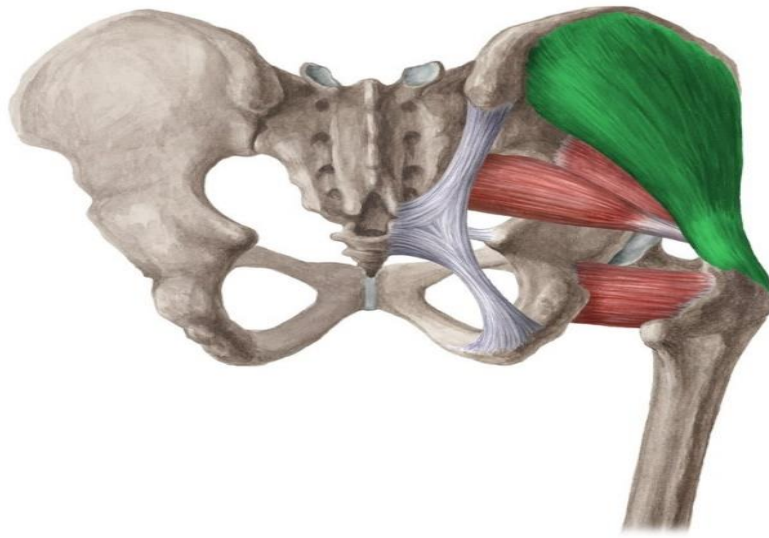
The iliopsoas has an extensive origin, with the **psaos major portion** originating from the twelfth thoracic and all five lumbar vertebrae, and the **iliacus portion** arising mainly from the iliac fossa of the pelvis. Their fibers merge into a single muscle belly around the pelvic brim. The iliopsoas ends in a tendon that inserts to the lesser trochanter of femur.

The iliopsoas muscle is the **main flexor of the hip joint**. The joint actions of psaos major and iliacus produce the flexion of the hip; however, due to its proximal attachments, the psaos major can produce additional movements of the lumbar vertebral column.

This article will discuss the anatomy and function of the iliopsoas muscle.

Key facts about the iliopsoas muscle	
Definition	Iliopsoas is the largest and strongest inner hip muscle extending from the vertebral column to the proximal femur
Muscular components	Iliacus, psaos major
Innervation	Iliacus: femoral nerve (L2-L4) Psoas major: anterior rami of spinal nerves (L1-L3)
Blood supply	Iliolumbar, obturator, external iliac and femoral arteries
Function	Flexion and external rotation of the thigh at the hip joint; flexion of the trunk at the hip, lateral flexion of the trunk (psaos major only)

Gluteus Medius:

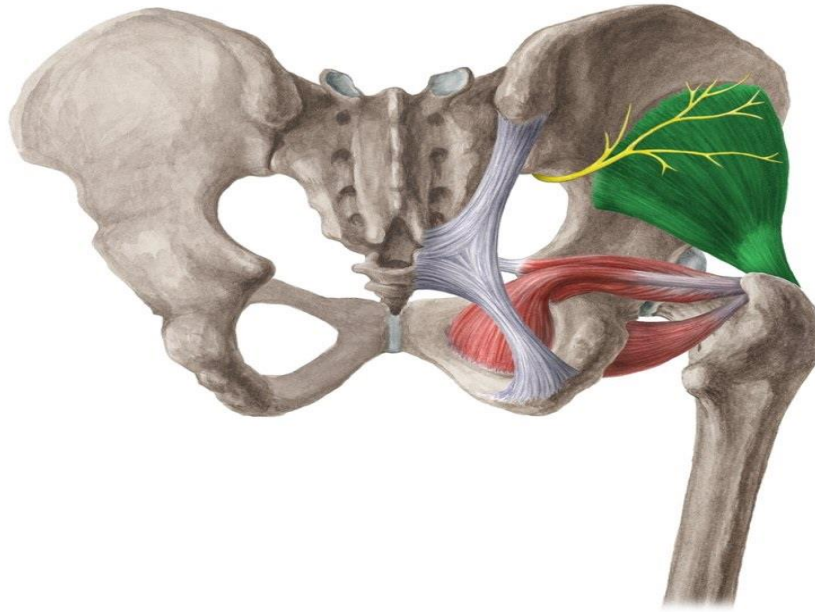


Gluteus medius is a large fan-shaped muscle located in the posterior hip, extending from the ilium to the proximal femur. Together with the gluteus maximus, gluteus minimus and tensor fasciae latae muscles, it belongs to the muscles of the gluteal region.

The gluteus medius muscle acts on the hip joint producing two movements; its anterior part internally rotates the thigh, while the contraction of the whole muscle abducts the thigh. Additionally, the gluteus medius muscle stabilizes the pelvis while standing or walking. This article will discuss the anatomy and function of the gluteus medius muscle.

Key facts about the gluteus medius muscle	
Origin	Gluteal surface of ilium (between anterior and posterior gluteal lines)
Insertion	Lateral aspect of greater trochanter of femur
Action	Hip joint: Thigh abduction, thigh internal rotation (anterior part); Pelvis stabilization
Innervation	Superior gluteal nerve (L4-S1)
Blood supply	Deep branch of superior gluteal artery, trochanteric anastomosis

Gluteus minimus:



The **gluteus minimus** is a small triangular muscle located deep in the posterior region of the hip, spanning from the gluteal surface of the ilium to the proximal end of the femur. It belongs to the group of gluteal muscles, along with the gluteus maximus, gluteus medius and tensor fasciae latae.

Gluteus minimus acts in synergy with the gluteus medius to produce the **movements on the hip joint**; the internal rotation and abduction of the thigh. Moreover, this muscle is an important stabilizer of the pelvis in the gait cycle.

This article will discuss the anatomy and function of the gluteus minimus muscle.

Key facts about the gluteus minimus muscle	
Origin	Gluteal surface of ilium (between anterior and inferior gluteal lines)
Insertion	Anterior aspect of greater trochanter of femur
Action	Hip joint: Thigh abduction, thigh internal rotation (anterior part); Pelvis stabilization
Innervation	Superior gluteal nerve (L4-S1)
Blood supply	Superior gluteal artery, trochanteric anastomosis

Gluteus maximus:

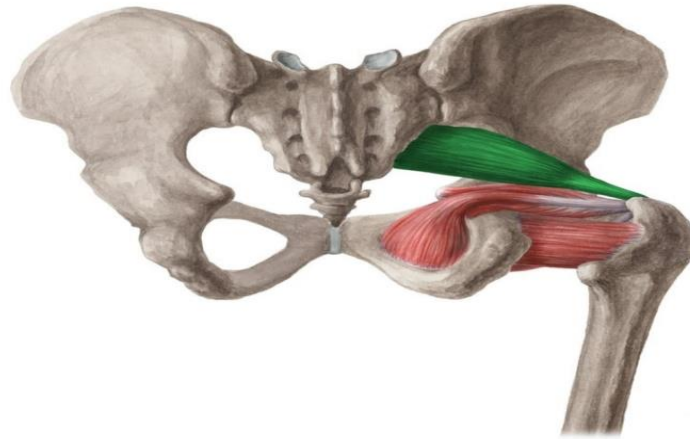


The **gluteus maximus** is the most superficial gluteal muscle that forms the prominence of the gluteal region. Along with the gluteus medius, gluteus minimus and tensor fasciae latae, it belongs to the gluteal group of the hip muscles. Gluteus maximus extends from the pelvis to the gluteal tuberosity of femur. It is innervated by the **inferior gluteal nerve** (L5-S2).

The main **functions** of the gluteus maximus muscle are the extension and external rotation of the thigh at the hip joint. Additionally, its superior part can produce thigh abduction, while the inferior part causes the thigh adduction.

Key facts about gluteus maxiumus muscle	
Origin	Lateroposterior surface of sacrum and coccyx, gluteal surface of ilium (behind posterior gluteal line), thoracolumbar fascia, Sacrotuberous ligament
Insertion	Iliotibial tract, gluteal tuberosity of femur
Action	Hip joint: Thigh extension, thigh external rotation, thigh abduction (superior part), thigh adduction (inferior part)
Innervation	Inferior gluteal nerve (L5, S1, S2)
Blood supply	Inferior gluteal and superior gluteal arteries

Piriformis:



Piriformis is a muscle of the **gluteal region** which lies deep to the gluteus maximus. Piriformis belongs to a group of six short external rotators of the hip, i.e. gemellus superior, obturator internus, gemellus inferior, quadratus femoris, obturator externus. Attaching to the sacrum on one end and the greater trochanter on the other one, this muscle is responsible for stabilizing the hip joint and moving the thigh in various directions.

In this article we will discuss the anatomy of piriformis, as well as its neurovascular supply and function. In addition we discuss the clinical relevance of the muscle.

Key facts about the piriformis muscle	
Origin	Anterior surface of the sacrum (between the S2 and S4), Gluteal surface of ilium (near posterior inferior iliac spine), (Sacrotuberous ligament)
Insertion	(Apex of) Greater trochanter of the femur
Action	Hip joint: Thigh external rotation, Thigh abduction (from flexed hip); Stabilizes head of femur in acetabulum
Innervation	Nerve to piriformis (S1-S2)
Blood supply	Superior gluteal artery, inferior gluteal artery, gemellar branches of the internal pudendal
Mnemonic	Structures passing through the greater sciatic foramen inferior to piriformis muscle: PIN & PINS (standing for: P osterior cutaneous nerve of thigh, I nferior gluteal vessels and nerves, N erve to quadratus femoris, P udendal nerve, I nternal pudendal vessels, N erve to obturator internus, S ciatic nerve)

Pectineus :



Pectineus is a flat muscle found in the superomedial part of the anterior thigh. Fascial compartments of the thigh muscles are specific in that each of them is innervated by a particular nerve. Due to having dual innervation, pectineus is one of a few muscles classified into **two compartments** at the same time; anterior and medial. The others being adductor longus and adductor magnus.

For didactic purposes, pectineus is usually described together with the five muscles of the medial compartment of the thigh. Namely, these muscles are the gracilis, adductor longus, adductor brevis and adductor magnus. This comprises the functional group of **thigh adductors**.

Besides adducting the thigh, the **functions** of pectineus include the additional movements of thigh flexion, external (lateral) rotation and internal (medial) rotation. Apart from being a prime mover, pectineus is also a postural muscle as it stabilizes the pelvis and balances the trunk on the lower extremity during walking.

Key facts about the pectineus muscle	
Origin	Superior pubic ramus (pectineal line of pubis)
Insertion	Pectineal line of femur, linea aspera of femur
Action	Hip joint: Thigh flexion, thigh adduction, thigh external rotation, thigh internal rotation; pelvis stabilization
Innervation	Femoral nerve (L2, L3) (Obturator nerve (L2, L3))
Blood supply	Medial femoral circumflex artery, obturator artery

Sartorius:



The sartorius muscle is the longest muscle in the human body. Together with the quadriceps femoris, it belongs to the **anterior (extensor) muscles of the thigh**.

The sartorius muscle has an oblique course throughout the thigh; it arises from the anterior superior iliac spine, crosses to the medial side of the thigh, and finally inserts to the medial side of the proximal tibia.

Due to its oblique course, the sartorius muscle doesn't extend the thigh, as one would expect from a muscle of the anterior compartment. Instead, the sartorius muscle flexes, abducts and externally rotates the thigh, and flexes and internally rotates the leg.

Key facts about sartorius muscle	
Origin	Anterior superior iliac spine (ASIS)
Insertion	Proximal end of tibia below medial condyle (via pes anserinus)
Innervation	Femoral nerve (L2-L3)
Blood supply	Proximal third: branches of femoral artery, deep femoral artery, artery of quadriceps, lateral circumflex femoral artery Middle third: branches of femoral artery Distal third: branches of femoral and descending genicular arteries
Function	Hip joint: thigh flexion, thigh abduction, thigh external rotation Knee joint: leg flexion, leg internal rotation

Tensor fasciae latae:



Tensor fasciae latae is a fusiform muscle located in the lateral aspect of the thigh. It belongs to the muscles of the gluteal region, along with the gluteus maximus, gluteus medius and gluteus minimus muscles.

Tensor fasciae latae is found superficial in the anterolateral aspect of the thigh, spanning from the anterior portion of the iliac crest to the superior portion of the tibia, onto which it inserts via the iliotibial tract. The main **function** of this muscle is to produce the extension and lateral rotation of the leg on the knee joint. Additionally, it contributes to the movements of the thigh, acting as a relatively weak abductor and medial rotator on the hip joint.

Key facts about the tensor fasciae latae muscle	
Origin	Outer lip of anterior iliac crest, Anterior superior iliac spine (ASIS)
Insertion	Iliotibial tract
Action	Hip joint: Thigh internal rotation, (Weak abduction); Knee joint: Leg external rotation, (Weak leg flexion/ extension); Stabilizes hip & knee joints
Innervation	Superior gluteal nerve (L4-S1)
Blood supply	Ascending branch of lateral circumflex femoral artery

Biceps femoris:



Biceps femoris is a long muscle of the posterior aspect of the thigh. Together with the semitendinosus and semimembranosus muscles, it makes the group of muscles commonly known as the hamstrings.

Biceps femoris muscle runs from the ischial tuberosity, all the way to the proximal part of the fibula. In doing so the muscle crosses **two joints**; the hip joint and the knee joint. Acting simultaneously on these joints, biceps femoris has many important functions; flexion and external rotation at the knee joint, extension and external rotation in the hip joint.

As its name suggests, this muscle consists of two heads, one lying deep to the other. Each head has a different origin and innervation but they share the **same** insertion.

In this article, we will discuss the anatomy and function of the biceps femoris muscle.

Key facts about the biceps femoris muscle	
Origin	Long head: (inferomedial impression of) ischial tuberosity, sacrotuberous ligament Short head: linea aspera of femur (lateral lip), lateral supracondylar line of femur
Insertion	(Lateral aspect of) head of fibula
Action	Hip joint: thigh extension, thigh external rotation; Knee joint: leg flexion, leg external rotation; stabilizes pelvis
Innervation	Long head: tibial division of sciatic nerve (L5-S2) Short head: common fibular division of sciatic nerve (L5-S2)
Blood supply	Inferior gluteal artery, perforating arteries, popliteal artery

Semitendinosus :



Semitendinosus is a fusiform muscle of the posterior compartment of thigh. Along with semimembranosus and long head of biceps femoris it comprises a group called the **hamstring muscles** with which it shares these three common features;

- They all attach between the ischial tuberosity of pelvis and bones of the leg
- They cross both hip and knee joints on their course and act on them
- They are all innervated by the tibial division of sciatic nerve

As a prime mover, semitendinosus extends and internally rotates the thigh, flexes and internally rotates the leg. It also has a postural role, stabilising the pelvic girdle.

Key facts about the semitendinosus muscle	
Origin	(Posteromedial impression of) Ischial tuberosity
Insertion	Proximal end of tibia below medial condyle (via pes anserinus)
Action	Hip joint: Thigh extension, thigh internal rotation, stabilizes pelvis. Knee joint: Leg flexion, leg internal rotation.
Innervation	Tibial division of sciatic nerve (L5-S2)
Blood supply	First perforating branch of deep femoral artery, medial femoral circumflex artery, inferior gluteal artery and inferior medial geniculate artery

Semimembranosus:



Semimembranosus is one of four **posterior thigh muscles** that are responsible for extending the hip. The other three muscles that belong to the hip extensor group are semitendinosus, biceps femoris and gluteus maximus. Collectively semimembranosus, semitendinosus and biceps femoris are referred to as the **hamstring muscles**.

Semimembranosus (along with semitendinosus) occupies the **medial aspect** of the **posterior thigh**. It is a relatively long muscle that spans the full length of the thigh – from the hip to the knee.

Key facts about the semimembranosus muscle	
Origin	Superolateral impression of ischial tuberosity
Insertion	Medial condyle of tibia
Action	Hip joint: thigh extension and internal rotation Knee joint: flexion and internal rotation of the leg Stabilizes pelvis
Innervation	Tibial division of sciatic nerve (L5 – S2)
Blood supply	Perforating branches of femoral and popliteal arteries

Adductor brevis:



Adductor brevis is a flat, triangular muscle that is found in the inner thigh. This muscle runs from the pubis to the medial aspect of the femur. Together with adductor longus, adductor magnus, gracilis and pectineus muscles, it comprises a group of muscles known as the **adductors of the thigh**.

These inner thigh muscles produce movements of the hip joint; primarily **thigh adduction**, but they also participate in flexion, internal and external rotation and stabilization of the pelvis while standing or walking. Adductor brevis, being one of the shortest muscles from this group, is a weak adductor of the thigh.

In this article, we will discuss the anatomy and function of adductor brevis muscle.

Key facts about the adductor brevis muscle

Key facts about the adductor brevis muscle	
Origin	Anterior body of pubis, inferior pubic ramus
Insertion	Linea aspera of femur (medial lip)
Action	Hip joint: thigh flexion, thigh adduction, thigh external rotation; pelvis stabilization
Innervation	Obturator nerve (L2-L4)
Blood supply	Arteria profunda femoris

Adductor longus:



The adductor longus is a large, fan-shaped muscle located in the medial aspect of the thigh. It belongs to the adductors of the thigh, together with adductor brevis, adductor magnus, adductor brevis, pectineus and gracilis muscles. Adductor longus is innervated by the anterior division of the obturator nerve (L2-L4).

The primary function of the is muscle group is to **adduct the thigh at the hip joint**. The muscles in this compartment are believed to be evolved from both extensor and flexor columns.

Key facts about the adductor longus muscle	
Origin	Body of pubis, inferior to pubic crest and lateral to the pubic symphysis
Insertion	Middle third of linea aspera of femur (medial lip)
Action	Hip joint: Thigh flexion, Thigh adduction, Thigh external rotation; Pelvis stabilization
Innervation	Obturator nerve (L2-L4)
Blood supply	Profunda femoris artery, obturator artery

Adductor magnus:



The **adductor magnus** muscle is a large triangular muscle of the lower limb, with its apex situated on the hip bone, and its base on the linea aspera of the femur. It is situated both in the posterior and medial fascial compartments of the thigh. The distribution of this muscle in two compartments is reflected in the fact that it receives **dual nerve supply**. Regardless of its position, the adductor magnus muscle is classified as a muscle of the medial compartment of the thigh.

The adductor magnus is the largest and strongest muscle of the medial compartment of the thigh, which also consists of adductor longus, adductor brevis, pectineus, and gracilis muscles. In respect to their function, these 5 muscles are collectively called the **adductors of the thigh**, even though their actions are a bit more complex than that.

Aside from thigh adduction, these muscles also participate in movements such as thigh flexion, extension, external and internal rotation, as well as pelvis stabilization during walking.

This article will teach you everything you need to know about the attachments, relations, neurovascular supply and functions of the adductor magnus muscle!

Key facts about the adductor magnus muscle	
Origin	Adductor part: Inferior pubic ramus, ischial ramus Ischiocondylar part: Ischial tuberosity
Insertion	Adductor part: Gluteal tuberosity, linea aspera (medial lip), medial supracondylar line Ischiocondylar part: Adductor tubercle of femur
Action	Adductor part: Hip joint - Thigh flexion, thigh adduction, thigh external rotation Hamstring part: Hip joint - Thigh extension, thigh internal rotation Entire muscle: Pelvis stabilization
Innervation	Adductor part: Obturator nerve (L2-L4) Ischiocondylar part: Tibial division of sciatic nerve (L4) Mnemonic: African Mouse Sneaks Out (refers to Adductor Magnus Sciatic Obturator)
Blood supply	Deep femoral artery; Femoral, popliteal and genicular arteries

Gracilis:



Gracilis muscle is a long and slender muscle located in the medial (adductor) compartment of the thigh. It forms part of the **adductor muscle group** together with adductor longus, adductor brevis, adductor magnus and pectineus muscles. Gracilis is the most superficial hip adductor, overlying the remaining four. It is also the weakest member but the only hip adductor that crosses and acts on two joints; the hip and knee.

Gracilis extends from the **coxal bone** to the tibia and thanks to it, you are capable of thigh adduction and flexion, as well as leg flexion and medial (internal) rotation. These actions have several important roles, for instance to balance the trunk during walking.

Key facts about the gracilis muscle	
Origin	Anterior body of pubis, inferior pubic ramus, ischial ramus
Insertion	Medial surface of proximal tibia (via pes anserinus)
Actions	Hip joint: Thigh flexion, thigh adduction; Knee joint: leg flexion, leg internal rotation
Innervation	Obturator nerve (L2-L3)
Blood supply	Deep femoral artery (via artery to the adductors)

Gastrocnemius:



Gastrocnemius is a large muscle located in the posterior leg. Posteriorly, is the most superficial of the muscles of the leg, and forms the bulk of the calf. It takes its name from the Greek words γαστήρ (gaster) meaning stomach or belly, and κνήμη (kneme) meaning leg; the combination of the two words means the “belly of the leg” or in other words the bulk of the calf.

In conjunction with the soleus muscle, it is a component of a composite, three-headed group of muscles referred to as triceps surae. Together, they act in many basic activities, such as walking, running and leaping. This article will outline the morphology of the gastrocnemius muscle, as well as its functional and clinical anatomy.

Key facts about the gastrocnemius muscle

Origin	Lateral head: Posterolateral aspect of lateral condyle of the femur Medial head: Posterior surface of medial femoral condyle, popliteal surface of femoral shaft
Insertion	Posterior surface of the calcaneus via the calcaneal tendon
Innervation	Tibial nerve (S1, S2)
Function	Talocrural joint: Foot plantar flexion Knee joint: Leg flexion
Mnemonic	Stand on your Soles. Gas explodes! (refers to the functions of posterior leg muscles; Soleus is for posture, while Gastrocnemius is for explosive movements)

Soleus:



The **soleus muscle** is a wide flat leg muscle found on the posterior leg. It runs from just below the knee to the heel and lays immediately deep to the gastrocnemius. These two muscles, along with the plantaris muscle, belong to the group of superficial posterior compartment calf muscles. Soleus' contraction results in strong **plantar flexion**. It also allows us to maintain an upright posture due to its important role as an antigravity muscle.

Together with the gastrocnemius, they form the three-headed group of muscles referred to as the **triceps surae**. They both insert on the calcaneus via the calcaneal tendon and act in many basic activities, such as walking, running, and leaping. The size and shape of the triceps surae muscle bellies determine the interindividual differences of human calves' appearance - from slim to rather robust. This article will outline the morphology of the soleus muscle, as well as its functional and clinical anatomy.

Key facts about the soleus muscle	
Origin	Soleal line, medial border of tibia, head of fibula, posterior border of fibula
Insertion	Posterior surface of calcaneus (via calcaneal tendon)
Innervation	Tibial nerve (S1, S2)
Vascularization	Posterior tibial artery and vein
Function	Talocrural joint: Foot plantar flexion
Clinical relations	Soleus tear, soleus pain
Mnemonic	Stand on your Soles . Gas explodes! (refers to the functions of posterior leg muscles; Soleis is for posture, while Gastrocnemius is for explosive movements)

Tibialis posterior:



Tibialis posterior is the most central and deepest muscle located in the posterior aspect of the leg. Together with popliteus, flexor hallucis longus and flexor digitorum longus, it forms the **deep group of muscles** of the **posterior compartment of leg**.

These muscles are located posterior to the tibia, fibula and interosseous membrane. Tibialis posterior is hidden from view by the large, superficial muscles of the leg; gastrocnemius and soleus. This muscle crosses the ankle joint to insert on the plantar surface of the foot. As a result, it helps with **plantar flexion** of the foot at the ankle joint. It is also a synergist of tibialis anterior in **inversion** of the foot.

Key facts about the tibialis posterior muscle	
Origin	Posterior surface of tibia, posterior surface of fibula and interosseous membrane
Insertion	Tuberosity of navicular bone, all cuneiform bones, cuboid bone, bases of metatarsal bones 2-4
Actions	Talocrural joint: Foot plantarflexion Subtalar joint: Foot inversion Supports medial longitudinal arch of foot
Innervation	Tibial nerve (L4, L5)
Blood supply	Branches of the posterior tibial artery

Flexor digitorum longus:



Flexor digitorum longus is a thin muscle that belongs to the deep posterior muscles of the leg. It runs from the posterior surface of the **tibia**, across the posterior compartment of the leg to the phalanges of the foot. On its course, the muscle receives functional support from the quadratus plantae muscle.

Due to its attachments, flexor digitorum longus performs its action in **four different joints**: talocrural (ankle joint), talocalcaneal (subtalar joint), metatarsophalangeal and interphalangeal joints. The main function of this muscle is the flexion of the foot and toes.

In this article, we will discuss the anatomy and function of the flexor digitorum longus muscle.

Key facts about the flexor digitorum longus muscle	
Origin	Posterior surface of tibia (inferior to soleal line)
Insertion	Bases of distal phalanges of digits 2-5
Action	Metatarsophalangeal and interphalangeal joints 2-5: toe flexion Talocrural joint: foot plantar flexion Subtalar joint: foot inversion
Innervation	Tibial nerve (L5, S1, S2)
Blood supply	Posterior tibial artery

Flexor hallucis longus:



Flexor hallucis longus muscle is a powerful muscle that comprises the deep layer of the **posterior compartment of the leg**. It belongs to a group called the deep flexors of the calf, which also include popliteus, flexor digitorum longus and tibialis posterior muscles.

According to the direction of its muscle fibers, flexor hallucis longus is described as an **unipennate muscle**. A pennate muscle is a muscle that attaches obliquely to its tendon. The uni- prefix means that its muscle fibers converge to attach on one side of the tendon. Flexor hallucis longus arises from the posterior two thirds of the fibula and descends on the fibular side of the leg. It then crosses over the posterior part of the ankle to emerge on the sole of the foot, where it passes anteriorly to insert on the great toe (hallux).

Key facts about the flexor hallucis longus	
Origin	(Distal 2/3 of) Posterior surface of fibula, interosseous membrane, posterior intermuscular septum of leg, fascia of tibialis posterior muscle
Insertion	Base of distal phalanx of great toe
Innervation	Tibial nerve (S2, S3)
Function	Metatarsophalangeal and interphalangeal joint 1: Toe flexion; Talocrural joint: Foot plantar flexion; Subtalar joint: Foot inversion
Blood supply	Posterior tibial artery, fibular artery

Fibularis (peroneus) longus:



Fibularis (peroneus) longus is a long muscle located superficially in the lateral compartment of the leg, together with the fibularis brevis muscle. Fibularis longus spreads from the proximal aspect of the fibula to the medial cuneiform and the first metatarsal bones.

It is innervated by the superficial fibular nerve (L5, S1). The main function of this muscle is to produce the plantarflexion and eversion of the foot on the ankle joint. In this article, we will discuss the anatomy and function of the muscle, as well as its clinical relevance.

Key facts about the fibularis longus muscle	
Origin	Head of fibula, proximal 2/3 of lateral surface of fibula, intermuscular septa
Insertion	Medial cuneiform bone, metatarsal bone 1
Action	Talocrural joint: Foot plantar flexion; Subtalar joint: Foot eversion; Supports longitudinal and transverse arches of foot
Innervation	Superficial fibular nerve (L5, S1)
Blood supply	Fibular artery

Fibularis brevis:



Fibularis brevis, or peroneus brevis, is a short muscle that together with the fibularis (peroneus) longus comprises the group of the lateral leg muscles.

Based on their function, both peroneal muscles belong to a larger group of leg muscles whose common function is to **plantarflex** the foot (point the toes downward). Other muscles of this group are the gastrocnemius, soleus, plantaris, tibialis posterior, fibularis tertius, flexor digitorum longus, and flexor hallucis longus. Besides plantarflexing the foot, fibularis brevis and longus also **evert** the foot while contracting.

Key facts about the fibularis brevis muscle	
Origin	Distal 2/3 of the lateral surface of fibula, anterior intermuscular septum
Insertion	Tuberosity of the 5th metatarsal bone
Action	Talocrural joint: Foot plantar flexion Subtalar joint: Foot eversion
Innervation	Superficial fibular nerve (L5, S1)
Blood supply	Anterior tibial artery

Tibialis anterior:



Tibialis anterior is a fusiform muscle found in the anterior part of the leg. Lying superficially in the leg, this muscle is easily palpable lateral to the anterior border of tibia. Along with fibularis (peroneus) tertius, extensor digitorum longus and extensor hallucis longus, it comprises the anterior (or extensor) compartment of the leg.

This muscle acts as the **main foot dorsiflexor** on the talocrural joint, but it also inverts the foot at the subtalar joint. Both actions play important roles in the gait cycle.

This article will discuss the anatomy and function of tibialis anterior muscle.

Key facts about the tibialis anterior muscle	
Origin	Lateral surface of tibia, interosseous membrane
Insertion	Medial cuneiform bone, base of metatarsal bone 1
Action	Talocrural joint: foot dorsiflexion; subtalar joint: foot inversion
Innervation	Deep fibular nerve (L4, L5)
Blood supply	Branches of anterior tibial artery: Anterior and medial muscular branches; anterior tibial recurrent, dorsalis pedis and anterior medial malleolar arteries
	Branches of posterior tibial artery: Medial malleolar and calcaneal arteries

Extensor digitorum longus:



Extensor digitorum longus (EDL muscle) is a feather-like muscle of the anterior (extensor) compartment of leg. Besides EDL muscle, this compartment also contains the tibialis anterior, extensor hallucis longus and fibularis (peroneus) tertius muscles.

Since all these muscles cross the dorsal aspect of the ankle joint, their common function is **dorsiflexion of the foot**. As the extensor digitorum longus also crosses the subtalar, metatarsophalangeal and interphalangeal joints of the foot, it also everses the foot and extends the toes.

This article will discuss the anatomy and function of the extensor digitorum longus muscle.

Key facts about extensor digitorum longus muscle	
Origin	(Proximal half of) medial surface of fibula, lateral tibial condyle, interosseus membrane
Insertion	Distal and middle phalanges of digits 2-5
Action	Metatarsophalangeal and interphalangeal joints 2-5: Toe extension; Talocrural joint: Foot dorsiflexion; Subtalar joint: Foot eversion
Innervation	Deep fibular nerve (L5, S1)
Blood supply	Anterior tibial, fibular, anterior lateral malleolar, lateral tarsal, metatarsal, plantar, and digital arteries

Erector spinae:



The erector spinae muscles comprise the **intermediate layer** of the **deep (intrinsic) muscles of the back**. They extend on either side of the vertebral column, between the base of the cranium superiorly and pelvis inferiorly. The erector spinae are divided into three groups, from medial to lateral:

- Spinalis muscles
- Longissimus muscles
- Iliocostalis muscles

The function of the spinal erectors is to move the vertebral column. Bilateral contraction of these muscles **extends the spine**, while unilateral contraction causes **lateral flexion** (ipsilateral). They also help to maintain **posture** by steadying the spine on the pelvis during walking.

Key facts about the erector spinae muscles	
Definition	Three groups of deep muscles located on either side of the vertebral column
Function	Bilateral contraction - extension of spine Unilateral contraction - lateral flexion of spine (ipsilateral)
Mnemonic	I Like S tanding (stands for I liocostalis, L ongissimus & S pinalis)

Multifidus:

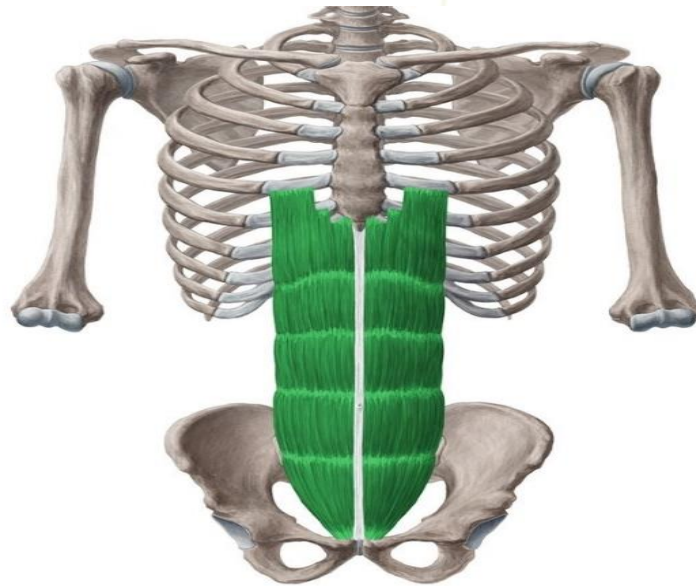


Multifidus is a group of short, triangular muscles that along with the semispinalis and rotatores comprises the **transversospinal group** of deep back muscles. They are the thickest muscles in the transversospinal group, and are shorter than semispinalis, but longer than rotatores. Multifidus is found on either side of vertebral column, extending from the cervical all the way to the lumbar spine. The group is regionally divided into **cervical multifidus**, **thoracic multifidus** and **lumbar multifidus**.

Each multifidus muscle bridges over three to six vertebral levels, spanning between the transverse and spinous processes of certain cervical, thoracic and lumbar vertebrae. Although small, multifidus muscles aid several movements of vertebral column; when contracting **bilaterally** they extend the spine, while **unilateral** contraction aids lateral flexion of the spine to the same side and rotation of the spine to the opposite side.

Key facts about the multifidus muscle	
Origin	<i>Multifidus cervicis</i> : Superior articular processes of vertebrae C4-C7 <i>Multifidus thoracis</i> : Transverse process of thoracic vertebra <i>Multifidus lumborum</i> : Mammillary processes of lumbar vertebrae, posterior aspect of sacrum, posterior superior iliac spine (PSIS) of ilium and posterior sacroiliac ligament
Insertion	Lateral aspect and tips of the spinous processes of vertebrae 2-5 levels above origin
Action	<i>Bilateral contraction</i> : Extension of spine <i>Unilateral contraction</i> : Lateral flexion of spine (ipsilateral), rotation of spine (contralateral)
Innervation	Medial branches of posterior rami of spinal nerves
Blood supply	Vertebral artery, deep cervical artery, occipital artery, posterior intercostal arteries, subcostal artery, lumbar arteries and lateral sacral arteries

Rectus abdominis:



Rectus abdominis, informally known as the *abs muscle*, is a long muscle of the anterior abdominal wall. In those with low body fat, it is clearly visible beneath the skin forming the 'six pack'. It extends from the rib cage all the way to the pubic bone.

Rectus abdominis belongs to the **anterior abdominal muscles** together with pyramidalis muscle. But taking the functional anatomy into account, these two muscles comprise the anterolateral abdominal wall along with the three lateral abdominal muscles; external oblique, internal oblique and transversus abdominis.

Key facts about the rectus abdominis muscle	
Origin	Pubic symphysis, pubic crest
Insertion	Xiphoid process, costal cartilages of ribs 5-7
Innervation	Intercostal nerves (T7-T11), subcostal nerve (T12)
Blood supply	Inferior epigastric and superior epigastric arteries; contributions from posterior intercostal, subcostal and deep circumflex arteries
Function	Trunk flexion, compresses abdominal viscera, expiration

Transversus abdominis :



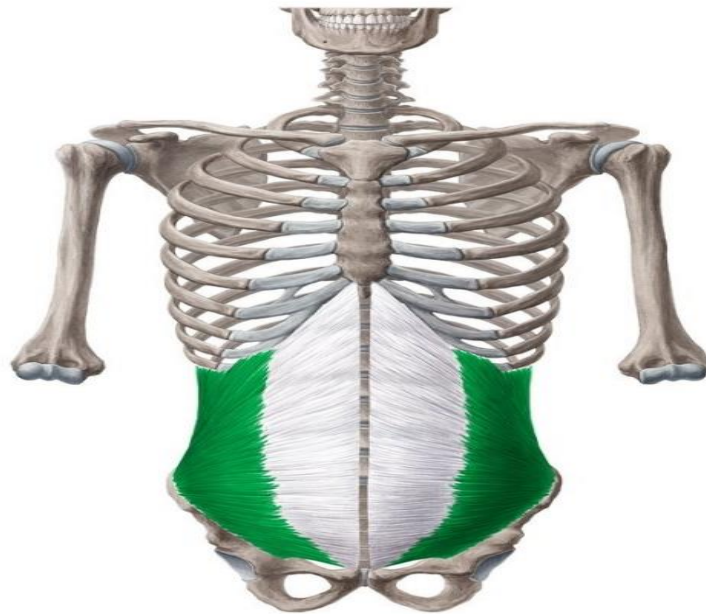
The **transversus abdominis** is a broad paired muscular sheet found on the lateral sides of the abdominal wall. Along with the external abdominal oblique and the internal abdominal oblique, it comprises the **lateral abdominal muscles**. Combined with the two anterior abdominal muscles (rectus abdominis and pyramidalis), these muscles make up **the anterolateral abdominal wall**.

As its name suggests, the fibers of transversus abdominis are oriented transversely, perpendicular to the linea alba. Together with the other abdominal muscles, transversus abdominis is important for maintaining normal abdominal tension and increasing intra-abdominal pressure.

This article will teach you all you need to know about the anatomy and function of the transverse abdominis muscle.

Key facts about the transversus abdominis muscle	
Origin	Internal surfaces of costal cartilages of ribs 7-12, thoracolumbar fascia, anterior two thirds of iliac crest, iliopectineal arch
Insertion	Linea alba, aponeurosis of internal abdominal oblique muscle; pubic crest, pectinal line of pubis
Action	Bilateral contraction - Compresses abdominal viscera, expiration Unilateral contraction - Trunk rotation (ipsilateral)
Innervation	Intercostal nerves (T7-T11), subcostal nerve (T12), iliohypogastric nerve (L1), ilioinguinal nerve (L1)
Blood supply	Lower posterior intercostal and subcostal arteries, superior and inferior epigastric arteries, superficial and deep circumflex arteries, posterior lumbar arteries

Internal abdominal oblique:



Internal abdominal oblique is a broad thin muscular sheet found on the lateral side of the abdomen. Going from superficial to deep, the external abdominal oblique, internal abdominal oblique and transversus abdominis comprise the three distinct layers of the lateral abdominal wall.

As its name suggests, the direction of its fibers are obliquely oriented, perpendicular to those of the external abdominal oblique. Together with the other abdominal muscles, the internal abdominal oblique is important for movements of the trunk, maintaining normal abdominal tension, and increasing intra-abdominal pressure.

This article will teach you all you need to know about the anatomy of the internal abdominal oblique muscle, including its origin, insertion, neurovascular supply and function.

Key facts about the internal abdominal oblique	
Origin	Anterior two-thirds of iliac crest, iliopectineal arch, thoracolumbar fascia
Insertion	Inferior borders of ribs 10-12, linea alba, pubic crest & pecten pubis (via conjoint tendon)
Action	Bilateral contraction - Trunk flexion, compresses abdominal viscera, expiration Unilateral contraction - Trunk lateral flexion (ipsilateral), trunk rotation (ipsilateral)
Innervation	Intercostal nerves (T7-T11), subcostal nerve (T12), iliohypogastric nerve (L1), ilioinguinal nerve (L1)
Blood supply	Lower posterior intercostal and subcostal arteries, superior and inferior epigastric arteries, superficial and deep circumflex arteries, posterior lumbar arteries

External abdominal oblique:



External abdominal oblique is a paired muscle located on the lateral sides of the abdominal wall. Along with internal abdominal oblique and transversus abdominis, it comprises the **lateral abdominal muscles**. In a broader picture, these muscles make up the anterolateral abdominal wall together with two anterior abdominal muscles; the rectus abdominis and pyramidalis.

Abdominal muscles work together to produce movements of the spine as well as to compress the abdominal viscera. External abdominal oblique in particular causes ipsilateral lateral flexion of the trunk and contralateral rotation of the trunk when it contracts unilaterally. Bilateral contraction flexes the trunk anteriorly, increasing intra abdominal pressure, which is useful in processes such as breathing, singing and defecation.

Key facts about the external abdominal oblique muscle	
Origin	External surfaces of ribs 5-12
Insertion	Linea alba, pubic tubercle, anterior half of iliac crest
Action	Bilateral contraction - Trunk flexion, compresses abdominal viscera, expiration Unilateral contraction - Trunk lateral flexion (ipsilateral), trunk rotation (contralateral)
Innervation	Motor: Intercostal nerves (T7- T11), Subcostal nerve (T12) Sensory: Iliohypogastric nerve (L1)
Blood supply	Lower posterior intercostal arteries, subcostal artery, deep circumflex iliac artery

Splenius cervicis:



Splenius cervicis is a paired back muscle found in the prevertebral space of the neck. Together with splenius capitis, it forms the **superficial layer** of the deep (intrinsic) back muscles, thus covering the other deep back muscles in the cervical region of the back.

The **function** of the splenius cervicis is to coordinate the position of the head and neck during various body movements, such as standing up from a sitting position. It does so by producing the lateral flexion, rotation and extension of the neck.

This article will discuss the anatomy and function of splenius cervicis muscle.

Key facts about the splenius cervicis muscle	
Origin	Spinous processes of T3 – T6 vertebrae
Insertion	Transverse processes of C1 – C3 vertebrae
Action	Unilateral contraction: lateral flexion and rotation of neck (ipsilateral) Bilateral contraction: extension of neck
Innervation	Lateral branches of posterior rami of lower cervical spinal nerves
Blood supply	Vertebral, occipital, superior intercostal, deep cervical, transverse cervical arteries

Splenius capitis:



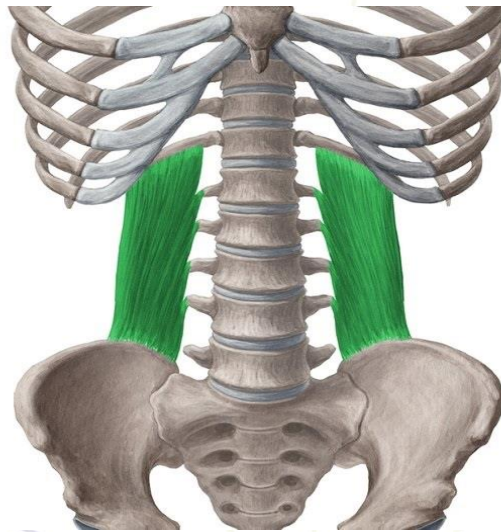
Splenius capitis is one of the **deep**, or **intrinsic muscles** of the back of the neck. It is located in the **superficial layer**, meaning that it is closer to the surface compared to the rest. The name originates from the Greek word 'Splenion' meaning bandage, and the Latin word 'caput' meaning head, hence it has a bandage-like appearance.

Splenius capitis extends from the spinous processes of the last cervical and first three thoracic vertebrae until the occipital and temporal bones of the skull. As a result, it acts as an **extensor** and **lateral flexor** of the **neck**, and assists with its **rotation**. The muscle also forms part of the floor of the posterior triangle of the neck.

In this article we will discuss the gross and functional anatomy of the splenius capitis muscle.

Key facts about the splenius capitis muscle	
Origin	Spinous processes of vertebrae C7-T3, nuchal ligament
Insertion	Lateral superior nuchal line of occipital bone, mastoid process of temporal bone
Action	Bilateral contraction: Extends head/neck Unilateral contraction: Lateral flexion and rotation of head (ipsilateral)
Innervation	Lateral branches of posterior rami of spinal nerves C2-C3
Blood supply	Muscular branches of occipital artery

Quadratus lumborum:



The **quadratus lumborum muscle** is a muscle of the posterior abdominal wall lying deep inside the abdomen and dorsal to the iliopsoas. It is the deepest muscle of the posterior abdominal wall, and it is often mistakenly referred to as one of the muscles of the back. Its shape is irregular, but is generally quadriangular, which is the reason why it is described as '*quadratus*' in latin.

Besides the spine, it also attaches to the twelfth rib, which makes it very important for **stabilization** of both vertebral column and the rib during various movements of the spine. In order to palpate the muscle one needs place the fingers above the posterior iliac crest at the level of the hip.

This article will discuss the anatomy and clinical importance of the quadratus lumborum muscle.

Key facts about the quadratus lumborum muscle

Origin	Iliac crest, iliolumbar ligament
Insertion	Inferior border of rib 12, transverse processes of vertebrae L1-L4
Innervation	Subcostal nerve (T12), anterior rami of spinal nerves L1-L4
Blood supply	Lumbar, median sacral, iliolumbar and subcostal arteries
Function	Bilateral contraction - fixes Ribs 12 during inspiration, trunk extension Unilateral contraction - lateral flexion of trunk (ipsilateral)

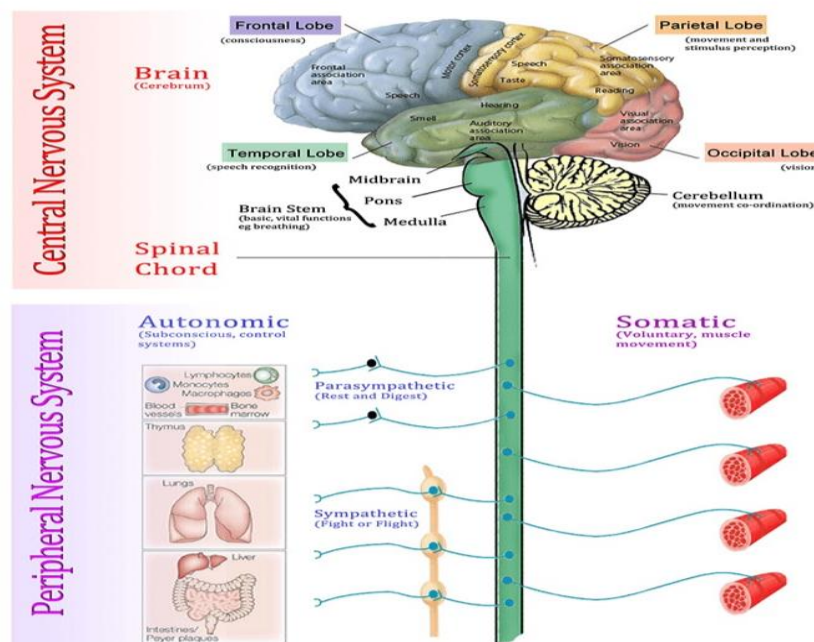
The Nervous System :-

Introduction and Goal :

The **nervous system** is the command center of the body. It plans, initiates and synchronizes our movement and is the first of our bodily systems to significantly adapt due to exercise. This work is performed by specialty cells (nerve cells or **neurons** located throughout the body) which are able to process and transmit information between the central nervous system and the body. When an individual reaches for a dumbbell to do a biceps curl, it is the nervous system response that activates the appropriate number and type of muscle fibers to perform the movement. Neuromuscular adaptations (the interaction and adaptations of the nervous and muscular systems) are primarily responsible for initial gains in strength, balance, agility, and speed during the first few weeks of training. Ironically, this essential system is often neglected by trainers in exercise program designs. Many trainers create “on the fly” or “cookie cutter” programs. This is NOT personal training. In order to create customized, personal training programs, the trainer must recognize the interactions of the nervous system and the entire kinetic chain. The goal of this section is to review the basic organization of the nervous system and to demonstrate the importance of a proper, “neurologically enriched” exercise program.

Structure and Function of the Nervous System

Central Nerv Training Adaptations to the Nervous System Central and Peripheral Nervous System



The type of neural adaptation which occurs depends upon the stimulus placed on it. This is what training is – the body's adapting to a new stimulus. In fact, the first results obtained from a new training stimulus are neural adaptations. Early strength gains (within the first few weeks) from resistance training are not due to increases in muscle mass. The body adapts to a new stimulus with more efficient movement or enhanced ability to move greater loads due to the improved communication between the nervous and musculoskeletal systems, if the training stimulus has been correctly applied. This method of adaptation is called the **Law of Facilitation**. The law of facilitation states that when an impulse passes through a given set of neurons to the exclusion of others, it will tend to do so again, and each time it transverses this path, the resistance will be smaller. The body will then adapt and be able to respond with greater ease and increasing ability.

Trainers must understand how an individual learns. Anytime an individual learns something new, several mistakes are often made. This is normal and called the **cognitive phase of learning**. In this phase the individual must think about the action, requiring the involvement of the central nervous system. Movement is usually gross and uncoordinated and relies on visual and verbal cues. However, with continued practice, less errors or mistakes are made. This is called the **associative phase of learning**. There is greater consistency in replicating the movement while less visual cues and greater **proprioception** are required. Proprioception is an awareness of one's position in space. Ultimately, with continued (quality) practice, an individual will perform movements "flawlessly." This is called the **autonomic phase of learning**. The skill is now performed automatically. Conscious thought is not required, which is the ultimate goal in performing efficient movement. This is known as the **Fitts and Posner Three-Stage Model of Motor Learning**.

What does this mean? The more an action is repeated, the more efficient the brain becomes in communicating with the muscles, and the message between the brain (central nervous system) and muscles (musculoskeletal system) is enhanced. Personal trainers should view this as the process of learning technique. If a person learns poor technique and repeats it continually, the client will only be good at poor technique. That's why teaching proper form from the beginning is the key. Trainers must not only teach good form, but the importance of good form. Proper form (technique) leads to the development of proper motor skills and proper motor skills lead to **mechanical efficiency** (see next section).

The more appropriately the nervous system is challenged, the better the overall adaptation. In other words, exercises which challenge overall balance and coordination are going to allow for better neuromuscular adaptation and improved bodily control. All other movements should see improvement. Movement starts with the brain. If a trainer is able to have their client "realize" and "become conscious" of what is happening, this will lead to more effective and efficient exercise execution and result in an increase in performance. This is no easy task. This is your challenge.

Lastly, when talking about neuromuscular development, we must mention power. Remember, power is the rate that work is performed and What is going to determine how fast and powerful someone is able to move? A major factor is how quickly their brain can tell their muscles to fire.

What is Mechanical Efficiency?

In the introduction chapter, the job of the personal fitness trainer is defined as maximizing your client's performance safely and efficiently. It was later pointed out that Maximum Performance will be determined by Mechanical and Physiological Efficiency.

In physics, mechanical efficiency measures the efficiency of a machine. Mechanical efficiency is a ratio of the work output to work input.

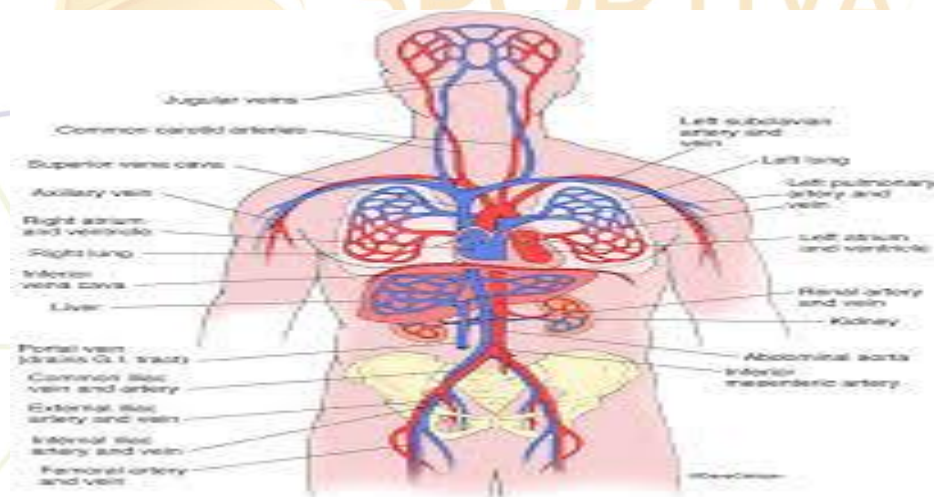
$$\text{Mechanical Efficiency} = \frac{\text{Work Output}}{\text{Work Input}}$$

So if a machine is 100% efficient, all the work going into it would be transferred to work output (an "ideal machine"). Well, there are no ideal machines and the human body is far from ideal. That said, we want to be as efficient as possible. We don't want "wasted" energy.

in and spinal cord

Peripheral nervous system

Circulatory System



The circulatory system serves as the body's transportation system. The heart, arteries, veins and blood vessels are parts of this system. The circulatory system consists of two subsystems: the cardiovascular system and the lymphatic system. In the cardiovascular system, the heart pumps blood through a vast network of blood vessels. The lymphatic system helps preserve fluid balance within the body and protects against disease.

How Respiratory and Circulatory Interaction Works

From the lungs, the oxygen goes directly into the bloodstream—the “assembly line” of the body. The lungs contain millions of tiny air sacs, called **alveoli**, around which the blood flows. These sacs are like tiny balloons filled with air dangling in the liquid of the bloodstream. The air is forced into these sacs by atmospheric pressure. Then, following the **law of gaseous diffusion**, the oxygen moves from the area of higher pressure in the alveoli to the red blood cells, where the pressure is lower. Going back to the dairy farm analogy, the red blood cells are now in effect “empty bottles,” having delivered their supply of oxygen and disposed of the returning wastes.

alveoli: Capillary-rich air sacs in the lungs where the exchange of oxygen and carbon dioxide takes place.

Limiting factors include the number of red blood cells, the amount of hemoglobin they carry, and total blood volume. Even if your lungs could process more oxygen, your body tissue still would not receive more oxygen unless there were more “bottles” to put it in for delivery.

This is another benefit of the training effect. It produces more blood, resulting in: more hemoglobin, which carries the oxygen; more red blood cells, which carry the hemoglobin; more blood plasma, which carries the red blood cells and, consequently, more total blood volume. Laboratory tests have repeatedly shown that people in good physical condition invariably have a larger blood supply than deconditioned people of comparable size. An average-sized person may increase his or her blood volume by nearly a quart in response to aerobic conditioning, and the red blood cell count increases proportionately. So, now there are not only more “bottles” to deliver the oxygen, but more “empties” to carry away the wastes. The removal of carbon dioxide and other waste products is just as important in reducing fatigue and increasing endurance as is the production of energy. It is like your home. Even if you stock your pantry with good food, you still have to clean out the garbage regularly to maintain a livable space.

law of gaseous diffusion: Law stating that a gas will move across a semipermeable membrane (e.g., alveolar, capillary) from an area of higher concentration to an area of lower concentration.

The process by which the “bottles” get to the tissue level, unload the oxygen, and pick up the wastes from the tissue cells is called **osmosis**. The oxygen and food particles, now in liquid form, pass through the cell membrane, and waste products exit the cell in the opposite direction. That basic life cycle is represented here: Materials for nourishment and energy go in, and leftover wastes go out. To complete the cycle, when the carbon dioxide and other wastes are carried away in the bloodstream through the veins and they reach the lungs, the law of gaseous diffusion now works in reverse. The pressure of the carbon dioxide in the veins is greater than it is in the alveoli, so it passes freely into the alveoli and is exhaled with the expired air.

osmosis: The scientific process of transferring fluid between molecules.

The efficiency of this cycle, and its capacity for gas exchange, is a function of the training effect. The more exercise the body does, the stronger the training effect will be; the less exercise the body does, the weaker it will be.

In a treadmill test, conditioned adult males would start with a diastolic pressure of 70 and experience only a slight increase during their run. Then, upon stopping, they would return to normal within a few minutes. However, deconditioned people—and especially the overweight types—might start with a diastolic pressure of 90, then shoot up to 105 during exercise, and take 10 minutes or more to recover.

diastolic pressure: Pressure exerted on the walls of the blood vessels during the refilling of the heart.

All of these processes are occurring with the body at rest and the heart beating at a normal rate. Physical activity and emotional stress raise the heart rate. They also raise the blood pressure because the heart is pumping more blood into the system at a faster rate. Excessive demands on the heart can cause trouble in people with pre-existing medical conditions.

Years ago, treatment for high blood pressure was rest and relaxation. However, recent reports, such as the Surgeon General's Report (1996) suggest that regular exercise can be an effective means of reducing high blood pressure. Most people, especially those with clinical conditions, reduce their blood pressure significantly after adhering to an exercise program for even a few weeks. The blood vessels make compensatory adjustments to handle the increased workload because of the exercise they get regularly. Almost all of the body systems do so in response to increased stress; this adaptive response is the training effect.

Another effect of conditioning on the blood vessels is an augmented blood supply made possible by the creation of new routes (called vascularization) supplying blood to the working muscles. This vascularization is the most essential factor in building endurance and reducing fatigue in the skeletal muscle. Saturating the tissue with oxygen and carrying away more waste is a crucial factor in the health of the heart, the most important muscle of all. Larger blood vessels supplying the heart tissue with energy-producing oxygen considerably reduce the chances of cardiac failure. Even if a heart attack were to occur, the improved blood supply would help to keep surrounding tissue healthy and improve chances for a speedy recovery.

One final problem involving the blood vessels is **fat metabolism**. As discussed in Unit 1, "metabolism" is a big word with a reasonably simple meaning: It means change. You have already been introduced to one type of metabolism: energy metabolism, where foodstuffs are burned by oxygen and converted into energy. Another form of metabolism is tissue metabolism, in which foodstuffs are changed to make new tissue.

fat/lipid metabolism: A metabolic process that breaks down ingested fats into fatty acids and glycerol and then into simpler compounds that can be used by cells of the body for general bodily function as well as energy production.

Fat is one of the foodstuffs; proteins and carbohydrates are others. Dietary fat is important because it is one of the major factors in the development of arteriosclerosis. It is also important in the development of cholesterol. The crust found on the inner walls of arteries in arteriosclerosis (hardening of the arteries) contains large amounts of cholesterol.

The body can tolerate and easily metabolize a moderate amount of fat. But as you will learn later in this text, high-fat diets strain its metabolic capabilities. When this happens, fat circulates in the bloodstream for prolonged periods following fatty meals, and how long it takes to get rid of it depends on the body's condition.

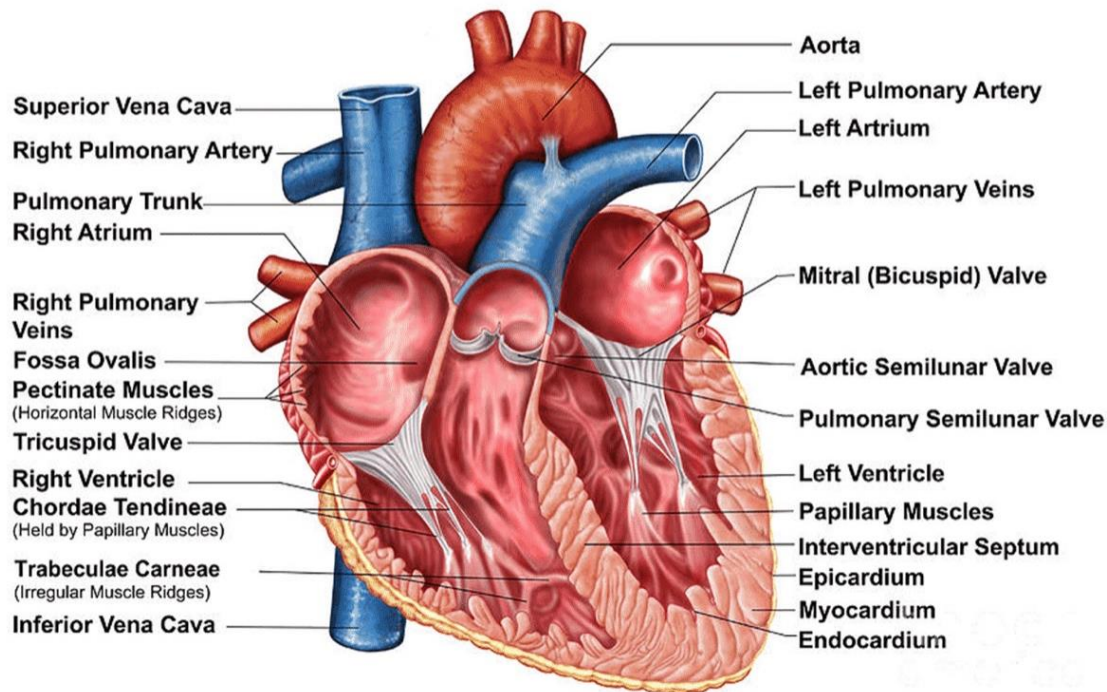
The training effect does three things for blood vessels:

1. It enlarges them and makes them more pliable to pressure.
2. It increases their number for saturation coverage.
3. It helps keep their linings clear of corrosive materials.

Tissues are the end of the assembly line, where the oxygen is turned over to the consumer and the waste products are picked up for carting away. Each cell is like a small factory; it has its own receiving and shipping facilities, storeroom, and power plant for creating energy, heat, and new protoplasm—the stuff of which all cells and all living things are made. All the food you eat and all the oxygen you breathe is meant to serve this one tiny little factory.

Heart:

Heart Anatomy



The heart is the magnificent engine that keeps the whole assembly line going. It takes oxygen-laden blood from the lungs and pumps it throughout the body, and it takes carbon dioxide-laden blood back from the body and pumps it into the lungs where it is exchanged for more oxygen.

The heart begins working before birth and continues to work until death. Ironically, the heart works faster and less efficiently when you give it little to do than it does when you make more demands on it. It is a remarkable engine. Both **anaerobically** and **aerobically** conditioned people who exercise regularly tend to have a **resting heart rate** of about 60 beats per minute (bpm) or less. A deconditioned person who does not exercise may have a resting rate of about 80 bpm or more. Women tend to have a slightly higher heart rate than men, as do children. Even though you may appear to be in great condition, obesity, stress, and many other factors can speed heart rate considerably.

anaerobic: Occurring without the use of oxygen.

aerobic: Occurring with the use of oxygen, or requiring oxygen.

resting heart rate: The number of times the heart beats in one minute: 72 beats per minute for the average adult.

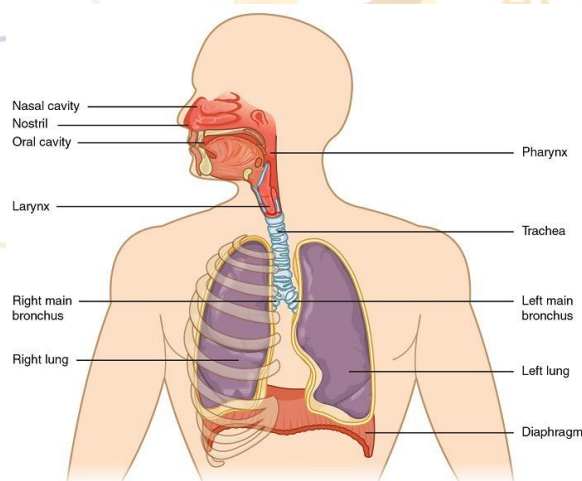
Suppose that the two people in the previous example were at complete rest for a full 24 hours. A comparison between their two resting heart rates would look like this:

- **Conditioned person**
 $60 \text{ bpm} \times 60 \text{ min} = 3,600 \text{ beats per hour (bph)}$
 $3,600 \text{ bph} \times 24 \text{ h} = 86,400 \text{ beats per day (bpd)}$
- **Deconditioned person**
 $80 \text{ bpm} \times 60 \text{ min} = 4,800 \text{ bph}$
 $4,800 \text{ bph} \times 24 \text{ h} = 115,200 \text{ bpd}$

Even at complete rest, a deconditioned person who does not exercise the heart forces it to beat nearly 30,000 times more during every day of life. But no one is at complete rest 24 hours a day, and for ordinary activity such as getting up from a chair, walking across the room, and climbing a flight of stairs, the deconditioned heart would beat proportionately faster than a conditioned heart for the same activity.

What have you done for your heart lately? In considering how healthy your heart is, you must look at two factors: the tissue itself and the number of times it beats during rest or exercise.

Respiratory system:



The respiratory system consists of the lungs and air passageways leading to and from them: mouth, throat, trachea, and bronchi. The respiratory system supplies oxygen and eliminates carbon dioxide to tissues in helping to regulate the acid-alkaline (pH) balance of the body. Respiration is the overall exchange of gases between the atmosphere, the blood, and the cells.

It all begins with the lungs. This is where the air you breathe is processed; the oxygen is removed and then transferred to the bloodstream for distribution throughout your body.

The amount of air that your lungs can process is the first limiting factor on your physical condition.

To understand how conditioned lungs can process more air, you need to understand how breathing works. Think of the lungs as a dairy in which bulk milk comes in and the cream is separated from it. The cream is then bottled and sent off for distribution. Empty bottles come back, get flushed out, and receive more cream—and the cycle begins again.

Think of oxygen as the “cream” of the air you breathe. When bulk air comes into your lungs, the oxygen is extracted from it, “bottled” in red blood cells (**hemoglobin**), and then sent off on the bloodstream assembly line for distribution. When they reach the tissue, the “bottles” exchange oxygen for carbon dioxide and water and then carry these wastes back to the lungs, where they are flushed out. The “bottles” are then ready to pick up more oxygen and begin the cycle again.

The air you breathe is approximately 21% oxygen and 79% nitrogen, with negligible traces of other gasses. This ratio rarely varies. What does vary is the amount of air you can process. If your lungs cannot process enough air, they cannot extract enough oxygen to produce enough energy. Two factors limit the lungs’ ability to process air.

First, the lungs have very little muscle of their own. Expansion and contraction of the lungs depends on the muscles of the rib cage and the diaphragm. As you inhale, the muscles surrounding the lungs create a larger area in the lung cavity, thereby creating a partial vacuum. Aided by this differential in atmospheric pressure, air then rushes in. When exhaling, the muscles, aided somewhat by the natural elasticity of the lungs and chest wall, contract to create greater atmospheric pressure inside the lungs than outside your body. Inhaling is the air is being “sucked” in; exhaling is the air being “pushed” out.

The process described occurs with the body at rest. Most bodies at rest consume basically the same amount of oxygen, and consequently they inhale and exhale just about the same amounts of air. Now, as you move into physical activity, the amount of air you can inhale and exhale is limited. The first limiting factor is the size of the vacuum your muscles can create for the lungs to expand into; the second is the size of the area they can be squeezed back into.

Conditioned athletes have the capability to inhale more air and sustain the process for longer periods. Conditioned athletes are also more capable of exhaling more waste because the muscles surrounding their lungs have been trained and thus are more efficient.

The second limiting factor on how much air the lungs can process is the condition inside the lungs. Lungs vary in size; a larger person naturally has proportionately larger lungs than a smaller person. In terms of sports performance, the concern is less about the size (total capacity) of the lungs than with how much of that capacity is usable. This usable portion is called the **vital capacity**, and it is measured in the laboratory by assessing the amount of air that can be completely exhaled in one deep breath.

Research has shown that a conditioned person has a vital capacity equaling approximately 75% of his or her

hemoglobin: An oxygen-transporting protein found in blood cells.

vital capacity: The usable portion of the lungs.

total lung capacity. However, a deconditioned person may match this percentage by virtue of genetics. To differentiate between the two individuals, you must look at the **maximum minute volume**, the amount of air that a person can process during 1 minute of vigorous exercise. The results of this test generally provide a clear indication of who is the conditioned individual and who is the deconditioned individual. Conditioned athletes may force as much as 20 times their vital capacity through their lungs in 1 minute, whereas deconditioned individuals might be hard-pressed to force even 10 times through. They simply lack the muscle strength and endurance to perform at any higher level.

maximum minute volume: The amount of air that a person can process during one minute of vigorous exercise.

After usable lung volume has been measured, the remainder of the air in the lungs is called the **residual volume**. This volume is fixed, and even a conditioned athlete cannot breathe it in or out. However, too much residual volume is unhealthy. If your body deteriorates from inactivity or disease, the unusable portion of the lungs may increase, providing less space for normal breathing, let alone vigorous exercise. Ultimately, shortness of breath results from even light activity, such as climbing a flight of stairs. When you need more oxygen in a hurry, the muscles controlling the lungs will not be in condition to force high volumes of air through them, and the usable space within the lungs may be seriously reduced.

residual volume: The remainder of the air in the lungs after the usable lung volume has been measured.

The training effect can reverse both trends. Exercising the muscles surrounding your lungs increases their strength and efficiency and helps open more usable lung space. It has the net effect of increasing your vital capacity and reducing the residual volume. In each instance, it makes your lungs more efficient organs to process more air and extract more of the essential oxygen. The oxygen supply to the blood at rest is only about 1 cup per minute. Extreme exercise in a trained athlete can step this number up to 1 gallon per minute. At rest, only about 12 percent of the stagnant air in the lungs is renewed during each breath.

A good way to test the breathing condition of your lungs is to take a deep breath and see how long you can hold it. Most adults in moderately good physical condition and with healthy lungs should be able to hold their breath for 50 seconds or longer. Most individuals in average condition have a respiration rate of 10 to 16 breaths per minute. Respiration rate is measured at rest with the subject breathing within a normal resting heart rate. The training effect can have beneficial effects on normal breathing, resulting in fewer breaths per minute. Exercising the muscles surrounding the lungs increases their strength and efficiency, which increases usable lung capacity. It has the net effect of increasing the vital capacity and reducing residual volume. The training effect thus transforms the lungs into more efficient organs that are capable of processing more air and extracting more essential oxygen.

Common Workout Injuries: How To Treat And Prevent Them:

Working out is a great way to take care of one's health. It keeps one physically fit as well as mentally and emotionally balanced. There are many ways that one can choose to exercise depending on their physical requirements and comfort level, but no matter which type of exercise you choose, it is crucial to take special precautions against workout injuries.

Major or minor, every injury is painful and can set a person back. While the causes for the workout injury may vary, what is important is that every injury should be given timely care and treatment to avoid the condition from getting more serious. For somebody who is passionate about fitness, an injury setback can be quite a frustrating deal. But with proper treatment, the injury can heal quickly, letting the person get back to their workout routine as soon as possible.

Below are **25 common workout injuries** that an individual can encounter, along with the treatment process and preventative measures that can be taken to avoid injuries.

Workout: Running

Injury: Runner's Knee

This is one of the most common injuries for runners. Irritation of the cartilage under the kneecap causes 40% of running injuries. Runner's knee happens when the kneecap is out of alignment. Over a period of time the cartilage under the kneecap wears down, which causes pain when climbing up or down the stairs, squatting, or sitting with the knees bent for a long time. The pain experienced underneath the kneecap is also known as patellofemoral knee syndrome, and the pain can get worse with an increase in the intensity of the pain.



Treatment: As a short term remedy, people suffering with runner's knee should hit the pause button on running to alleviate the pain. Putting an ice pack for 20-30 minutes in 3-4 hours interval for few days can help cut down the pain. Use patellar straps or an elastic bandage to give the knees extra support, and anti-inflammatory drugs like Advil, Aleve, and Motrin can take care of the pain and swelling. Individuals could also practice stretching and strengthening exercises for the hips, glutes, and quad muscles. Severe cases may need surgery to correct the position of the kneecap so that stress is distributed evenly.

Avoidance Tactics: Avoid putting excess weight on the knee, and avoid kneeling or squatting repeatedly. Keep the knee in an elevated position when sitting or lying down. Switching up the surfaces you run on, using short strides, wearing the proper footwear, and striking the ground directly underneath can help avoid this injury.

Injury: Shin Splints

This injury is common in runners but is also found in people who participate in running sports or jumping. It is an umbrella term that indicates various ailments involving pain in the shin area. It is characterized by inflammation of the muscle which causes pain along the inner edge of the shinbone. But in the worst case scenario it can turn into a stress fracture along the tibia, causing searing pain with every stride. This injury is commonly found among new runners and at the start of high school track or cross-country season. If the muscles surrounding the shin are tight, it puts more stress on the entire area.



Treatment: Doctors will take X-rays and bone scans to rule out any fracture. Icing the shin will help ease the pain. Painkillers and anti-inflammatory drugs like Ibuprofen and Naproxen, or an aspirin also help relieve the pain and reduce swelling. Physical therapy is usually recommended to treat the issues in the legs and running mechanics that cause shin splints. In some rare, severe cases, surgery is needed to take care of the stress fracture that led to shin splints.

Avoidance Tactics: A sudden spike in the intensity and frequency of workouts is one of the greatest causes of shin splints. So, the frequency and intensity of the exercise should be increased gradually and methodically. New runners' lower legs aren't strong enough, so they need to develop a firm base before increasing mileage or introducing speed work. Always do warm-ups like jumping jacks to get the blood flowing and the muscles warm and ready, and also stretch the leg muscles after the workout. Wear proper footwear with good support and padding. If you are getting back to running after this injury, you need to ease back into it gently.

Injury: Achilles Tendinitis

Inflammation of the large tendon that attaches the calf to the back of the heel, the Achilles tendon, is called Achilles Tendinitis. This injury is characterized by pain and stiffness at the back of the lower leg, just above the heel at the Achilles tendon. Tight calves put a lot of strain on the Achilles tendon, and hard training over a period of time leads to this injury. Wrong, poorly supported footwear, or a sudden increase in volume and intensity of training also puts a lot of stress on that area, causing the same effect more quickly. Individuals with this injury often complain of sharp pain and swelling close to the heel.



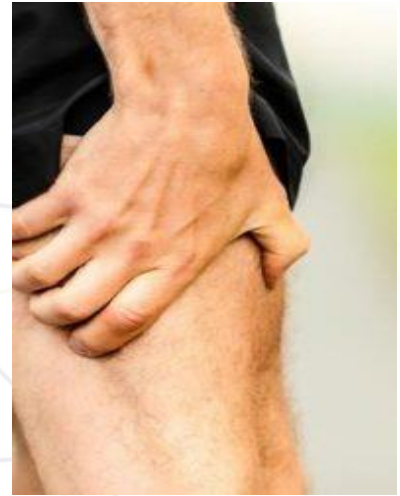
Treatment: Individuals suffering from this injury should focus on strengthening the lower legs by doing exercises like calf raises, single-leg dead lifts, single-leg squats, and box jumps. Icing the area and taking aspirin or Ibuprofen helps with relieving the pain and decreasing the swelling. If done right, self-massage also aids in easing the pain. If the injury doesn't respond to home-based treatment in two weeks, you need to consult a physical therapist or orthopedic surgeon. Surgery to scrape the scar tissue is employed as a last resort treatment.

Avoidance Tactics: Pay good attention to the footwear used for running, as the wrong choice of footwear can aggravate the injury. A dramatic and gradual increase in training, and incorporating exercises that strengthen the calf muscles into the workout routine will reduce the risk of this injury. Pay close attention to the training intensity and frequency. Don't do too much or too quickly. Avoid weight bearing exercises and hill work.

Injury: Pulled Hamstring

This is quite a common injury among athletes, including runners. The hamstring is a group of four muscles that run along the back of the thigh which allows the leg to bend at the knee. When one or more of those muscles get overloaded it leads to hamstring strain, and the muscles are also likely to tear. If the strain is mild, the pain might be just bearable. But in certain cases the pain could be unbearable, making it nearly impossible to walk or even stand.

Individuals could also experience a snapping or popping feeling while exercising. They also experience tenderness around that region and pain in the back of the thigh and lower buttock while walking, bending over, or straightening the leg.



Treatment: Ice the leg and use an elastic bandage around the leg to keep the swelling down. Anti-inflammatory drugs like Ibuprofen, Advil, Aleve, and Motrin give relief from pain and swelling. Ensure not to put any weight on the leg as much as possible, and keep the leg elevated while sitting or lying down. If the pain is severe, crutches may be recommended to ease the stress on the legs. You could also consult a physical therapist, who can recommend some strengthening exercises for the hamstring to protect against the strain. In cases of severe injury, surgery may be required to repair the muscles and reattach them.

Avoidance Tactics: Do warm-up exercises before and after running. Increase the intensity and volume gradually and in a systematic manner. Incorporate strengthening exercises for the leg muscles, especially the hamstrings as a *preventive* measure. Stop exercising if you feel any kind of pain. While healing the strain, work the hamstring with a new activity that won't aggravate the strain, as pushing too much before a full recovery could re-injure the hamstring, leading to permanent muscle dysfunction.

Injury: Plantar Fasciitis

Plantar Fasciitis is inflammation of the bottom of the feet, and one of the peskiest problems that runners encounter. This condition is characterized by a sharp, tight painful sensation at the base of the heel like stepping on a nail. The pain can vary from moderate to excruciating. Over-training, overuse, and wrong choice of footwear can lead to this injury. The tight and weakened muscles in the foot cause the heels to take on an excessive load. Repeated strain can cause tiny tears in the ligament, causing weakness, pain, and inflammation.



Treatment: The treatment aims to relieve the pain and inflammation, heal the small tears in the ligament, and correct foot problems like excessive pronation, so that there is no stress on the plantar fascia ligament. Limit, and if possible, stop the daily activities causing heel pain, like running or walking on hard surfaces. Orthotics and shoes with good shock absorption and the right arch support for the feet help eliminate the symptoms in the short run. Rolling the foot on a golf ball, icing the injured area, and stretching and strengthening the calf muscles will help provide relief. Toe stretches, calf stretches, and towel stretches after getting up in the morning can help ease the pain.

Avoidance Tactics: Maintain a healthy weight for your height. Make sure to wear supportive shoes with good arch support and heel cushioning. Wear supportive shoes as soon as you get out of bed. Avoid being barefoot for long periods of time as it puts stress on the feet. Also alternate running with other sports that don't put stress on the heels.

Workout: Cycling

Injury: Iliotibial Band Syndrome

This is an overuse injury common among cyclists. This injury occurs when the ligament that runs along the outside of the thigh from the hip to the shin becomes tight and inflamed. Cycling causes a flare-up of this symptom due to incorrect cycling position or a saddle that is too tall. Pain is felt when bending and straightening the knee, and weakness is felt in hip abduction or moving the leg out sideways. Naturally wide or tight IT bands or poor foot bio-mechanics makes one more prone to this injury.



Treatment: Individuals with this syndrome need to give their body complete rest, as cycling with this syndrome will only make it worse. Icing the affected area during the injury alleviates pain and inflammation, and icing after exercise is a good way of ensuring the pain doesn't return. Anti-inflammatory medicine like Ibuprofen provides relief from acute pain and swelling. Doing strengthening exercises to strengthen the *tensor fasciae latae muscle* and gluteus medius help reduce the friction on the IT band tendon at the knee. Sports massage by a professional therapist can help relax and loosen the tissues.

Avoidance Tactics: Bike set-up is the major key in avoiding this syndrome. Saddle height dictates knee position. The knee over-straightens if it is too high, or over bends if it is too low. The seat height should be such that your torso is in an ideal position, and you are able to reach the bar without being bunched up. Over training or increasing the mileage too quickly should be avoided.

Injury: Lower Back Pain

Long hours in an aggressive riding position, and the repeated and prolonged position are the major causes of lower back pain from cycling. The flexed position while cycling helps maintain a good aerodynamic performance and generate the force to pedal, this causes cyclists to suffer from lower back pain. Excessive flexion in the lumbar region of the spine can lead to herniated lumbar discs, nerve entrapment, and sciatica. The spinal column inserts into the pelvic bone, which means having weak muscles can prevent cyclists from generating the power needed from the core for aggressive riding positions. Hence, the supporting muscles are forced to compensate for the weakness leading to lower back pain.



Treatment: Treatment for this depends on the individual's history and the severity of the pain. Putting a stop to all activities, especially cycling, will give time for the injured tissue and the nerve roots to begin healing, which in turn will relieve the back pain. Heat and/or ice packs give relief from the nagging pain and reduces inflammation. There are also many over-the-counter and prescription medicines available that help with the pain. There are medications that help with inflammation which causes pain, and some treatments inhibit the transmission of pain signals from reaching the brain. Exercising to build strength and flexibility, usually under the guidance of a trained physical therapist, is an essential element of any treatment plan.

Avoidance Tactics: A proper bike set-up is the most crucial aspect to avoid over-reaching if the frame is too large, or a hunched posture if the frame is too small. Watch the posture as slumping makes it harder to support the weight. Also, be extremely careful of the posture when lifting heavy objects. Do not bend over from the waist. Bend and straighten from the knees. A good core-strengthening exercise should be part of a cyclist's regimen that will help them maintain riding positions for longer periods, while increasing balance and power.

Injury: Foot Numbness

Quite common among the athletes, foot numbness is a loss of feeling in the feet. It is mainly caused due to ill-fitting cycling shoes squeezing the metatarsal heads and cleats placed too far forward causing increased pressure around the ball of the foot. Cycling technique involving low cadence and excessive hill riding also causes this occurrence in cyclists. Damage, blockage, infection, or compression of a nerve that travels to the foot causes the numbness. The main symptom of this injury is loss of sensation in the foot along with additional symptoms like prickling, pins-and-needles, and a tingling sensation.



Treatment: Treatment for foot numbness depends on an accurate diagnosis and treatment of the underlying cause. Following a medically-supervised exercise routine can help with the numbness. Correctly fitting shoes will ensure proper flow of blood to the feet, thus preventing the numbness. Proper positioning of the cleats ensures the pressure is not focused on the balls of the feet. Gentle massage of the numb area and warming the feet will aid proper circulation, thus reducing the numbness. However, if numbness occurs frequently, you should consult a doctor to rule out any other underlying causes of pain.

Avoidance Tactics: Avoiding tight clothing and ill-fitting shoes while cycling will go a long way in preventing numbness. Ensure the cleats are properly positioned so that there isn't excessive stress on the balls of the feet. Hill climbing involves excessive push phases of cycling, which increases the pressure on the feet, causing numbness. Hence, hill climbing should be tapered to alleviate the problem.

Injury: AC Joint Sprain



The Acromioclavicular (AC) joint is a part of the shoulder, which consists of the collarbone connecting to the front of the shoulder blade with strong ligaments. An injury to one of these ligaments causes AC joint sprain. The sprain ranges from very mild injury (grade 1) to a severe case (grade 6). Individuals will experience symptoms like pain right at the end of the collarbone, localized to a bony point on top of the shoulder. Usually a large force like a forceful fall or a monstrous dive into a pothole or off-road obstacle causes the connective tissue and ligaments of the AC joint to tear. The elbows and wrists are generally locked in position when holding the handlebars on a bike. So, when a large force is applied while cycling, these forces are transferred up to the shoulder joint.

Treatment: Applying ice or cold therapy and compression wrap immediately to the affected area will reduce the pain and swelling. A doctor may prescribe Ibuprofen to take care of the pain and inflammation. AC joint taping fixes and supports the joint into the correct position to assist the healing process. The taping may be needed for 2-3 weeks. Once the pain subsides, exercises specifically recommended for this injury should be followed. Grade 4, 5, and 6 may not respond to treatment and may require surgery. If an AC joint sprain doesn't heal adequately, it may require pharmaceutical intervention, corticosteroid injection, or a review by a specialist who can advise on the optimal procedure to treat the injury, which might also include surgery.

Avoidance Tactics: The basic preventive method for an AC joint sprain is to avoid falling. This joint is vulnerable to injury during falls. The shoulder needs complete rest, so wear a sling that will immobilize the shoulder and ease the pain by taking the weight off the arm. In case of bumps, potholes, or any other obstacle, use the elbows as a shock absorber while biking. Since the sprains have varying degrees, the severity of the injury dictates the rehab procedure.

Injury: Saddle Sores

For majority of the cyclists out there, especially the ones competing in cycling competitions, saddle sore is a common occurrence. It generally refers to problems occurring in the area where the cycling shorts' chamois contacts the body due to friction from the saddle. A saddle sore is essentially a tender spot that is usually raised, pink, or red in the area that rubs against the saddle. For a lot of cyclists, the sore looks like a pimple or an ingrown hair. These sores can hurt really bad, which makes riding a very painful experience.



Treatment: Once you get saddle sores, the best remedy is to stay off the bike for few days. The delicate skin needs time to heal without being subjected to more friction and sweat. A cool bath with Epsom salts speeds up the healing process. Once the area has been cleaned, apply tea tree oil to the affected area using cotton swabs, followed by Vaseline. A few days off the bike should calm down the inflamed area, but if it keeps coming back, it is time to see a dermatologist to see if there is any underlying problem. Sometimes if the skin breaks, bacteria gets inside, which could infect the sore. In this case antibiotics might be prescribed.

Avoidance Tactics: The two keywords for prevention is to keep everything clean and dry. Keep the area clean by taking a cold shower and allowing the area to air out completely. Lowering the saddle prevents side-to-side motion of the pelvis, which can cause friction against the seat. Find saddle shorts that work for you, and always wear clean shorts. Increase the distance biked gradually, and also use chamois cream to reduce friction and ease the discomfort of the skin rubbing against the saddle.

Workout: Gym

Injury: Carpal Tunnel Syndrome



This syndrome occurs as a result of a combination of health conditions and activities that put pressure on the median nerve as it passes through the carpal tunnel in the wrist, and this pressure leads to the various symptoms. Gym activities like push-ups, plank pose, and other exercises that involve bending the wrist forward or backward excessively decreases the amount of space in the carpal tunnel, increases the amount of tissue in the tunnel, or increases the sensitivity of the median nerve. Repeated hand and wrist movements cause the membranes surrounding the tendons to swell. Even broken wrist bones, dislocated bones, or bone spurs take up space in the carpal tunnel, putting more pressure on the median nerve.

Treatment: The goal of the treatment is to allow the individuals to prevent nerve damage and loss of muscle strength in the fingers and hands, which will enable the individual to resume normal activities. Wear a wrist splint and avoid or change up the activities, especially the gym routines. Anti-inflammatory drugs, and in some cases, oral corticosteroids or injections are also prescribed. A **carpal tunnel syndrome** surgery is also recommended when other treatments fail, or if the condition persists for a long time, leading to nerve damage.

Avoidance Tactics: Avoid exercises like push-ups and planks that involve excessive bending of the wrist. Use hand and wrist movements that spread the pressure and motion evenly. Switch hands and positions often when doing repeated motions. Use correct posture, and reduce the salt intake that leads to fluid retention. Staying at a healthy weight and keeping the arm, hand, and finger muscles strong and flexible can help avoid this injury.

Injury: Sternal Fracture



The sternum is the bone that attaches the two clavicles on either side of the upper chest and the upper seven ribs. While doing the bench press, if the bar comes crashing down on the chest, the sternum bears the brunt of the impact. A direct fall on the chest can result in a bruised sternum, or if the impact is beyond what the bone can bear, a break in the sternum may occur. A sternum fracture may be associated with damage to other structures and organs of the body like the ribs, vertebrae, lungs, and blood vessels. The fracture causes sharp and intense chest pain which increases during breathing, coughing, laughing, or sneezing. This condition may also cause swelling and pain with certain movements such as pushing, pulling, or heavy lifting.

Treatment: Individuals with a sternum fracture should undergo a thorough medical examination to rule out damage to the heart, lungs, blood vessels, or any other organs. The treatment consists of rest from strenuous activities, along with pain-relieving medicines. Surgical intervention is required in cases of bone displacement to restore the bone alignment and to fix the bones using pins or plates. Icing the fractured area in the initial stage helps ease pain and swelling. Exercises that improve posture, strength, and flexibility will help prevent stiffness and weakness from developing.

Avoidance Tactics: It is important that individuals with a sternum fracture take rest from any heavy, strenuous activities that will increase the pain and aggravate the injury. Activities that place stress on the sternum, like lying face down and applying direct pressure to the chest, should be avoided. Even after the sternum has healed, there should be a gradual return to the gym routine under the guidance of the treating physiotherapist or surgeon. Use of protective padding or chest guards may be required when returning to gym workouts.

Injury: Bicep Tendon Rupture



The bicep muscle has two tendons; one that attaches to the shoulder, and one tendon that attaches at the elbow. The tendon at the elbow is called distal biceps tendon.

When one attempts to curl a weight that is too heavy or suddenly drops the bar or dumbbell, it risks a tear in the bicep tendon. It can either be a partial tear that damages the soft tissue, but does not completely sever the tendon, or a complete tear that detaches the tendon from its attachment point at the bone. The rupture is accompanied by intense pain and swelling in the front of the elbow. There is a visible bulge in the upper part of the arm created by the recoiled, shortened biceps muscle, and a gap in the front of the elbow due to the absence of the tendon.

Treatment: The doctor will conduct an examination of the elbow, testing the strength of the forearm. Non-surgical treatment includes anti-inflammatory medications to reduce pain and swelling, and using a sling to give the arm rest from any kind of activities. Once the pain subsides, rehabilitation exercises to strengthen the surrounding muscles to restore movement and flexibility are recommended. If surgery is required, a **hand surgeon** may recommend performing it within the first 2-3 weeks, as over time the tendon and biceps muscles begin to scar and shorten, and surgery might not be effective. The surgery aims to reattach the distal biceps tendon to the forearm bone.

Avoidance Tactics: Any kind of heavy lifting should be avoided to help speed up the healing process. After the arm has healed, and the pain has subsided, the individual should gradually ease back into physical activities, including exercising. As you ease back into your gym routine, work with a weight you can handle, and work with the help of a spotter.

Injury: Pectoral Strain



This condition is characterized by a partial or complete tearing of one or more of the pectoral muscles. The pectoral muscles are a group of two muscles at the front of the chest.

During stretching or contraction of the pectorals, tension is placed on the muscles and tendons. When there is excessive tension due to repetitive movements or high force, one or more of the pectoral muscles can tear. Usually while attempting to perform dumbbell flys with a weight that is heavier than one is accustomed to lifting, it leads to a tear of the chest muscle. This injury results in losing control of the dumbbell or barbell while bench pressing. Individuals experience pain that is centered around the chest, and front of the shoulder or arm pit. Individuals also experience bruising, swelling, muscle spasm, and weakness.

Treatment: Resting from intense physical activities, and icing the strained pectoral muscle reduces inflammation and keeps the pain under control. Applying compression bandages to the chest and torso helps in further containing the swelling. Massages, dry needling, trigger point release, and electrotherapy helps to improve range of movement, pain, and function. Individuals with pectoral strain should indulge in pain-free flexibility and strengthening exercises to ensure optimal recovery. If the strain does not improve, it may require pharmaceutical intervention, corticosteroid injection, or even a surgery may be needed in case of a complete rupture of the major muscle.

Avoidance Tactics: Stretching the pectorals before working out loosens up the muscles, which helps prevent muscle strain. Once the pain and swelling has been taken care of, you need to gradually get back into your workout routine. Make sure to work only with a load that is within your control. If it wobbles or feels like it is going to drop, chances are it is too heavy. In such cases, use a spotter for heavy sets. Use protective taping to reduce the likelihood of injury.

Injury: Joint Dislocations



When a joint experiences an unexpected or unbalanced impact, dislocation occurs. It is a bone slipping out of a joint, and it can be any joint in the body, such as the knee, hip, ankle, or shoulder. Not paying attention to one's form or loading the bar with too much weight can result in a dislocated hand, shoulder, elbow, or knee. The excess pressure on the ligaments forces the bone out of its normal position. The dislocated area swells up and looks bruised. The area becomes discolored and may also have a strange shape as a result of the dislocation. It also results in loss of motion, pain during movement, numbness, and tingling sensations.

Treatment: The choice of treatment depends on which joint has been dislocated. Resting the area from any exhausting activities, and icing it or using a compression wrap gives relief from pain and swelling. The doctor repositions the joint back into place, and advises to wear a sling, splint, or cast for few days. This prevents any movement of the dislocated joint, speeding up the healing process. A pain killer or a muscle relaxant may be prescribed to help deal with the pain. If the dislocation is not given timely treatment, it can cause damage to the ligaments, nerves, or blood vessels, and surgery might be required in such extreme cases.

Avoidance Tactics: Practicing safe behavior helps prevent joint dislocation. Wearing protective gear or clothing, removing rugs from the floor or replacing them with non-skid rugs can help protect from dislocations. Do not try to move a dislocated joint back into place, except by a trained medical professional, or else it can damage the joint and the surrounding ligaments, blood vessels, and nerves. Alternate exercises for any one muscle group from one workout to the next. Ensure that the weight is light enough, such that you are able to press or pull in a smooth, steady motion to complete the set.

Workout: Sports

Injury: Cubital Tunnel Syndrome



This is also known as ulnar neuropathy, and is caused by excessive pressure on the ulnar nerve. Abnormal bone growth in the elbow or intense physical activity increases the pressure on the ulnar nerve. This syndrome is often found in baseball pitchers because of the twisting motion required to throw a slider that causes damage to the delicate ligaments in the elbow. Repeatedly leaning the elbow on hard surfaces or bending the elbow for sustained periods gives rise to this symptom. Sometimes the ulnar nerve snaps back and forth as the elbow is moved, which irritates the nerve. The symptoms of cubital tunnel syndrome include pain and numbness in the elbow, tingling in the ring and little fingers, muscle wasting in the hand, and decreased overall hand grip.

Treatment: The doctor may use electromyography as part of the **cubital tunnel syndrome treatment**, in which electrodes are placed into the muscles and on the skin to identify the area of nerve damage and determine the severity of the condition. If the electromyography reveals just minimal pressure on the ulnar nerve, then the syndrome can be treated conservatively. Use of a splint is also recommended to restrict the movement. If the nerve compression is serious, surgery might be recommended to simple decompression of the nerve, shift the nerve to the front of the elbow, or move the nerve under a layer of fat.

Avoidance Tactics: The first act of prevention is to avoid actions that cause the symptoms. A **hand specialist** can help find ways to avoid pressure on the nerve in the elbow. Restrictions are placed on lifting and elbow movements. Individuals suffering from this syndrome should avoid leaning on the funny bone. Ergonomic education like nerve-gliding exercises, stretching and strengthening exercises, and the effects of stress on the area can help avoid the injury.

Injury: **Tennis/ Golf Elbow**



Tennis elbow is the inflammation of the tendons on the outer bony prominence of the elbow and the muscles of the forearm. The pain is felt because the muscles and tendons become inflamed from overuse. In the game of tennis, a one-handed backhand with poor form, a late forehand swing that results in bending the wrist, or constantly turning the wrist for more spin on the serve results in tennis elbow symptoms. Golfers also tend to suffer from tennis elbow as pulling the club through the swing causes irritation in the elbow. Medial epicondylitis, which causes pain on the inside of the elbow, is common among golfers, baseball pitchers, and tennis players who hit topspin forehands, and weight lifters.

Treatment: Tennis elbow doesn't usually lead to serious complications, unless the condition is left untreated, which could result in loss of motion or loss of function of the elbow and forearm. Resting the elbow and tendon, and putting a cold compress or a heat pack on the elbow helps it to heal. It is also advised to wear a strap, splint, or brace around the forearm to restrict the movement, and speed the healing process of the tendon. Painkillers like Ibuprofen pills, creams, and gels give relief from pain. Gentle exercises are also recommended to increase the muscle's strength without any damage.

Avoidance Tactics: Avoiding overuse is the simplest prevention tactic for tennis elbow. Avoid using the wrong equipment like a golf club or tennis racket that is too heavy to handle or has a very large grip. Warming up before playing any sport is crucial, especially the ones that will exercise the elbow or arm. Ice the elbow immediately if there is any discomfort after the sports, and stop if you feel any pain during an activity. Avoid or minimize activities that tend to increase and aggravate the pain.

Injury: Rotator Cuff Tear

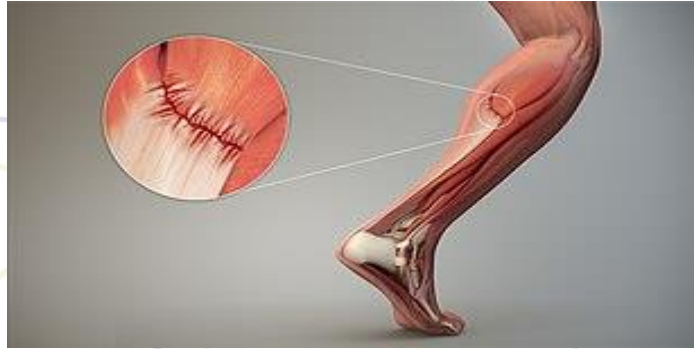


A rotator cuff tear is a common fear for pitchers, tennis players, swimmers, and football players. A severe case of this injury could cause a premature end to a player's career. The rotator cuff is a group of four tendons and muscles that converge around the shoulder joint at the top of the humerus. These tendons and muscles together form a cuff that holds the arm in place, allowing its movement in different directions. The shoulder is a weak joint, so too much stress, or too many fastballs can lead to partial tears and swelling in the tendons of the rotator cuff. Excessive stress could also cause one or more tendons to pull away from the bone or tear in the middle. This injury is characterized by pain in the shoulder and arm which depends on the severity of the tear, snapping sounds while moving the shoulder, weakness and tenderness in the shoulder.

Treatment: The **hand specialist** will conduct a thorough diagnosis to understand the cause of the pain, and formulate the appropriate treatment. The joint needs complete rest for a speedy recovery. Doctors recommend wearing a sling so as to restrict any movement, and give the joint complete rest. Even anti-inflammatory pills or corticosteroid injections are prescribed to help with the intense pain. Individuals can also do range-of-motion exercises that will strengthen the rotator cuff muscles. Strengthening the muscles will help hold the shoulder firmly in place. Physical therapy, ultrasound, moist heat, and electrical muscle stimulation are also recommended as part of the rotator cuff tear treatment. A physical therapist may suggest rehabilitative exercises to strengthen the joint. More serious cases may be referred to an orthopedic surgeon and may require surgery.

Avoidance Tactics: The long-term changes that occur in the shoulder joint because of everyday wear and tear cannot be avoided, but the serious rotator cuff tear can be avoided with daily exercise, which will give strength and flexibility to the shoulder muscles. Avoid sports and other activities that involve forceful falls till the shoulder muscles heal completely. Take frequent breaks in between the physical activities, and ice the shoulder at night to give it some relief.

Injury: Muscle Pull

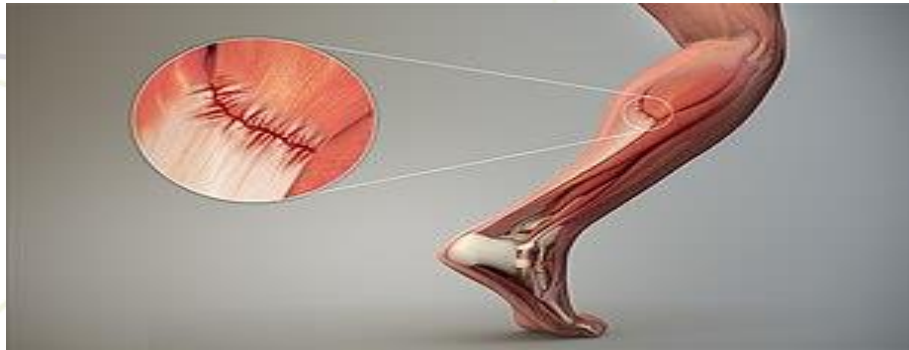


A muscle strain or pull, or even a muscle tear occurs when the muscle is overstretched or torn. A strain or pull causes damage to the muscle or its attaching tendons. A muscle pull usually happens as a result of fatigue, overuse, or improper use of a muscle. The muscles are subjected to undue pressure while playing sports like rowing, baseball, golf, and tennis, or even with sudden heavy lifting and performing other tasks. The muscle tear can be a tearing of all the muscle fibers and the tendons attached to the muscle, it can also cause damage to the small blood vessels, causing local bleeding, bruising, and pain due to irritation of the nerve endings. A muscle pull is characterized by swelling and weakness of the muscle or tendons. Pain is experienced when that particular muscle or the joint relating to the muscle is used.

Treatment: The swelling of the affected area or the local bleeding into the muscle can be treated early on by applying ice packs to close the blood vessels, and maintaining the strained muscle in a relaxed and stretched position. Heat can also be used to treat the muscle pull once the swelling has reduced. Anti-inflammatory medication like Ibuprofen or Ecotrin are prescribed to help deal with the pain and improve one's ability to move around. An elastic bandage can be used to provide support and decrease the swelling. The injured area should be kept elevated to decrease the swelling. In some extreme cases, crutches or braces may be needed as part of the treatment process.

Avoidance Tactics: Make sure to do stretching exercises and warm up before playing any sport. This helps increase the blood flow and decrease the risk of injury. Avoid activities that put excessive strain on the muscles, or activities that are painful. Make stretching a daily habit to avoid such injuries. A safe exercise program can be chalked out in consultation with the doctor that will make the muscles strong and flexible, and reducing the risk of sports injuries.

Injury: Concussion



A concussion is caused by a blow to the head or body, a fall, or any injury that shakes the brain inside the skull. It is a type of mild traumatic brain injury that results in an altered mental state that also includes becoming unconscious. Apart from cuts or bruises on the head or face, there may be no other visible signs of a brain injury. Some of the common symptoms of concussion include disorientation, vision disturbance, severe headaches, amnesia, nausea, and loss of balance. This type of injury is most common in contact sports like football, boxing, hockey, and soccer, and sometimes in skiing and gymnastics.

Treatment: Concussions are not usually life-threatening, but can lead to grave consequences if proper and timely treatment is not given. The treatment depends on the severity of the symptoms. In most cases, individuals recover fully with complete rest. Since the brain is very sensitive after a concussion injury, individuals are advised to take a complete break from any strenuous activities or exercises. In case of headaches, painkillers may be prescribed. In extreme cases with internal bleeding, swelling of the brain, or any serious impact to the brain, surgery or other intensive medical procedures might be required.

Avoidance Tactics: The best preventative measure is to avoid any type of contact sports or any aggressive physical activity for a few weeks or months, depending on the severity of the condition. Avoid alcohol and any other drugs as it slows down the recovery time. You also need to stay away from mentally exhausting tasks, as the brain is in a delicate stage post the injury. The brain is more sensitive to damage, so it is important to avoid any kind of situation that may aggravate the condition or lead to serious complications.

Workout: Dance Exercises

Injury: Muscle Spasms



When a muscle is overstretched or torn, inflammation occurs, which results in muscle spasms. Spasms can affect different types of muscles, exhibiting different symptoms. Muscle spasms occur due to overuse, dehydration, or electrolyte abnormalities, or if the muscle has been held in the same position for a prolonged period of time. A dance workout involves muscular effort that can overstretch or tear the muscle, causing sustained and involuntary contraction. Muscle spasm symptoms include an acute onset of pain, along with a severe decrease in the efficiency of the muscular group that is affected. A bulging, tight muscle may be felt underneath the skin.

Treatment: Most muscle spasm cases resolve within a couple of days or weeks. Rest and regulation of the dance workouts help treat the condition. Wet heat or deep heat like ultrasound or short waves, professional massage, and soft mechanical stretching are also some of the treatment options to treat muscle spasms. There is a circular behavior in spasms, in which the contraction causes pain, and the pain leads to further contraction. So painkillers are prescribed to eliminate the pain and to break the circle and release the muscular fiber

Avoidance Tactics: Most of the muscle spasms are associated with dehydration and electrolyte disturbances. So, it is essential to keep hydrated during the dance routines. If the dance workout is taking place in a hot environment, it is important to fuel the body with enough fluids prior to the workout.

Muscles need to be prepped for the physical activity, hence it is essential to maintain a warm-up routine before each dance session. Regular exercise stretches the muscle and keeps it well conditioned so as to avoid any such occurrence.

Injury: Snapping Hip Syndrome



A snapping hip syndrome is also known as dancer's hip, in which one hears a snapping sound or feels a snapping sensation in the hip while walking, running, or swinging the leg around. Snapping is caused by the movement of a muscle or tendon over a bony structure in the hip. For most people the only symptom is the snapping sound or the sensation, but for dancers and athletes, this syndrome is also accompanied by pain and weakness that interfere with their performance. Iliotibial (IT) band tightness, weakness along the outside of the hip and lordosis are some of the causes of this syndrome. Dancers could experience a snapping sound in the frontal hip joint, as the IT band glides over the upper-leg bone.

Treatment: In most cases, the snapping hip syndrome can be treated with home remedies. Icing the area reduces swelling, while over-the-counter pain medications are used to relieve the snapping pain. Corticosteroid injections to the affected area are prescribed to reduce the inflammation. Reducing or modifying the activity is recommended to help in the healing process. Physical therapy is an effective treatment method, with emphasis on stretching, strengthening, and alignment. Surgery is recommended in very rare cases if the condition doesn't improve.

Avoidance Tactics: Strengthening the lower abs, abductors, adductors, and hip flexors help prevent this syndrome. So, once the symptoms have stabilized, a maintenance program involving stretching and strengthening is initiated. Light warm-up exercises should be done before a dance workout, which warms and preps the muscles for the activity. Dancers should avoid turning out the feet, which causes stress to the knees and hips.

Injury: Meniscus Tear



Dancers commonly suffer from meniscus tears. The medial and lateral menisci are two large C-shaped cartilages within the knee joint that protect the joint from the stresses placed on it from walking, running, climbing, and bending. A tear in the meniscus occurs because of forceful twisting or hyper-flexing of the knee joint. A quick turn or twist, often with the foot planted while the knee is bent, like in many dance forms, causes a torn meniscus. It can also occur while playing sports. Its symptoms include knee pain, swelling, popping, and giving way.

Treatment: The doctor conducts a physical examination, checking both the knees, looking for tenderness, range of motion, and the stability of the knee. The treatment procedure for a torn meniscus depends on various factors like the type of tear, location of the tear, and how serious it is. Resting the knee, icing the affected area, wrapping it with an elastic bandage to give it a break from any kind of movement, and propping up the legs are some of the treatment procedures used to heal the tear. If the conservative measures fail, surgery may be needed to repair or remove the damaged cartilage.

Avoidance Tactics: It is recommended to maintain an ideal body weight that helps avoid such tears from occurring. Specific recommendations regarding weight-bearing are suggested by the surgeon and therapist to prevent the painful meniscus tear. Activities that put a lot of stress on the knee should be avoided. There should be a gradual return to dancing activities to prevent a recurrence of the tear.

Injury: Ankle Sprain



Even though it is quite a common injury, it is by no means a minor injury. A rapid shifting movement with the foot planted, like in dancing, or even in soccer or getting tackled in football, causes the sprain. In a sprain, the ankle rolls outward, while the foot turns inward, which causes ligaments of the ankle to stretch and tear. In some cases the ankle rolls inward and the foot outward, damaging the ligaments inside the ankle. An ankle sprain ranges from mild to severe. It depends on how many ligaments are injured and how badly they are injured. The common symptoms are pain, tenderness, and swelling. In case of severe sprains, a tear is experienced along with a pop or snap.

Treatment: The sprained ankle needs rest for complete healing. Icing the area frequently gives relief from swelling. An elastic compression wrap also reduces swelling, but as compression wraps don't offer protection, a brace might be needed if any weight is being put on it. The ankle should be elevated above the level of the heart for two to three hours a day. Either prescription painkillers or over-the-counter pain relievers are used to relieve the pain. If the sprain doesn't heal correctly, the joint may become unstable and turn into chronic pain. In rare cases, arthroscopy or reconstruction surgery is performed to repair the torn ligament.

Avoidance Tactics: The risk of an ankle sprain can be avoided by warming up before a dance workout, and using proper footwear. Wearing a brace gives it added support, thus avoiding the occurrence of a sprain. Stretching, strength training, and balance exercises keep the ankle strong and flexible, preventing further injury.

Injury: Neck strain



Dancing and choreography which incorporates a lot of head movements causes a strain on the dancers' neck muscles. The cervical spine provides support for the head, and allows mobility and range of motions, but it also makes this area of the spine so flexible that it becomes vulnerable to injury. If dancers do not use the full spine properly when arching the head or neck, it causes strain to the neck muscles. A neck strain is usually characterized by dull aching pain, numbness, tingling, tenderness, and difficulty swallowing. Individuals can sometimes also experience a sharp pain, dizziness, and swelling in the lymph nodes.

Treatment: The treatment of a neck strain is based on the precise cause of the injury. For moderate pain, bed rest is advised. A cervical collar or putting a pillow to support the neck helps with the discomfort. The cold from an ice pack reduces the inflammation, and applying heat brings nutrient-rich blood flow to the injured area, stimulating a healing response. Anti-inflammatory medications, or pain relievers are used to get relief from the pain. In many cases, a gentle massage helps stimulate the blood flow in the area.

Avoidance Tactics: Individuals should avoid stretching the neck beyond their normal limit to prevent stress on the neck muscles. For dance moves requiring a lot of head movements, try to lengthen the neck rather than collapsing it. Avoid any additional strain to the neck, so that the symptoms subside quickly. Make sure to warm up before the dance routine, so that the muscles are strengthened and flexible enough for the workout.

Address your Sports Injury Immediately

Ignoring an injury is not a wise move. An untreated injury only gets worse with time, and will end up requiring surgery or other intense medical procedures. But if the injury is treated right away, it will heal quickly, allowing the person to get back to enjoying their workouts, and take all necessary precautions to avoid a recurrence of the injury. So, the next time you feel any aches and pain during the course of your workout, address it immediately by contacting a orthopedic specialist to eliminate the issue, as you continue working your way to tremendous body and health.



Bioenergetic

Metabolism & Energy Systems:

The following part explains how the body obtains and utilizes energy. Why is this important to a personal fitness trainer? All movement and performance come down to energy. By understanding how the body utilizes energy, we can better understand the fuel sources (carbohydrates, fats, or proteins) utilized during training, which muscle fiber types (Type I, IIA or IIB) are involved, and how to monitor energy systems using heart rate measurement.

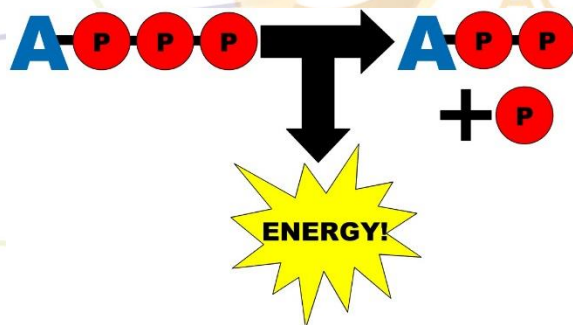
What are Energy Systems?

While not considered one of the eleven bodily systems, these are the processes the cells that the body uses to convert the chemical energy we get from food to a form chemical energy they body can use

(ATP). By understanding how we utilize energy

(ATP), we will better understand the how to make the body's systems work better with less effort - Physiological Efficiency.

Energy:



Energy is required by all cells to perform work. Cells have the capacity to convert food (carbohydrates, proteins, and fats) into usable energy. All cells must have a constant supply of energy or work will cease. If a muscle cell doesn't have adequate energy supply (the mechanical energy necessary to perform movement), there won't be any muscular contractions. Work is one form of energy, known as mechanical energy.

In order to achieve a constant energy supply, muscle cells are constantly breaking down molecules into usable forms of energy and utilizing the energy released from these chemical reactions to build and repair tissue. **Catabolism** is the breakdown of molecules and **anabolism** is the synthesis of molecules. Anabolic reactions utilize the energy created from the breakdown of molecules in catabolic reactions for the building process. Both reactions are normal and necessary, but if catabolic reactions are so substantial that repair and synthesis from anabolic reactions cannot occur, injury and illness/disease occur. This is what happens when training is too intense or the right nutrients or rest/recovery is not achieved (**overtraining**).

Energy transfer (since energy cannot be created, merely transferred or transformed) occurs when energy is released from the chemical bonds of molecules (food molecules in this case). The process of removing an electron from a molecule is called **oxidation** (which will often include the addition of oxygen to a substance). The speed of cellular chemical reactions such as oxidation is regulated by catalysts called **enzymes** (cell proteins which are crucial to the metabolic process).

What is a calorie?

There are a variety of ways to express energy. However, the most common measurement in the system, used in the United States, is the calorie. Everyone knows that food contains calories, but few know how to define a calorie. A calorie is a measure of heat. It is the amount of heat needed to raise the temperature of one gram of water one degree Celsius. A calorie is such a small unit of energy, it is often referred to as a kilocalorie (which is equal to 1,000 small calories because kilo = 1000) and is commonly abbreviated “kcal”. The “calories” on food labels are actually kilocalories.

What is a kilojoule (kJ)?

The term kJ is not often used in the United States, but is more common in other parts of the world and is actually the proper unit for energy in the International Unit System. 1 kcal is equivalent to 4.2 kJ.

Adenosine Triphosphate:

the immediate source of energy for muscular contractions is **adenosine triphosphate**

(**ATP**). Without sufficient amounts of ATP most cells die quickly. ATP is formed when **adenosine diphosphate (ADP)** and an **inorganic phosphate** are combined. This process of combining these phosphates requires a large amount of energy. For this reason, approximately 60% of the energy from food is converted to thermal energy during the process of ATP formation and only about 40% of the actual energy from food is converted to ATP.

ATP is the muscle’s primary energy source, just as gasoline is the energy source for an automobile. How quickly and efficiently a muscle cell produces ATP will be a determining factor of how much work the cell can perform before it fatigues.

While there is some ATP stored within a muscle cell, the supply is limited (enough for 2-4 seconds of muscle action). Therefore, muscle cells must be able to produce more ATP in order to continue working. Energy is released when an ATP molecule is broken into an ADP molecule and an inorganic phosphate.

Since muscle cells can only store a limited supply of ATP, and they require a constant supply of ATP to function, various metabolic pathways must exist to produce ATP quickly or cell death, not just poor functioning, will occur. ATP must be regenerated by reactions that add the phosphate back to ADP (the reverse reaction of the image above). Muscle cells replenish the

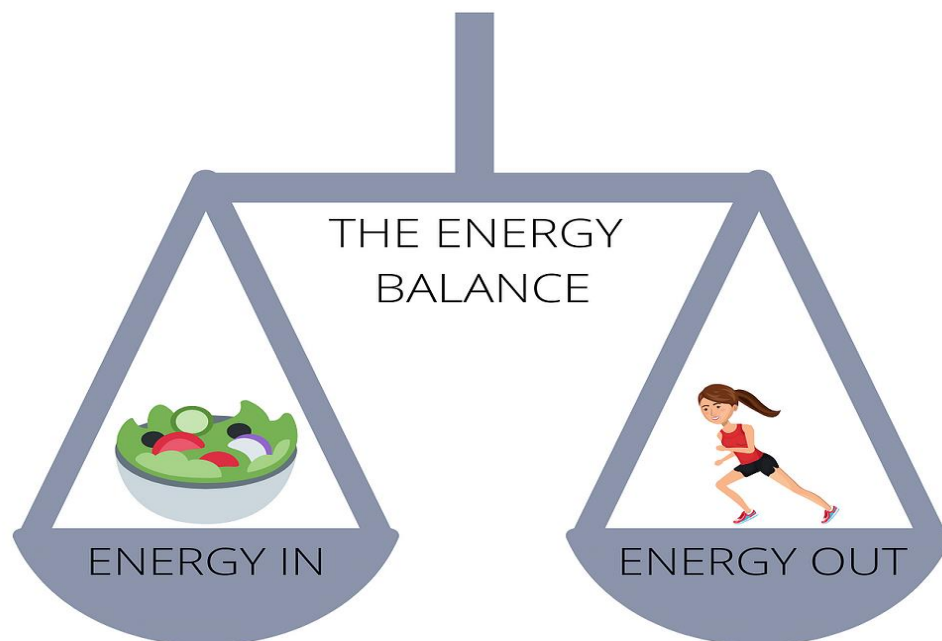
ATP supply using three distinct biochemical pathways. ATP in muscle cells can be formed through a combination of phosphocreatine (PC) breakdown, glycolysis (glucose or glycogen breakdown), and oxidation (which is inherently aerobic as oxidation requires oxygen).

Anaerobic processes do not require oxygen for reactions while **aerobic** processes utilize oxygen in the processes of breakdown and synthesis. All anaerobic processes occur in the sarcoplasm (the cytoplasm/cell fluid of the muscle cell). All aerobic processes (utilizing the oxidative/aerobic energy system/metabolic pathway) occur in the mitochondria of muscle cells. The *intensity* and *duration* of activity will determine which energy system/metabolic pathway is dominant at any particular time.

Aerobic and Anaerobic Energy Systems:

When muscles need energy, ATP is **hydrolyzed** (broken down with H₂O) rather quickly to provide the needed energy, and ADP and inorganic phosphate (Pi) are produced. In order to continue to supply muscles with energy, ATP must be reformed by combining ADP and Pi. The following sections explain the various systems whereby the body regenerates ATP. ALL of these systems are working in the body at ALL times. It is important for the personal fitness trainer to understand which system is **dominant** at a particular time.

The aerobic energy system for producing ATP is dominant when adequate oxygen is delivered to the cell to meet energy production needs, such as when the muscle is at rest. However, as work by the body increases (such as during exercise), oxygen cannot be delivered quickly enough to the working muscles. The muscles must then rely on other processes which do not require oxygen (anaerobic) to gain their ATP (energy).



Anaerobic Pathway I: ATP/ PCr System

The first anaerobic (“without oxygen”) source of ATP is known as Phosphocreatine (PCr, also known as Creatine Phosphate). Phosphocreatine functions as a site of storage of high energy phosphates in muscle (the amount of PCr is proportional to the muscle mass). PCr itself cannot be used as an immediate source of energy, but can rapidly replenish ATP.

ATP stored in the muscle is the primary provider of energy for muscular contraction at the onset of exercise (for less than 5 seconds) and during short-term, high intensity exercise (for less than 10-15 seconds at maximal intensities of effort). Like ATP, only a small supply of PC is stored within muscle cells (approximately 80-100 grams of ATP can be stored at one time with concentrations of PC being about 4-6 times greater). PC stores can supply energy at near

maximal intensities of effort for up to about 8 seconds. After about 8 seconds of maximal to near maximal effort, the ATP- PC energy production system is no longer the dominant energy system.

This process does not involve oxygen so the ATP-PC energy system provides an anaerobic energy supply. Once PC stores are depleted (ATP cannot be completely depleted or the cell(s) would die), the reformation of PC requires ATP and can only occur during rest.

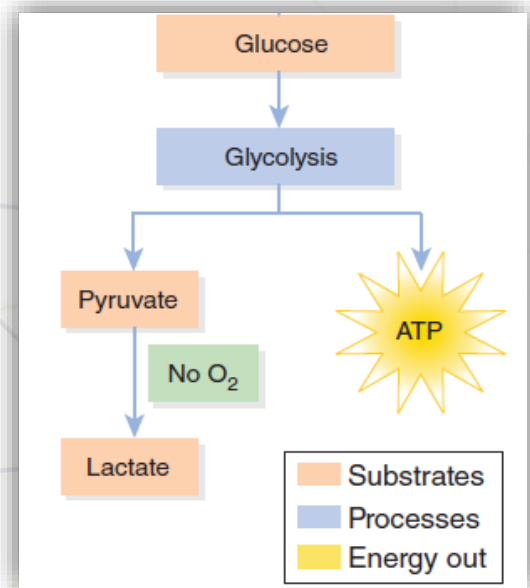
Approximately 70% of ATP is reformed after 45- 60 seconds of complete rest from high intensity activity and 100% of ATP is reformed after 3-5 minutes of complete rest. The greater the effort that is required, the more rest is needed between efforts in order to allow time for the reformation of ATP and PC within muscle cells.

As an example, PC provides much of the energy for a 100m sprint or nearly all of the energy during a single explosive jump or burst of speed. The ability to perform repeated exercises or movements at near maximal effort is largely dependent on PCr stores. For some clients, enhancing PCr stores through creatine monohydrate supplementation can increase the amount of work that can be performed in repeated efforts at high intensity.

Beyond the first few seconds of near maximal exertion or after the first few seconds of activity, the phosphagen and glycolysis energy systems share in the energy production process (between 6-45 seconds of intense exertion).

Anaerobic Pathway II: Glycolysis:

The second metabolic pathway is known as glycolysis (anaerobic), and is capable of producing ATP quickly without oxygen (glyco = “relating to the breakdown of glucose (a sugar, specifically a monosaccharide) or glycogen (polysaccharide) in a multi-step process into pyruvate. Pyruvate can then either be converted to lactate for more immediate energy usage at higher intensities or be transferred into the mitochondria of the cell during lower intensities of exercise. The process of glycolysis actually requires ATP (2 molecules) in order to complete the transfer of energy from glucose to rejoin an inorganic phosphate with ADP and results in more ATP (with a net gain of two more ATP if the conversion involves the simple sugar glucose and three ATP if it involves the starch glycogen) being made available for use.



muscular contraction during moderately high intensities of effort and moderate duration (from about 45 seconds to about 2 minutes) efforts. The glycolytic energy system becomes involved at about 9-10 seconds into an exercise and will actually have some involvement in energy production until about 10 minutes of exercise duration has elapsed. This is the predominant energy system involved during most strength training workouts, as most sets will last between the 45 seconds to 2 minute time frame.

(fast glycolysis) and aerobic (slow glycolysis) to differentiate the fast resynthesis of ATP when pyruvate is converted to lactate (anaerobic or fast glycolysis) from the slow conversion when pyruvate is shuttled into the mitochondria of the cell before undergoing the Krebs Cycle (aerobic glycolysis). However, as glycolysis does not involve oxygen, these terms are not practical for describing the process.

Lactate Threshold and the Onset of Blood Lactate Accumulation during Glycolysis

As the level of intensity of exercise increases, the body reaches a point where the level of oxygen within the cell's mitochondria is not sufficient. Since there is not enough available oxygen, the pyruvate produced by glycolysis cannot be oxidized. Furthermore, since energy production must continue, ATP production from glycolysis must increase.

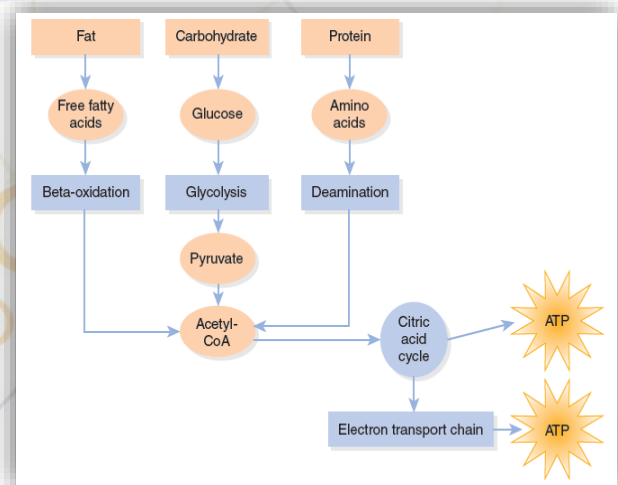
Lactate accumulation in the blood is signal that there is not enough oxygen getting to the working muscles (i.e. “going anaerobic”). During energy production, there are several reactions that cause a release of a hydrogen ion (H^+). This causes a decrease in blood pH (i.e. making the blood more acidic). Without going into detailed biochemistry, when the body is “aerobic” it uses buffering capabilities so that the pH of the blood does not go down. However, when there is not enough oxygen being supplied to the working muscles, muscle acidosis can begin. The “burning” sensation in muscles is attributed to this muscle acidosis and *correlates* to the accumulation of lactate. The point where so much lactate accumulates that exercise intensity is forcibly decreased (due to muscle acidosis induced fatigue) is known as the

Aerobic Pathway: Aerobic Glycolysis and Oxidative Phosphorylation:

Most cells, including muscle cells, contain structures called **mitochondria**. The mitochondria are the sites of cellular aerobic energy production (also known as **cellular respiration**, **aerobic oxidation** and **oxidative phosphorylation**). Larger numbers of mitochondria in a cell lead to a greater capability for aerobic energy production. Therefore, cells with larger numbers of mitochondria have a greater aerobic capacity and are more resistant to fatigue.

The third pathway for producing ATP involves oxygen. This oxidative phosphorylation (utilizing oxidation and adding a phosphate) or aerobic metabolic pathway utilizes both the Krebs cycle and the electron transport chain.

The Krebs cycle involves a series of chemical reactions to help convert macronutrients (carbohydrates, proteins, and fats) into usable energy (ATP). Oxygen is not involved in the Krebs cycle but it is involved at the end of the electron transport chain (a chain of molecules involved in the aerobic production of ATP). The Krebs cycle removes hydrogen molecules from the ingested macronutrients so that these hydrogen molecules can be used to complete the electron transport chain along with oxygen, an inorganic phosphate, and ADP to form ATP. This process will form a net gain of two more ATP molecules to be used for energy (the process also involves two carrier molecules known as NAD and FAD which transport the hydrogen molecules). During the activation of the electron transport chain in the aerobic production of ATP free radicals are formed.

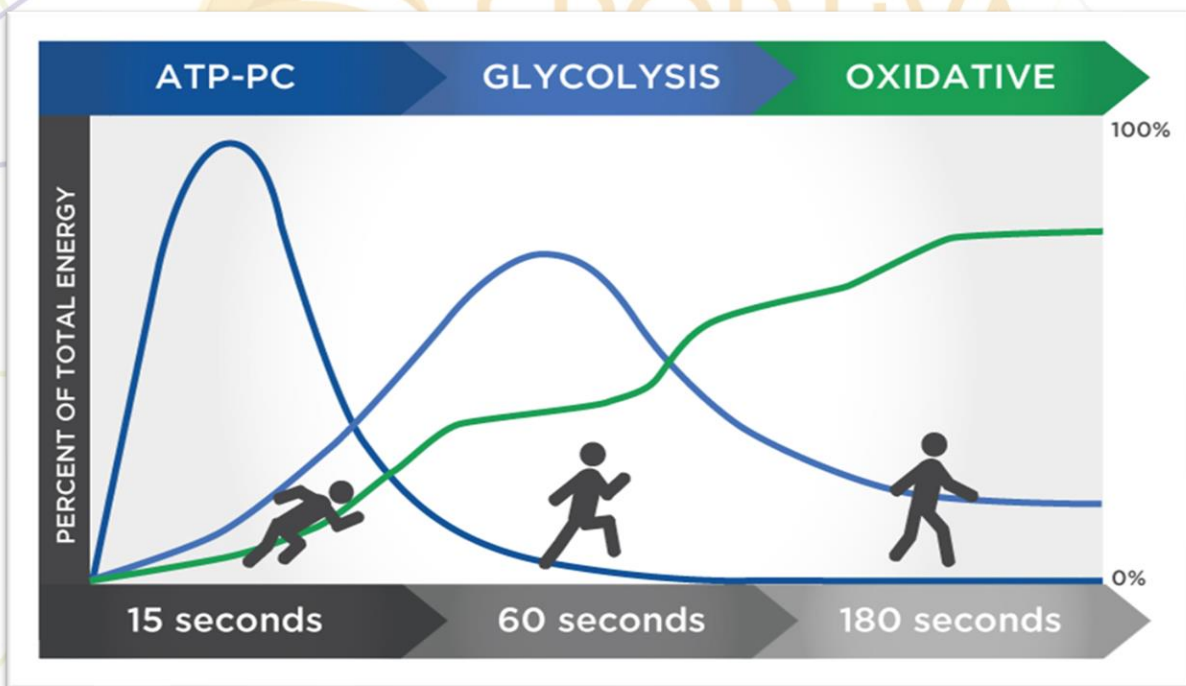


Free radicals are molecules that now have an unpaired electron as a result of the oxidative phosphorylation or from the aerobic production of ATP. Free radicals are highly volatile and reactive and will bind quickly to other molecules and damage the cells they contact. The number of free radicals produced is directly linked to the rate of aerobic metabolism (which is dependent upon the conditioning level of the individual combined with the current exercise intensity and duration).

The breakdown of fat to yield ATP is known as *lipolysis*. Fat is the most abundant energy source available to the muscle, and the predominant energy source used during the aerobic oxidation phase of energy production. Although there is a virtually unlimited supply of fatty acids in humans, the rate at which the body can metabolize them is the limiting factor in obtaining ATP.

Lipolysis is the primary source for energy production during rest and low level activity. But, its input to the overall muscular energy supply will decrease as the intensity of muscular contraction increases. When lipolysis is unable to meet the increasing energy requirement of exercise (it is too slow to supply enough energy), stored glycogen from both the liver and muscles is used. Once glycogen depletion occurs, exercise intensity will decline.

All exercise intensities (including rest or no activity) that can be maintained for 2 minutes or longer utilizes the oxidative system for energy production. Once approximately 10 minutes of continuous exercise has elapsed, glycolysis is no longer significantly involved in energy production and the oxidative energy system supplies almost all cellular energy for the muscles.



Nutrition

Nutrition and physical activity should be addressed in conjunction with one another. Focusing on one at the exclusion of the other will yield less than optimal results for clients. Personal trainers can enhance their overall effectiveness by maintaining a core knowledge of nutrition and by individualizing their nutrition advice. Nutrition assessment and recommendations should match the needs and goals of the client and will vary accordingly. Lastly, personal trainers should know when they need to refer their clients to a dietitian because the client's needs are outside of the trainer's scope of expertise or practice

Role of the Personal Trainer Regarding Nutrition

Television, newspapers, magazines, and websites are the major sources of nutrition information for most people. Nutrition information communicated as sound bites and advertisements can lead to consumer confusion. Personal trainers have the opportunity to help clear the confusion by serving as a source of credible nutrition information.

It is well within the personal trainer's scope of practice to address misinformation and to give general advice related to nutrition for physical performance, disease prevention, weight loss, and weight gain. A personal trainer would be conveying general nutrition knowledge if he or she said, for example, "According to the American Heart Association, omega-3 fatty acids from fatty fish like salmon

or mackerel may benefit those who are at risk of developing cardiovascular disease." An important part of the core knowledge, from the standpoint of both ethics and safety, is the ability to recognize more complicated nutrition issues and know who to refer clients to. Referral to a nutrition professional is indicated when the client has a disease state (i.e., diabetes, heart disease, gastrointestinal disease, eating disorder, osteoporosis, elevated cholesterol, etc.) that is affected by nutrition. This type of nutrition information is called *medical nutrition therapy* and falls under the scope of practice of a licensed nutritionist, dietitian, or registered dietitian (RD) (depending on the country and in the United States, on the state's licensure laws) (3). Referral is also indicated when the complexity of the nutrition issue is beyond the competence of the personal trainer, which will vary. Personal trainers should find a nutrition professional they feel comfortable referring their clients to and with whom they can communicate about clients. In the United States and Canada, registered dietitians can be located through state dietetic organizations; the American Dietetic Association (ADA) website, www.eatright.org; the Sports, Cardiovascular and Wellness Nutritionists website (SCAN, a dietetic practice group of the ADA), www.scandpg.org; and Dietitians of Canada, www.dietitians.ca. The European Federation of the Association of Dietitians, www.efad.org/everyone, provides links to each country's dietitian organization. Sports Dietitians Australia has a website where people can locate a sports dietitian: www.sportsdietitians.com.au/ find a sports dietitian. In other countries, personal trainers will want to consult local dietetic organizations or national Web sites. To facilitate communication, the client should sign a release of information form so that the personal trainer and the nutrition professional can communicate about the client's training program and general nutrition needs.

or to their country's organization for dietetic regulations to find out the laws within their particular state that govern the provision of nutrition advice. In the United States, each state regulates the provision of nutrition information through licensure, statutory certification, or registration. According to the American Dietetic Association, definitions for these terms are as follows:

Licensing: Statutes include an explicitly defined scope of practice, and performance of the profession is illegal unless a license has been obtained from the state.

Statutory certification: Limits use of particular titles to persons meeting predetermined requirements, while persons not certified can still practice the occupation or profession.

Registration: This is the least restrictive form of state regulation. As with certification, unregistered persons are permitted to practice the profession. Typically, exams are not given and enforcement of the registration requirement is minimal.

As an example, at the time of this writing, the scope of nutrition practice in Louisiana is clearly defined, and specific guidance on a person's diet is allowed only by a registered dietitian or nutritionist (25). But in Arizona, no licensure law exists, and any professional can offer nutrition advice (23). Various states and countries have different regulations governing whether or not personal trainers can provide dietary advice, and personal trainers should follow these guidelines.

Dietary Assessment

Should a client seek nutrition information that is within the scope of the personal trainer's practice, the personal trainer may want to assess the client's diet. If this is out of the scope of his or her practice, the personal trainer can work alongside a dietitian who assesses the client's diet.

A complete nutrition assessment includes dietary data, anthropometric data, biochemical data (lab tests), and a clinical examination (condition of skin, teeth, etc.). Although personal trainers are usually not involved in the comprehensive assessment, they may want to be familiar with the individual components of a comprehensive dietary assessment so they can work with the dietitian to provide their client the best service possible. (*Note:* The term *diet* as used throughout this chapter refers to the usual eating pattern of the individual, not a restrictive weight loss plan.)

Dietary Intake Data

Before the personal trainer can give valid nutrition advice, glean some information about the client's current diet is imperative. How balanced is the client's current diet? Is the client allergic to certain foods? Is the client vegetarian? Restricting food groups? Dieting to lose weight? Is he or she a sporadic eater? Has the individual just adopted a new way of eating? The answers to all of these questions and others may influence the personal trainer's advice to the client.

Gathering dietary intake data is a simple concept, but it is extremely complex to do. Most people have difficulty recalling fully and accurately what they ate in a given day. Research shows that there is a tendency to underestimate or underreport actual intake, especially in persons who are overweight. Keeping in mind these general shortcomings, personal trainers, again if under the scope of practice as allowed in their state, have three methods for gathering dietary intake data to choose from:

- Dietary recall
- Diet history
- Diet records

In a diet recall, clients report what they have eaten in the past 24 hours. With a diet history, clients answer questions about usual eating habits, likes and dislikes, eating schedule, medical history, weight history, and so forth. The diet record is typically a log, filled out for three days, in which the client records everything consumed (foods, beverages, and supplements).

The three-day diet record is considered the most valid of the three methods for assessing the diet of an individual. However, a valid record requires scrupulous recording as well as scrupulous analysis. The pitfall of this method is that recording food intake usually inhibits regular eating patterns, and recorded intake thus underestimates true intake. To get useful data, the personal trainer should ask only the most motivated clients to complete this process. The diet recall or diet history is more appropriate for many clients.

The personal trainer should never make assumptions about a client's eating habits. Assessing the client's diet is essential before one makes dietary recommendations.

Evaluating the Diet

When the personal trainer has successfully gathered dietary intake data, several options exist for evaluating the information. One way to evaluate the client's diet is to compare a client's diet to the

recommendations given in the country's general dietary guidelines. In the United States, the U.S. Department of Agriculture (USDA) created **MyPlate** (46). For clients who are keenly interested in nutrition, a more detailed analysis of the diet using diet analysis software may be indicated. Both methods are reviewed here.

MyPlate

The USDA MyPlate (figure 7.1) is a reminder for healthy eating. The “my” in “MyPlate” signifies the importance of personalizing the recommendations to one’s lifestyle, while the familiar plate symbol provides a visual representation of how much of one’s diet should be made up of the following food groups:

1. Grains
2. Vegetables
3. Fruits
4. Protein
5. Dairy

Balancing Calories

Enjoy your food, but eat less.

Avoid oversized portions.

Foods to Increase

Make half your plate fruits and vegetables.

Make at least half your grains whole grains.

Switch to fat-free or low-fat (1%) milk.

Foods to Reduce

Compare the sodium in foods like soup, bread, and frozen meals and choose the foods with lower numbers.

Drink water instead of sugary drinks.

The USDA has also identified specific goals that relate to each of the main food groups. For instance, consumers are told to eat at least half their grains as whole grains, vary their vegetable intake, focus on fruits, eat more calcium-rich foods, and go lean with protein.

Though the older MyPyramid contained guide- lines for “discretionary calories,” MyPlate has replaced this with an “empty calories” category. Empty calories are calories from solid fats or added sugars that have no nutrients. Solid fats (i.e., fats solid at room temperature) include butter, beef fat, and shortening. Added sugars are sugars or syrups that are added during food processing or preparation.

foods and beverages that provide the largest number of empty calories for Americans:

- Cakes, cookies, pastries, and donuts (contain both solid fat and added sugars)
- Sodas, energy drinks, sports drinks, and fruit drinks (contain added sugars)
- Cheese (contains solid fat)
- Pizza (contains solid fat)
- Ice cream (contains both solid fat and added sugars)
- Sausages, hot dogs, bacon, and ribs (contain solid fat)

Along with providing useful educational material, the MyPlate website (www.ChooseMyPlate.gov) contains tools that allow visitors to create customized meal plans, analyze their diets, and track their physical activity. This website is an excellent starting point for a personal trainer to use to help educate clients. Each food group in MyPlate provides key nutrients that are more difficult to acquire in the diet if that group is omitted.

The MyPlate website is an interactive tool that clients can use on their own or with a personal trainer's assistance. The Plan a Healthy Diet function allows users to enter data concerning their age, weight, height, gender, and physical activity

level and in turn provide them with individualized directions for meeting their daily nutrition goals. If consumers click on a food category, they'll be given more specific information on how to include that food group into their diet (46).

Computerized Diet Analysis

Computerized analysis can provide a snapshot of a client's diet, including vitamin and mineral intake. However, it is very important that the client accurately and completely record usual intake for at least three days. The client should input the amount of each food and beverage, specify how it was cooked, and give the brand name versus the generic term (e.g., "Wheaties" vs. just "bran flakes").

Even if the diet is recorded perfectly, the analysis will not be completely accurate because all software programs have shortcomings. For some foods in the database, values for certain vitamins and minerals are missing, meaning that analysis for those nutrients is also missing, which will result in an erroneously low intake value. Additionally, without fail, some foods that the client eats are not in the database, so it is necessary to make substitutions or type in the actual food data (and for processed foods, the results may not include values for all vitamins and minerals since this is not a requirement for food labels).

Before asking clients to assess their diet, it is helpful for personal trainers to complete a computerized diet analysis on themselves to recognize the bias that recording can impose on true habits. Additionally, analyzing one's own diet makes one aware of the level of detail needed to accurately assess a diet.

In many cases, the personal trainer does not have the training, time, knowledge, or resources to complete a computerized dietary analysis. This is an area in which many personal trainers turn to dietitians for assistance. Another option for motivated clients is to refer them to websites where they can enter their own diet and receive feedback (see the following list). These websites are excellent resources because they put the responsibility on the client. Additionally, some clients feel more comfortable asking questions and reporting intake in private situations. One drawback is that most of these websites do not have the extensive database that comes with food analysis programs, and they typically do not analyze food intake for all vitamins, minerals, types of fat, and so on. Instead, most tell the user only how many calories he or she has consumed, along with grams of fat, carbohydrates, and protein. A second drawback is that these sites cannot calculate calorie needs with the precision that a professional dietitian can.

Energy

Energy is commonly measured in kilocalories (kcal). A kilocalorie is a measure of energy that equates to the heat required to raise the temperature of 1 kilogram of water 1 degree Celsius (or 2.2 pounds of water 1.8 degrees Fahrenheit). The general public refers to this as a *calorie*.

Factors Influencing

Energy Requirements

Three factors make up the energy requirement of adults: resting metabolic rate, physical activity, and the thermic effect of food. Each of these factors can be affected directly or indirectly by age, genetics, body size, body composition, environmental temperature, training conditions, nontraining physical activity, and calorie intake. For infants, children, and teens, growth is another variable that increases the energy requirement.

Resting metabolic rate (RMR) is the largest contributor to total energy requirement, accounting for approximately 60% to 75% of daily energy expenditure. It is a measure of the calories required for maintaining normal body functions such as respiration, cardiac function, and thermoregulation (i.e., the energy a person would expend lying in bed all day and doing nothing). Factors that increase RMR include gaining lean body tissue, young age, growth, abnormal body temperature, menstrual cycle, and hyperthyroidism. Factors that decrease RMR include low caloric intake, loss of lean tissue, and hypothyroidism. All things equal, RMR can vary up to 20% between individuals due to normal genetic variations in metabolism.

The second largest component of energy requirement is physical activity. Of all the components, this one is the most variable. The amount of energy needed for physical activity depends on the intensity, duration, and frequency of training. It also depends on the environmental conditions; that is, extreme heat or cold increases calorie expenditure. When estimating how physically active a client is, the personal trainer needs to remember to ascertain how physically active the client is aside from structured exercise. Even if people have an exercise routine, those with the sedentary lifestyle of a desk job and sedentary leisure activities may be considered only lightly active. The **thermic effect of food** is the increase in energy expenditure above the RMR that can be measured for several hours following a meal. The thermic effect of food is the energy needed to digest and assimilate foods, approximately 7% to 10% of a person's total energy requirement.

Estimating Energy Requirements

A true estimation of energy requirement (i.e., energy expended in a day) is difficult to obtain directly. Therefore surrogate methods are often used. One such method is to measure calorie intake. This method is valid if the client is maintaining a stable body weight, because a stable body weight indicates that energy intake generally equals energy expenditure. For the motivated client who accurately records intake, the best way to determine energy requirement using this method is to assess the calorie intake from the three-day food log. If that is not possible, one can use mathematical equations that roughly estimate caloric expenditure. However, it is difficult to calculate energy needs because of the many variables affecting caloric requirements and the significant inter- and intraindividual variation. It is essential to stress that these

equations are only estimates and are meant to serve as a frame of reference. Actual energy expenditure of individuals will vary widely. Table 7.2 lists factors that can be used for energy requirement estimation. For example, for a male who weighs 170 pounds (77 kg) and is highly physically active, the requirement would be 3,910 kcal (23 × 170).

Another method for calculating energy expenditure is to first calculate resting energy expenditure (REE), then multiply it by a factor based on activity level. Several equations for estimating REE exist. One set of REE equations, developed by the World Health Organization (14), is shown in table 7.3.

The result is the number of calories that are likely expended by the person in an average day. Clients wishing to maintain current body weight would need to consume the same number of calories that they expend.

Daily Calories Need

- Step 1. Weight (Kg) X 2.2 = Weight (lbs)
- Step 2. Weight (lbs) X 10 = RMR
- Step 3. RMR X Activity Factor = TEE

Sedentary

Activity more than 30 minutes, 2-3 Times/Week

Activity more than 45 minutes, 4-5 Times/Week

Activity more than 60 minutes, 5 Times/Week

Very Light	1.2–1.3
Low Active	1.5–1.6
Active	1.6–1.7
Heavy	1.9–2.1

Nutrients

Once the personal trainer knows a client's dietary intake and energy requirements, he or she can assess general nutritional needs. To understand the relationship between the body and food, as well as to provide nutrition guidance, it is important to have an understanding of the six nutrients: protein, carbohydrate, fat, vitamins, minerals, and water.

- **Macronutrients:**

Macronutrients are: proteins, fats carbohydrates. They are the building blocks that comprise the food we eat for fuel, growth, endurance, satiety and recovery. All

three fuel the body, but they each have critical functions that extend beyond energy. Every food we eat falls into at least one of the above categories. Some foods are a combination of two, or even three. Edamame, for example, contains protein, fat and carbs. Turkey is just protein. Looking at a nutrition panel can help decipher which categories foods are in.

Protein

Protein is found throughout the body—in muscle, bone, skin, hair, and virtually every other body part or tissue. It makes up the enzymes that power many chemical reactions and the hemoglobin that carries oxygen in your blood. There is a minimum of 10,000 different proteins that make up, and maintain us.

The Institute of Medicine recommends that adults get a minimum of 0.8 grams of protein for every kilogram of body weight per day (or 8 grams of protein for every 20 pounds of body weight). In the United States, the recommended daily allowance of protein is 46 grams per day for women over 19 years of age, and 56 grams per day for men over 19 years of age.⁷

Many athletes, weight lifters and exercisers require, or prefer, a lot more protein than that. Many are closer to 1-2 grams per pound of their body weight. Most people will take a stance on how much protein they believe is right for them after reading an article or just discussing it with people whose bodies they admire. They think, “she looks good, so I’ll do what she does.” Everyone is built differently, and demands for all the macronutrients depend on an array of variables, including: body weight, sex, activity, goals, liver function, blood sugar and overall health. There’s no one size fits all when it comes to macros.

Protein is built from building blocks called amino acids. Our bodies make amino acids in two different ways: Either from scratch, or by modifying others. A few amino acids (known as the essential amino acids) must come from food, meaning we cannot make them

Animal sources of protein tend to deliver all the amino acids we need.

Other protein sources, such as fruits, vegetables, grains, nuts and seeds, lack one or more essential amino acids to be considered a “complete protein,” which is why food combining is important in some instances. Vegans and vegetarians (or anyone who doesn’t eat meat, fish, dairy or eggs) should eat a variety of protein foods each day to ingest the amino acids needed to make new protein.⁸

Carbohydrates

Not all carbs are created equal. Not by a long shot. Chocolate is a carbohydrate, aka, carb, and so is broccoli. The amount, and types you choose, are critical. People often slash carbs to lose weight. It's true; cutting carbs very well may be a quick way to drop unwanted pounds, but unless it's a lifestyle, people almost always regain the weight they lost by "going low-carb." It's far more important to, "eat carbohydrates from healthy foods than to follow a strict diet limiting or counting the number of grams of carbohydrates consumed," says The Harvard School of Public Health."⁹

Carbohydrates are found in both healthy and unhealthy

foods: bread, beans, milk, popcorn, potatoes, cookies, pasta, rice, French fries, soft drinks, corn, and cake are all forms of carbs. They look vastly different from food to food. "The most common and abundant forms are sugars, fibers, and starches." Broccoli would be a fiber, while gummy bears would be sugar.

The Harvard School of Public Health states, "Carbohydrates provide the body with glucose, which is converted to energy used to support bodily functions and physical activity. But carbohydrate quality is important; some types of carbohydrate-rich foods are better than others. The healthiest sources of carbohydrates—unprocessed or minimally processed whole grains, vegetables, fruits and beans—promote good health by delivering vitamins, minerals, fiber, and a host of important phytonutrients."

"Unhealthier sources of carbohydrates include white bread, pastries, sodas, and other highly processed or re- refined foods. These items contain easily digested carbohydrates that may contribute to weight gain, interfere with weight loss, and promote diabetes and heart disease."

Most fibrous vegetables are relatively low in calories with a high nutritional punch in the form of fiber (to keep you full) and vitamins and antioxidants for disease prevention and fighting free-radicals. It stands to reason that the more veggies on a plate means less room for empty calories, while staying nice and satisfied.

Fats:

For a few decades, we were told that fat made us fat. Food companies swapped fat for sugar, creating Franken-foods that made Americans fatter than ever. Low-fat diets are often high in carbohydrates, often from quickly-digest- ed foods like white flour, white rice, potatoes, sugars and processed snacks. Eating lots of these simple carbs' spikes blood sugar and insulin levels, which increases the risk of diabetes and heart disease. "High-carbohydrate, low-fat diets also have a negative effect on the fats and cholesterol in our blood: They raise "bad" blood fats (tri- glycerides) and they lower the "good" blood cholesterol (HDL), both of which can increase the risk of heart dis- ease. These diets also tend to increase blood pressure,"

per The Harvard School of Public Health. Furthermore, low-fat diets are unsatisfying. Low-fat fanatics finish a meal and within a few hours, they are hungry again, or have sugar cravings. The quickest way to get glucose into the system to squash the hunger is more carbs. This cycle of eat, craving, eat, craving leads to weight gain, high blood pressure, low HDL, cancer, and more.

Within the last decade, the tides turned and fat is no longer the villain it was once painted to be. Just like carbs, there are healthy and unhealthy fats. It's also important to remember, that even though some fats are healthy and aid in lower cholesterol, there are 9 calories per gram of fat, which makes it more calorie-dense than protein and carbohydrates. So, portions must be adjusted.

The Harvard School of Public Health states, "Unsaturated fat is the healthiest type of fat. Plant oils, such as olive, canola, corn, peanut and other nut oils; nuts, such as almonds, peanuts, walnuts, and pistachios; avocados; and fish, especially oily fish such as salmon and canned tuna, are excellent sources of unsaturated fat. Eating unsaturated fat in place of refined grains and sugar can improve blood cholesterol profiles and lower triglycerides, and in turn, lower the risk of heart disease. It is essential to include a special kind of unsaturated fat, called omega-3 fats, in the diet; good sources include fish, walnuts, flax seeds, and canola oil. Keep in mind that omega-3 fats from marine sources, such as fish and shellfish, have much more powerful health benefits than omega-3 fats from plant sources, like walnuts and flax seeds. But omega-3 fats from plant sources still are a good choice, especially for people who don't eat fish.

Saturated fat is less healthy, since it raises "bad" cholesterol in the blood. We can't eliminate saturated fat from our diets, though, because foods that are rich in healthy fats also contain a little bit of saturated fat. The best strategy is to limit foods that are very high in saturated fat, such as butter, cheese, and red meat, and replace them with foods that are high in healthy fats, such as plant oils, nuts, and fish. An alternative approach is to just use a very small amount of full-fat cheese, butter, cream, or red meat in dishes that emphasize plant foods such as vegetables, whole grains, and legumes."¹⁰

The Mayo Clinic states, "Trans fat is considered by many doctors to be the worst type of fat you can eat. Unlike other dietary fats, trans fat — also called trans-fatty acids — both raises your LDL ("bad") cholesterol and lowers your HDL ("good") cholesterol.

A diet laden with trans fat increases your risk of heart disease, the leading killer of men and women."

Trans fat in your food

The manufactured form of trans fat, known as partially hydrogenated oil, is found in a variety of food products, including:

Baked goods. Most cakes, cookies, pie crusts and crackers contain shortening, which is usually made from partially hydrogenated vegetable oil. Ready-made frosting is another source of trans fat.

Snacks. Potato, corn and tortilla chips often contain trans fat. And while popcorn can be a healthy snack, many types of packaged or microwave popcorn use trans fat to help cook or flavor the popcorn.

Fried food. Foods that require deep frying — French fries, doughnuts and fried chicken — can contain trans fat from the oil used in the cooking process.

Refrigerator dough. Products such as canned biscuits and cinnamon rolls often contain trans fat, as do frozen pizza crusts.

Creamer and margarine. Nondairy coffee creamer and stick margarines also may contain partially hydrogenated vegetable oils.¹¹

The US government's "Healthy Eating Plate" came out in 2011 and replaced The Food Pyramid, which had been the guideline since 1991. Major changes were made, including the number of servings of grains, more vegetables, limiting dairy and emphasizing no sugary drinks. Conceptually, the plate idea and portions are much easier to understand, and hopefully, follow. People still need to be cautious of portions and the quality of food within each category.

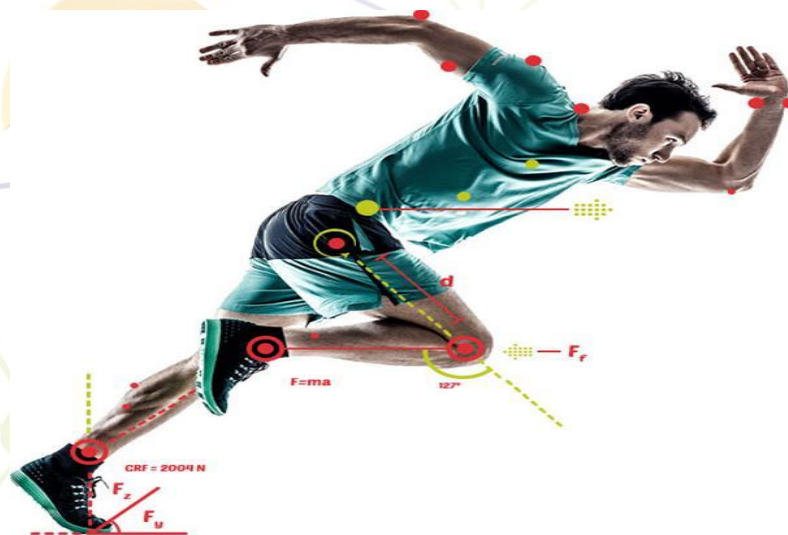


Kinesiology

kinesiology: The science or study of movement, and the active and passive structures involved.

Kinesiology is the study of human motion, and it deals primarily with the muscles and muscle functions. It describes movement, which muscles are involved in the movement, and how they are involved. Kinesiology explores the muscular involvement in strength exercises and sport-specific techniques, while **biomechanics** looks at the physical factors involved in the movement. By applying basic scientific laws, it is possible to come up with accurate descriptions not only of what should take place in the exercise, but also the role that each key joint action and muscle plays. Thus by studying the physical characteristics of the human body and the principles of mechanical physics, you can better guide workouts. You will also have the basis for selecting and using specific exercises to produce desired results.

biomechanics: The study of the mechanical aspects of physical movement, such as torque, drag, and posture, that is used to enhance athletic technique.



Biomechanics is the study of movement more specifically, the movement involved in a strength exercise or in the execution of a sport skill. It deals mainly with physical factors such as speed, mass, acceleration, levers, and force, and with the physical functions of the movement. So, you can think of biomechanics as the science of movement based on principles derived from physics and anatomy. It explains the why of a movement and how the movement can be improved through science-based modifications.

Kinesiology tells you exactly which muscles are involved in the particular actions that take place in an exercise, and biomechanics shows you the way to do exercises most effectively. Thus, a basic understanding of kinesiology and biomechanics helps you to determine what exercises a person should do, how the workouts should be conducted, how effective exercise execution is, and whether the exercises are safe.

KINESIOLOGY



Proper exercise selection plays an important role in the overall process of program development. However, before going into a detailed analysis of exercises and the muscles involved, you should have a good understanding of how muscles function. It is important to have a solid comprehension of the various types of muscular contractions, as well as the different dynamic and static regimes in which the muscles must operate during execution of strength and explosive exercises. This basic understanding will enable you to effectively evaluate exercises and exercise execution.

TYPES OF MUSCLE CONTRACTIONS

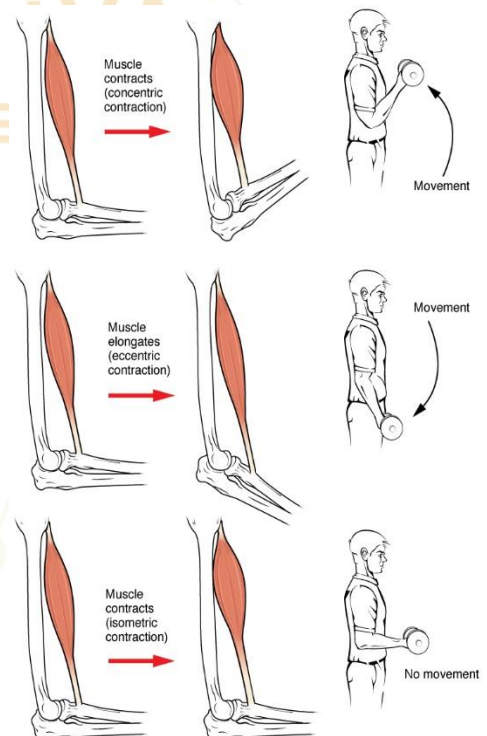
Muscles perform three types of contractions: concentric, eccentric, and isometric. These three types are defined next and depicted in Figure 4.1. When executing a strength exercise, all three of the muscle contractions are involved. As you perform a movement, the main muscles undergo a **concentric contraction** while the opposite muscles undergo an **eccentric contraction**. The adjacent parts of the body that are not in use are stabilized via the **isometric contraction**. Thus, all three operate simultaneously, each with a very important purpose.

concentric contraction: A type of muscle activation that increases tension on a muscle as it shortens.

eccentric contraction: A type of muscle activation that increases tension on a muscle as it lengthens.

isometric contraction: A muscle activation in which the muscle fires but there is no movement at the joint and no change in length of the muscle.

stabilization: The act of being stable or balanced.



Concentric Contraction

In a concentric contraction, the muscles shorten to produce movement. It is sometimes known as overcoming strength. In other words, when the muscle contracts, it overcomes the resistance and the cause of the resistance is set into motion. An example is the biceps curl. When you contract the biceps and other elbow flexor muscles, you get movement of the forearm, which raises the weight held in the hand. Concentric strength is usually measured by the maximum amount of weight that can be overcome in one repetition (also called one-repetition maximum, or 1RM).

Eccentric Contraction

In an eccentric contraction (often known as a yielding contraction), the muscle lengthens (stretches) as it contracts. The more the muscle lengthens or the faster it is stretched, the greater the tension that is developed. The eccentric contraction plays a very important role in controlling and stopping movement and in preparing the muscles for an explosive type contraction. For example, in the biceps curl exercise, when you return to the initial position, the same muscles are involved and they remain under contraction as they lengthen when you lower the weight. Because gravity is the force involved in lowering the weight, the eccentric contraction counteracts the pull of gravity to guide the movement. The intensity of the contraction depends on the resistance being handled. In a ballistic movement, as the muscle lengthens, it increases in the intensity of its contraction. When it is strong enough, it stops the movement. The eccentric contraction can generate up to 50 percent greater tension than the concentric. This is why the eccentric contraction is so powerful not only in controlling and stopping movement, but also in generating sufficient tension in the muscles in order for them to contract explosively.

Isometric Contraction

In an isometric contraction, the muscle exhibits strength but the limbs do not move. The muscle develops tension and some shortening of the muscle fibers and tendons occurs, but the limbs and body do not move. This type of contraction is seen in the **stabilization** of a joint or body as when you hold a particular position to execute an exercise. You can generate approximately 20 percent greater strength in an isometric contraction than you can in a concentric contraction.

Anatomical Terms of Movement

Anatomical terms of movement are used to describe the actions of muscles upon the skeleton. Muscles contract to produce movement at joints, and the subsequent movements can be precisely described using this terminology.

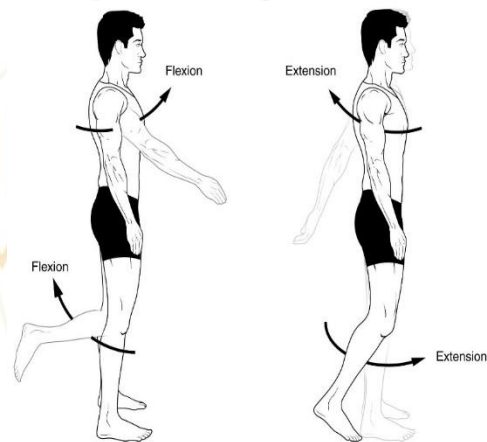
The terms used assume that the body begins in the **anatomical position**. Most movements have an opposite movement – also known as an antagonistic movement. We have described the terms in antagonistic pairs for ease of understanding.

Flexion and Extension

Flexion and extension are movements that occur in the sagittal plane. They refer to increasing and decreasing the angle between two body parts:

Flexion refers to a movement that decreases the angle between two body parts. Flexion at the elbow is decreasing the angle between the ulna and the humerus. When the knee flexes, the ankle moves closer to the buttock, and the angle between the femur and tibia gets smaller.

Extension refers to a movement that increases the angle between two body parts. Extension at the elbow is increasing the angle between the ulna and the humerus. Extension of the knee straightens the lower limb.

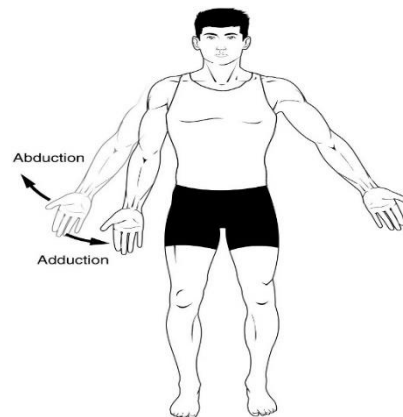


Abduction and Adduction

Abduction and adduction are two terms that are used to describe movements towards or away from the midline of the body.

Abduction is a movement away from the midline – just as abducting someone is to take them away. For example, abduction of the shoulder raises the arms out to the sides of the body.

Adduction is a movement towards the midline. Adduction of the hip squeezes the legs together.

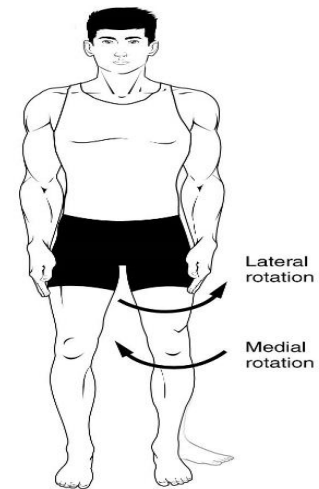


In fingers and toes, the midline used is not the midline of the body, but of the hand and foot respectively. Therefore, abducting the fingers spreads them out.

Medial and Lateral Rotation

Medial and lateral rotation describe movement of the limbs around their long axis:

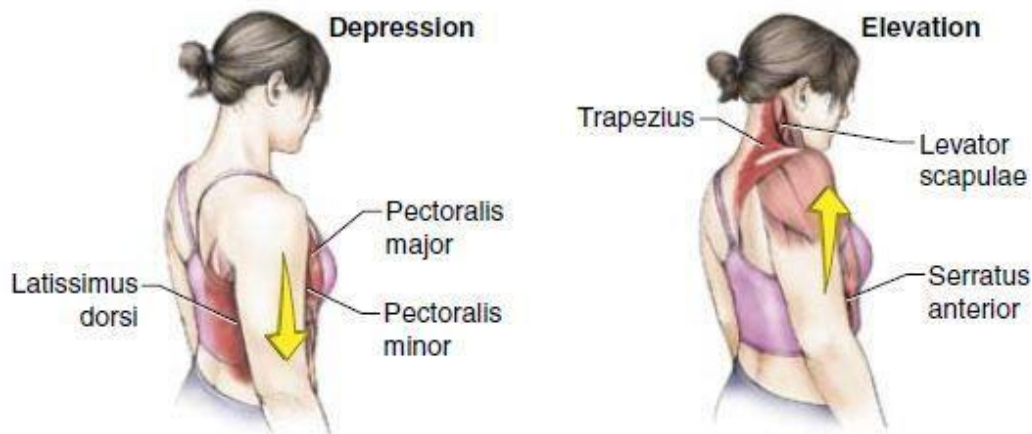
Medial rotation is a rotational movement towards the midline. It is sometimes referred to as internal rotation. To understand this, we have two scenarios to imagine. Firstly, with a straight leg, rotate it to point the toes inward. This is medial rotation of the hip. Secondly, imagine you are carrying a tea tray in front of you, with elbow at 90 degrees. Now rotate the arm, bringing your hand towards your opposite hip (elbow still at 90 degrees). This is internal rotation of the shoulder.



Lateral rotation is a rotating movement away from the midline. This is in the opposite direction to the movements described above.

Elevation and Depression

Elevation refers to movement in a superior direction (e.g. shoulder shrug), **depression** refers to movement in an inferior direction.

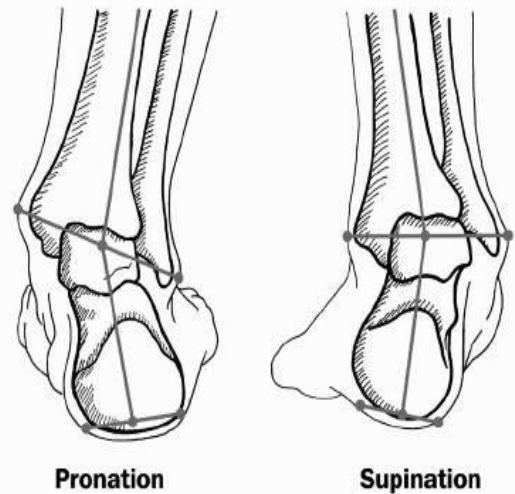


Pronation and Supination

This is easily confused with medial and lateral rotation, but the difference is subtle. With your hand resting on a table in front of you, and keeping your shoulder and elbow still, turn your hand onto its back, palm up. This is the supine position, and so this movement is **supination**.

Again, keeping the elbow and shoulder still, flip your hand onto its front, palm down. This is the prone position, and so this movement is named **pronation**.

These terms also apply to the whole body – when lying flat on the back, the body is supine. When lying flat on the front, the body is prone.

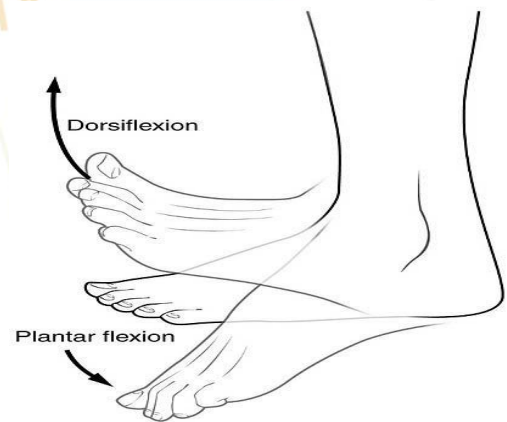


Dorsiflexion and Plantarflexion

Dorsiflexion and plantarflexion are terms used to describe movements at the ankle. They refer to the two surfaces of the foot; the dorsum (superior surface) and the plantar surface (the sole).

Dorsiflexion refers to flexion at the ankle, so that the foot points more superiorly. Dorsiflexion of the hand is a confusing term, and so is rarely used. The dorsum of the hand is the posterior surface, and so movement in that direction is **extension**. Therefore we can say that dorsiflexion of the wrist is the same as extension.

Plantarflexion refers extension at the ankle, so that the foot points inferiorly. Similarly there is a term for the hand, which is palmarflexion.

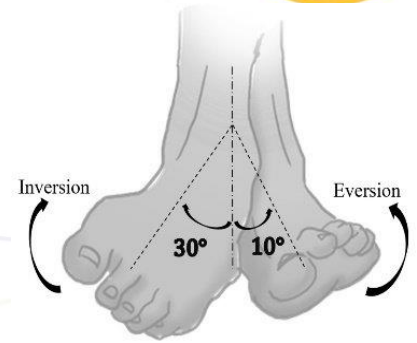


Inversion and Eversion

Inversion and **eversion** are movements which occur at the ankle joint, referring to the rotation of the foot around its long axis.

Inversion involves the movement of the sole towards the median plane – so that the sole faces in a medial direction.

Eversion involves the movement of the sole away from the median plane – so that the sole faces in a lateral direction.

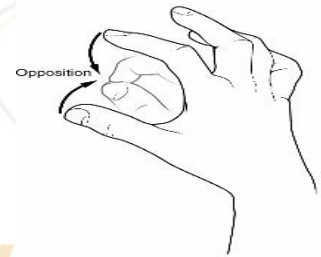


Opposition and Reposition

A pair of movements that are limited to humans and some great apes, these terms apply to the additional movements that the hand and thumb can perform in these species.

Opposition brings the thumb and little finger together.

Reposition is a movement that moves the thumb and the little finger away from each other, effectively reversing opposition.



Circumduction

Circumduction can be defined as a conical movement of a limb extending from the joint at which the movement is controlled.

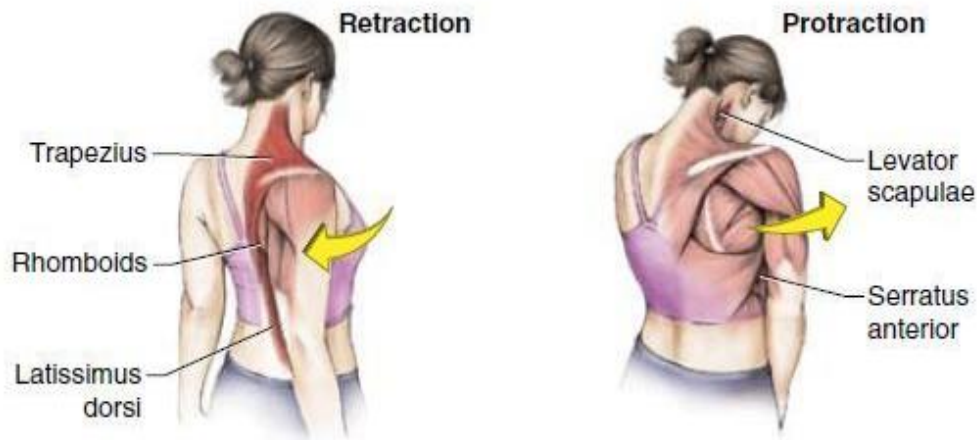
It is sometimes talked about as a circular motion, but is more accurately conical due to the 'cone' formed by the moving limb.



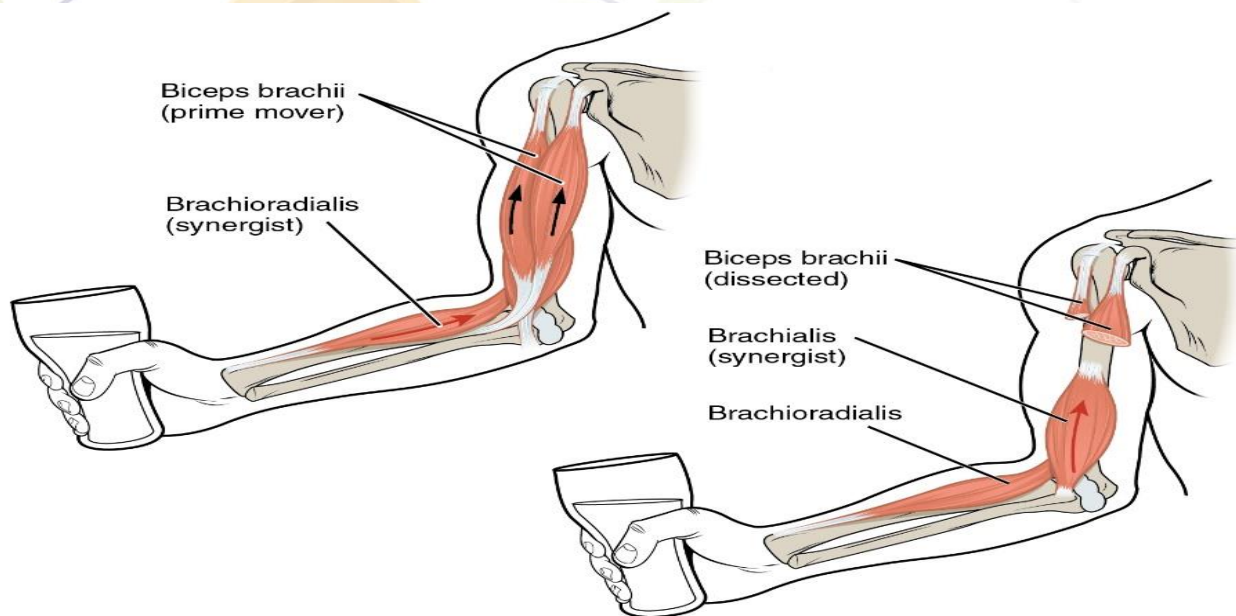
Protraction and Retraction

Protraction describes the anterolateral movement of the scapula on the thoracic wall that allows the shoulder to move anteriorly. In practice, this is the movement of 'reaching out' to something.

Retraction refers to the posteromedial movement of the scapula on the thoracic wall, which causes the shoulder region to move posteriorly i.e. picking something up.



ROLES OF MUSCLES



Prime Mover (Agonist)

A muscle is called a **prime mover** or **agonist** when it is the main muscle involved in a concentric contraction. Thus in the biceps curl, the biceps brachialis and brachioradialis are agonists for elbow flexion. Also, many muscles are prime movers in more than one action. For example, the biceps is also the prime mover in forearm supination.

prime mover (agonist): Denoting a muscle in a state of contraction, with reference to its opposing muscle, or antagonist.

Assistant Mover

An **assistant mover** usually plays a secondary role to the prime mover muscles involved. However, secondary muscles sometimes play a main role in certain ranges of motion or certain exercises. An example is the pronator teres in elbow flexion. It is a prime mover in pronation and an assistant in elbow flexion. Usually secondary or assistant muscles are not as powerful in the movement as the main agonists (prime movers).

assistant mover: Muscle that plays a secondary role to the prime mover involved.

Antagonist



An **antagonist** muscle has an action directly opposite to that of the agonist. When an agonist undergoes a concentric contraction, an antagonist undergoes an eccentric contraction to guide the movement and to stabilize the joint. As the movement goes through the full range of motion (ROM), the antagonist muscle develops greater tension and stops the movement before it goes beyond the normal ROM.

antagonist: Something opposing or resisting the action of another.

Keep in mind that the role of antagonist and agonist can change depending on the action that's taking place. For example, in the biceps curl illustrated in Figure 4-2 the biceps is a prime mover while the triceps is an antagonist. When the triceps is involved in elbow extension, however, it becomes a prime mover and the biceps becomes its antagonist.

During a muscular contraction, especially when the weights are very heavy, both the agonist and antagonist undergo contraction (known as **co-contraction**). This simultaneous contraction is needed to stabilize (hold in place) the joint while the action occurs. When the resistance is very light, the agonist and antagonist are not strongly contracted. The antagonist undergoes a strong eccentric contraction mainly to slow down and stop movement injury from occurring. When the weights are very heavy and both agonist and antagonist are under contraction, the antagonist contracts eccentrically, lengthening in order to make the movement possible.

co-contraction: When both the agonist and antagonist undergo contraction.

Stabilizer

Usually a **stabilizer** muscle holds a body part in place. It anchors the bone so that the prime mover has a firm base against which to contract (i.e., for the muscle to pull against). In order to create precise movements, stabilization of the limb or body part is important in all movements.

stabilizer: Muscle that steadies or holds a body part in place.

The stabilizer muscle usually undergoes an isometric contraction to hold the bone in place. At times, there may be slight movement of the body part, but it is still considered stabilization. For example, in the knee extension exercise with rubber tubing (in which you hold the thigh at a 45-degree angle and then extend the leg), there may be some natural movement of the thigh. However, the thigh is still considered as stabilized by the muscles around the hip joint.

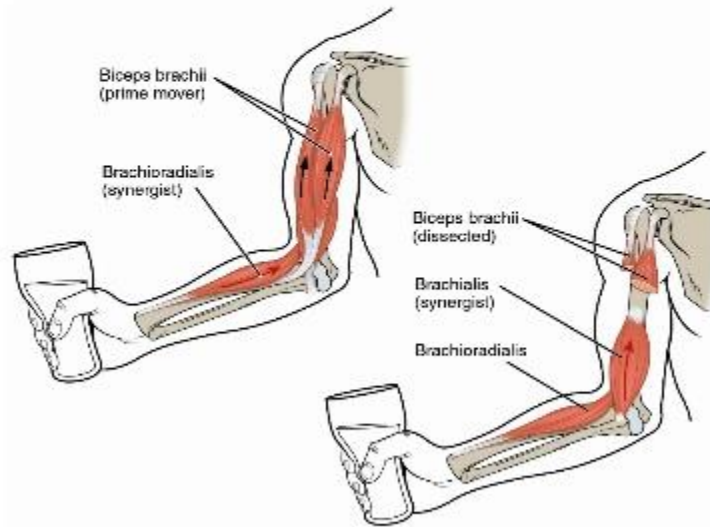
When doing an overhead press, the quadriceps and especially the erector spinae of the lower back contract to hold the trunk erect as you raise the weights overhead. If these muscles did not contract, you would have a loose spine that was very susceptible to injury. You would also lack a firm base against which the muscles could contract. Breath holding at this time also contributes greatly to stabilizing the trunk. Thus the muscles and breathing play an important role in stabilization when doing strength exercises, especially when the weights are heavy.

Because the isometric contraction is used in stabilization, the muscles involved can become fairly well developed. However, you should not ignore other exercises and exercise regimes to strengthen the muscles involved. Keep in mind that the isometric contraction is not as effective as the concentric contraction in developing strength, mass, or definition.

Synergy

Synergy Muscle

- Some cause movements by working as pairs or groups (**synergy**)
- Synergy muscle assist prime mover



Because this term has been used in many different ways, its meaning has become somewhat diffused. Most often it is used in two ways. First is **helping synergy**, in which two muscles contract simultaneously to produce one movement while their other actions cancel each other out. For example, in the sit-up exercise, when the internal and external oblique muscles contract, they have a tendency to not only perform spinal flexion but to rotate the shoulders. In order to prevent the rotation, the internal and external oblique muscles cancel out their rotational action and the resultant force is used for spinal flexion.

helping synergy: When two muscles contract together to create one movement.

Second is **true synergy**, in which a different muscle contracts to stop the secondary action of another muscle. For example, the biceps brachii is both a supinator of the forearm and a flexor of the elbow joint. Thus, in elbow flexion, the pronator quadratus comes into play to cancel the supinating tendency of the biceps so that only flexion occurs. The pronator teres also comes into play, but because it is a flexor of the elbow, it acts as a helping synergist.

true synergy: When a muscle contracts to stop the secondary action of another muscle.

Synergy can also be used synonymously with the term **neutralizer**. In other words, a muscle acts as a neutralizer when it contracts to counteract or neutralize an undesirable action of another muscle during its contraction.

neutralizer: When a muscle contracts to counteract an undesirable action of another muscle.

BIOMECHANICS AND PERSONAL TRAINING

Now that you understand the basics of kinesiology, you can move on to **biomechanics**, which is the physical study of movement. Knowledge of joints, muscles, and their possible actions leads to a better understanding of what occurs during the execution of strength and other exercises. As a trainer, you must understand the mechanical and physical factors involved in exercise and movement. These factors determine how effectively and safely an exercise is executed. When you combine your knowledge of anatomy with kinesiology and biomechanics, you have powerful tools for creating personalized programs for your clients and executing them in effective, efficient ways.

biomechanics: The study of movement

KEY CONCEPTS OF BIOMECHANICS

Your knowledge of biomechanics will help you to create safe, effective, efficient personal training programs for your clients. The following sections provide brief descriptions of key biomechanical concepts. Examples of specific exercises are used where applicable but will be discussed in detail in Unit 8.

STABILITY

Stability is the ability to maintain a balanced state. In order to ensure safety during exercise execution, stability in the body is necessary. Stability helps produce the desired results when using free weights; you must stabilize yourself in order to isolate the desired movement and perform it correctly. With machine weights, when you assume the necessary position, there is little need to balance your body as you execute the exercise; the machine stabilizes itself along the movement path so that its user can perform the exercise correctly.

stability: The ability to maintain a balanced state.

For example, when doing an overhead press, the muscles of the legs and trunk must contract in order to hold the body in place. The trunk must be rigid in order to provide a stable base for effective contraction of the shoulder muscles. If not, any change in the balance of the weights overhead may lead to a loss of balance, which in turn could cause injury, especially if you lose control of the weights.

The basic principles of stability are simple. One such principle is that the larger the base of support, the greater the body's stability. This is why you should most often assume a position with the feet at approximately shoulder width or wider. Standing with your feet together results in a narrow base of support, which does not provide the foundation needed for stability when doing heavy lifts, especially overhead lifts.

Another basic principle of stability is that the lower the body is, the more stable it becomes. Bending the knees helps to lower the body's center of gravity (where the bodyweight is concentrated). For example, in order to prevent lower-body movement and keep the spine vertical during shoulder (upper-body) twisting, it is important to bend the knees to stabilize the lower body and hips. This position keeps the spine from falling out of alignment and limiting the movement to the shoulders. The bent-knee position also helps to prevent knee injuries.

Foot placement also plays an important role. If the feet are parallel and shoulder-width apart (A), the weight should be close to you or overhead. This is the preferred stance in most exercises because you have good stability in a left to right direction. In a stride position (B), you can better balance the weight in a forward-backward direction.

When lying on a bench, always place the feet on the floor (C) in order to increase sideward stability. Keeping the feet on the bench creates an unstable position, especially when using heavy weights or a barbell. It becomes even more important to keep the feet on the floor when doing explosive or throwing actions.

FORCE

Force is the interaction that creates work, action, or physical change. Muscular force is exhibited in a push or pull type motion. Only the muscles (or, more accurately, muscle strength) can create the force needed in order to put an object into motion. When moving a weight in a strength exercise, you must take four components of force into consideration: magnitude, direction, point of application, and line of action.

force: The interaction that creates work, action, or physical change.

Magnitude refers to how much force is applied to the dumbbells, barbells, or machine handles. For example, if you wish to lift a barbell weighing 100 pounds, you must apply more than 100 pounds of force to lift it. Keep in mind that additional force is needed in order to overcome the weight of the limbs and body involved and to overcome resting inertia.

magnitude: How much force is applied to the dumbbells, barbells, or machine handles.

Direction refers to the way in which the force is applied. For example, is it applied horizontally, vertically, or a combination of both? This information is especially important in sports such as running, swimming, and in the throwing events.

direction: The way in which the force is applied.

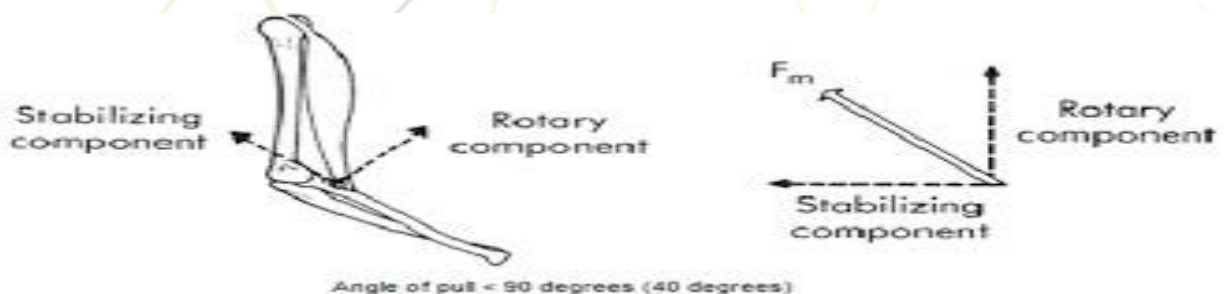
Point of application refers to where the force is applied on the body or implement being used. It plays a role in many exercises including the overhead press and the squat. For example, if you hold a barbell in the middle of the bar with your hands close to one another, the force applied is close to the center of mass of the bar and the exercise becomes more effective. But in this situation, greater balance is also needed, especially if the barbell is long (such as an Olympic bar). In this case, you must assume a wider grip, in which you apply force at two points to raise one barbell. This position loses efficiency but enhances safety and enables you to do the exercise. In sports, the point of application of force is where the hand or fingers are in contact with a ball when shooting (basketball) or throwing (baseball). In hitting, it is where the ball and hitting implement make contact; for example, a golf ball on the club head, a baseball on a bat, or a tennis ball on a racquet.

point of application: Where the force is applied on the body or implement being used.

Line of action (also line of force) refers to an imaginary straight line drawn from the point of application of force through the direction of force. The more directly the force is applied in exactly the same direction as the intended movement, the greater the amount of force that goes in this direction. It is important to understand that you can push in one direction to get motion in another direction, usually a side component. For example, if you have a very wide stance in the squat, as in the sumo style, you have less distance to rise up when doing the squat. However, the force generated by each leg does not go straight upward; it is at an angle to the body. As a result, only a portion of the total force raises you. To raise all of the applied forces that go through the body's center of gravity, you must keep the feet under the hips. In sports, the legs must drive the hips upward or forward or a combination of both in various jumping and running actions. Thus, it is important that the feet remain directly under the hips when the force is applied in the intended direction.

line of action (line of force): An imaginary straight line drawn from the point of application of force through the direction of force.

ANGLE OF MUSCLE PULL



When you do a strength exercise, the strength exhibited at different points in the range of motion varies because of the angle at which the muscle pulls. For example, if you do a biceps curl beginning with fully extended arms, it is more difficult to generate sufficient force in order to start moving the weight than when you start with the arms bent. When your arm is straight, the biceps muscle inserts at an angle of approximately 10 degrees on

the radius bone of the forearm. When the muscle shortens (i.e., when you begin the curl), most of the muscle's force goes into the joint

angle of pull: The angle at which a muscle pulls relative to the long axis of the bone on which it pulls.

in order to stabilize the elbow rather than to raise the forearm with the weight. When the angle of insertion approaches 90 degrees, all the force of the muscle is used in raising the weight (all the strength generated is used to rotate the forearm). So, you are much stronger when there is approximately a 90-degree angle in the elbows than when the arms are extended. This is known as having a mechanical advantage, which means that you can do more work at this angle of muscle pull.

If you use a weight that is the heaviest you can overcome in the early range of motion, it may appear light when you approach the 90-degree angle in the elbow. To overload the muscle in this range you must use more weight or go through a shorter ROM resulting in your arms not fully straightening in the bottom position. Doing only this over a long period of time, however, results in loss of flexibility.

The body is best suited for speed, not force. Thus, even though you are weak at the beginning of a straight-arm elbow flexion movement, you possess the ability to develop great speed in the hand (if the resistance is not too great). Very little shortening of the muscles produces a large movement of the hand. This is known as having a physiological advantage, which is very important in speed and quick movements.

WORK

In physics, **work** refers to what happens when a force is applied to an object. The actual amount of work that you do is measured by the formula $W = F \times D$, where W = work, F = force, and D = distance or displacement of the object being moved. The greater the force and the greater the distance over which the force is being applied (the weight is moved), the more work is done. When you hold a weight in the hands with an isometric contraction—even though the muscles are generating great tension and you may use a lot of energy to hold the position, you are not doing any work because you are not moving the weight any distance. For work to be done, there must be movement. If not, you are only expending energy, which is not work. Energy is more physiological, while work is more mechanical.

work: Force times distance. Measured in foot-pounds and similar units. Example: Lifting a 200-pound barbell 8 feet and lifting a 400-pound barbell 4 feet each require 1,600 foot-pounds of work.

POWER

The term **power** is often misused in fitness and sports literature. Power is often equated with the amount of force one generates, but this is only partially correct. In physics and in most sports, power is defined as the work done in a unit of time. To calculate it, you must first consider the time involved in executing the movement. For example, if you do a squat

with a 300-pound barbell and you move the barbell three feet from the bottom position to the top position, you will have done 900 foot-pounds of work ($W = 300 \text{ lb} \times 3 \text{ ft} = 900 \text{ ft} \cdot \text{lb}$). In reality, you actually lifted more. But for simplicity, your body weight and external factors are not taken into consideration.

power: The work done in a unit of time.

To calculate power, you can use the same example, but consider the amount of time it took to lift the weight 3 feet. For simplicity, assume it took 3 seconds. Therefore, 900 foot-pounds of work divided by 3 seconds equals 300 foot-pounds of work per second. If you executed the squat in approximately 2 seconds, then the amount of power generated would be 450 foot-pounds per second. Thus you can see how the amount of power generated depends very much on the amount of time it takes to accomplish the work. The faster the work is done, the greater the amount of power; the slower the work is done, the lesser the amount of power.

Much confusion has arisen in this area because of powerlifting. In this sport, maximal weights are lifted slowly, so it should be considered a pure strength sport (rather than a power sport). The amount of power is not great in comparison to that of a weightlifter, who lifts maximal weights as quickly as possible. The weightlifter exhibits much greater power than the powerlifter, but the powerlifter exhibits greater strength than the weightlifter.

NEWTON'S LAWS OF MOTION

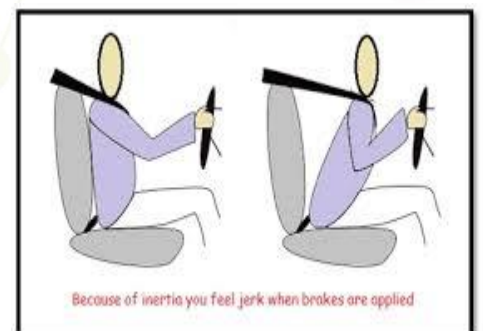
Isaac Newton's three laws of motion contribute to the key principles of biomechanics by describing inertia; the relationship between force, mass, and acceleration; and the relationship between equal and opposing forces. Understanding these three laws and how they apply to training will help you create thoughtful training sessions.

Newton's First Law: Inertia

Newton's first law states that **an object at rest tends to stay at rest, and an object in motion tends to stay in motion**; this is the essence of **inertia**. There are two types of inertia: resting inertia and moving inertia. **Resting inertia** means that **when an object is at rest, it will stay at rest unless acted upon by some outside force**. For example, a loaded barbell or dumbbell lying on the ground has resting inertia. In order to lift it, you must apply a force greater than the weight of the implement itself.

inertia: The tendency for an object to remain in its current state (in motion or at rest).

resting inertia: An object is at rest, it will stay at rest unless acted upon by some outside force



Moving inertia means that **when an object is in motion, it will remain in motion unless acted upon by some outside force**. Thus, once you put a barbell or dumbbell into motion, it will continue on its own accord without additional application of force to keep it moving. This can easily be seen with lighter weights. For example, when doing lateral arm raises with straight arms and light weights, you will experience the weights “flying” upward without your effort if you apply a vigorous thrust in the bottom position.

moving inertia: An object that is in motion will stay in motion unless acted upon by some outside force.

When you use heavy weights in a strength exercise, the movement is slow so that you must apply force through the entire **range of motion (ROM)** in order to keep the weights in motion. In this case, although the heavy weights are moving, they will quickly stop because of the constant force of gravity. Gravity pulls down and thus creates the slow speed. However, if the weight moves too fast (like when you “throw” a heavy weight, such as a medicine ball), it may require great force to stop it at the end of the ROM. If your muscles are not capable of generating this stopping force, injuries can occur if you do not release the weight. The role of the eccentric contraction is especially important here.

range of motion (ROM): The movement of a joint from full flexion to full extension.

The further away the mass of the object is during weightlifting, the greater the inertia, especially on the return. This means that when the weight is moving, it is more difficult to control regardless of whether it is moving up or down, although it is especially difficult on the down return when gravity pulls. Because of this fact, you should always position yourself as close as possible to the weight you are lifting. The closer the weight, the easier it becomes to control; it has less lateral rotational inertia. The further away the mass, as in straight-arm front or lateral arm raises, the more difficult it is to move the weights with control.

Note also that with straight arms, you can use a lighter weight and it will feel the same as a heavy weight held at half the distance. Because of this fact, most bodybuilders bend their elbows to bring the weights closer to their bodies. A bent-elbow position also allows them to use more weight. However, it is important to understand that when you bend the elbow, you may be putting the arm in a different position, which will change the muscular involvement. In the front-arm raise when you bend the arms, the elbows may turn out approximately 45 degrees. You will therefore be doing a combination of front and lateral arm-raise exercises.

The use of heavy weights is not always best for precise strength development. By using a long lever arm to create greater rotational inertia, you can create great resistance with lighter weights. Also very important to consider is that the stress on the spine is considerably less when lighter weights are used. This consideration is especially important for beginners.

Newton's Second Law: $F = MA$

Newton's second law of motion deals with force and its relationship to **mass** and **acceleration**. In essence, **in order to create a force, you must place a mass into motion with acceleration and a change in velocity**. Note that mass multiplied by velocity is known as **momentum**. Thus in the previous weightlifting example, when a weight (mass) is moving (has velocity), it has momentum.

mass: A body of coherent matter.

acceleration: The rate of change of velocity per unit of time.

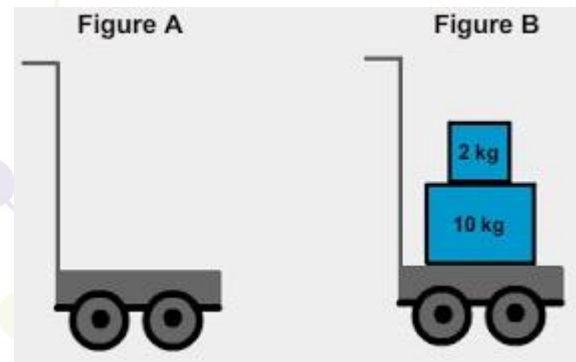
velocity: The speed of something in a given direction.

momentum: The quantity of motion of a moving body, measured as a product of its mass and velocity.

Momentum is seen in many exercise machines that use weight stacks attached to a cable. When you do the exercise, you must adjust the speed of your movement to how fast the weights move up and down. If you move the resistance levers at a very fast rate, the weight stack will continue to move when you stop. Thus, at times, the weight may be going up when it should be going down, or vice versa. When this happens, the cables may bind or snap off the pulley. Therefore, to do an exercise quickly, you must use machines that allow for quick movements, or you must use free weights or rubber tubing (resistance bands).

A key point to keep in mind while training is that when the muscle generates a force, there must be acceleration of the weight. In other words, the speed of the object must be increasing in order to be a true force. When first starting an exercise, you must generate force. When the weight is stationary and you begin movement, you accelerate the weight. However, once the weight is in motion, it only has velocity. You are no longer creating force unless you are changing the speed of the object.

In essence, you must place the weight into acceleration in order to get it moving. Once the weights are in motion, they have momentum. For some reason, acceleration and momentum of weight have become unacceptable to many trainers. They consider acceleration and momentum dangerous—to be avoided at all costs. Rather than avoiding it, though, you should understand what is taking place and thus prepare the body to handle the forces that are encountered. Force, acceleration, and momentum can be your allies rather than your enemies. The key is to learn how to work *with* them in order to make the best use of them.



If a force is maximal or if a weight is accelerated at a maximal speed, then you must release it in a throwing or pushing action because it will be impossible for the muscles of the body to stop the weight or the limb from continuing its movement. The heavier the weight or the greater the acceleration of the weight, the more difficult it is to stop it. It is because of this that most strength exercises are performed at a slow to moderate speed.

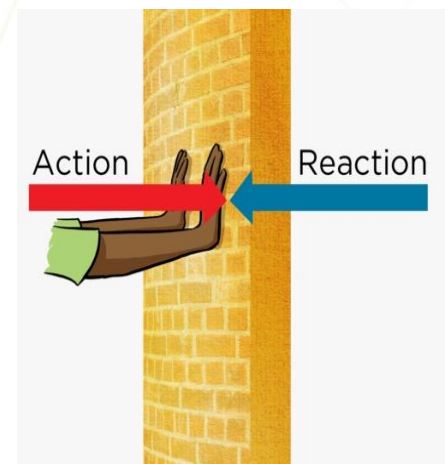
Note that even lighter weights can generate a lot of force when they are accelerated, as in some aerobic routines. A 5-pound weight with extended arms moving rapidly can generate a tremendous amount of force that must be stopped by the eccentric contraction of the antagonist muscles. If the muscles are not sufficiently strong to stop the movement, an injury may occur. In general, you can increase speed of execution in some exercises (usually at the beginning of the movement) so that momentum will carry the weight through the remaining range before damage is done to the joints. However, a strong eccentric contraction of the antagonist muscles is still needed in order to stop the movement.

In addition to creating force to move a weight against the pull of gravity, it is also necessary to create a force to counteract gravity when lowering weights. In this case, gravity is the major force pulling the weight down. The muscles contract eccentrically in order to create the force needed to control the downward movement of the weight. Keep in mind that gravity causes a weight to go into motion at increasingly greater rates of speed. Gravity exhibits an acceleration of approximately 32.2 feet per second squared (32.2 ft/sec^2). Thus for every second of downward fall, a weight gains greater speed.

Care must be taken when handling weights, especially heavy weights. If they drop from a high height, the amount of force upon landing can be extremely high. Even the height of a few feet can cause a weight to have a force of several times its actual weight. Because of the acceleration produced by gravity, you must always control the weights on the (negative) return with the eccentric contraction.

Newton's Third Law: Action-Reaction

Newton's third law of motion is also known as the *equal and opposite reaction principle* and it applies to weightlifting and bodybuilding exercises. The law states that **objects in contact exert equal and opposite forces on each other**. When doing an exercise such as the push-up, you must push against the floor with your hands. The floor, in turn, pushes against you and as a result you raise the trunk. This is known as reactive force. The same concept applies when jumping. When you land on the ground, you apply a force against the ground; in turn, the ground applies an equal and opposite force against you to propel you into the air.



LEVERS

A **lever** is a rigid bar that turns about an axis of rotation, called a **fulcrum**. In the body, bones represent bars, joints represent fulcrums, and muscle contractions represent force. The lever rotates about the fulcrum as a result of the force being applied to it, causing its movement against a resistance (e.g., a weight). The amount of resistance can vary from maximal to minimal. In fact, the bones themselves or the weight of the body segment may be the only resistance applied. All lever systems have each of these three components in one of three possible arrangements. The arrangement of these points and the direction in which the force is being applied will determine the type of lever being used.

lever: Rigid bar that turns about an axis of rotation or a fulcrum.

fulcrum: The point on which a lever rests or is supported and on which it pivots.

First-class Lever

The **first-class lever** (see Figure 5.5A) is similar to the seesaw, in that it has its fulcrum between the force and the resistance. When you sit on one end, you apply a downward resistance on one side, of the fulcrum and an upward force on the other side. If someone else sits on the opposite side another downward resistance is brought into play. And if the weights are equal and the distance from the fulcrum is equal, you will be in balance with no movement occurring. In order to get movement, you will have to make your body heavier (or lighter) to place it into motion. Nodding the head is an example of a first-class lever. The head is the resistance, and the contraction of the neck muscles lifts the weight around the fulcrum (the joints of the neck). First-class levers do not produce a great amount of force. They do produce a maximum ROM and speed of movement.

first-class lever: Fulcrum in the middle, the effort is applied on one side of the fulcrum and the resistance on the other.

Second-class Lever

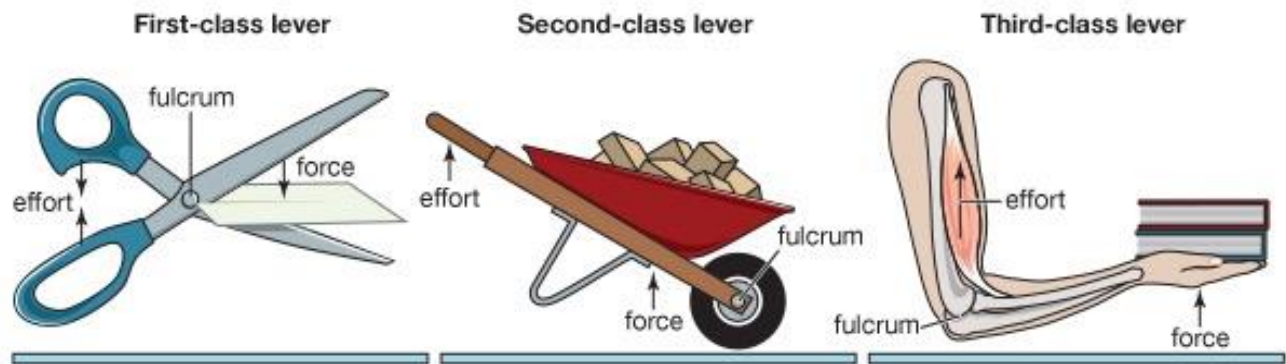
In the **second-class lever** (see Figure 5.5B), the weight (resistance) is distributed between the axis of rotation (fulcrum) and the application of force. This type of lever is most suited for a gain in force. Picture a wheelbarrow: The fulcrum is the wheel, and the weight is in the bucket located in the middle; the action of your arms pulling upward on the handles generates the force. The second-class lever is exemplified when you rise up on your toes. This type of leverage allows you to walk and run, and is effective for overcoming resistance. One weight-training exercise that utilizes the second-class lever is the push-up. In this case, the fulcrum is the balls of the feet in contact with the floor, the weight is the center of gravity of your body mass, and the force is in the arms, pushing you upward.

second-class lever: A lever in which the load lies between the fulcrum and the effort.

Third-Class Lever

In the **third-class lever** (see Figure 5.5C), the force is applied between the fulcrum and the resistance. This is the most common type of lever found in the body. For example, in the biceps curl, the biceps inserts approximately 1 inch below the elbow joint. The point of attachment is known as the point of application of force. The elbow is the axis of rotation (fulcrum), and the resistance is the forearm and weight held in the hand. Thus the distance from the point of application of force to the fulcrum (called the force arm) is very short and mechanically inefficient. The key reason for this inefficiency is that the resistance arm (the distance from the fulcrum to where the weight is located) is quite long. This configuration places the weight far from the application of force. A short force arm and a long resistance arm is a

third-class lever: A lever in which the effort is placed between the fulcrum and the load



By definition, **kinesiology** is the study of movement. However, this definition is too general to be of much use. Kinesiology brings together the fields of anatomy, physiology, physics, and geometry, and relates them to human movement. Thus, kinesiology utilizes principles of mechanics, musculoskeletal anatomy, and neuromuscular physiology.

Mechanical principles that relate directly to the human body are used in the study of **biomechanics**. Because we may use a ball, racket, crutch, prosthesis, or some other implement, their biomechanical interaction must be considered as well. This may involve looking at the static (nonmoving) and/or dynamic (moving) systems associated with various activities. Dynamic systems can be divided into kinetics and kinematics. **Kinetics** are those forces causing movement, whereas **kinematics** are those time, space, and mass aspects of a moving system.

This text will give most emphasis to the musculoskeletal anatomy components, which are considered the key to understanding and being able to apply the other components. Many students are subject to negative thoughts at the mere mention of the word kinesiology. Their eyes glaze over, and their brains freeze. Perhaps, based on past experience with

anatomy, they feel that their only hope is mass memorization. However, this may prove to be an overwhelming task with no long-term memory gain.

As you proceed through this text, you should keep in mind a few simple concepts. First, the human body is arranged in a very logical way. Like all aspects of life, there are exceptions. Sometimes the logic of these exceptions is apparent, and sometimes the logic may be apparent only to some higher being. Whichever is the case, you should note the exception and move on.

Second, if you have a good grasp of descriptive terminology and can visualize the concept or feature, then strict memorization is not necessary. For example, if you know generally where the patella is located and what the structures are around it, you can accurately describe its location using your own words. You do not need to memorize someone else's words to be correct.

By keeping in mind some of the basic principles affecting muscles, understanding individual muscle function need not be so mind-boggling. If you know what motions a particular joint allows, that a muscle must span a particular joint surface to cause a certain motion, what that muscle's line of pull is, then you will know the particular action(s) of a specific muscle. For example, the elbow allows only flexion and extension. (A muscle must span the joint anteriorly to flex and posteriorly to extend. The biceps brachii is a vertical muscle on the anterior surface of the arm. Therefore, the biceps flexes the elbow.

Yes, kinesiology can be understood by mere mortals. Its study can even be enjoyable. There is no natural or human-made law that says otherwise. A word of caution should be given: Like exercising, it is better to study in small amounts several times a week than to study for a long period in one session before the exam.

POSTURAL DEVIATIONS:

UNDERSTANDING GOOD POSTURE

Posture is the way the body holds itself when sitting, standing, lying down, or moving. In the past, the analysis of posture and exercises to correct posture were strongly emphasized. In the present, posture appears to be mostly ignored. Proper musculoskeletal alignment creates good posture. When you have good posture, your muscles and bones are basically in balance and your body is symmetrical. When a deviation in posture exists, it is often due to a lack of strength of particular muscles to hold the body in the needed position. For example, weak erector spinae muscles of the lower back are the main culprits in not being able to maintain an erect trunk in standing and walking, or maintain proper back posture when lifting weights. As a trainer, you must learn to recognize good posture and postural deviations in your clients and in yourself.

posture: The way the body holds itself when sitting, standing, lying down, or moving.

Most people with postural deviations are usually unaware of their posture. More startling is that they show little concern about having good posture. However, for athletes and fitness-minded people, posture should be of great concern. The reason is quite simple: Posture can determine the outcome of performance and well-being. Before relating how this can happen, it is necessary to first examine some of the benefits of good posture.

BENEFITS OF GOOD POSTURE

Good posture is important to health. It is needed in order to keep the organs in place and to allow them to work efficiently and effectively. For example, if you have swayback, the intestines press against the floor of the abdominal cavity instead of being held in place. This position interferes with their normal work. If you have rounded shoulders and an excessively rounded upper back, constriction occurs in the chest cage. Because of this constriction, it becomes impossible to completely fill the lungs with air, which is vital in athletic performance and fitness activities.

Posture affects how you walk, run, jump, lift weights, and execute other skills. For example, if you have rounded shoulders, your arms may be slightly in front of your body instead of hanging alongside your body. As a result, you may find that instead of lifting the arms sideways directly overhead, you are lifting them up and in front of the body. This adaptation changes the muscular movement and the movement pathway.

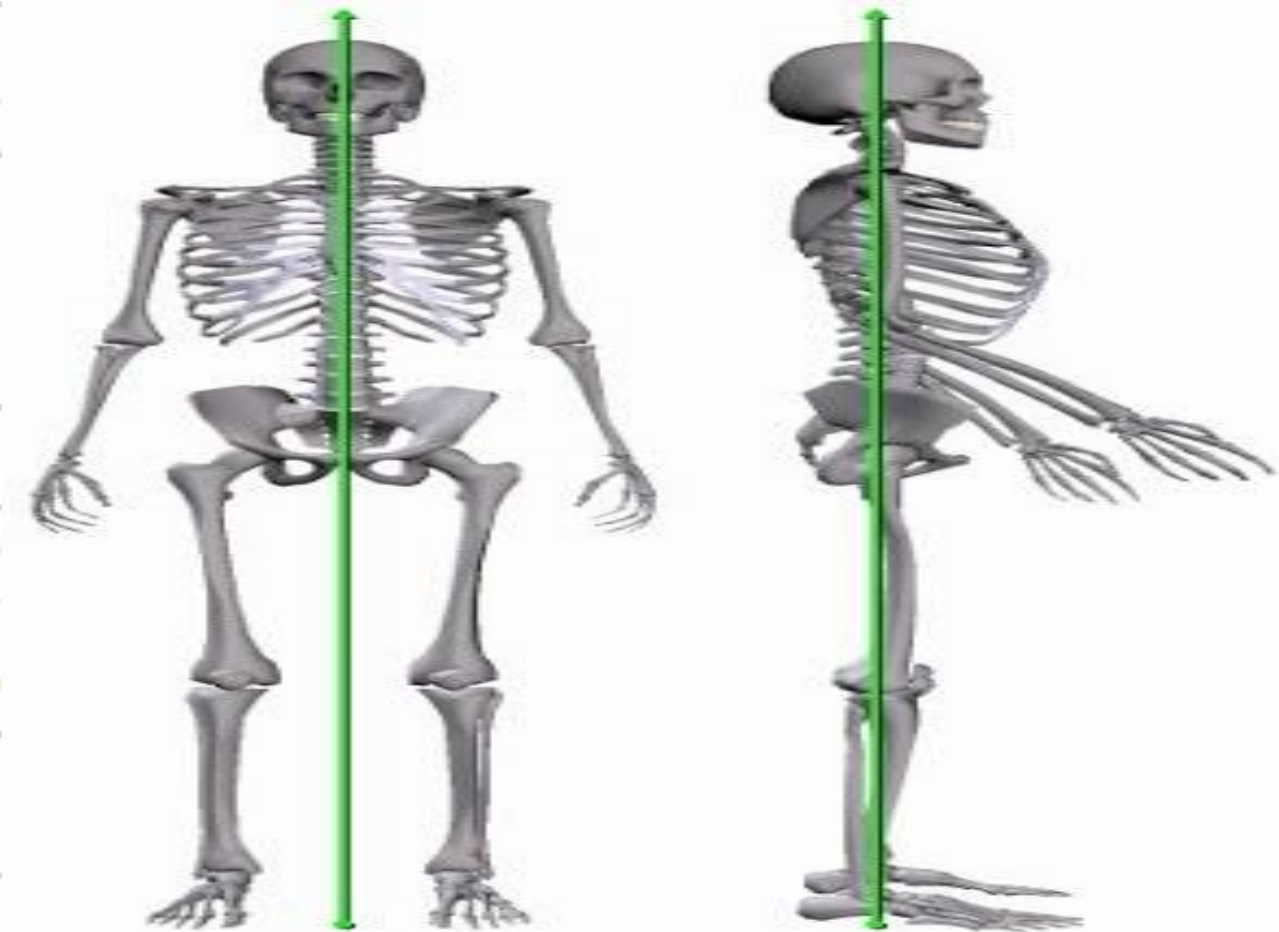
If you cannot hold your trunk erect during running, you will not have an effective push-off or knee drive for a long stride length. Even in walking, if your feet or thighs are excessively rotated outward, greater stress will be placed on the hip and knee joints. If walking in this manner is carried on for a long period of time, injuries to these joints can occur.

Posture plays an important role in the prevention and rehabilitation of back problems. For example, tight hip flexors may keep your pelvis tilted forward, causing swayback. If the hip flexors are too weak while the abdominal muscles are strong, it may cause flattening of the curve in the lower spine. If the upper portion of the hamstrings is too tight, they do not allow you to hold the arch in your lower back when doing exercises such as the squat, or when bending over to lift something. When your abdominal muscles are too tight, they may flatten the spine, which places excessive pressure on the anterior aspects of the spinal discs. If they are too weak, they may cause swayback.

By strengthening and stretching the necessary muscles to create good posture, you not only prevent injuries but also rehabilitate them. Merely correcting posture is often all that is needed in order to relieve back pain. For example, pulling the head back into proper alignment is often sufficient to produce the normal curvature of the vertebral column. By lifting the head and looking forward, you can activate the low-back muscles to hold the spine in place and alleviate the problem.

Good posture makes you feel good. Because of its many benefits, such as ease of movement, good balance of muscle strength and flexibility, proper positioning of the spine, and proper functioning of the internal organs, your body feels good and you feel good. You feel vibrant, confident, and ready to perform. Thus, posture should be of prime focus in all fitness activities.

Posture is dynamic. Good posture is relatively easy to attain and maintain. Part of creating good posture is learning new habits of sitting, standing, and walking. However, the major factor is strengthening the key muscles that hold you in the proper posture.



POSTURAL SELF-CHECK

Knowing the status of your own posture will help you to assess the posture of your clients. To assess your posture, perform this self-check (see Figure 6.1):

Stand with your back against a wall. Your heels, backs of the calves, buttocks, upper back, and head should comfortably touch the wall. If you must strain to make all points of contact, then you probably have some deviations.

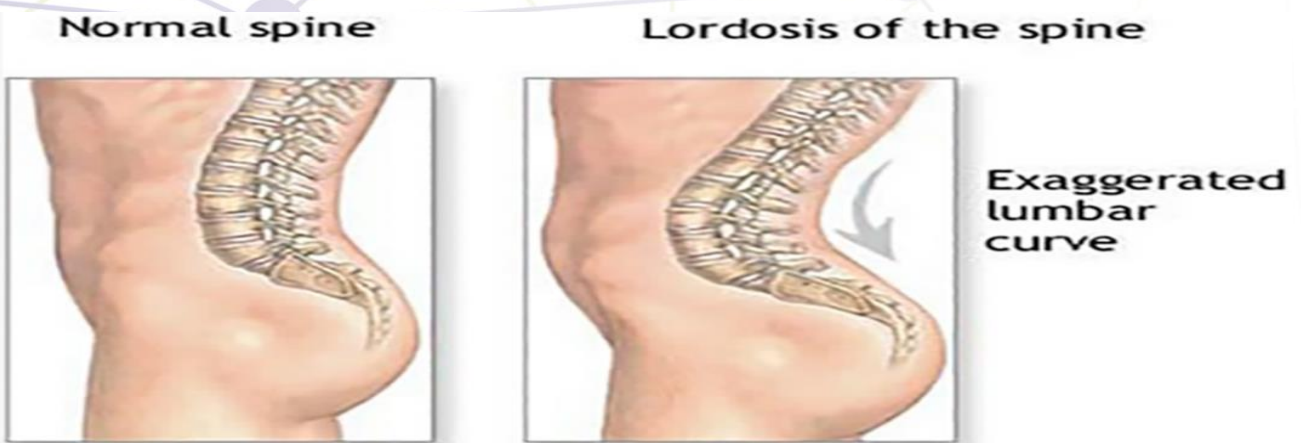
Also effective is to secure a string to the ceiling and hang a weight at the end of the string. Stand so that the string is lined up with your nose, and then have a front-view picture taken (or look in a mirror). Notice whether your shoulders are leaning to one side or another or if more of your body is on one side of the line. With good posture, you should be symmetrical on both sides of the string.

To get a graphic representation of how your weight is distributed in front of you and behind you, try lining up the string in the middle of your shoulder down to the floor. This method of postural self-check also shows whether you have any major deviations in spinal curvature or positioning of the hips.

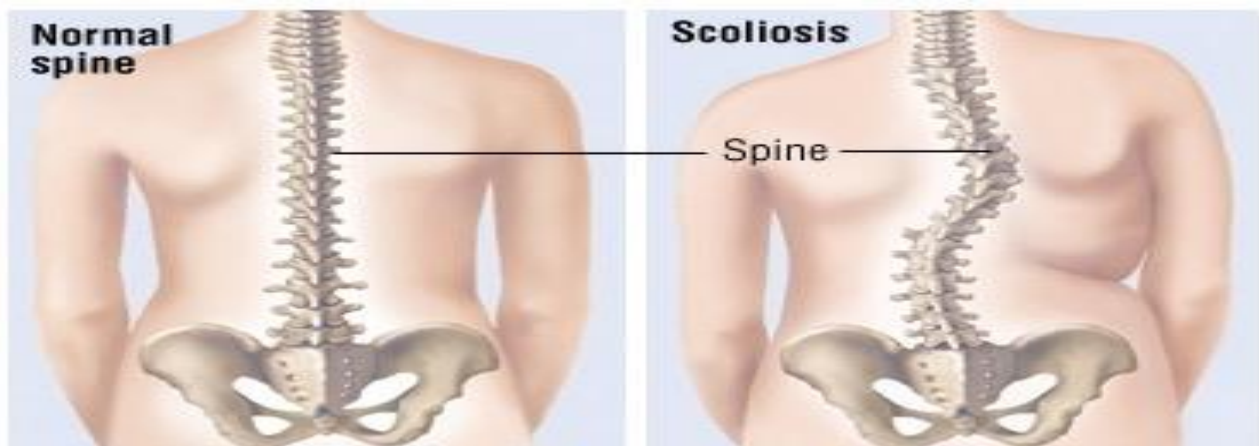
RECOGNIZING POSTURAL DEVIATIONS

The relative strength and flexibility of the spinal muscles play a role in the alignment of the trunk and pelvis. When imbalances exist, three abnormal conditions result: **lordosis**, **scoliosis**, and **kyphosis**. Table 6.1 includes these conditions as well as other muscle misalignments and summarizes the muscles involved in them. Figure 6.2 illustrates these conditions as well as proper posture for comparison. When you learn to recognize these imbalances in your clients and where they come from, you can help them take steps to improve their posture.

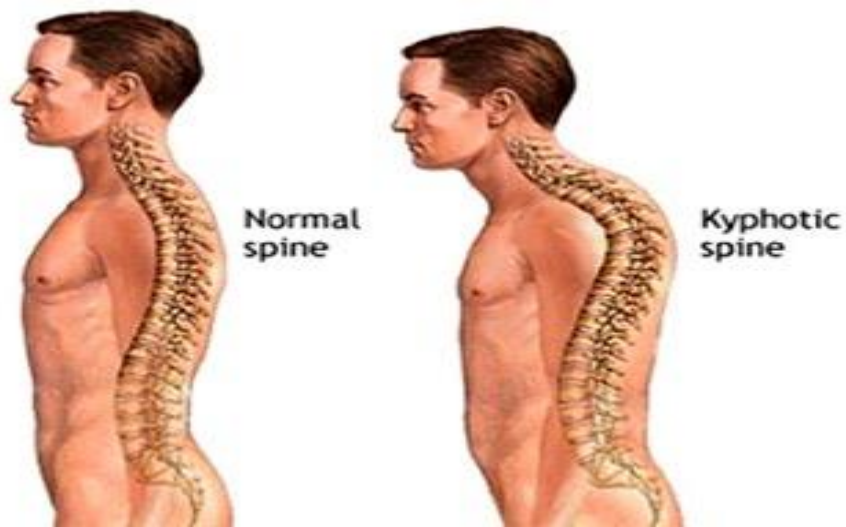
lordosis: A spinal disorder in which the spine curves significantly inward at the lower back. Also called 'Swayback.'



scoliosis: A spinal disorder in which there is a sideways curve to the spine. The curve is often S-shaped or C-shaped.



kyphosis: A spinal disorder which is characterized by an abnormally rounded upper back (more than 50 degrees of curvature).



In lordosis, the superior iliac crests of the pelvis move forward and downward from the normal anatomical position. This position is known as anterior tilt of the pelvis. In most cases, the hip flexor muscles are shortened and the abdominal muscles are lengthened or severely relaxed.

In posterior pelvic tilt, the hip flexor and low-back muscles stretch while the abdominal and hamstring muscles shorten. Posterior tilt is not as common as anterior tilt and is rarely brought about by lack of muscular strength. Both anterior and posterior pelvic

tilt place the lumbar vertebrae in potentially dangerous positions because of increased disc pressure and a change in the line of gravity of the trunk.

In scoliosis, excessive lateral curvature of the spinal column exists. If the curvature is relatively minor, you can do exercises to stretch the concave side and strengthen and shorten the convex side of the curve. These exercises usually bring about straightening of the spine. If rotation of the vertebral column also exists, the affected abdominal oblique or erector spinae muscles must be strengthened.

Kyphosis is an exaggerated anterior-posterior curvature of the spinal column. It occurs most frequently as excessive forward bending of the thoracic area and is seen most frequently in older adults. It is usually associated with osteoporosis and osteoarthritis and can result in the hunchback position. It also appears to occur more frequently with younger adults as a result of practicing poor posture and performing an excessive number of crunches through a shortened range of motion.

The term *flatback* is used frequently in conjunction with a kyphotic condition because the exaggerated curvature of the thoracic spine creates a reduction in the natural lumbar curvature. The flattening creates a posterior tilt condition of the pelvis.

Rounded shoulders are sometimes associated with kyphosis of the thoracic vertebrae, but it is not the same condition. A round shoulder condition is technically abduction or protraction of the scapulae. This position creates a “hollow” chest condition. You can have abducted scapulae without having a kyphotic condition, or you can have both conditions.

ROLE IN ATHLETIC PERFORMANCE

Most people have some alignment deviation. As a result, the body does not work at maximum efficiency. It is analogous to a machine. When a machine is properly aligned, the working parts act efficiently. The machine will last much longer than one that is out of alignment. In a misaligned machine, wear and tear on the bearings increases and stress and strain on the working parts produce general depreciation.

As with any machine, when optimal performance is desired in the body, you must pay attention to the alignment of the body parts. The balance of all the muscles acting on any joint or body part affects proper maintenance of alignment. Faulty posture indicates a shift of a body segment in relation to the other segments. In addition, a shift of joint positions (or alignment) occurs in relation to the normal gravitational line. Under optimal conditions, all body segments are lined up properly so that undue stress does not fall on any one particular joint. When misalignment exists, stress is placed on particular joints. For example, if your shoulders drop forward, your head goes back and your pelvis rotates to the rear. If you constantly lean to one side, your pelvis tilts sideways and your spine curves to the opposite side, sloping one shoulder. Therefore, if you assume and maintain an out-of-line position, your body must adjust the controlling ligaments and muscles. In other words, if one body part is out of alignment, another body part must likewise get out of alignment in order to balance it. Keep in mind that approximately 75 to 80 percent of the work of the muscles is involved in merely obtaining and maintaining joint stability.

A high development of the agonists–antagonist function is essential to the development of coordinated, skilled movement since antagonists control the speed, range, and force of the action of agonists. This is true of the main muscles involved and of all the stabilizers of the joints that are activated. This includes the stabilizer muscles that hold the joint of the non-moving part in place to allow for movement at the other end of the muscle. Good balance must exist between the opposing muscles.

When you have faulty posture, the normal length of the opposing muscles is changed so that if one is shortened, its opposing muscle must be stretched or lengthened.

Therefore, any skill that you execute is affected by the performance of these muscles. When faulty posture exists, movement is abnormal.

For example, if you have round shoulders, you may have adaptive shortening of the pectoralis major and the serratus anterior, as well as tight anterior shoulder joint ligaments. The opposing muscles (the mid-lower trapezius and rhomboids) are overstretched. Thus the scapulae will not only swing apart but also rotate, resulting in the lowering of the tips of the shoulders. Consequently, the more muscle mass is developed, the greater the force that is applied to the joint. Because of this relationship, the stress on the joint increases and the imbalance is increased even more.

Two factors operate in round shoulders. First, the arm weight and head weight fall forward of the line of gravity. Therefore, an increase in the dorsal curve of the spine must compensate. In turn, this increase must be balanced by a forward position of the pelvis and increased lumbar lordosis (arching). Such a shift of body weight onto the forefoot tends to increase pronation and depression of the foot arch.

Second, all arm work (shot put, discus, javelin, baseball pitching, golf swing, or exercise execution such as the bench press or dumbbell fly) will show decreased efficiency, because the weakened rhomboids have become long and the pectorals short. As a result, the arm cannot be moved back to the maximum shoulder joint range, because the contractile length of the pectoralis major will not allow it.

Deficiencies can also be seen in the shoulder girdle rotation (e.g., when twisting with a barbell on the shoulders). This is because the spinal column cannot rotate on a “straight” axis, as the spine is now bent. Thus more attention should be given to the antagonist musculature instead of only trying to develop the agonist. You must restore the muscle balance. This is essential to arm and shoulder girdle performance.

In the lumbar area, you must strengthen the abdominal muscles, especially the internal and external oblique muscles that are mainly responsible for forward shoulder rotation and flattening the abdominal wall. In addition, you must stretch and strengthen the erector spinae by doing back raises to a position above level. You can also do the reverse sit-up and reverse trunk twist to stretch the erectors even more, and at the same time strengthen the abdominal muscles. Other muscle group pairs should be corrected in a like manner if there is any imbalance.

The key to having a well-aligned and balanced body is to proportionally develop the muscles (agonists) on one side of the joint with the muscles on the other side of the joint (antagonists). This way the muscles will keep the joints in their natural state and prevent deviations from occurring, allowing for more efficient and sustainable athletic performance.

TONUS

Muscular tonus is associated with blood circulation and economy in movement. Improper alignment results in additional muscular effort and strain, especially since it creates rotary movements at the various joints. If excess muscular effort is sufficient to produce fatigue, it can eventually affect your health. In more severe cases, the strain on the joints can be sufficient to alter structure. There is also evidence to indicate that chronic strain contributes to the development of arthritic types of ailments in later life. Such alterations mean limited use of body parts and continued fatigue and strain.

muscular tonus: A state of partial contraction present in a muscle in its passive state which, in skeletal muscles, aids in the maintenance of posture and in the return of blood to the heart.

Perhaps the most important component of a sound exercise recommendation for fitness is the level of **exercise intensity**. The prescribed level of intensity must be sufficient to overload the cardiovascular system, but not so severe that it overtaxes any of the systems of the body. For the apparently healthy individual who wants to develop and maintain an adequate level of cardiorespiratory fitness, the ISSA—as does the ACSM—recommends that the intensity level of the exercise be 55% to 85% of the person's $\dot{V}O_2$ max. For most participants, exercise intensities between 60% to 80% of $\dot{V}O_2$ max are prescribed. It is generally believed, however, that the appropriate intensity threshold to elicit a training effect is at the low end of the recommended continuum for those who have been relatively sedentary and at the high end of the scale for those who are physically active.

exercise intensity: Refers to how much energy is expended when exercising.

Unfortunately, many adults — particularly those just starting an exercise program — have difficulty estimating the intensity of exercise needed to produce improvements in their aerobic fitness levels. This difficulty in properly gauging exercise intensity often results in a lack of improvement or overexertion which, in turn, leads to excessive muscle soreness and/or injury. The key you want to remember is that to maximize the benefits from your client's aerobic workouts, he/she needs to exercise within an appropriate intensity range. In order to identify what intensity range is appropriate for your client, use one or more of the several techniques which exist for monitoring acceptable levels of exercise intensity. The most widely used methods are **target heart rate (THR)**, **ratings of perceived exertion (RPE)** and **metabolic equivalent units (METs)**.

target heart rate (THR): A specific age-based pulse rate to be maintained during aerobic exercise to ensure optimal cardiovascular function.

ratings of perceived exertion (RPE): A rating scale ranging from six to 20 that gives an indication of your workout intensity level.

RPE Scale	
6	No exertion
7	
8	
9	
10	
11	Light
12	
13	Somewhat Hard
14	
15	Hard
16	
17	Very Hard
18	
19	
20	Maximal Exertion

metabolic equivalent units (METs): A unit of measurement that refers to the relative energy demands of an activity in comparison to your energy demands in a resting state.

Using Target Heart Rate (THR)

When an individual exercises, the heart beats faster to meet the demands of the muscles for more blood and oxygen. The more intense the exercise bout, the faster the heart beats. Thus, all factors considered, monitoring heart rate during exercise provides a very accurate reflection of the metabolic intensity of the exercise. Several techniques can be used to determine an appropriate target heart rate.

Exercise intensity can be prescribed in terms of heart rate (HR) by using specific heart rate values that are equal approximately to 60% to 80% of $\dot{V}O_2\text{max}$. One method involves monitoring HR at each stage of a maximal graded exercise test (GXT). Heart Rate is plotted on a graph against the $\dot{V}O_2\text{max}$ equivalents (METs)

for each stage of the test to define the slope of the heart rate response to exercise. From this relationship, the exercise heart rate associated with a given percent of $\dot{V}O_2$ max can be obtained. While this method is generally the most accurate approach for determining what constitutes an appropriate level of exercise intensity, it is oftentimes impractical since it requires you to complete a maximal GXT with its associated logistical and safety considerations.

Another method for assessing exercise intensity is based on the observation that 70% and 85% of maximal heart rate is equal to approximately 60% and 80% of functional capacity $\dot{V}O_2$ max. For example:

THR range = maximal HR x 0.70 and 0.85

Where:

THR range = target heart rate

maximal HR = 220 - age (*low estimate*)

or 210 - (0.5 x age) (*high estimate*)

The third method for determining the exercise heart rate is to calculate the heart rate reserve (HRR). Developed by Dr. M. Karvonen, the HRR method of determining training or target heart rate range requires a few simple calculations. (See Karvonen Formula)

Whenever possible, use an accurate measurement of maximal heart rate rather than a predicted or estimated one. Estimated maximal heart rates have a distinct disadvantage in that they are based on population averages and, as a result, have a relatively large standard deviation of plus or minus 10 to 12 bpm.

Karvonen Formula	
	EXAMPLE
220	220
- ____ Subtract age	<u>-30</u>
____ Predicted, age-adjusted maximum	190
- ____ Subtract resting heart rate	<u>-70</u>
____ Heart rate reserve	120
x ____ Multiply by exercise intensity of 70%	<u>x</u> <u>0.70</u>
____	84
+ ____ Add resting heart rate back in	<u>+70</u>
= ____ Target heart rate (beats/minute)	154

The HRR or Karvonen method adjusts THR so that a given percentage of THR calculated by using the Karvonen method is the same as the HR equivalent of that specific percentage of $\dot{V}O_2$ max. In other words, a THR calculated at 70% of maximal heart rate is roughly the same as the heart rate corresponding to 70% of $\dot{V}O_2$ max.

Using Rate of Perceived Exertion (RPE)

Another method for prescribing and monitoring exercise intensity involves using ratings of perceived exertion (RPE). Perceived exertion refers to the physical strain individuals believe they are experiencing while exercising. Perceived exertion feedback is important because it provides a practical means for individuals to become sensitive to what constitutes appropriate exercise intensity. During exercise, perception of effort is influenced by a variety of cues

— some local in nature (e.g., sensations of muscular discomfort or strain) and some central in nature (e.g., heart rate, breathing rate, etc.).

The most common measurement tool for assessing perceived exertion is a ratings scale named after its developer, Swedish psychologist Gunnar Borg. Borg's pioneering concepts involving perceived exertion have evolved dramatically over the years. Originally, Borg proposed a scale (also referred to as the RPE Scale) that utilized 15 categories (6 to 20) of sensations — seven of which were identified by descriptors.

Borg's original efforts were based on the premise that his numbering system closely correlated to heart rate. When a "0" was added to each category number, a range of numbers quite comparable to the heart rate range in

young, trained individuals (60 to 200 bpm) was created. Borg's initial RPE scale followed the patterns of heart rate and oxygen consumption during incremental aerobic exercise. As exercise intensity increased, heart rate, oxygen consumption, and RPE increased. Subsequently, however, Borg and other exercise scientists recognized that perceived exertion is extensively influenced by additional variables, some of which have a nonlinear relationship to exercise intensity (e.g., lactic acid production, ventilatory responses, etc.).

In the more than 20-year period since Borg introduced his original work on perceived exertion, RPE scales have been found to have a wide range of applications. In using RPE to monitor exercise intensity, the individual who is exercising provides a verbal description of how (relatively) difficult the exercise is based on categories assigned to each of the numbers. For example, a perceived exertion rating of 4 of the Category/Ratio RPE scale has been observed to correspond to approximately 60% of $\dot{V}O_2$ max. On the other hand, a rating of 16 corresponds to approximately 80% to 85% of $\dot{V}O_2$ max. Therefore, an RPE range of 14 (somewhat hard) to 16 (hard) is recommended for most healthy adults.

Assessment for Cardiovascular Endurance

Testing for cardiovascular endurance includes a variety of methods and equipment, reflecting the wide differences in capabilities found among clients. This section will provide protocols for submaximal tests in swimming, leg cycle ergometry, and walking/running. There are several simple methods used to assess cardiovascular endurance in older persons. The two most common methods are the 12-minute run-walk and the step test, both of which are included in this unit.

12-Minute Run-Walk Test

12-Minute Running/Walking Assessment				
CLASSIFICATION	UNDER 30 YR.	30-39 YR.	40-49 YR.	50+ YR.
Very Poor	Under 1.0 mi.	Under .95 mi.	Under .85 mi.	Under .80 mi.
Poor	1.00-1.24	0.95-1.14	0.85-1.04	0.80-0.99
Fair	1.25-1.49	1.15-1.39	1.05-1.29	1.00-1.24
Good	1.50-1.74	1.40-1.64	1.30-1.54	1.25-1.49
Excellent	Over 1.74	Over 1.64	Over 1.54	Over 1.49
SOURCE: COOPER, K., <u>THE NEW AEROBICS</u>				

The goal of this test is to have the participant cover as much distance as possible in 12 minutes. It is best to perform the test on a track for ease of calculations. It is useful to place cones every 100 yards to facilitate measurement of distance. This test can be used to compare pre-training and post-training levels of aerobic fitness. Some older people will not perform well and may receive a poor classification. It is more important to focus on the progress they achieve through training than the classification they might receive. Remember to focus on the positive!

12-Minute Swimming Test

12-Minute Swimming Assessment				
CLASSIFICATION	UNDER 30 YR.	30-39 YR.	40-49 YR.	50+ YR.
Very Poor	Under 440 yd.	Under 418 yd.	Under 374 yd.	Under 352 yd.
Poor	440-545	418-502	374-458	352-435
Fair	546-655	503-612	459-568	436-545
Good	656-765	613-722	569-678	546-655
Excellent	Over 765	Over 722	Over 678	Over 655
SOURCE: COOPER, K., <u>THE NEW AEROBICS</u>				

This test can be used to assess cardiovascular endurance in people who are not able to run or walk comfortably enough to complete the run/walk test. This is also an excellent assessment for asthmatics who seem to experience fewer incidents while performing water exercise. Be sure to find out the length of the lap lanes of the pool that you use for the testing site. Although most pool lanes measure 25 yards or meters, you need to be sure of the distance in order to obtain accurate results.

12-Minute Cycling Test

This test can be used to assess cardiovascular endurance in the client that prefers cycling to the other previous assessments listed. This test can be done outside (using appropriate distance markers) or inside on a stationary cycle.

12-Minute Cycling Assessment				
CLASSIFICATION	UNDER 30 YR.	30-39 YR.	40-49 YR.	50+ YR.
Very Poor	Under 2.5 mi.	Under 2.38 mi.	Under 2.12 mi.	Under 2.0 mi.
Poor	2.50-3.10	2.38-2.85	2.13-2.60	2.00-2.47
Fair	3.11-3.72	2.86-3.47	2.61-3.22	2.48-3.10
Good	3.73-4.35	3.48-4.10	3.23-3.85	3.11-3.72
Excellent	Over 4.35	Over 4.10	Over 3.85	Over 3.72
SOURCE: COOPER, K., <u>THE NEW AEROBICS</u>				

Assessment for Muscular Strength

The participant's posture, range of motion, and starting position for selected activities should be considered in the following strength assessment. It is also important to note any compensatory mechanisms or muscle substitutions that occur during a movement.

Prior to engaging in any physical assessments, be sure to have your client warm up. Stretching or challenging the muscle without a proper warm up is a surefire way to injure your client. Let him/her warm up thoroughly and be sure to explain the procedures you are about to perform. Scores on the following tests, which gauge arm and leg strength, reliably reflect strength throughout the upper and lower body, respectively. Remember, all client information is confidential!

Upper Body Test

Have the client lie on a flat bench and with the barbell racked. When you are ready to begin, instruct the client to place his/her hands on the bar slightly wider than shoulder width. After lifting the bar off the rack, instruct the client to lower the bar to chest level and then push it back up to the starting position. Be sure to stress the importance of breathing properly and keeping the low back pressed into the pad of the bench.

When attempting to determine a client's **one repetition maximum (1RM)**, a good starting place is the client's body weight multiplied by the minimum value listed in the "fair" or "average" category found under the appropriate age-group heading. This will give you a good idea of the amount of weight to start with when testing for 1RM.

one repetition maximum (1RM): The maximum resistance with which a person can execute one repetition of an exercise movement.

For example, say your client is a very sedentary 58-year-old man who weighs 170 pounds. To estimate a weight to begin testing his 1RM, multiply 170 (his weight) by 0.65 (minimum value for fair in his age group) to arrive at 110.5. The number 110.5 represents the amount of weight in pounds that your client should

be able to bench press for a 1RM, taking into consideration his age and assuming that he is in fair condition.

Additional testing of this client shows that he can actually perform a 1RM bench press with 130 lbs. You would determine his rating by dividing 130 lbs. by his body weight of 170 lbs to arrive at 0.76. When you look on the chart in the 50 to 59 age group, you will see that 0.76 would give this client a rating of average.

Values for Bench Press Strength in 1RM/body weight in pounds					
RATING	AGE IN YEARS				
MEN	20-29	30-39	40-49	50-59	60+
Excellent	>1.26	>1.08	>0.97	>0.86	>0.78
Good	1.17-1.25	1.01-1.07	0.91-0.96	0.81-0.85	0.74-0.77
Average	0.97-1.16	0.86-1.00	0.78-0.90	0.70-0.80	0.64-0.73
Fair	0.88-0.96	0.79-0.85	0.72-0.77	0.65-0.69	0.60-0.63
Poor	<0.87	<0.78	<0.71	<0.64	<0.59
WOMEN	20-29	30-39	40-49	50-59	60+
Excellent	>.78	>0.66	>0.61	>0.54	>0.55
Good	0.72-0.77	0.62-0.65	0.57-0.60	0.53-0.59	0.51-0.54
Average	0.59-0.71	0.53-0.61	0.48-0.56	0.43-0.50	0.40-0.50
Fair	0.53-0.58	0.49-0.52	0.44-0.47	0.40-0.42	0.37-0.40

SOURCE: THE INSTITUTE FOR AEROBIC RESEARCH. 1985 FITNESS NORMS.

Lower Body Test

A 1RM seated leg press can be used to assess your client's leg strength. Through a process of trial and error, you will estimate a 1RM max for your client and then divide that value by the client's body weight to come up with a number. Locate the number you came up with on the chart below to rate your client's leg strength.

active range of motion (ROM): The range of movement through which an individual can actively (without assistance) move a joint using the adjacent muscles.

passive range of motion (ROM): The range of motion through which a therapist or equipment can move the joint through the range of motion with no effort from the patient.

Values for Seated Leg Press in 1RM/body weight in pounds					
RATING	AGE IN YEARS				
MEN	20-29	30-39	40-49	50-59	60+
Excellent	>2.08	>1.88	>1.76	>1.66	>1.56
Good	2.00-2.07	1.80-1.87	1.70-1.75	1.60-1.65	1.50-1.55
Average	1.83-1.99	1.63-1.79	1.56-1.69	1.46-1.59	1.37-1.49
Fair	1.65-1.82	1.55-1.62	1.50-1.55	1.40-1.45	1.31-1.36
Poor	<1.64	<1.54	<1.49	<1.39	<1.30
WOMEN	20-29	30-39	40-49	50-59	60+
Excellent	>1.63	>1.42	>1.32	>1.26	>1.15
Good	1.54-1.62	1.35-1.41	1.26-1.31	1.13-1.25	1.08-1.14
Average	1.35-1.53	1.20-1.34	1.12-1.25	0.99-1.12	0.92-1.07
Fair	1.25-1.34	1.13-1.19	1.06-1.11	0.86-0.98	0.85-0.91
SOURCE: THE INSTITUTE FOR AEROBIC RESEARCH. 1985 FITNESS NORMS.					

Muscular Endurance Assessment

Push-ups — measures upper-body strength and endurance.

Directions: Men should use the standard “military style” push-up position with only the hands and the toes touching the floor. Women have the option of using the bent-knee position, if needed. With knees on the floor, hands on either side of the chest and back straight, do as many push-ups as possible with good form until exhaustion. Count the total number of push-ups performed.

Push-Up Norms (men)							Push-Up Norms (women)						
%	AGE					RATING	%	AGE					RATING
	20-29	30-39	40-49	50-59	60+			20-29	30-39	40-49	50-59	60+	
99	≥100	≥86	≥64	≥51	≥39		99	≥70	≥56	≥60	≥31	≥20	
95	62	52	40	39	28	S	95	45	39	33	28	20	S
90	57	46	36	30	26		90	42	36	28	25	17	
85	51	41	34	28	24		85	39	33	26	23	15	
80	47	39	30	25	23	E	80	36	31	24	21	15	E
75	44	36	29	24	22		75	34	29	21	20	15	
70	41	34	26	21	21		70	32	28	20	19	14	
65	39	31	25	20	20		65	31	26	19	18	13	
60	37	30	24	19	18	G	60	30	24	18	17	12	G
55	35	29	22	17	16		55	29	23	17	15	12	
50	33	27	21	15	15		50	26	21	15	13	8	
45	31	25	19	14	12		45	25	20	14	13	6	
40	29	24	18	13	10	F	40	23	19	13	12	5	F
35	27	21	16	11	9		35	22	17	11	10	4	
30	26	20	15	10	8		30	20	15	10	9	3	
25	24	19	13	9.5	7		25	19	14	9	8	2	
20	22	17	11	9	6	P	20	17	11	6	6	2	P
15	19	15	10	7	5		15	15	6	4	4	1	
10	18	13	9	6	4		10	12	8	2	1	0	
5	≤13	≤9	≤5	≤3	≤2	VP	5	≤9	≤4	1	0	0	VP
Rating Key: S, Superior; E, Excellent; G, Good; F, Fair; P, Poor; VP, Very Poor.							Rating Key: S, Superior; E, Excellent; G, Good; F, Fair; P, Poor; VP, Very Poor.						
DATA PROVIDED BY THE INSTITUTE FOR AEROBICS RESEARCH, DALLAS, TX (1994).							DATA PROVIDED BY THE INSTITUTE FOR AEROBICS RESEARCH, DALLAS, TX (1994).						

Sit Ups — measures overall muscular strength and endurance.

Directions: Have the client lie on the floor with knees bent, feet flat, and hands resting on his/her thighs. Have the client contract the abdominals (squeeze the abdominals and have him/her raise high enough for his/her hands to touch the tops of his/her knees. Do not allow the client to pull at the neck or head, and instruct him/her to keep the lower back flat on the floor. If the client has back problems, this test should be avoided.

1 Minute Sit-Up Norms (men)								1 Minute Sit-Up Norms (women)							
%	AGE						RATING	%	AGE						RATING
	<20	20-29	30-39	40-49	50-59	60+			<20	20-29	30-39	40-49	50-59	60+	
99	>62	>55	>51	>47	>43	>39		99	>55	>51	>42	>38	>30	>28	
95	62	55	51	47	43	39	S	95	55	51	42	38	30	28	S
90	55	52	48	43	39	35		90	54	49	40	34	29	26	
85	53	49	45	40	36	31		85	49	45	38	32	25	20	
80	51	47	43	39	35	30	E	80	46	44	35	29	24	17	E
75	50	46	42	37	33	28		75	40	42	33	28	22	15	
70	48	45	41	36	31	26		70	38	41	32	27	22	12	
65	48	44	40	35	30	24		65	37	39	30	25	21	12	
60	47	42	39	34	28	22	G	60	36	38	29	24	20	11	G
55	46	41	37	32	27	21		55	35	37	28	23	19	10	
50	45	40	36	31	26	20		50	34	35	27	22	17	8	
45	42	39	36	30	25	19		45	34	34	26	21	16	8	
40	41	38	35	29	24	19	F	40	32	32	25	20	14	6	F
35	39	37	33	28	22	18		35	30	31	24	19	12	5	
30	38	35	32	27	21	17		30	29	30	22	17	12	4	
25	37	35	31	26	20	16		25	29	28	21	16	11	4	
20	36	33	30	24	19	15	P	20	28	27	20	14	10	3	P
15	34	32	28	22	17	13		15	27	24	18	13	7	2	
10	33	30	26	20	15	10		10	25	23	15	10	6	1	
5	27	27	23	17	12	7	VP	5	25	18	11	7	5	0	VP
1	<27	<27	<23	<17	<12	<7		1	<25	<18	<11	<7	<5	<0	
Rating Key: S, Superior; E, Excellent; G, Good; F, Fair; P, Poor; VP, Very Poor. DATA PROVIDED BY THE INSTITUTE FOR AEROBICS RESEARCH, DALLAS, TX (1994).								Rating Key: S, Superior; E, Excellent; G, Good; F, Fair; P, Poor; VP, Very Poor. DATA PROVIDED BY THE INSTITUTE FOR AEROBICS RESEARCH, DALLAS, TX (1994).							

Goal-setting is an essential preliminary step when designing an effective training program.

repetitions or for an extended period of time. A common example is what the muscles do during an aerobic workout: the lower body muscles contract and relax thousands of times during a 20-minute run.

Muscular strength. Many clients will state that they “want to be stronger or ‘lift heavier.’” These types of clients generally want to exercise at heavier workloads to enhance their strength and power. Typically, these are athletes who are looking to improve their performance and are already familiar with resistance-based training programs. However, a client who is just starting out should start with a program that emphasizes hypertrophy or a muscle endurance training program first to acclimate his or her body to this type of training. **Cardiovascular Training**

Cardiovascular training is often referred to as aerobic endurance training or cardiovascular exercise or even more commonly as “cardio” or “aerobics.” These terms are synonymous because they all encompass exercise that recruits cardiovascular and respiratory systems (heart, blood vessels, and lungs). This mode of training is an integral part of any exercise session. To effectively and safely design a client’s cardiovascular training program, the Personal Trainer must be familiar with the client’s current level of fitness, previous fitness history, and, of course, established goals. Commonly, the goal of many people who begin or continue a cardiovascular pro-gram is to “burn fat” (i.e., fat loss).

As with a resistance training program, the same principle of specificity applies to designing a cardiovascular exercise program. Therefore, the results of a cardiovascular-based program will be more specific to aerobic-based training. In other words, resistance training will not significantly improve maximal aerobic power. In addition, training that involves one mode of aerobic exercise will not necessarily improve a different mode. For example, a client who has a high level of aerobic endurance as a runner may not be able to achieve that same level of endurance as a cyclist. The muscle activation patterns and oxygen requirement vary greatly among different modes of exercise. Therefore, the responses and adaptations will not be equal.

As with all types of training programs, cardiovascular training programs are composed of different components. These components are meant to be manipulated in a variety of ways to ultimately produce the desired outcome. These components include the mode of exercise, intensity of exercise, frequency of exercise sessions, and duration of each session. The first logical step is to decide on the mode or type of exercise. Exercise selection plays a major role in all types of training programs. Cardiovascular exercise modes consist of machine- and non-machine-based exercises such as swimming, jump rope, jumping jacks. There are several factors that need to be considered when choosing the mode of cardiovascular exercise for the client. Some of those factors include, but are not limited to, the following:

- Availability
- Client's ability to perform the specified exercise ➤ Client's preference
- Client's goals

It is also important to note that when training athletes, the athletes should choose modes of cardiovascular exercise that most closely mimic their sport or activity. Some of the more popular modes of machine-based cardiovascular exercises include stair stepper, treadmills, rowers, step mills, cycle ergometers, and elliptical trainers. If the Personal Trainer is in a facility that does not have access to equipment, activities such as walking, jogging, running, boxing, swimming, and jumping rope that encompass non-machine-based cardiovascular exercise modes can be utilized. As with resistance training, the mode of exercise should also be based on the client's goals. For example, a client who is preparing to run his or her first 10K will spend the bulk of the session focusing on running, whether on the treadmill or on the track. This may include, but is not limited to, speed and agility drills, running techniques, and running intervals. If you have a client whose goal is to lose body fat, you should implement a cardiovascular program that uses a variety of exercise modes (e.g., 15 minutes on the treadmill; varying intervals, 15 minutes of boxing drills, and 10 minutes of cycling). This type of training is referred to as "cross training" and is very effective in fat loss and caloric expenditure. This also helps the client fight boredom with a specific mode of cardiorespiratory equipment.

EXERCISE INTENSITY

Before the Personal Trainer can establish frequency and duration of the exercise sessions, he or she must determine the appropriate intensity level to attain the client's goals. To design the best aerobic training program for a client, the Personal Trainer must be able to monitor and regulate the exercise intensity. As explained in Chapter 6, a certain threshold of oxygen consumption or heart rate (HR) reserve, which is the difference between a client's resting HR and his or her maximal HR, must be attained during an aerobic exercise session before improvements in the cardiorespiratory system are seen. Ultimately, the necessary aerobic exercise threshold depends on the client's initial fitness level. In apparently healthy adults, that threshold for continuous training is generally between 50% and 85% of HR reserve (1).

- **Heart rate.** Heart rate is the number of times the heart beats per minute. Heart rate increases as the work rate and oxygen uptake during dynamic exercise increase. The actual magnitude of the HR increase is related to age, fitness level, medications, body position, blood volume, presence of heart (and other chronic) disease, and environmental factors such as temperature and humidity. Maximum attainable HR decreases with age.
- **Stroke volume (SV).** Stroke volume is the amount of blood ejected per heart beat. As explained in Chapter 6, during exercise, SV increases in a curvilinear pattern with the work rate until it reaches near-maximal level equivalent to approximately 50%–70% of aerobic capacity, increasing only slightly thereafter.
- **Cardiac output Q.** The product of SV and HR determines cardiac output. In apparently healthy adults, the cardiac output increases as the work rate increases. However, like HR, the maximal level of cardiac output depends on factors such as age, posture, body size, level of physical conditioning, and presence of cardiovascular disease.
- **Blood pressure.** With increasing levels of exercise comes an increase in systolic blood pressure. Maximal values typically reach 190–220 mm Hg and should not exceed 260 mm Hg. Diastolic blood pressure should decrease slightly or remain unchanged.

Flexibility Training

Proper flexibility and stretching techniques are essential when designing a program to fit a client's needs. Everyone can learn to stretch and improve flexibility, regardless of age, level of fitness, or initial flexibility. The way that the body responds to stretching depends on how the sensory organs respond to the stretch stimulus. When educating the client on the importance of proper stretching techniques, refer to Table 15.6 to discuss the many benefits that are associated with increased levels of flexibility. When starting a stretching or flexibility routine with a client, good communication is essential. Always make sure the client knows that this program will require a lot of hands-on work and always ask questions before the start of the session and throughout the stretching session. This will help build the client-trainer rapport, as well as allow the trainer to understand the client's limitations and preferences.

It is important that the Personal Trainer learn proper stretching techniques when assisting with or assigning stretching exercises to a client. There is no right or wrong answer as to when is the best time to stretch; it is based on individual preference. Stretching may be performed just before or after exercise (and in between sets as well). When designing an exercise program, the Personal Trainer should allow for stretching exercise to be part of both the warm-up and cool-down phases of the session. As with all exercise programs, there are contraindications and precautions to flexibility training.

HOW TO STRETCH?

TYPES OF STRETCHES

There are various ways to gain flexibility through proper stretching programs. The following techniques focus on a different stimulus to elicit a response.

Static stretching. This type of stretching is slow and sustained to increase movement at a particular joint when one segment is manipulated relative to another. The benefits of this type of stretching include the following:

- Decreased possibility of exceeding normal range of motion (ROM)
- Lower energy requirements
- Lower instance of muscle soreness and muscle fatigue • Decrease in muscle tension

Passive stretching. Passive stretching requires assistance from another person. This form of stretching is widely used by Personal Trainers during the client exercise session. To obtain optimal results when performing a passive stretch, the person being stretched must remain relaxed and refrain from any reflexive movements.

Active/dynamic stretching. During an active stretch, the muscle being stretched is actively moved through its ROM. This particular technique requires greater energy than passive or static stretching.

Active assistive stretching. In this type of stretch, the muscle being stretched may require some assistance to go through its ROM, because of muscular weakness or restricted movement due to injury or intervention. This is what is considered a “partner stretch.”

Proprioceptive neuromuscular facilitation. Proprioceptive neuromuscular facilitation stretching is an effective and fast way known to increase static–passive flexibility. Technically, this method utilizes a technique of combining passive stretching and isometric stretching to achieve maximum static flexibility. PNF refers to any of several post isometric relaxation

stretching techniques in which a muscle group is passively stretched and then contracts isometrically against resistance while in the stretched position and then is passively stretched again through the resulting increased ROM. PNF stretching is usually performed with a partner to provide resistance against the isometric contraction and then later to passively take the joint through its increased ROM. It may be performed, however, without a partner, although it is usually more effective with a partner’s assistance.

There are different types of PNF stretching that can be performed to elicit the best response to the stretch stimulus. These types include the contract–relax (hold–relax) and the contract–relax–contract (hold–relax–hold). When performing the contract–relax stretch, the muscle is contracted, relaxed, and then further stretched into its available ROM during the brief relax phase. Contract–relax–contract follows the same procedure; however, a subsequent contraction of the antagonist muscle gains slightly more ROM and should be done only by trained practitioners and used with caution due to increased intensity and risk of client injury from overstretching.

Dynamic, phasic, or ballistic stretching. These terms, when associated with stretching, refer to quick jerking and often bounce-like movements such as bouncing when trying to touch the toes. Generally it is thought that the disadvantages of this type of stretching far outweigh the benefits. Performing these jerking movements can predispose the muscles to injury. When educating the client on the different types of stretches, the Personal Trainer should provide some flexibility exercises that will help the client gain more flexibility.

Plyometrics and Sports Performance

Plyometric exercises are exercises that engage the body in jumping movements or motions. Examples of these exercises can include skipping, trampoline bounding, jumping rope, hopping, split-squats, clap push-ups, and box jumps. Plyometric exercise programming is very similar to resistance and cardiovascular training. For the Personal Trainer to safely design a plyometric-based training program that will enhance the client's sports performance, the client's goals, needs, and preferences

must be established. Plyometric training is not for everyone. This method of training is sport-specific and requires the client to be well conditioned. For athletes to achieve optimal performance in their chosen sport, they must train for speed, strength and power, endurance and flexibility, and coordination.

Plyometrics was first introduced as jumping exercises for the lower extremities. Now, however, there are a variety of upper-body (and lower-body) plyometric exercises that athletes and trainers are using to improve skills in almost every sporting platform. These upper-body exercises are performed using a multitude of small fitness tools such as weighted balls, bands, and bars. Plyometrics has been described as beginning with rapid stretching (eccentric muscle actions), followed by a shortening of the same muscle (concentric muscle action). This is known as the stretch-shortening cycle. Plyometric training is based on the principle of using the elastic properties of the muscle. These conditions will create a greater force.

Balance and Stability

Another, less obvious form of sports performance training is balance or stability training. "Balance" is defined as the ability to maintain a position for a given period of time without moving. There are many sports performance drills that require the athlete to jump, slide, or step in nontraditional movement patterns (i.e., single leg hop or squat hops). These drills rely on the client's proprioception and balance to safely and successfully complete the exercise. Working on clients' core muscles will enhance their ability to maintain the balance and stability needed for their sport. There are several balance drills that can be performed, at various fitness levels, to help develop core strength. Some of these drills include standing on one leg at the beginner level or single leg half squat at the advanced level.

PROGRAM DESIGN

The process of exercise programming can be divided into three steps. The first step is the pre-screening stage, where the Personal Trainer assesses all health and fitness information. The second step is interpreting that information. The third step is combining the information with the interpreted results as well as the client's goals, to formulate an exercise program. When designing the actual client program, there are certain variables that the Personal Trainer must consider, such as what exercises to include in the program, the sequence of the selected exercises, how often the client wants to train, the load, repetition, and set assignment, as well as rest periods.

Each individual potential client has specific goals related to health and fitness. Being able to understand and identify this range of goals is an essential step in program design and exercise selection. To most efficiently and effectively design a client exercise program, the Personal Trainer must fully assess the needs of the client based not only on goals but also on the results of the fitness assessment. The measurement or assessment of health-related physical fitness is a common practice by most fitness professionals. Most professionals rely on a PAR-Q (Physical Activity Readiness Questionnaire) or HRA (Health Risk Assessment) form to gauge the health and physical activity of their clients. The Personal Trainer should be able to identify and understand the different components of fitness assessment results to do the following (2):

- Educate individuals about their current health-related physical fitness
- Use data from the assessments to design individualized exercise programs
- Provide baseline and follow-up data to evaluate exercise programs
- Motivate individuals toward more specific exercises
- Assist with clients' risk stratification
- Discuss clients' preferences
- Discuss clients' goals
- Identify need for physician follow-up or further clearance necessity

Before the initial exercise selection process begins, the Personal Trainer must consider the client's physiological capacity for beginning and maintaining an exercise program. "Exercise training" is defined as planned, structured, and repetitive bodily movement done to improve or maintain one or more components of physical fitness. In Chapter 6, the body's response to exercise and the princi-

ples of adaptation are fully explained. Each of these principles guides the design of an exercise program. In exercise training, the mode of exercise and the frequency, duration, and intensity of

training are critical in achieving fitness, athletic, or health and lifestyle outcomes (3). When designing a client's exercise program, the mode must be specific to the targeted component of fitness, and the frequency, duration, and intensity must be combined in a systematic overload that will result in physiological adaptations. This concept is commonly referred to as the S.A.I.D. (Specific Adaptations to Imposed Demands) principle.

Anatomy of an Exercise Session

The three basic components to any personal training session are the warm-up, the conditioning (stimulus), and the cool-down. Workouts should always begin with some warm-up exercises so the body is better prepared to meet the challenges that will be presented by the succeeding conditioning

stimulus phase.

Warm-up and cool-down phases are the periods

of metabolic and cardiorespiratory adjustment from rest to exercise and exercise to rest, respectively. Therefore, the most appropriate types of warm-up and cool-down are activities similar to the conditioning stimulus activities, performed at approximately 50% of the stimulus intensity.

An appropriate warm-up can improve performance and decrease the risk of cardiac disease events

as well as aid in injury prevention. Cool-down has these benefits as well as helping clear metabolic waste from skeletal muscle and prevent exercise-induced hypotension. Older individuals and those at risk of cardiac disease events benefit from longer periods of warm-up and cool-down. The conditioning stimulus may contain a period of aerobic conditioning, muscle conditioning, or both. Depending on the allotted session time, this may be as short as 20 minutes or as long as 60 minutes.

exercise Selection:

The exercises selected, the resistance or level of stimulus, as well as the order in which the exercises are selected, will determine the intensity of the client's workout. An advanced training program with a client who is an experienced athlete may include as many as 20 exercises, focusing on sports-specific, plyometric, and/or functional exercises as well as total-body conditioning. However, a beginning or basic exercise program may include only one exercise for each large muscle group of the body. As previously stated, the order of the exercises affects

the intensity of the training session and is therefore a very important consideration. For instance, alternating upper- and lower-body exercises does not produce as high an intensity level as performing all lower-body exercises first. Exercises that involve multiple joints and muscles (referred to as multijoint exercises) are more intense than those that involve only one joint (referred to as single-joint exercises)

(8). Below you will find the formulas that are most commonly used to determine sequence of exercise selection.

- ➤ Large muscle groups (i.e., bench press, leg press squats) before smaller muscle groups (i.e., wrist curls, calf raises). This method of exercise arrangement is the most widely used.
- ➤ Multijoint exercises before single-joint exercises, alternate push/pull or upper/lower for total body (extension = push, flexion = pull). An example of this would be a triceps extension (push) followed by a biceps curl (pull).
- ➤ Exercise weak areas before exercises for strong areas
- ➤ Perform exercises that are most intense before those that are least intense

The next step to designing a client exercise program, after determining exercise selection and arrangement, is determining workload. Load refers to the weight used or intensity of the exercise. To best determine the load at which the client should be working



Resistance training design

This chapter provides the Personal Trainer with the key elements needed to design a safe and effective client exercise program. It provides the Personal Trainer with the tools needed to structure the client exercise program by incorporating a review of basic anatomy, applied physiology, exercise protocol, and program design. This process begins with the initial consultation and the fitness assessment, this chapter applies the fundamentals of exercise programming, through the actual personal training session. The core concepts of this chapter are program design, client objectives, and assessment results, the various phases of an exercise program, and methods of strength, resistance, balance, flexibility, and functional training. Other chapters in this book explain, in detail, the physiological responses to the specific types of training programs detailed in this chapter. This chapter briefly addresses the musculoskeletal system, cardiovascular system, and respiratory system, as well as posture and body alignment.

Resistance Training:

GENERAL TRAINING PRINCIPLES:

Resistance training has become an integral part of fitness, wellness, and physical activity programs. Resistance training has been shown to have a positive impact on many health measures, and the collective impact may significantly improve functionality, well-being, and quality of life. When mapping out a plan for an effective resistance training program, the Personal Trainer must follow the following three general training principles: specificity, over- load, and progression.

Specificity. The specificity principle dictates that training a client in a specific way will produce a specific result and that to reach a specific goal the client has to follow a specific type of training program.

For example, a client whose goal is to strengthen the muscles of the chest must perform exercises such as the bench press or dumbbell fly's, to target that area, rather than a lat pull-down, which will work the muscles of the back. The term specificity is also important when the Personal Trainer is designing a "sports-specific" program for a client with specific training goals. For this training method, exercise selection should be based on movements that mimic movements and systems used in competition. For example, a golf-specific training program should focus on exercises that will strengthen the core and trunk muscles, such as swing drills and trunk twists, because they carry over to the specified activity.

Training that incorporates this principle challenges the body to meet and adapt to greater than normal physiological stress. As it does, a new threshold is established that requires an even greater stress to produce an overload. The amount of overload clients' needs to attain their goal is based on their level of muscular fitness. For example, a hockey player requires a different level of overload than a sedentary person. To best determine the load at which your client should be working (either starting from or progressing to).

Progression. Progression is defined as an increase in workload to maintain overload. The concept of progression may also include the practice of using very modest weights during the initial sessions of an exercise program. Guidelines for the progression of exercise become a factor in the success of individuals who are beginning exercise programs or who are engaging in specific types of exercise programs. The recommended frequency, intensity, time, type (FITT) framework for the frequency, intensity, and time of aerobic exercise for apparently healthy adults and an example of exercise progression for beginning exercisers.

1. GOAL-SETTING

When developing a resistance training program, the most valuable tool that a certified Personal Trainer can have is knowing the client's primary goals. Clients should set short-term and long-term goals. Goal-setting is an essential preliminary step when designing an effective training program. An acronym that can be used to define and quantify goals is S.M.A.R.T. (Specific, Measurable, Attainable, Relevant, Time-bound).

Goals should be realistic and attainable and focus on attempting to meet the client's medical, emotional, and functional needs, within the limitations of time, interest, and actual physical ability. The three primary resistance training goals are hypertrophy, muscular strength, and muscular endurance. Explaining the specifics of each of these three primary resistance training goals to the client

will not only educate them but also help establish strong client-trainer rapport.

Resistance training design:

Designing a safe and effective resistance training program is a multifaceted process in which the personal trainer must consider and manipulate specific training variables to achieve specific goals. While many personal trainers consider the resistance training program separate from the aerobic exercise or sprint interval program, it is important to consider all elements of the training plan when constructing the resistance training program. Regardless of the client's goals, the best approach begins with an initial consultation to determine the client's goals, an evaluation of the client's health or medical history, and a fitness assessment (see chapters 9, 10, and 11). At this time the personal trainer can find out about the client's experience with resistance training and perform a cursory evaluation of his or her technical capabilities. The information gathered at this time will assist the personal trainer in the development of the overall training program, which will include factors such as how often the client will train, the types of exercises selected, the training loads, the repetition and set schemes, the order of exercises, and the rest intervals. To facilitate continued improvements, the personal trainer should consider a periodized training plan. Periodization can ensure that appropriate training variation and progression are used so as to minimize the potential for overtraining while maximizing improvements in the targeted training outcomes.

General Training Principles:

Four basic principles should guide the development of effective resistance training programs: specificity, overload, variation, and progression. Any program that does not address all of these principles can result in a failure to meet client goals, poor adherence to the exercise plan, and potential litigation as a result of an increased risk of injury.

Specificity:

The principle of **specificity** is a foundational aspect of every effective training program. *Specificity of training* refers to training a client in a specific way to produce a targeted change or result. The personal trainer can accomplish this by targeting specific muscle groups, energetic systems, movement velocities, movement patterns, or muscle action types (31, 107, 117). For example, if a client wants to strengthen his or her leg muscles, the personal trainer would select exercises such as the back squat because of its emphasis on lower body development. This type of specificity is often referred to as *muscle group specificity*.

Another method of applying specificity to a client's training program is to target specific movement patterns. Movement pattern specificity is generally used in work with athletes or clients who wish to develop strength that translates to a specific activity or sport. For example, if the client is a volleyball player, the personal trainer takes into account the fact that volleyball involves repeated jumping movements. The personal trainer could target these movements by incorporating exercises such as back squats (or front squats), power cleans, and power snatches into the program because they mimic the jumping movements in volleyball. The more similar the resistance training exercise is to the movement pattern, the greater the likelihood of translation to the targeted activity (104). Exercises such as the Olympic lifts (i.e., cleans, snatches, etc.) appear to offer the most translation to sports because their movement patterns are similar to those used in many sports and activities.

Overload

Overload refers to a training stress or intensity that is greater than what a client is used to. If a program fails to adhere to the principle of overload, it will produce limited results. The most common methods of inducing overload include increasing the weight lifted, having the client perform more repetitions or sets of a given exercise, shortening the rest interval between sets, and increasing the number of training sessions in a week. While overload is an essential component of a resistance training program, the personal trainer should apply overload systematically and should adhere to the principles of progression and variation. Overload must be progressive to allow sufficient time for the client to adapt to the new training stimulus.

Variation

Variation refers to the manipulation of specific training variables (4, 117) such as volume, intensity, exercise selection, frequency of training, rest interval, and speed of movement (117). The application of appropriate training variation is essential when one is attempting to ensure long-term adaptation (65, 72, 113, 117). The best way to apply training variation is through use of the principles of periodization (117), which refers to the logical phasic manipulation of training factors to optimize specific training outcomes at specific time points (90). If appropriate training variation is not incorporated in the program, the rate of improvement will plateau (4, 117) or decrease, resulting in what is termed “monotonous program overtraining” (106). The variation in the resistance training plan should not be haphazard, and the sequencing of training factors should be appropriate (45, 89, 117). For example, clients who want to target power and strength development may undergo periods of training designed to target muscular strength followed by a period of power development. This type of sequencing has been shown to result in significantly greater improvements in performance compared to a program that lacks planned or sequenced variation

Progression

Regardless of how effective a training program is, it should not continue indefinitely without modification. As the client adapts to the training plan, the training stress or intensity must be altered to continue to induce positive adaptations. The process of altering training stress as a client adapts is termed **progressive overload**. The stress is changed as the client becomes better trained (31), allowing him or her to continue advancing toward a specific training goal (4). Progression in resistance training must be applied systematically (4) and in proportion to the client’s training status. Often the effective application of progression in a program requires appropriate variation strategies and the use of periodized training models.

Sequential Steps for Designing a Resistance Training Program:

To design an effective resistance training program, the personal trainer must make specific decisions about the manipulation of the program design variables, such as the training frequency, exercises used and how they are arranged, the structure of the program, the rest periods between sets, and the overall progression of the program. To facilitate the decision-making process, the personal trainer should consider a sequential approach. The culmination of this process is the establishment of a periodized training plan that increases the likelihood of the clients’ achieving their predetermined training goals. The first step is to conduct an initial fitness consultation and evaluation, which serves as the foundation for the subsequent steps that are generally associated with the sequential approach.

1. Initial consultation and fitness evaluation
2. Determination of training frequency
3. Exercise selection
4. Arrangement of exercises (exercise order)
5. Training load: resistance and repetitions
6. Training volume: repetitions and sets
7. Rest periods
8. Training variation
9. Sequencing the training plan
10. Progression

NB. Successful resistance training programs must incorporate specificity, overload, variation, and progression. Ignoring any of these factors can limit the program's ability to stimulate the desired outcome, decrease adherence to the program, and increase the probability of injury and legal risk.

Initial Fitness Consultation and Evaluation:

Before any exercise program begins, it is essential that the personal trainer conduct an initial client consultation to assess compatibility, establish a client-trainer agreement, discuss the client's exercise goals, and determine the client's level of commitment. After this discussion, the personal trainer needs to evaluate the client's exercise history and current level of fitness, identify strengths and weaknesses, identify areas of potential risk for injury, determine if contraindications to exercise exist, and refine the goals for the program

The initial consultation is an essential component of the resistance training program design process in that it gives the personal trainer valuable information about the client's training status. More specific information will be collected as the personal trainer assesses the client's initial resistance training status and resistance exercise technique experience, conducts a fitness evaluation and analyzes the results, and determines the client's primary goals.

Initial Resistance Training

Status and Experience

The client's current training status and experience with resistance training will exert a significant impact on the training program to be developed. General information about the client's experience with resistance training can be gathered during the initial meeting (or soon afterward) when the client's exercise history is discussed.

The first step in determining the client's resistance training status is to have the client answer the general exercise history questions listed on the Health/ Medical Questionnaire on page 171. The personal trainer then needs to obtain more specific information about the client's resistance training status and experience. A way to do this is to ask the client five basic questions (26):

1. Do you currently participate in a resistance training program?
2. How long have you been following a regular (one or more times per week) resistance training program?
3. How many times per week do you resistance train?
4. How intense (or difficult) are your resistance training workouts?

What types of resistance training exercises do you perform and how many of them can you perform with proper technique?

These questions can be used to establish a basic understanding of the client's resistance training experience and as a tool for establishing a general classification of the training status one approach for classifying resistance training status on a continuum from beginner to advanced. When the client's answers to the five questions match those shown in at least three of the five columns in one row, the estimated or predicted resistance training status is the one shown in the right-hand column. While this basic classification system is useful, it is important to realize that pigeonholing a client into a generalized training classification is difficult. Therefore, the personal trainer must also make decisions about the client's responses based on professional knowledge and experience.

A Method for Classifying Resistance Training Status:

Do you currently participate in a resistance training program?

If no, this would be beginner. If Yes, then go to the following question.

How long have you been following a regular (one or more times per week) resistance training program?

- NO.
- If less than 2 months.
- From 4 to 6 months.
- From 8 to 10 months.
- From 1 year.
- from 1 to 1 1/2 years.
- from ≥2 years.

How many times per week do you resistance train?

- No.
- 1-2
- 2-3
- 3
- 4
- 4
- ≥5

How intense (or difficult) are your resistance training workouts?

- No
- Low intensity
- Low to medium intensity
- Medium intensity
- Medium to high intensity
- High intensity
- Very high intensity

What types of resistance training exercises do you perform and how many of them can you perform with proper technique?

- No
- 3 to 5 machine exercises
- 6 to 10 machine core and assistance exercises; 3 to 5 free weight assistance exercises
- 11 to 15 machine core and assistance exercises; 6 to 10 free weight assistance exercises; 3 to 5 free weight core exercises
- >15 free weight and machine core and assistance exercises
- >15 free weight and machine core and assistance exercises; 3 to 5 power/explosive exercises
- >15 free weight and machine core and assistance exercises; most power/explosive exercises

Estimated resistance training classification?

If less than 6 months = Beginner.

From 6 to 10 months = intermediate.

More than 1 year = advanced.

Fitness Evaluation

Typically the fitness evaluation involves an assessment of the client's resting heart rate and blood pressure, body composition, height, weight, girth, muscular strength and endurance, cardiorespiratory fitness or endurance, and flexibility. In the context of this chapter, the fitness evaluation focuses on assessing or estimating the client's muscular strength and endurance.

Primary Resistance Training Goal

Establishing the client's training goals is a very important part of the resistance training program design process. In keeping with the specificity principle, the client must train in specific ways in order to achieve desired results. Four primary resistance training goals include muscular endurance, hypertrophy (muscular size or tone), muscular strength, and muscular power.

Often the client's goals are not clearly identifiable. For example, it is uncommon for clients to say "I want to follow a resistance training program for hypertrophy"; they may say something more like "I want to have flatter abs." When the training goals are established during the initial consultation and fitness evaluation, the personal trainer will have to match the resistance training goals with the desires expressed by the client. Frequently during the consultation, the personal trainer needs to educate clients about the various goals that resistance training can address.

Muscular Endurance

A client may express a desire to improve muscular endurance with statements like “I want to have better endurance” or “I want to increase my stamina.” Resistance training that targets muscular endurance, which is often termed *strength endurance training*, would address these goals by enhancing the ability of the targeted muscles to perform at a submaximal level for many repetitions or an extended period of time. The appropriate application of a resistance training program has great potential to improve muscular endurance.

Muscular endurance is commonly viewed as a part of aerobic exercise, as the muscle may contract thousands of times during a 20-minute activity such as running.

Hypertrophy

Statements like “I want my arms to be bigger,” “I want more size,” “I want to be more sculpted,” or “I want to change my body shape” suggest that the client wants to follow a program that will result in muscular hypertrophy or increased tone. *Hypertrophy* refers to an increase in muscle size, and hypertrophy training typically leads to an increase in fat-free mass and a reduction in percent body fat.

Muscular Strength

Of the four resistance training goals, muscular strength is the easiest to establish. Clients often state this goal directly—“I want to get stronger.” While this is a common goal for athletes interested in a resistance training program designed to enhance their athletic performance, it may also be a goal for other clients. For example, older clients may say things like “I want to be able to carry my golf bag” or “I want to be able to get up and down the stairs better.” Current scientific literature suggests that older adults can improve their ability to engage in activities of daily living by increasing their muscular strength through performance of appropriate training programs.

In comparison to resistance training programs that target muscular endurance or hypertrophy, a program that targets the development of muscular strength uses heavier training loads. Therefore, it is prudent for people to develop muscular endurance and hypertrophy before engaging in a program that targets muscular strength improvements.

Muscular Power

Traditionally, resistance training programs that target **muscular power** have been used only with athletes or clients who want to improve their sport performance abilities. Typically, statements like “I want to jump higher” or “I need to improve my speed and agility” suggest that the client would like to improve muscular power. Training programs that target muscular power have great potential to improve sport performance (. To maximize the benefits of these types of programs, it is prudent to sequence the client’s training program. For example, it has been shown that improvements in speed and jumping ability are greater when a program targeting muscular strength is performed for a period of time prior to a program targeting muscular power.

While power training has generally been used with athletes, contemporary literature suggests that it also has a place in work with older adult and nonathletic populations. Recent evidence indicates that older adults who engage in power training experience improvements in their ability to engage in activities of daily living and in functional performance.

Determination

of Training Frequency

Training frequency refers to the number of workouts a client will undertake during one week. Many factors contribute to the determination of the optimal frequency for an individual client. Factors such as the types of exercises used, the number of muscle groups trained per session, the structure of the program (volume and intensity), and the client's training status and overall fitness level dictate the training frequency (4). The client's work schedule, social schedule, and family obligations will also strongly influence how frequently the client can train.

Influential Factors

The primary factor that the personal trainer should consider when determining training frequency is the client's training status and overall level of fitness. Lesser-trained clients usually require more rest between workouts, which lowers the frequency, and highly trained clients will be able to tolerate more frequency. However, the personal trainer may need to reduce the frequency of resistance training if the client's overall amount of physical or psychological stress is high as a consequence of other demands (e.g., work, social, or academic schedule; other forms of exercise; or some combination of all demands). For example, if the client is a construction worker who performs repetitive lifting tasks in his or her job, he or she may not want or be able to tolerate more than two or three days per week of resistance training.

An influential factor that many personal trainers overlook is how the different components of the training program interact. Many personal trainers plan resistance, endurance, agility, and plyometric training without considering how each factor affects the overall workload. It is essential to examine how the various training activities interact and to take the client's overall workload into account. For example, if the client is running 30 minutes a day, five days per week, he or she may be able to tolerate only two days per week of resistance training.

Guidelines for Determining

Training Frequency

When determining the training frequency, it is important to plan sufficient recovery into the program. A general rule that many personal trainers follow is to allow at least one day (but no more than three) between workouts that stress the same muscle group or groups. More specific guidelines depend on a client's overall resistance training status. Most novice or beginner clients can experience the benefits of resistance training with as few as two or three days per week. However, individuals who are already accustomed to resistance training can only maintain their strength gains and cannot increase strength levels with one or two days per week. In general, the more frequent the sessions, the greater the strength gains.

Novice or beginner

Resistance Training Status

The recommendation for the novice or the beginner to resistance training is to use frequencies of two or three days per week when training the entire body. With this frequency, resistance training days should be nonconsecutive (i.e., Monday and Thursday; Tuesday, Thursday, and Saturday):

General Guidelines for Resistance Training Frequency:

Resistance training status:

Recommended number of sessions per week:

Novice or beginner	2-3	
Intermediate	3 if using total body training	4 if using a split routine
Advanced	4 to 6	

NB. Advanced resistance trainers may perform multiple sessions in one day.

or Monday, Wednesday, and Friday) to allow for appropriate recovery between sessions. As a general rule novice client should have 1 to 3 days between resistance training session, but never more than 3, in order to facilitate recovery. If, for example, the client was to train on Monday and Wednesday, the amount of time between the Wednesday and the next Monday training session would be greater than three days and result in a less effective training program. As the client progresses from the novice to the intermediate level, a change in frequency is not always necessary. However, increasing the frequency to three or four days per week allows for greater program flexibility.

Intermediate Resistance

Training Status

A general recommendation for the client who has achieved an intermediate training status is to increase the training frequency to three or four days per week. However, this increase will mean that the client trains two or more days in a row. A common strategy is to use a split routine, which spreads four or more workouts evenly across the week. With this structure, the client can train only one part of the body (i.e., upper back or lower body), certain muscle areas (e.g., chest, back, or legs), or certain movement patterns (i.e., “push” or “pull”) during a session. This structure allows for an increase in frequency while still allowing sufficient time for recovery between sessions.

A common example of a four time per week split routine for the intermediate client includes upper body exercises on Monday and Thursday and lower body exercises on Tuesday and Friday. Even though the client is training two days in row, changing the targeted muscle groups ensures adequate recovery. Additionally, there are two days of rest between sessions that target the same muscle groups, which allows for greater recovery between similar sessions. With this type of frequency split, the nonresistance training or rest days are consistently Wednesday, Saturday, and Sunday. More importantly, when training specific muscle groups or body parts, this split allows for maximization of results because the client can use higher volumes.

advanced Resistance Training Status

Depending on their goals, intermediate clients may need to increase their training frequency as they become more experienced and are reclassified as advanced in status. The general recommendation is for advanced individuals to resistance train four to six days per week to allow for an increase in training stimulus. It is common for these clients to use double split routines, performing two sessions on the same day, which increases the number of training sessions from 8 to 12 per week. Strong evidence in the scientific literature supports the concept of performing multiple short training sessions in one day.

Another method for increasing the frequency from five to six days is to use a “three days on, one day off” **split routine**. With this structure, three distinct workouts target specific muscle groups, and the client completes one workout on three consecutive days and rests on the fourth day. A common strategy is to divide the program into upper body “push” exercises (i.e., chest, shoulders, triceps), lower body exercises, and upper body “pull” exercises (i.e., upper back, trapezius, and biceps). In this type of structure, the workouts are on unspecified days; that is, the rest day is not the same each week.

Exercise Selection:

The selection of exercises to be incorporated into the client’s training program is influenced by the principle of specificity, the equipment available, the client’s resistance training experience, and the amount of time the client has to dedicate to training. Once the personal trainer considers these issues, he or she can make exercise selections that maximize the training adaptations and increase the chances of achieving the client’s specific training goals.

Influential Factors

When selecting exercises, the personal trainer should make decisions based on the specific needs of the client and the target goals established in the initial consultation with the client. Many factors will affect this decision-making process, including how much time a client has to dedicate to training. The time available can have a large impact on the number and complexity of exercises chosen for a given training session. For example, performing the back squat to target lower body development takes significantly less time than performing leg curls and leg extensions.

An additional factor is the equipment available to the client. Even if an exercise is effective, goal specific, and efficient it cannot be a major part of a client’s training plan if the equipment is not available. To account for this before planning the program, the personal trainer needs to gather an equipment inventory for the facility in which the client intends to train. This important step can help avoid wasting time in constructing the training plan. Another factor that will dictate the selection of exercises is whether or not the client can perform the exercise correctly. If the client is inexperienced in or unfamiliar with proper technique for an exercise, or if the extent of the client’s experience is unknown, the personal trainer must provide a complete demonstration, explanation, and familiarization period in which to teach the client appropriate technique. This may require beginning the client with remedial exercises that can build a foundation for more complex exercises in future programs. Inexperienced clients are often taught machine exercises and free weight assistance exercises first because these require less skill than many of the core exercises. (In this context, *core exercises* does not refer to exercises targeting the abdominal core. See the next section on types of resistance training exercises.) Following these recommendations can be effective in reducing the potential for litigation, decreasing injury risk, promoting adherence, and improving the overall effectiveness of the program.

Types of Resistance

Training Exercises

There are a plethora of exercises from which to construct a resistance training program. These exercises can be classified as either core or assistance exercises based on the size of the muscles recruited, the complexity of the movement pattern, and the degree of contribution toward the client's training goals.

Core Exercises

Core exercises should form the bulk of a program because they are more effective than assistance exercises in helping clients achieve their specific training goals. Generally, an exercise is classified as a core exercise if it

- involves two or more primary joints, which would make it a **multijoint exercise**, and
- engages large muscles while activating synergistic muscles.

One multijoint large-muscle exercise has the potential to activate as many muscles or muscle groups as four to eight small-muscle, single-joint assistance exercises (112). A program that uses core exercises appropriately is more efficient than one that uses many small-muscle exercises.

The personal trainer has many core exercises to choose from when constructing a program (see chapter 13). An example is the bench press, which involves movement at the shoulder and elbow joint while recruiting the chest muscles with synergistic help from the anterior deltoids and the triceps brachii.

If a core exercise loads the axial skeleton (places a load on the spine) (i.e., power clean, squat, front squat, shoulder press, etc.), it is further classified as a structural exercise. Structural exercises require the muscles of the torso to maintain an erect or near-erect position. For example, during the squat, the barbell loads the axial skeleton, and the musculature of the torso must maintain a near-erect position as the client descends and ascends. Structural exercises that are performed very quickly, such as the power clean or power snatch, are also classified as **power** or **explosive exercises** (e.g., push press, snatch or clean pull, high pull, push jerk). These types of exercises are extremely effective because they provide a multidimensional training stimulus through the engagement of a large amount of muscle mass and result in a large caloric expenditure.

N.B: In the context of resistance training, core exercises refer to exercises that engage large muscles and multiple joints.

assistance Exercises

Assistance exercises supplementary exercises that are performed to maintain muscular balance across a joint, help prevent injury, rehabilitate a previous injury, or isolate a specific muscle group or muscle. Exercises are classified as assistance exercises if they

- are **single-joint exercises**, engaging only one primary joint, and
- recruit a small amount of muscle mass (i.e., a small muscle group or area).

Assistance exercises should not be the major components of the program; they should be considered a secondary emphasis in relation to the more effective core exercises.

One of the most popular assistance exercises is the barbell biceps curl. This exercise engages a small amount of muscle mass (biceps brachii, brachialis, and brachioradialis) and involves movement of only a single joint (elbow joint). The “pec deck” and dumbbell fly are also classified as assistance exercises even though they engage the muscles of the chest. They involve movement only at the shoulder and place primary emphasis on the chest musculature.

Guidelines for Choosing Exercises

Selection of the exercises for a training program should meet the individual client’s needs, whether the client is an elite athlete or a novice, is severely detrained, or has been recently injured.

For a novice or untrained individual, the personal trainer should pay more attention to developing a training base, probably with assistance exercises or basic core exercises. In this situation one may decide to target specific muscle groups or train each muscle group. This strategy involves choosing one exercise per muscle group: chest, shoulders, upper back, hips and thighs, biceps, triceps, abdominal muscles, and calves (6, 85). As the client becomes better trained, the number of exercises per muscle group can be increased (85). Typically, this type of program uses small-muscle or single-joint exercises but programs should progressively incorporate more multijoint large-muscle or core exercises.

Employing multijoint large-muscle exercises can magnify the adaptive response and increase the overall metabolic cost of the training program. An example is the squat and press. This complex exercise engages a large amount of muscle mass, substantially increases the metabolic cost of training, and produces a substantial training effect. Other examples are exercises such as power cleans, power snatches, and pulls. These all train the entire body and appear to be extremely effective training methods for athletes and clinical populations. With athletes, the more closely the exercise relates to the sporting movement pattern, the more likely the strength gains developed will translate to the sport. For example, when working with a volleyball or basketball player, the personal trainer should consider using exercises such as the power clean or power snatch since they involve a jumping movement, which is important in these sports.

In work with clients who have special needs, such as lower back problems or recent injuries, it is important to adapt the training program to address these issues. The exercises should be selected with guidance from the appropriate medical professional. It is essential that a training program for these clients avoid exercises that are contraindicated or not recommended. For example, if a client was recently released from the care of a physical therapist for a shoulder impingement, the physical therapist may prescribe the dumbbell lateral raise in place of the overhead shoulder press to work the deltoid muscle group.

Exercise Order

The exercise order or arrangement refers to the order in which the exercises are performed during the workout. The exercise order is dependent on many factors but is most strongly influenced by the type and characteristics of the exercises selected.

Influential Factors

Factors that influence the order of exercises include the goals of the client, the fatigue-generating potential of the exercise, and the type of exercise (core or assistance).

One method for ordering exercises is to place them in a descending order of priority or application to the client's goals, activity, or, in the case of athletes, sport. With this type of structure, the client performs the exercises that target his or her individual goals earlier in the workout when fatigue is lowest, performing less goal-specific exercises toward the end of the session.

A second method for arranging exercises is based on their type (core or assistance). With this method, core exercises are performed first, and assistance exercises are performed later in the session. This arrangement allows the client to perform the more complex, multijoint core exercises under minimal levels of fatigue. In general, the personal trainer should attempt to maximize the client's ability to tolerate the training loads and complete all the exercises in one session by arranging the exercises to manage fatigue.

Guidelines for arranging Exercises

There are many ways to arrange exercises in a training session (31). The ordering of exercises can be categorized into several primary methods, such as placing power and core exercises before assistance exercises, alternating "push" and "pull" exercises, and alternating upper and lower body exercises. There are also combination methods and secondary arrangement methods (tables 15.3 and 15.4).

Placing Power and Core Exercises

before assistance Exercise

One of the most commonly used guidelines is to order the exercises as follows:

Power exercises → Core exercises → Assistance exercises

Conceptually, since power exercises and core exercises are often multijoint exercises and assistance exercises are typically single-joint exercises, the following is another possible order:

Multijoint exercises → Single-joint exercises

Another way to think about exercise order is to consider the amount of muscle mass activated with the exercise. Large-muscle exercises, which are generally core or multijoint exercises, should be performed prior to single-joint exercise:

Large-muscle exercises → Small-muscle exercises

Regardless of the way one conceptualizes these exercise sequences, the basic structures noted here should be considered to be the most effective exercise sequencing methods for the majority of clients.

These arrangements of exercises are effective because power or multijoint exercises require more effort, skill, and focus than single-joint assistance exercises (31) and should be performed when the client is fresh (31, 30). Historically it has been recommended that only athletes use power exercises, such as the Olympic lifts and their derivatives; but contemporary literature suggests that all populations can use power exercises to achieve their training goals (20, 57, 64). In fact, one study in adults showed that transfer of training effects to activities of daily living was greater with power training than with strength training (46). Therefore, the personal trainer should consider the use of these types of exercises not only with athletes, but with all clients.

alternating “Push” and “Pull” Exercises

Another method of arranging exercises is to alternate “push” exercises (e.g., vertical chest press and triceps press-down) with “pull” exercises (e.g., seated row and dumbbell biceps curl) (8). It has been suggested that this system allows rest between exercises and guarantees that the same muscle group is not used for two exercises in a row (8). While the push-pull system is commonly used, it is important to remember that core exercises can activate a vast array of muscles and that many will use the same muscle groups. For example, if a client performs the back squat (a “pushing” exercise) followed by the leg curl (a “pulling” exercise), the hamstrings will be activated in both exercises.

alternating Upper and lower body Exercises

A traditional approach to ordering exercises requires the client to alternate between upper and lower body exercises (31). This type of ordering is typically used with a circuit weight training program and short rest intervals. As with the push-pull system, this ordering may be best suited for machine-based training or when the predominant exercises selected are those classified as small-muscle or assistance exercises.

Combination arrangement

Methods

It is possible to combine the most common methods of arranging exercises. Two or three of the previously mentioned methods can be combined: for example, core exercises and then assistance exercises that alternate push and pull. Often with a combination arrangement, a lower body exercise precedes an upper body exercise.

Secondary arrangement

Methods

Two popular secondary methods involve completing a set of two different exercises in succession without an intervening rest interval. If the two exercises coupled together train the same muscle group (e.g., incline bench press and incline dumbbell fly), the set is considered a compound set (8). Compound sets are often used by bodybuilders in an attempt to induce muscular hypertrophy (31).

Another secondary arrangement, which is termed a super set, uses the performance of two exercises that activate opposing or antagonistic muscle groups (e.g., biceps curl and triceps press) with no rest between each exercise (8, 31, 107). Super setting is popular among bodybuilders, individuals who are attempting to increase muscular endurance, and individuals with limited time for training (31).

Training load:

Resistance and Repetitions

The training load or the amount of weight to be used in a resistance training program is one of the most important factors to consider in the design of a training program. There are many ways to determine the training load. The assigned load will exert a large impact on the number of repetitions that can be performed and ultimately the types of physiological and performance adaptations stimulated. Ultimately the interplay between the load and volume of training (repetitions $\times$ sets $\times$ resistance) is dictated by the type of training program established and the intended goals of the program.

Before assigning the training load and the repetitions to be performed in each set, the personal trainer must test the client. The purpose of testing is to determine the client's abilities to handle specific loads in a series of selected exercises. Once the personal trainer establishes these abilities, he or she can assign training loads.

Influential Factors

The load, the number of repetitions, and the targeted outcomes of a resistance training program are strongly related. For example, higher loads ($>80\%$ of 1-repetition maximum [1RM]) performed for fewer repetitions (three to five) target the development of muscular strength. Lifting lighter loads ($<70\%$ of 1RM) for higher repetitions (>10 repetitions) results in improvements in muscular endurance. In its most basic form, a client's training program would address specific goals and target them with selected repetition and load relationships. This, however, is a gross simplification of the training process. Regardless of the client's goals, the training plan should include periods in which muscular endurance, strength, and potentially power are developed. Sequencing different repetition and loading schemes will magnify training adaptations and provide a greater chance of accomplishing the client's goals (45, 90, 117).

Percentage of 1-Repetition Maximum Relationship

The **repetition maximum (RM)** is the maximum load that the client can handle in a specific exercise for specific number of repetitions. As the load becomes heavier, the number of repetitions that the client can perform will decrease. Eventually the load will become so heavy that the client can perform only one repetition; this is the **1-repetition maximum (1RM)**. Conversely, the lighter the load, the more repetitions the client can perform. This association between 1RM and repetitions has been termed the percentage of 1RM ($\%1RM$)–repetitions relationship.

Load assignments are best accomplished with the use of percentages of the 1RM or of a specified targeted maximum repetition range. For example, table 15.5 indicates that if a client's 1RM is 200 pounds (91 kg), he or she should be able to perform eight repetitions with around 160 pounds (73 kg). If his or her 10RM is 150 pounds (68 kg), we can estimate his or her 1RM as around 200 pounds (91 kg). By using these relationships, the personal trainer can estimate the training loads to prescribe for the client. It is important to note that the numbers in table 15.5 are only estimates and that these values can vary slightly depending on the training status of the client and the exercises used in the program.

An alternative method is to use an intensity that allows for performance of a specific number of repetitions, or what is known as a **repetition maximum**

(RM) target. With use of this method, the client lifts the heaviest load he or she can for the selected repetition schema. For example, if the client were to perform three sets at a 12RM load, he or she would use the heaviest weight that would allow him to perform three sets with no less or more than 12 repetitions. A similar method, using what is called the RM target zone, is to assign a range such as 3- to 5RM (31). When a **repetition maximum zone (RM target zone)** is prescribed, the client uses the heaviest weight he or she can to perform the exercise for the number of repetitions within the range. The RM target and RM target zone methods for assigning load are problematic because in both instances the client is required to train to muscular failure. The scientific literature shows clearly that training to failure results in reduced training adaptations and may increase the client's risk for over-training and experiencing injuries. Therefore it is prudent for personal trainers to avoid both the RM target and the RM target zone methods when establishing the client's training load.

limitations in the Percent 1-Repetition Maximum Relationship

While the %RM–repetitions relationship is an excellent tool for prescribing intensities in resistance training, it is important to realize a number of limitations that may affect its accuracy.

1. While there appears to be a relationship between the %1RM and the number of repetitions that can be performed, several studies suggest that the relationship is not as robust as was once thought
2. Training status appears to influence the relationship between repetitions and %1RM, with trained individuals being able to perform more repetitions at a given %1RM (52, 53, 79, 100, 118).
3. When working with the %1RM–repetitions relationship, it is important to remember that the number of repetitions performed at a specific %1RM is related only to a single set and not to multiple sets. With multiple sets, fatigue will cause a reduction in the number of repetitions that can be performed in later sets, thus altering the %1RM–repetitions relationship.
4. The %1RM–repetitions association is for the most part based on research on the bench press, back squat, and power clean. The relationship between %1RM and repetitions does not likely apply to all exercises and should be applied with care in work with clients.
5. The mode of resistance training appears to affect the number of repetitions that can be performed at a given %1RM. Generally, more repetitions can be performed at any given %1RM with machine-based exercises (e.g., vertical chest press) than with free weight exercises (e.g., bench press).
6. More repetitions can be performed at any given %1RM for core exercises compared to assistance exercises.
7. The order of exercises may also affect the number of repetitions that can be performed at any given %1RM. Whether it is a core or assistance exercise, placing an exercise toward the end of the workout results in a decrease in the number of repetitions.

Guidelines for assessing

load Capabilities

Before assigning training loads, the personal trainer must perform some form of assessment in order to estimate the client's capabilities. The following are methods for accomplishing this:

1. Directly assessing the 1RM
2. Estimating the 1RM
3. Using a percentage of the client's bodyweight for testing
4. Repetition maximum testing

One or more of these methods can be used depending on the client's training status, technical proficiency, and the type of exercise being tested.

assessing the 1-Repetition

Maximum

The personal trainer must determine the 1RM in order to use the %1RM–repetitions relationships presented in table 15.5. As a general rule, the 1RM test poses minimal risks to both clinical and athletic clients (98, 99); and it is considered the gold standard of muscular strength assessments (56). The biggest issue with the 1RM test is whether or not the client has the technique needed to perform the exercise correctly with increasing loads. If technique is lacking, it may be best to use other methods. While use of a test-established 1RM is the most accurate way to determine loading, it is also possible to use a submaximal load to estimate the 1RM. It is important to note that using estimations from submaximal loads may result in overestimation of the 1RM which could lead to a problem in prescribing training intensities. Use sound judgment when selecting exercises for a 1RM test.

As one example, even though the lunge and step-up are large-muscle multijoint exercises, they are not typically used with 1RM testing because of the uneven loads they place on the lower body. The uneven loads can increase the potential for injury and accidents. The bent-over row is another exercise that would not be assessed with 1RM testing. While this exercise does activate the large muscle joints of the upper back and functions across several joints, it is likely that during a 1RM test the weaker muscles of the lower back may not be able to maintain appropriate body positions, which would increase the risk of injury and result in an inaccurate assessment of strength. Once an exercise has been deemed acceptable for testing and if the client is of appropriate training status,

As a rule, clients who have never lifted, who are classified as untrained, have been recently injured, or are under medical supervision should not perform a 1RM test. It may be prudent to reserve this type of testing for more advanced clients who have developed the appropriate technical base and can perform the test exercise using appropriate technique with various loads.

In selecting the exercises to be tested with a 1RM, the personal trainer should choose only those exercises that can be performed safely, accurately, and consistently (8). In general, the larger multijoint core exercises are best suited for 1RM testing because heavy loads are better tolerated with these exercises. As a rule, 1RM testing should not be used with assistance exercises because of the large physiological stress placed on the smaller muscle groups across a single joint. Use sound judgment when selecting exercises for a 1RM test.

As one example, even though the lunge and step-up are large-muscle multijoint exercises, they are not typically used with 1RM testing because of the uneven loads they place on the lower body. The uneven loads can increase the potential for injury and accidents. The bent-over row is another exercise that would not be assessed with 1RM testing. While this exercise does activate the large muscle joints of the upper back and functions across several joints, it is likely that during a 1RM test the weaker muscles of the lower back may not be able to maintain appropriate body positions, which would increase the risk of injury and result in an inaccurate assessment of strength. Once an exercise has been deemed acceptable for testing and if the client is of appropriate training status.

Estimating the 1-Repetition Maximum

If the client is unable to perform an actual 1RM test, there are several ways in which the 1RM can be estimated. Such estimation procedures can be used to develop the loading structure and resistance training plan.

1. Use of repetition maximum tests 2. Use of prediction equations

One way to estimate a client's 1RM is to use a repetition maximum (RM) test, which then becomes an estimate of the 1RM. The best means of ensuring accuracy is to use lower numbers of repetitions (5-10) such as a 6RM or 10RM (i.e., the heaviest load the client can lift 6 or 10 times with proper technique). Once the 6- or 10RM is established, table 15.6 can be used to estimate the 1RM. As a basic rule, the RM should be determined within three testing sets.

The 6RM test is very similar to the test for determining the 1RM; the main difference is that each testing set includes six repetitions (67). Since the number of repetitions is greater, the load changes across testing sets should be smaller (~50% of the amounts suggested for 1RM testing in chapter 11).

Once the client's 6RM has been determined, table 15.6 can be used to estimate the 1RM. The personal trainer will look at the row titled "Max repetitions" and move across that row until they find the notation for 6 repetitions, which is "~85% 1RM". Once this column is located, the trainer can move down this column looking for the number closest to (but not greater than) the client's 6RM. For example, if the client's 6RM for the leg press is 140 pounds (64 kg), the estimated 1RM will be 165 pounds (75 kg). This load will be used as the basis of the client's actual training loads.

The results of the RM test can also be used with prediction equations to determine the 1RM. A number of these equations, which use repetitions to failure to estimate 1RM, have been published (table 15.7) (1, 18, 27, 73, 74, 79, 80, 84). In general, the accuracy of these equations is greater when heavier loads are used (1). Therefore the recommendation is to use heavier loads performed for fewer repetitions than the 10RM with prediction equations (1, 17, 21, 81, 122). The phase of the training plan also affects the accuracy of these equations (9). If the client has been training with high volumes such as sets of 10 to 15 repetitions, the equations will become less accurate; conversely, a prediction equation becomes more robust if the client is using lower volumes and heavier weights.

Percent of body Weight Testing

Another method for estimating strength in either core or assistance exercises is to test using a certain percentage of a client's body weight (%BWT) (10). This method of determining training loads is best suited for untrained or inexperienced clients because the loads are typically relatively light. The %BWT method should not be used with more experienced clients because they will have a greater strength to body weight ratio and thus their training loads will be grossly underestimated.

The %BWT method uses an exercise-specific numerical factor to determine a **trial load** for testing based on a percentage of the client's body weight (tables 15.8 through 15.10). Table 15.8 provides basic guidelines for performing an assessment based on %BWT. A maximum body weight of 175 pounds (79 kg) for male clients and 140 pounds (64 kg) for female should be used with this test. This accounts for individual differences in body composition and ensures test safety. The goal of this type of assessment is to derive a trial load that the client can lift for only 12 to 15 repetitions. Using the factors presented in tables 15.9 and 15.10 will accomplish this goal; but it is important to realize that individual differences, variations in technical proficiency, and the number of different types of equipment make it impossible to calculate the testing load perfectly. Nevertheless this method appears to be a valuable tool for establishing training loads.

As an example, to use the %BWT method to test a 130-pound (59.1 kg) female on the leg press on a pivot-based machine, one would use a trial load of 130 pounds (BWT $\square$ 1.0; table 15.9). After an appropriate warm-up, the client would perform as many repetitions as possible with the 130-pound load, which in theory should be somewhere between 12 and 15.

Repetition Maximum Testing

Another method for determining the client's capabilities is to decide on a number of **goal repetitions** for the client to perform in the actual training program. For example, if an advanced client will perform four repetitions of the back squat in the workout, the personal trainer uses a 4RM test.

While this is probably not the best method for determining training loads, it can be used to assess all core exercises, although higher RMs (i.e., eight or more repetitions) can result in a large amount of accumulated fatigue if multiple trial sets are performed (107). With assistance exercises, only an 8RM load or lighter should be used (10), for the same reasons these exercises should not be 1RM tested. Repetition maximum testing is usually appropriate for most intermediate to advanced clients, while novice or untrained clients should be limited to lighter loads with higher repetitions (≥ 8 RM). Regardless of the client's training status, it is important to monitor the client to help avoid excessive fatigue that could result in an increased risk of injury; the general recommendation is to limit testing to three sets.

Repetition maximum testing is very similar to assessment of the 1RM (see chapter 11), but the client performs the number of goal repetitions for each trial set with lower load ranges (~50-75% of 1RM) than recommended for a 1RM.

Guidelines for assigning loads

When determining the loads to be used in a program, the personal trainer must consider the primary goals of the client (muscular endurance, hypertrophy, muscular strength, or muscular power). Different load and repetition schemes can have different training effects (table 15.11). It is recommended that various loading schemes be employed in a program (4, 117) to avoid overtraining, allow for progression, and maximize training adaptations.

assigning load based

on 1-Repetition Maximum

The training load can be established directly based on the 1RM test or from an estimation of the client's 1RM. The repetition scheme will be based on the goals of the training phase—muscular endurance, hypertrophy, muscular strength, or muscular power (table 15.11). Having established the repetitions, the personal trainer can multiply the 1RM load (directly tested or estimated) by the percent of the 1RM associated with the goal repetitions to establish the training load.

If, for example, a client's goal is hypertrophy (increase in muscle size), he or she will perform 6 to 12 repetitions with a load between 67% and 85% of 1RM. If his or her 1RM back squat is 220 pounds (100 kg) and he or she is to perform 10 repetitions, the load could range from 147 to 187 pounds ($220 \times 0.67 = 147$; $220 \times 0.85 = 187$). To properly load the barbell, he or she will set the resistance at 150 to 185 pounds (68 to 85 kg) depending on the focus of the workout. The load will be adjusted according to the goals for the individual training session and the targeted training intensity for the session.

assigning load based

on Percent body Weight Testing

Another method for determining the training load is based on the %BWT assessment. Once the personal trainer has established the training plan and determined the repetition range (tables 15.9 and 15.10), he or she can compare the number of repetitions performed during the assessment with repetition ranges in table 15.11. Although the %BWT assessment is designed to produce 12 to 15 repetitions, it is possible that the client will perform less than 12 or more than 15 repetitions. Thus the trial load will need to be adjusted in order to better target the client's goals. Table 15.12 provides one possible method for making this adjustment, increasing the load if the client performs too many repetitions or reducing the load if he or she performs too few. These adjustments are only guides and are not without error (e.g., multijoint exercises may require larger load adjustments), but they provide a valuable starting point for structuring the training plan.

For example, if a client wants to target muscular hypertrophy, a load and repetition scheme of 67% to 85% 1RM for 6 to 12 repetitions is recommended (table 15.11). If the client weighed 170 pounds (77 kg) and performed the %BWT assessment with 100 pounds on the bench press ($BWT \times 0.60 = 102$ pounds or 46 kg) and completed 10 repetitions, the personal trainer could adjust the load according to the program goals. If the repetitions had been set at eight or nine, the load would have to be adjusted based on the information provided in table 15.12. The rows in table 15.12 show "Goal repetitions" and the columns show "Repetitions completed with the trial load." This client had a goal of eight or nine repetitions and completed 10 repetitions. The 8-9 goal repetition row and the 10-11 repetitions completed column intersect at the +5 cell, indicating that the trial load should be increased by 5 pounds (2.5 kg). Thus the

assigned load would be 105 pounds ($100 + 5 = 105$ pounds = 48 kg). This method can be used to determine the training loads for each of the exercises listed in table 15.9 and 15.10, where they can be recorded.

An additional option is to use this adjustment to modify the training load during a workout session. For example, if the client cannot complete two or more sets of eight or nine repetitions with the 105-pound load in the bench press, the load can be adjusted according to the information in 15.12; if the client can perform only six repetitions with the 105-pound load, the load could be reduced by 5 pounds. Although this method is easy to apply, it should be used with caution. The use of repetition ranges with this type of loading can create an over-training scenario due to a lack of intensity variation. Conceptually this type of model requires the client to train to failure on each set to achieve the pre-scribed repetition ranges, and training to muscular failure has been clearly established as a poor way to train (86, 105).

assigning load based

on Repetition Maximum Testing

Another method sometimes used to assign training load is the RM method. There are no calculations, and the load established in the RM test is used as the training load. The number of repetitions used in the RM test is identical to that in the actual training plan; therefore the RM is the training load.

For example, in table 15.11, six or fewer repetitions with loads $\geq 85\%$ of 1RM are suggested for improving strength with advanced clients. If the personal trainer assessed a 4RM for the bench press and the client was able to bench press only 200 pounds (91 kg) for four repetitions, the load would then be established as 200 pounds. While this method is straightforward, it involves training to muscular failure, which as noted earlier is a poor method of developing strength and in fact has been shown to mute strength gains when compared to not training to failure (86, 105). If clients are directed to always use maximal loads (RMs) to establish their **Training Volume: Repetitions and Sets**

Volume in resistance training is best defined as the *volume load* (112, 117), which is calculated as follows:

$$\text{Volume load} = \text{Total repetitions} \times \text{Load}$$

where total repetitions = sets $\times$ reps.

The volume load is a reasonable indicator of

the amount of work that is accomplished during resistance training (83). The inclusion of the load in the calculation of volume is important because it contributes to the total amount of work performed. If only repetitions are used to determine volume, the depiction of work will be inaccurate. For example, three sets of 10 repetitions and five sets of six repetitions both total 30 repetitions. This might suggest that the work accomplished in the two set and repetition schemes is identical, but it is not. A client whose maximum back squat is 300 pounds (136 kg) would be able to perform 10 repetitions with 225 pounds (102 kg) and 255 pounds (116 kg) for six repetitions (table 15.6). It is easy to see that the volume is substantially higher for five sets of six repetitions with 255 pounds (7,650 pounds or 3,477 kg) than for three sets of 10 repetitions with 225 pounds (6,750 pounds or 3,068 kg).

Influential Factors

Many factors can affect the volume of training. The targeted goal of the client or training plan exerts a primary influence on the number of sets (series of repetitions linked together) or repetitions placed into the plan. For example, if the client wants to develop muscular endurance, higher repetitions (≥ 12 repetitions) should be planned. Conversely, if muscular strength is targeted, lower repetition schemes would be employed (six repetitions or less). The personal trainer can also manipulate the training volume by modifying the number of sets used for a particular exercise.

Single-set protocols have been shown to be inferior to multiple-set protocols (4, 87, 95) in untrained individuals (94, 95) and trained populations (86, 95, 129). Single-set protocols may be effective with the novice or untrained client when employed in the initial stages of the training program; but as the client becomes more trained it is essential that volume of training be increased, and this is probably best accomplished with use of a multiple-set training plan

training loads, overtraining and increased occurrence of injuries are possible. A far better approach to loading is to use percentages of 1RM and to vary intensity within the training week.

Guidelines

for assigning Volume

The volume of training that clients undertake is dependent on their training goals and more importantly on their training status.

Resistance Training Goals

The ability to induce specific adaptations is dependent on the ability to induce an overload and on targeting the specific training outcomes by manipulating volume and intensity of training (tables 15.11 and 15.13). Additionally, it is essential that progression and variation be structured into the training plan to facilitate the adaptive responses necessary to achieve the client's goals. The design of the training plan must address one of four goals: muscular endurance, hypertrophy, muscular strength, or muscular power (table 15.13).

- ■■ *Muscular endurance*: The plan generally uses more repetitions (i.e., ≥ 10 repetitions per set) with varying numbers of sets depending on training status (4, 9, 26, 31, 66, 86, 95, 111, 117).
- ■■ *Hypertrophy*: The plan uses a higher training volume (i.e., 6-12 repetitions) with moderate to high loads (67-85% 1RM) depending on training status (4, 9, 26, 31, 66, 86, 95, 111, 117).
- ■■ *Muscular strength*: For core exercises, the plan uses six or fewer repetitions with multiple sets (three or more) per exercise. For assistance exercises, it uses eight or more repetitions with one to three sets (4, 9, 26, 31, 66, 86, 95, 111, 117).
- ■■ *Power exercises*: Power exercises should not be used with untrained or novice clients. These clients should target the development of strength before focusing on training for power (66). With intermediate clients, programs that target power development should use one to three sets of three to six repetitions, while advanced clients should use three to six sets of one to six repetitions (4, 9, 26, 31, 66, 86, 95, 111, 117).

Resistance Training Status

The training status of the client can affect the number of sets he or she can perform with a resistance training exercise. Generally, untrained or novice clients will perform fewer sets than either intermediate or advanced clients (4, 66). Initially it may be acceptable for novice clients to use a single-set program, but a multiple-set training plan is eventually necessary in order for them to progress (4). Table 15.13 presents a volume progression that can be used as the client moves from a novice to intermediate or advanced training status (4, 66).

Rest Intervals

The time period between multiple sets, or **rest interval**, can exert a strong impact on the physiological responses to a training session (4, 8, 9). The term *rest interval* can also refer to the time period between exercises. Overall the rest interval, whether between sets or exercises, is largely predicated on the goals of the client and the structure of the training plan.

Influential Factors

In general, there is a direct relationship between the load and the need for rest between sets; the heavier the load, the longer the rest needed between sets or exercises (9, 31, 77, 107). An additional consideration is the client's training status; untrained or novice clients may require longer times between sets and exercises in order to achieve adequate recovery to maintain exercise technique.

Guidelines for Establishing

Rest Intervals

The overall goals of the client play a large role in dictating assignment of the resistance training loads and volumes, both of which exert a large influence on the rest intervals chosen.

Resistance Training Status

Table 15.14 presents recommended rest interval lengths, but it is important to note that these are merely recommendations. Typically they will not work for novice or untrained clients, who will require twice as much rest as a more trained individual. With these clients it may be best to use a rest interval of 2 to 5 minutes to facilitate recovery between sets and exercises. This is particularly important when the client is learning new exercises or attempting to master a technical exercise (4). As the client becomes more trained and has mastered the technical aspects of each lift, the recommendations presented in table 15.14 may become more appropriate.

Resistance Training Goal

To tailor the training plan to the individual client's goals, the personal trainer must craft a program that allows enough rest between sets and exercises to allow the client to lift the selected loads for the assigned number of repetitions. The basic rest interval lengths for muscular endurance, hypertrophy, power, and strength, presented in table 15.14, are as follows:

- ■ **Muscular endurance:** A relatively short rest interval, typically ≤ 30 seconds, is recommended for circuit-based training with exercises using dissimilar muscle groups (96, 126), while up to 3 minutes' rest may be warranted between exercises using similar muscle groups (126).
- ■ **Muscular hypertrophy:** Rest intervals for hypertrophy should be short to moderate in duration (4, 126), ranging from 30 seconds to 1.5 minutes (49, 85, 119). Longer rest intervals with multijoint large-muscle exercises may be warranted because of their high metabolic demands (107).
- ■ **Muscular power:** Longer rest intervals on the order of 2 to 5 minutes are warranted (107, 126) between maximal-effort sets.
- ■ **Muscular strength:** Longer rest intervals are necessary for exercises targeting strength development (126). This is especially true for lower body or whole-body exercises (4, 128). The general recommendation is 2 to 5 minutes

Variation

Regardless of how effective or individualized a training program is, it will eventually need to be varied to ensure continued adaptation (4, 117). The concept of variation involves the appropriate alteration of training variables to produce adaptations over the long term (65, 72, 112, 113, 117). One key component of variation is a purposeful sequencing of training factors (45, 90, 117). If training is sequenced correctly, the adaptations stimulated by one period of training can exert a powerful effect on the next period of training, resulting in a summation of training effect (45, 58, 59). If the resistance training plan includes appropriate and correctly sequenced variation, it will be incrementally more effective.

Much less training variation is needed with novice or untrained clients, as any reasonable training plan will produce results (117). As clients become more trained, their progress may begin to slow if the plan does not include variation and progression. Maintenance of the same training plan without variation can induce what has been termed “monotonous overtraining” (106). If the training stimulus is unvaried for a long period of time (i.e., several months), even more experienced or trained clients can show decreases in muscular strength and neuromuscular activation (40, 41, 42, 43, 45) and an increased occurrence of overtraining symptoms (32, 33, 45, 106).

The best way to avoid stagnation is to include purposeful variation of the program design variables used to expose the client to training stimuli (23, 117). The many levels at which the personal trainer can alter the training stimulus include frequency, intensity, and volume of training, as well as the rest intervals between sets and exercises (23, 29, 45, 85, 90, 108, 109, 117). Variation in the training plan can occur within a workout, during a week, or over a period of several weeks. Only a few of the possible methods for introducing variation into the training plan are presented here. (See chapter 23 for a detailed discussion of periodization, or a systematic method for inducing training variation.)

Within-Session Variation

During an individual training session, there are a number of ways to vary the training stimulus. One often employed method is to vary the intensity at which individual exercises are performed—some exercises are performed at higher intensities and others at lower intensities (117). A single session may employ some longer rest intervals and some shorter rest intervals. Typically, longer rest intervals are used with core exercises and shorter rest intervals are used with assistance exercises (4, 65, 88, 97, 127).

Another way to vary the training stimulus during a session is to alter the set configuration (36, 37, 39). Traditionally sets are performed in a continuous fashion with little or no rest between each repetition. Recent research suggests that one can modify the focus of a set by altering the interrepetition rest interval to construct what has been termed a “cluster set” (39). A cluster set uses a rest interval of 5 to 45 seconds between each repetition (36). For example, if endurance or conditioning is the focus, a shorter rest interval may be employed (~5 seconds). If power generation is the main focus, rest intervals may be longer (~45 seconds). When using the cluster set paradigm, the personal trainer can add an additional layer of variation by introducing changes to the loading used during each repetition of the set. There are three categories of cluster sets: (1) the standard cluster set, in which the load is not altered across the set; (2) the undulating cluster, in which the load is increased and then decreased within the set; and (3) the ascending cluster, in which the load increases across the set (table 15.15) (36, 37, 39). Cluster sets offer another level of variation that can affect the focus of an individual training session.

Within-Week Variation

One of the best ways to incorporate variation into a training plan is to vary the intensity of exercises selected across the week (117). It may be difficult for clients to undertake high training loads across an entire week. As the week progresses, the client’s cumulative stress (i.e., training, work, home, personal, etc.) is likely to increase steadily and his or her ability to tolerate the training stress will decline (107, 117); this increases the potential for overtraining (106). One strategy to combat this is to use light and heavy days of training; the light days allow the client to recover, thereby reducing the risk of overtraining.

There are a multitude of strategies for varying the training load, but one that is particularly easy to employ is to vary the percentages of 1RM during the sessions across the week. For example, if a client’s maximum back squat is 300 pounds (136 kg) and the targeted number of repetitions for the training day is five, a very heavy load for this client would be around 260 pounds (118 kg), which would be roughly a 5RM. Let’s say the client is squatting two days a week (Monday and Thursday). Monday would be the heaviest day, in this case 240 pounds (109 kg, ~92% of 260 pounds), and Thursday would be around 208 to 220 pounds (95 to 100 kg, ~80-85% of 260 pounds). The second session would use ~10% to 15% less resistance, thus accounting for the cumulative fatigue associated with the training week, while still supplying adequate training stimulus to induce adaptive responses. As mentioned previously, training to failure (100% of RM values) is not necessary (105), produces suboptimal results (87), and increases the risk of overtraining (32). A far superior strategy is to plan specific load variations while maintaining the targeted repetitions. Table 15.16 offers an example of how intensity could be varied across a training week and across a period of four weeks. Another example of weekly variation is presented in table 15.17. In this example, the client achieves a 225-pound (102 kg) back squat during the 1RM test. This value can be used in conjunction with table 15.6 to estimate the 5RM (195 pounds or 89 kg) or 10RM (170 pounds or 77 kg). The estimated 10RM can then be used to determine percentage zones, which can be used for creating light and heavy days. This is an excellent method for programming and has been shown to be very effective in maximizing training adaptations while decreasing the risks of overtraining.

between-Week Variation

The training program should vary between weeks, through alterations in volume, intensity, frequency, exercise selection, and training focus. For example, if a four-week period of training targets muscular endurance, the plan may call for three sets of 12 repetitions based on the information presented in table 15.11. According to the principles of periodization, the average intensity could be increased for three weeks and then reduced during the fourth week, creating what is termed a 3:1 loading paradigm (table 15.16) (15, 90). In table 15.16, the weekly volume load (e.g., week 1 = 79,500 pounds; week 2 = 82,650 pounds; week 3 = 85,950 pounds; week 4 = 79,950 pounds) and average training intensity (i.e., week 1 =

134 pounds; week 2 = 140 pounds; week 3 = 145 pounds; week 4 = 135 pounds) increase across the first three weeks. This type of loading is considered the most basic loading structure for resistance training, but there are other possibilities, for example, a 2:1, 4:1, 3:2, or 4:2 structure. If these types of loading are coupled with daily variations as discussed previously, the client will be able to maximize training adaptations while minimizing the chances of overtraining.

Another way to incorporate variation between training weeks is to alter the training density. For example, week 1 may include four resistance training sessions, two high-intensity interval training sessions, and two aerobic training sessions. In this week, the main focus is the resistance training. In the second week, the resistance training sessions could be reduced to three, the high-intensity interval training sessions increased to three, and the frequency of aerobic training maintained, thus changing the focus of the week to interval training. By playing with the various training modes over the training week, one can selectively target specific outcomes and manage fatigue more efficiently, ultimately allowing the client to experience greater training adaptations.

Sequencing Training

One of the classic mistakes made in the design of a resistance training program is not appropriately sequencing the training factors (45, 90, 117). Appropriately sequenced training programs result in superior adaptations and performance gains (45). Conversely, an inappropriately sequenced training plan will mute the physiological and performance adaptations that are expected from the training plan.

It appears that traditionally, personal trainers have targeted only the primary training goals (i.e., muscular endurance, hypertrophy, muscular strength, or muscular power) when designing a program. While at first glance this seems logical based on the principles of specificity, scientific evidence strongly supports the idea of interdependence among the goals. For example, when maximal strength is the main desired training outcome, a period of training

that targets hypertrophy should precede training that targets maximal strength.

Training sequence targeting maximal strength
= Hypertrophy training → Maximal strength training

By sequencing training in this fashion, greater levels of strength can be developed because of the underlying muscular adaptations stimulated by increasing muscle cross-sectional area.

Similarly, it is very clear that if maximal power-generating capacity is the target outcome, one needs to first develop overall strength levels.

Training sequence targeting power development
= Hypertrophy training → Maximal strength training → Muscular power training

The increases in muscle cross-sectional area would thus increase the ability to increase maximal strength, which would then contribute to maximizing muscular power development when the training focus shifts to power-based training.

As a general rule, all clients should begin resistance training with a plan based on either muscular endurance or hypertrophy. These phases are important because they establish the training base, which lays the foundation for the development of other targeted goals (90, 107). Most clients can tolerate these phases of training for two to four weeks with appropriate variations in volume and intensity. After about

four weeks of this type of training, the client's adaptive responses become asymptotic and training adaptations will slow dramatically. At this time the personal trainer should alter the training goals to obviate the problem by shifting the focus of the program toward the development of muscular strength, typically targeting strength development for two to five weeks. Most clients would then shift back to a muscular endurance or hypertrophy focus, but athletes would likely progress toward a muscular power-based plan. Regardless of the goals, the sequencing of training appears to result in a phase potentiation effect, whereby the attributes developed in one phase facilitate the physiological and performance adaptations seen in the subsequent phase (15, 58, 59, 107). When sequencing training, the personal trainer should consider the basic guidelines for the length of each phase presented in table 15.18.

The client's overall training goals, as established early on by the personal trainer and client, will determine the sequence of the training plan. As noted previously, the goals will also help the personal trainer establish the basic volume and loading parameters. However, no client can keep targeting the same goals indefinitely. Therefore the program should vary the targeted goals by periodically altering the training target. For example, a client who is targeting muscular hypertrophy might perform a four-week period of hypertrophy training (e.g., three sets of 10) followed by three weeks of muscular strength work, then return to four weeks of hypertrophy training. Alternating between muscular strength and hypertrophy will produce greater increases in hypertrophy and strength. Three of many methods for sequencing training.

Progression

An essential component of any training program is the concept of progression. Progression is a process in which the client continues to move toward a predetermined goal (4), while maintenance is a time point at which the goal has been achieved and the client is attempting to maintain a specific level of fitness (4, 66). Progression should be considered a general training principle and is a function of training variation. Progression can be accomplished through increasing loads, volume, or frequency or making any number of alterations that modify the training stimulus. The most frequently used method for progressing clients is to change the load and volume lifted in each exercise.

The best method for determining when to increase training load is to schedule periodic RM assessments. These assessments can use any of the methods previously described to either directly measure or estimate the client's capabilities. If, for example, the client is assessed on the back squat

for a 10RM and lifts a 165-pound (75 kgE)

personal trainer can use table 15.6 to estimate the 1RM (i.e., 220 pounds or 100 kg) and then determine the training loads by taking percentages of this value. It is easy to integrate this method into the weekly and daily variation schedules described previously in this chapter. This appears to be the best way to establish training loads because of the inferiority of training to failure as a method of loading (95, 105). Generally, when progressing training loads, one can use a conservative increase in load of 2.5% to 10%, depending on the type of exercise, client training status, and the part of the body being trained.

Another method that is somewhat popular is the **2-for-2 rule** (8, 9, 10), which is based on the achievement of goal repetitions for a training session. The rule states that if the client can complete two more repetitions than the repetition goal in the final set of an exercise for two consecutive training sessions, the load in all of the sets for that exercise during subsequent training sessions can be increased (see table 15.20 for example). While some find this method effective, it may not be the best because it requires consistently training to muscular failure.

Sample Programs for Targeted Training Outcomes:

Numerous combinations of training programs can be constructed based on the sequential approach training plans. Remember that before focusing on strength development, the client should complete a muscular endurance or hypertrophy phase. Additionally, a strength phase should precede a phase of training that focuses on muscular power. Each program should be individualized to meet the client's needs and should meet all the general principles of training addressed at the beginning of this chapter. There are numerous possibilities; the programs presented in tables 15.21 through 15.24 on pages 377 to 383 are simply examples rather than plans to apply to all clients.

Table 15.21 presents a plan that targets **muscular endurance training** for a novice client. This program uses the body weight method for determining training loads because the client is a beginner. The methods are primarily machine based, and the rest intervals are short to facilitate the development of muscular endurance. This program can be used for a short period of time until a training base has been established; at this point the program should be modified so that training intensities can be based on percentages of either predicted or actual maximal lifts in order to maximize the client's adaptive responses. After completing this program, the client should enter into a two- to four-week phase that targets muscular strength. Subsequently the focus can be reoriented toward muscular endurance, and more advanced training techniques can be employed.

The program presented in table 15.22 is a four- week training block that targets muscular hypertrophy. This example is based on the client's stated

goals of muscular hypertrophy, a fitness evaluation that uses a 10RM test to estimate the individual's 1RM, and the client's training status. From this 1RM, the actual training loads are used to determine the client's training zones (based on table 15.17). Training takes place four times per week, with most of the exercises targeting multijoint movements performed on free weights. The program employs inter- and intrainensity variation as indicated by the inclusion of heavy and light training days. The overall block exhibits a 3:1 loading paradigm: The load increases over the first three weeks, while the fourth week is an unloading week as indicated by the overall decrease in training intensity.

A sample training program that targets muscular strength is presented in table 15.23. This example includes a four-week block of training. The plan is for an advanced client who is targeting the maximization of muscular strength. Based on the client's level of development and experience, the plan calls for a frequency of four times per week (based on table 15.2). The majority of the exercises are performed with free weights, with three sets of five repetitions for core exercises and three sets of 10 repetitions for assistance exercises. Intensities are determined based on 1RM and 10RM testing, which is used to estimate 5RM loads. From these loads training zones are established. The program has a 3:1 loading structure, with heavy and light days to facilitate recovery and adaptation.

Finally, table 15.24 presents a sample muscular power training plan for an advanced client. This program uses training intensities based on estimated 1RM and 3RM loads for the core exercises. The majority of the exercises are free weight exercises. Because this client is advanced, the training frequency is set at four days per week. In keeping with the recommendations presented in tables 15.13 and 15.14, the client will perform three sets of three repetitions with 3 minutes' rest between sets. Additionally, the power clean and power snatch will use ascending or traditional clusters as shown in table 15.15. A 3:1 loading paradigm with daily and weekly fluctuations in training intensity is used.

Conclusion

Designing a resistance training program requires the personal trainer to understand the concepts of specificity of training, overload, variation, progression, and sequencing. Many training factors can be altered, such as the exercises selected, frequency of training, overload, volume, intensity, rest interval, and sequence, to optimize outcomes. Once the program is established, the personal trainer must vary and advance the plan in order to continue to help clients achieve their training goals.



gym equipment

Why you should learn to identify gym equipment

Gym related stresses and strains are the most common reason why people don't turn up to workouts the next day.

An easy way out – learn what each piece of equipment does.

This can help you turn them to your benefit rather than work against your efforts.

Here's how it can help:

- Avoid injuries
- Maximize benefits
- Get fit quicker

Benefits of proper form and technique

Proper form while exercising is a lot more critical than the weight you're pushing.

This is especially important during strength training when you deal with a huge amount of weight. There are immediate and long-term problems if you do not use proper form to deal with heavy weights.

Let's look at a few:

Strength Training Injuries

Short Term

- Fractures
- Sprains
- Strains
- Dislocations

Long Term

- Muscle overload and fatigue
- Nerve damage
- Rotator cuff injuries
- Bone-related stress injuries

The key to avoiding MOST of these is just to be mindful of the type of movement you're executing and not forgetting to warm up properly.

Warming up is more important than you know – it increases blood flow to muscles, improves oxygen delivery and preps your muscles for the impending workout.

So why don't we get ourselves familiarized with these exercise machines and get fit the right way without injuries or any problems?

Here we go!



. Dip station



An incredibly effective piece of equipment on which you can perform a variety of exercises. It has 2 arms and a large base which increases stability and prevents it from toppling over.

A dip bar or station is one of the best options for increased upper body muscularity. [Check out our dip bar review](#) which can give you the lowdown on the best equipment in the market.

How often should you use it?

Dips work your chest and arms so you need to ideally supplement this with your arm and chest workouts. You can do this on chest and arm days but just make sure you don't hit them too hard consecutively.

Muscles worked: [Chest](#), triceps, front shoulders

Tips for optimal performance

If you want to work your chest during dips just make sure that you lean your body forwards a bit. When you want to work your triceps, you need to stay upright throughout the movement.

Best places to buy: Etailers or specialist fitness brands like Rogue Fitness, Westside Barbell.

What we recommend: [Monster Lite Matador](#)

2. Battle Ropes



Battle ropes give you full body strength training, enhances your fatigue resistance and helps your twitch muscles. It has become a mainstream feature in gym these days and adds an element of cardio to resistance workouts.

You can add battle ropes to plenty of regular exercises like burpee slams, single arm plank waves, double wave/ jump squat combos and the like. [Read here](#) to get more ideas about how battle ropes can tone your body.

How often should you use it?

On days when you are looking to maximize your workout you can start with battle ropes. It can warm up and loosen your muscles and joints for more strenuous exercises later.

Muscles worked: Full-body workout!

Tips for optimal performance

When shopping for a battle rope, make sure it's at least 45' in length and your exercise area is at least 30' in length. This will give you ample space to effectively workout using these ropes.

Best places to buy: Etailers like Walmart or Tesco, specialist fitness brands like Rogue

What we recommend: [Rogue Bulldog Ropes](#)

3. Bench Press



A device which can help you get that perfect rock hard chest you've always dreamt of. Mainly used in upper body strength training exercises along with a barbell.

This is one of those gym cornerstone pieces of equipment that everyone has to go to one time or another. It is vital for building upper body strength and musculature. Find out the right way to bench from the experts.

<https://garagegymplanner.com/bench-press-for-beginners/>

How often should you use it?

If you are pushing heavy, have rest days in between so your chest can recover.

Muscles worked:

Chest, triceps, deltoids, traps and back

Tips for optimal performance

If you want to build a good chest, remember that form is more important than poundage.

Best places to buy: Etailers and specialist fitness brands like Rogue fitness and York

What we recommend: F2C Pair of Adjustable Rack Sturdy Steel Squat Barbell Free Bench Press Stands

4. Incline Bench Press



The inclined bench press is a variation of the bench press, using which you can perform strength training exercises at an elevated height. The shoulders and upper chest area can be targeted.

Incline bench presses develop your upper chest musculature while also being a bit more joint-friendly than the other variations. Take a look at our page which highlights the benefits of incline bench presses as well as the decline variation.

How often should you use it?

Inclined bench is part of the chest exercise, so the best answer would be when you have to work on your chest.

Muscles worked: Upper chest, triceps, deltoids.

Tips for optimal performance

Most of the rules of straight bench press apply here like not hyperextending the back or not bouncing it off your chest.

Best places to buy: Etailers like eBay

What we recommend: Body Champ Olympic Weight Bench

5. Hammer Strength machine



A plate loaded device which focuses on the body's natural path of motion. You can fully explore the advantages of converging and diverging arcs of movement.

A hammer strength machine lets you lift a lot more than when you lift free weights. You need to remember not to overexert yourself.

How often should you use it?

It's a good movement to supplement your biceps/arms workout. Remember to not overdo it and give your arms about 2 days rest before you work the same muscle group.

Muscles worked: Lats, chest, middle back, shoulders and triceps

Tips for optimal performance

The action is similar to a shoulder press but a bit more at an angle. The key is to maintain your form and lift smooth rather than jerk it.

Best places to buy: Etailers like eBay and specialist fitness brands like lifefitness

6. Lat Pull-Down Machine



The lat pulldown machine is a strength training device with a padded seat, thigh support and a long bar hanging from an upper rod. You can work your lats using this machine.

If you are not able to handle pull-ups, this could be a good alternative. You can vary your grips on the pull-down rod to work with different parts of your upper body.

How often should you use it?

You can do lat pulldowns twice every week without fatiguing your muscles.

Muscles worked: Lats, deltoids, trapezius and rhomboids

Tips for optimal performance

When executing this movement, make sure that you don't hyperextend your back too much.

Best places to buy: Etailers Like eBay and specialist brands like Valor Fitness

What we recommend: [Powerline PLM180X Lat Machine](#)

7. Pec Deck Machine



The best machine to isolate your pectoral muscles and give them a good workout. You can perform many exercises such as chest flys, butterfly, etc. using a pec deck machine.

This machine is particularly beneficial to build chest and shoulder muscles, and also enhance arms strength and stability. The upper body muscles are squeezed together, causing the pectoralis major to expand and contract. This is what builds the muscle and also toughens the tissue fibers. When the exercise on this machine is done with appropriate weights, it does prove to be quite effective. If you're particularly working on your chest, this might help:

<https://garagegymplanner.com/build-big-chest-at-home/>

How often should you use it?

Can be used once a week, with varying weights.

Muscles worked: The pectoralis major, pectoralis minor and serratus anterior.

Tips for optimal performance

Pair up the pec deck fly along with bench press and bent over cable crossover flys for better results.

Best Places to Buy: E-tailers like Walmart and specialist fitness brands like Body Solid

What we Recommend: Body-Solid GPM65 Plate Loaded Pec Machine

8. Pull-up Bar (Chin-up bar)



As the name suggests, this device will help you do your pull-ups better. It's also called a chin-up bar because you pull yourself up with your chin above the bar.

One of the best upper body exercises, pull-ups actually work the whole body to some extent. To find out what you should be buying, take a look at [this article](#) which has the best pull up bars review by GGP.

How often should you use it?

Pull-ups can generally be done up to 3 times a week.

Muscles worked: Lats, biceps, also engages the entire body to a degree.

Tips for optimal performance

For increased difficulty, try pulling up from a dead hang with straight elbows. You can also add more weight through chains and barbell harnesses.

Best places to buy: Etailers like eBay and specialist fitness brands like Rogue fitness

What we recommend: [Ultimate Body Press XL Doorway Pull Up Bar](#)

9. Dumbbells



An essential piece of equipment in weight training with a bar and weights attached at each end. You can change the weight plates in certain dumbbells for greater versatility.

The tech of dumbbells has grown and now we even have adjustable varieties that do away with having to need a rack full of dumbbells. Take a look at our [best dumbbell guide for 2019](#) to learn more.

How often should you use it?

Varies depending on the type of workout.

Muscles worked: Varies depending on the type of workout.

Tips for optimal performance

A dumbbell is one of the most effective and nifty pieces of fitness kit ever. And if you know how, you can literally exercise every part of your body just using a pair of dumbbells.

Best places to buy: Etailers like eBay and specialist fitness brands like Rogue Fitness

What we recommend: [Bowflex SelectTech 552 Adjustable Dumbbells](#)

10. Tricep bars



If you are looking to work your triceps, then this is the specialized bar you need. When you use a barbell, there are limits on how much you can do and the grip is restricted as well. This bar will let you get a comfortable grip as well as perform tricep focused movement with ease.

A tricep bar not just gives your triceps a good workout but you can also use it for hammer curls or front raises which target the bicep and deltoids respectively.

How often should you use it?

You can use this when you need to work your triceps through a greater degree of movement with much better control.

Muscles worked: Triceps, biceps, deltoids

Tips for optimal performance

Don't load too much weight onto it before you get used to the grip.

Best places to buy: Etailers or specialist fitness brands like Rogue Fitness, Muscle Driver.

What we recommend: Rogue MG-3 Multi Grip Bar

11. Preacher Bench



The ideal machine to build your biceps. You can increase muscle mass by lifting the barbell up and down. The machine has an elbow budding, a bar rest, and a seat.

Using a preacher bench is a great way to target your biceps without putting pressure on your wrists. Another way to use this device is with a reverse grip to target the forearms.

How often should you use it?

It is best to train your biceps using a preacher bench once a week for the best results.

Muscles worked

Preacher curls are used for isolating the muscles of the upper arm. They improve wrist and grip strength while targeting your biceps and forearms

Tips for optimal performance

It is important not to overextend your elbows at the bottom portion of the lift. Always use a controlled motion and don't lift the bar too fast.

Best Places to Buy: E-tailers Like Walmart and specialist fitness brands like Valor Fitness

What we Recommend: [Valor Fitness CB-6 Adjustable Arm Curl Bench](#)

12. Handgrip Exerciser



Mechanical handgrips are inexpensive training devices that can help you build hand strength. They are thick springs with handles on them that isolate the muscles associated with grip strength.

When using mechanical handgrips, always warm up and vary your training to avoid overdoing it. For more tips on using handgrips take a look at this [post](#):

How often should you use it?

You can do this workout 3 times a week. Start with 2 warm-up sets, followed by 3 intense sets of moderate-to-low reps.

Muscles worked

A good handgrip exerciser has the ability to produce quick gains in hand health and grip strength.

Tips for optimal performance

A good tip to follow is to try reaching a goal of 12 consecutive reps on one gripper. Once you do this, you can start working on reaching the next level.

Best Places to Buy: E-tailers like Walmart and eBay

What we Recommend: [Kootek Hand Grip Strengtheners](#)

13. Reverse hyper



The reverse hyperextension combines strengthening and rehab into one effective machine. The spine is gently stretched and the lumbar area is strengthened.

This is definitely one of the best therapeutic machines for the back. One of the major reason for putting your back out is during heavy lifting or squatting. If you're doing squats, you might want to [check how you can do a proper squat here](#).

How often should you use it?

If you are recovering from an injury (foot, knee, or ankle), you can regularly use this machine till you get better. For strengthening your back, this can be added to your queue during back workouts.

Muscles worked: Lower lumbar, glutes, hamstrings, and hips.

Tips for optimal performance

Have a firm grip on the front handles and make sure that there are no wobbles when you use the machine. Make sure that you don't swing the weighted ends too hard.

Best places to buy: Etailers or specialist fitness brands like Rogue Fitness, Westside Barbell.

What we recommend: [RH-2 Rogue Reverse Hyper](#)

14. Plyo Boxes



It's a solid box/construct on which you can perform a variety of movements that can recruit several muscle groups at one time. You can do box jumps, burpees, pistol squats, jackknives, elevated crunches, decline side planks, and many more types of exercise.

A plyo box when used properly can increase your vertical jump height in a few sessions. Read up on our plyo box round up which can give you a good idea about what you need to choose.

How often should you use it?

You can use a plyo box every day because of the huge array of exercises that you accomplish on them.

Muscles worked: Full-body workout!

Tips for optimal performance

The basic exercise on the plyo box is box jumps. Box jumps are a full body exercise that can enhance your fast twitch muscles and build explosive strength. The key to good technique is to make sure that you nail the takeoff right along with proper landing.

Best places to buy: Online e-tailers or fitness vendors like Rogue

What we recommend: Rogue Flat Pack Games Box

15. Air resistance Exercise Bike



An exercise bike with a twist, the air resistance bike uses exponential resistive force generated by air to give you a thorough cardio workout.

One of the most grueling pieces of kit in the gym, an Assault AirBike can really turn the screws when it comes to pushing your limits.

How often should you use it?

You can incorporate into your current cardio regime. On days when you are looking to push your cardio limits, this can definitely help you get that burn.

Muscles worked: Full-body cardio workout!

Tips for optimal performance

This is just like a regular exercise bike but only really hard. Try not to overexert yourself when working out on this.

Best places to buy: Etailers like Walmart or Tesco, fitness brands like Rogue, Schwinn

What we recommend: Assault AirBike

16. Stair Stepper



This is an inexpensive machine which can simulate climbing stairs which is considered a really good cardio workout.

While the stair stepper may be an older machine compared to the newfangled pieces of equipment available, it can still give you that lower body burn. Learn more about its benefits from our [stepper machine article](#).

How often should you use it?

You can use this for leg days when you are really keen to get that burn going. This machine works the thighs and calf muscles considerably.

Muscles worked: Hips, quads, hamstrings, calves, and lower shin.

Tips for optimal performance

You need to remember to keep your body upright and hips centered over your legs. Keep your movements gentle and remember to push your heel into the step. Squeeze your glutes through each repetition.

Best places to buy: Etailers like Walmart or eBay

What we recommend: [Sunny Health & Fitness Twisting Stair Stepper](#)

17. Ankle Weights



A well-cushioned weight-bracelet for your ankle which is used to add resistance to your exercises.

This is a very useful and portable method to add more to your resistance to your morning walks or jogging routine. We have a [section](#) dedicated to helping you find the best ankle weights in the market.

How often should you use it?

You can use this whenever you want to add more resistance to lower body exercises or cardio sessions.

Muscles worked: Lower body muscles, depending on the type of movement

Tips for optimal performance

Sometimes with really heavy ankle weights, you might experience a bit of abrasion against your skin. You can wear tall socks to circumvent this.

Best places to buy: Etailers like Walmart and Tesco or specialist fitness brands like Rogue fitness

What we recommend: [Nordic Fitness Ankle/ Wrist Weights](#)

18. Leg Curl Machine



The ideal machine for toning your quadriceps is the leg extension machine. You will have to sit on the machine with your legs under the pads and lift weights using your quadriceps.

A lot of people have gone past leg extensions to compound exercises, but they still find them useful in certain situations.

How often should you use it?

Leg extensions are best done once a week because you shouldn't tax your knees too much.

Muscles worked: Quadriceps, rectus femoris.

Tips for optimal performance

You should remember never to hyper-lock your legs because that can cause stress on your knees.

Best places to buy: Etailers Like eBay and specialist fitness brands like Valor Fitness

What we recommend: Valor Fitness CC-4 Adjustable Leg Curl Machine

20. Leg Extension Machine



The ideal machine for toning your quadriceps is the leg extension machine. You will have to sit on the machine with your legs under the pads and lift weights using your quadriceps.

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Tips for optimal performance

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Best places to buy: Etailers Like eBay and specialist fitness brands like Valor Fitness

What we recommend: Valor Fitness CC-4 Adjustable Leg Curl Machine

21. Hack Squat Machine



Another fitness device that can give your legs a good workout. It is essentially a combination of leg press and squat machines. It works your quadriceps in a much more efficient way.

This along with a leg press machine can help develop your lower body especially the thighs and calf sections. Find our review of hack squat and leg press machines.

How often should you use it?

You can use hack squat machines for leg days, meaning about 3 times a week, max.

Muscles worked: Quadriceps, gluteus maximus

Tips for optimal performance

Like squats, always ensure that you're pushing against the weight with all of your foot rather than just the heel or toe.

Best places to buy: E-tailers Like Walmart and eBay

What we recommend: Body-Solid Leg Press & Hack Squat

22. Calf Machines



The calf machine can isolate the soleus muscles of the calves and give it a good workout. Your calves get a more muscular appearance when you work the soleus muscles.

This machine is basically used to do seated calf raises. Weights can be varied and the frequency depends on the person. A lot of athletes and people who require extensive legwork prefer to use calf machines to strengthen their lower legs and also increase speed and agility.

How often should you use it?

Can be used up to 4-6 times a week. Unless the calf muscles are sore, the machine can be used every day as well.

Muscles worked: Gastrocnemius, tibialis posterior and soleus muscles of the lower leg.

Tips for optimal performance

Keep increasing the weights from one alternative day to the other. This helps with pushing the core strength of the calf muscle. Avoid cardio exercises right after calf raises. Cardio before is a better option.

Best Places to Buy: E-tailers like Walmart and Tesco

What we Recommend: [Powerline PSC43X Seated Calf Machine](#)

23. Leg Adduction / Abduction Machine



A seated exercise machine which can provide resistance when closing and the opening the legs. You can tone your glutes with abduction and inner thigh muscles during adduction using this machine.

Adduction machines are mostly used by people who have been working out for longer, continuous periods of time. Beginners should avoid using this machine too much as it could cause stiffness in the legs and always pelvic aches. Even women are usually advised to not start off with this machine as it could cause muscle pulls and minor tears to someone who isn't well-versed with its usage. Check out these [interval training workouts](#) that would work well for beginners:

How often should you use it?

Not more than twice a week with appropriate weights.

Muscles worked:

It works the adductor brevis, which runs from the upper femur down to the pelvis; the adductor longus, which stretches from the pelvis to the middle of the femur; and the adductor magnus, which also begins at the pelvis and runs down to the lower part of the femur.

Also works on the pectineus and gracilis muscles.

Tips for optimal performance

Always better to use this machine towards the end of a workout session. Weights should not exceed threshold of the person as it could become injurious.

Best Places to Buy: E-tailers like Walmart

What we Recommend: [Body Solid Inner and Outer Thigh Machine](#)

24. Prowler/Dragging Sled



A prowler or sled is a flat surface with a simple projection onto which you can add weights. These come in various shapes from simple frames to flat pan shovel like profiles.

If you are on the lookout for cardio but are not able to run because of some constraints, then you should take a look at sleds. They expend more energy per time spent than running and are a lot easier on the knees as well. If you're looking for footwear for sled dragging, take a look at our CrossFit shoes review and find what suits you.

<https://garagegymplanner.com/best-crossfit-shoes-men-women/>

How often should you use it?

If you are into endurance training, you can use this every day. You can also use this to break up your regular weight training regime.

Muscles worked: Complete lower body workout!

Tips for optimal performance

When using a sled/prowler the key is loading the right weight for the surface you're going to be pushing on. You should be pushing it at a brisk walking pace.

Best places to buy: Etailers or specialist fitness brands like Rogue.

What we recommend: Rogue Dog Sled 1.2

25. Mini Trampoline



A fun little piece of equipment which you can use to strengthen your body, hone your balance and increase aerial body control.

There is nothing more fun than bouncing around on your own trampoline while improving your balance and your fitness levels. Doing this early in the morning can jumpstart your metabolism and you'll feel refreshed.

How often should you use it?

You can use it every day if you'd like but since it has to do more with agility and finesse but not classical strength building, people who are focused on increasing musculature most likely will not prefer this on a regular basis.

Muscles worked: Gluteal muscles, hip adductors and hip flexors

Tips for optimal performance

Some things you should keep in mind before you opt for a trampoline is obviously the height of your roof. Trampoline fitness should ideally be carried out at the start of the day rather than the end.

Best places to buy: Etailers like Walmart or Tesco

What we recommend: Zupapa TUV Approved Trampoline

26. Inversion Table



Inversion tables are specialized pieces of equipment that are mostly used for therapeutic purposes. It relies on the effect of gravity to pull and decompress joints of your body that are below the anchor. This effect stretches out your body and you can adjust being at a small angle or fully vertical.

When you are buying an inversion table, you need to make sure that you invest that bit of extra money for a durable piece of equipment because you're going to be in a vulnerable position when you use it. We have a review of the best inversion tables that you can get, [right here](#).

How often should you use it?

Inversion tables are good for people in certain situations and not a generic therapeutic piece of equipment. If you have conditions like heart disease, high blood pressure, or acid reflux, inversion tables can do more harm than good.

Muscles worked

The inversion table can strengthen your abdominal muscles and your back muscles, stretch thigh and hamstring muscles. It can alleviate neck and back pain.

Tips for optimal performance

If this is the first time you are using this, you need to consult your physician before you start. You also should make sure that you have ample room around the inversion table to rotate without obstruction.

Best places to buy: Etailers like Walmart or Tesco, specialist fitness brands like Rogue

What we recommend: [Ironman Gravity 1000 Inversion Table](#)

27. Vibration Plate



The machine looks like a weighing machine with handles but it has a base that vibrates at varying frequencies so people get a low-impact workout without the physical exertion normally associated with regular exercises.

Vibration plates work on the principle of muscle resonance, the frequency at which your muscles begin to vibrate, activating it and boosting your ability to exercise.

How often should you use it?

You can use it every day but it'd be more effective when you use in conjunction with your exercise routine.

Muscles worked: Full body vibration therapy.

Tips for optimal performance

The key to best using a vibration plate machine is to find the right frequency that your body responds to. The ideal frequency you need to use is between 20-45Hz.

Best places to buy: E-tailers like Walmart and Tesco

What we recommend: [Hurtle Fitness Vibration Platform Workout Machine](#)

28. Power Rack (Squat Rack/ Squat Station)



A piece of weight training equipment which helps you perform barbell exercises by functioning as a mechanical spotter with saddles that can rest at varied heights and bars that can go across such heights.

Power racks are the cornerstone of strength building and deserve their place in every gym, commercial or indoor.

How often should you use it?

You engage a lot of muscles when you use a power rack because all movements are compound exercises. Generally, if you are doing heavy movements involving a muscle group, then you rest it for at least 3 days.

Muscles worked: Varies with the type of exercise – squats, deadlifts, bench presses

Tips for optimal performance

Make sure you invest in a good quality power rack which has safeties and is built to last.

Best places to buy: E-tailers or specialist fitness brands like Rogue Fitness

What we recommend: PowerLine PPR200X Power Rack

29. Maxi Climber



Using a maxi climber is a fast and easy way to get into shape. It engages all the major muscle groups to build strength and burn calories.

A maxi climber is ideal for anyone looking for a compact piece of equipment. To know more about the maxi climbers and other vertical climbers on the market checkout our extensive reviews.

How often should you use it?

3 solid workout a week using a maxi climber is ideal.

Muscles worked

A maxi climber increases your cardiovascular fitness. It will also tone and strengthen your thighs, glutes, calves, shoulders, triceps, biceps and core.

Tips for optimal performance

Since your body weight provides resistance, the more effort you put in, the more calories you will burn.

Best Places to Buy: E-tailers like Walmart and Tesco

What we Recommend: Maxi Climber Vertical Climber

30. Stretching Machine



Stretching machines are ergonomically designed devices to make stretching movements easier and more efficient. They are ideal for athletes and also useful for preventing sport-related injuries.

There are both full body stretching machines and leg stretching machines available, so it's important to decide which one of these is better suited to your workouts. Always do your research and check out our detailed reviews of the best stretching machines.

<https://garagegymplanner.com/best-stretching-machines/>

How often should you use it?

Stretching before any workout is highly recommended to avoid injuries. Spend ten minutes stretching out your muscles before beginning your exercise regime.

Muscles worked: Some machines target specific areas while others work on the entire body.

Tips for optimal performance

When stretching, it is important to focus on using proper form and the complete range of motion

Best Places to Buy: E-tailers like Walmart and eBay

What we Recommend: Precor 240i Commercial Series StretchTrainer

31. Smith Machine



A weight training machine which will assist you in lifting weights and in performing squats. The machine has a barbell which is fixed within steel rails allowing vertical movements.

This is a great piece of equipment for intermediate and advanced lifters. But it is important to understand its limitations and use it accordingly. Take a look at our detailed home smith machine reviews for [more information](#).

How often should you use it?

It is a good idea to use a smith machine when lifting weights, as it can give you a more isolated and safer workout.

Muscles worked: Using a smith machine bolsters the upper back, improves core strength, and stabilizes the shoulders.

Tips for optimal performance

It is not advisable to use a smith machine for anything other than partial-range short duration training or calf raises.

Best Places to Buy: Etailers Like eBay and specialist fitness brands like Valor Fitness

What we Recommend: [Valor Athletics Smith Machine](#)

32. Suspension Trainer



It consists of several nylon bands and plastic buckles and is used to leverage your body against gravity to create resistance. Using a suspension trainer, you can perform hundreds of different exercise movements which work on various parts of the body.

This was an idea by a US navy seal to come up with TRX range of suspension trainers. All you needed with this training equipment was anchor and gravity, both of which are accessible anytime.

<https://garagegymplanner.com/best-suspension-trainer/>

How often should you use it?

You can use this everyday if you aren't focusing on weight training or alternate it with cardio or resistance training.

Muscles worked: Full-body workout!

Tips for optimal performance

One concern with suspension trainers is that you need to make sure that you are hanging the straps from stable points rather than wonky support.

Best places to buy: E-tailers or specialist fitness brands like Rogue Fitness, Westside Barbell.

What we recommend: TRX Suspension Trainer Basic Kit

33. Cable Pulley Machine



Also known as the cable crossover machine, it has a framework with grips that are attached to cables which are further connected to weights. A variety of exercises can be performed on this.

This has to be one of the most versatile pieces of gym equipment apart from a dumbbell. You can find our reviews of the best cable crossover machine you can get.

<https://garagegymplanner.com/cable-machine-reviews/>

How often should you use it?

Since there are a lot of variations you can perform on this, if you target different parts, you can use it every day.

Muscles worked: Full-body workout!

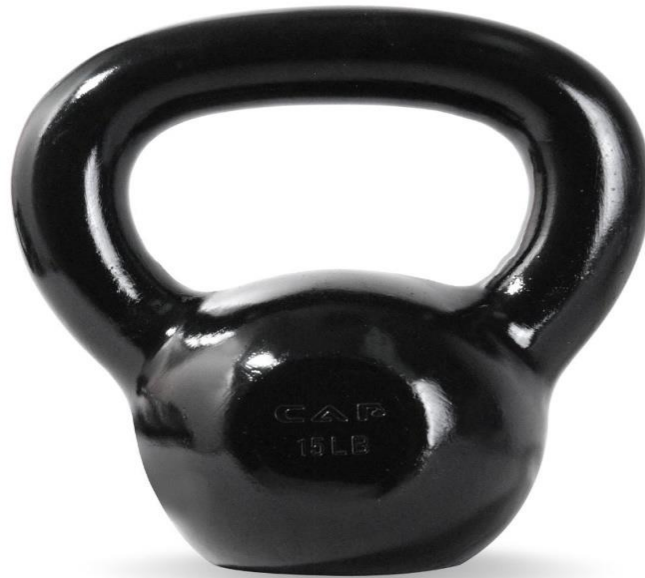
Tips for optimal performance

From lunges to squats to rows, there are a lot of things that you can accomplish with a cable crossover machine.

Best places to buy: Etailers like eBay and specialist fitness brands like Rock Solid fitness

What we recommend: XMark Functional Trainer Cable Machine

34. Kettlebells



One of the most ancient and efficient pieces of strength training equipment is the kettlebell. Consisting of an iron ball with a handle, there are hundreds of exercises which you can do using a kettlebell.

It's important to be careful and precise about posture as these work on multiple core muscles. There are a number of exercises and movements that can be paired with the kettlebell as it is quite diverse on its own. One can easily do a full body workout with just kettlebells and different weight modulation.

How often should you use it?

Twice a week

Muscles worked: Works on the core, shoulders, quads, glutes, hamstrings and back muscles.

Tips for optimal performance

Use a balanced weight and make sure your legs are wide apart while squatting to do the exercise.

Best Places to Buy: E-tailers like Walmart and specialist fitness brands like Rogue Fitness

What we Recommend: [Yes4All Solid Cast Iron Kettlebell](#)

35. The Roman Chair



The Roman chair is a piece of equipment that you can use to primarily tone your lower back or strengthen your abs, hamstrings and gluteal muscles.

Roman chairs are really flexible when it comes to working out, because you can either workout your back or your abs depending on your schedule. If you are looking for exercises for roman chairs, we have just the page for you.

<https://garagegymplanner.com/roman-chair-workouts/>

How often should you use it?

It can form a part of your back/abs workout routine, so use the machine accordingly.

Muscles worked: Back, core, abdominals, hamstrings

Tips for optimal performance

A lot of people prefer the roman chair for its no-nonsense back-strengthening performance. One thing you need to remember though is that you should avoid rounding the spine. Be mindful that you keep it straight throughout the movement.

Best places to buy: Etailers like Walmart and Tesco

What we recommend: BodyCraft F670 Hyperextension/Roman Chair

36. Abdominal Bench



Similar to hyperextension bench, the abdominal bench targets your abs. It is mainly used in performing squats and weight training. It is highly recommended that you maintain optimal posture while working out.

While it's alright to experience some muscular fatigue or burning when using an abdominal bench, the movements should not cause intense pain. Always start slow and consult a professional before beginning a new fitness routine.

How often should you use it?

The abdominal bench can be used 2-3 times a week. It is recommended to take rest days between workouts.

Muscles worked

This is a very versatile piece of exercise equipment that can be used to strengthen various unique muscle groups, not just in the abdomen but throughout the body.

Tips for optimal performance

Whatever exercise you are doing on the abdominal bench, it is important to maintain the proper form to maximize your movements and avoid injuries.

Best Places to Buy: E-tailers Like Walmart and eBay

What we Recommend: Universal UB100 Decline Bench

37. Ab Coaster



The Ab coaster machine works your abs from the bottom up, by targeting those hard to reach ab muscles. It can maximize your core workout by working your abdominal muscles.

Definitely enhances regular floor ab exercises and is more effective on working on the whole ab muscle, from bottom to top. It works especially on the abdominal muscles and creates core strength in weaker portions of the abs. It is also great to create core definition for aesthetic purposes and is used by athletes for during sports training. If you're still not sure about the ab coaster, [here's a guide](#) that will tell you everything you need to know:

How often should you use it?

Thrice a week

Muscles worked: Upper abs, lower abs and side obliques

Tips for optimal performance

Not recommended for people with weaker abdominal muscles. Definitely need followed up by stretches and periods of rest to let the muscles relax.

Best Places to Buy: E-tailers like Walmart and eBay

What we Recommend: [Ab Coaster Max](#)

38. Stationary Bike



This is the granddaddy of all cardio machines, the simple stationary bike. It has been around since forever but you'll still find them in all gyms over the country because it is that effective.

Stationary bikes have had many variations over the years, some of which have been more effective, and some less so. But it has not stopped it from being a staple on every gym floor ever. Here's our [review of the best stationary bike](#) for your home.

How often should you use it?

You can safely use this every day because it's a non-impact exercise and when used consistently over a longer period of time can burn flab and tone your body.

Muscles worked: Full body cardio workout, but more towards the lower body.

Tips for optimal performance

When starting out, find your comfort zone in the rpm range and stick with it. Find the optimal seat height so that your legs do not fully stretch when pedaling. Remember to pedal with the balls of your feet rather than the middle or the heel part.

Best places to buy: Etailers like Walmart and specialist fitness brands like Exerpeutic Therapeutic Fitness

What we recommend: [Exerpeutic Folding Magnetic Upright Bike With Pulse](#)

39. Cruiser Bike



Cruiser bikes are perfect for short rides and are available in an array of styles. They combine an upright seating position, balloon tires and a straightforward steel construction to deliver a comfortable ride.

Cruiser bikes are ideal for people with bad posture or back pain because they teach you to maintain an upright position and take it slow.

How often should you use it?

These can be ridden every day to get a healthy dose of exercise. However these are not recommended for long rides.

Muscles worked

Biking is a full body workout. It also builds strength and offers cardiovascular benefits.

Tips for optimal performance

Ensure that you maintain an optimal upright posture when using a cruiser bike. It is recommended to check the settings and structure of your bike before you start biking as a form of exercise.

Best places to buy: Retailers like Walmart and Target and specialist brands

What we recommend: Sixthreezero Around The Block Men's 26-Inch Single Speed Cruiser Bike

40. Fixie Bike



A bike without a freewheel is called a fixie or fixed gear bike. They are affordable with solid lines and a clean build. They can also be customized to suit your needs.

Fixie bikes are growing in popularity due to their uniqueness and elegant simplicity. Before opting for a fixie bike, read online reviews and talk to avid bikers. Also, [check out this comprehensive review by GGP](#):

How often should you use it?

Fixie Bikes are a great way to get from one place to another, while staying fit. They can be ridden every day.

Muscles worked: Fixie Bikes offer a full body workout. They also develop speed and strength.

Tips for optimal performance

These bikes can be customized to suit your fitness needs. For instance, adding disc wheels, flat bars, and motorcycle grips can improve performance and design.

Best places to buy: Retailers like Walmart or Target and specialist brands like mrbikeshop.com

What we recommend: [Streaker Single Speed Fixie Bike](#)

41. Mountain Bike



This is a specially designed bicycle with multiple greens and deep-treaded tires for extra grip. It is designed for riding on mountainous or uneven terrain.

When it comes to choosing a mountain bike, it is important to pay attention to factors such as durability and replacement costs. For all you need to know about mountain bikes, take a look at our detailed reviews.

How often should you use it?

It is recommended to use a mountain bike every time you ride on rough or uneven terrain.

Muscles worked

While the leg muscles get quite a workout during a ride, other muscle groups such as your biceps, Gluteus Maximus and Gastrocnemius also get activated.

Tips for optimal performance

Ensure that your bike is well maintained to avoid any accidents or distractions you when you are riding. Also, ensure that you are familiar with the bike and its features well before riding it.

Best Places to Buy: E-tailers like Walmart and Tesco

What we Recommend: Diamondback Mountain Bikes

42. Recumbent Bike



Recumbent Bikes are designed to give you great cardiovascular workout. They differ from other stationary bikes as the rider is in a reclining position when working out. This allows the rider's weight to be distributed comfortably over a larger area.

Riding a recumbent bike gives you resistance training along with cardiovascular exercise. Ensure that you choose a reputed brand. Check out our detailed [post](#) on recumbent bikes for more information.

How often should you use it?

Aim for 250 minutes of cycling per week at moderate-intensity or 100 minutes at a vigorous-intensity for weight loss.

Muscles worked: The quadriceps and glutes get worked when you exercise on a recumbent bike

Tips for optimal performance

Ensure that you maintain the right posture when using a recumbent bike.

Best Places to Buy: E-tailers like Walmart, eBay and Tesco

What we Recommend: Exerpeutic 900XL Extended Capacity Recumbent Bike

43. Road Bike



Road bikes are known for their sleek aerodynamic design and are primarily intended for use on tarmac. They are designed to move at high speeds on paved roads and can be customized to meet your riding needs.

A road bike can be your ideal exercise companion if you're planning to ride mostly in urban areas. Choosing the right bike will depend on a number of different factors.

<https://garagegymplanner.com/best-road-bikes-reviews/>

How often should you use it?

Road bikes can be used whenever you are riding on paved roads and are ideal for navigating through cities.

Muscles worked

Road bikes provide a full body workout and muscles such as the quadriceps, hamstrings, calf muscles and hip flexors are worked.

Tips for optimal performance

Always practice on your bike before heading out onto open roads and stay alert at all times

Best Places to Buy: E-tailers like Walmart or specialist fitness brands like purecycles.com

What we Recommend: Alchemy Eros

44. Spin Bike



A spin bike delivers an effective indoor workout. These bikes come with a heavyweight flywheel and cycling shoe compatibility. They also come with varying resistance levels to adjust the intensity level of the workout.

Spin bikes are growing in popularity as they are an excellent way to burn calories by sweating it out. It is important not to overdo it and choose the right bike for your needs from the numerous options available.

How often should you use it?

An hour of spinning 5 times a week will burn those calories and keep you fit.

Muscles worked

In addition to the leg muscles worked on a spin bike, the abdominal muscles also get a workout.

Tips for optimal performance

Remember not to set the seat too low or position it too close to the handlebars. Maintaining the right posture is key to getting maximum benefits from your workout.

Best Places to Buy: E-tailers like Walmart and eBay

What we Recommend: Sunny Health & Fitness Pro Indoor Cycling Bike

45. Comfort Bike



Comfort bikes are built mainly for recreational and short distance rides. They offer an easy and affordable way to get a workout while enjoying the great outdoors.

Comfort bikes offer a great fusion of the features of road bikes and touring bikes. They are great for daily commutes as well as occasional cruising.

How often should you use it?

Comfort bikes can be used every day, provided you maintain the right posture and don't overdo it.

Muscles worked

Riding a bike provides a full body workout. The number of calories you burn depends on the amount of time you spend riding and the terrain.

Tips for optimal performance

There are some simple adjustments that can make riding much more comfortable. Always check your saddle and evenly distribute your weight for an easier ride.

Best Places to Buy: E-tailers like eBay and Walmart

What we Recommend: Schwinn Sierra 26" MTB

46. Treadmill



When you think of cardio workouts, treadmills are the first to come to mind. They are used to help you achieve a walking or running motion while staying in one place. Using a treadmill regularly can help you lose weight and build strength.

Purchasing a treadmill can be quite an investment as these tend to be expensive. It is important to understand your fitness objectives and be clear about what you're looking for before taking the plunge. Our comprehensive reviews can help you choose the right one for your needs.

How often should you use it?

The frequency will depend on the level of impact of your workouts. Treadmills can be used every day, when opting for low-intensity exercise. For high-intensity runs, it is better to stick to alternate days.

Muscles worked

Running or walking on a treadmill primarily involves the cardiovascular system and the lower body. The other muscles involved are the hamstrings, quadriceps, calves and glutes.

Tips for optimal performance

Always make sure that you warm up before using the treadmill. Start with a slight incline and ensure that it is not too steep. With practice, you can work on improving your stride count.

Best Places to Buy: E-tailers like Walmart, Tesco and specialist fitness brands like NordicTrack

What we Recommend: NordicTrack T 6.5 S Treadmill

47. Mini Exercise Bikes



A mini exercise bike is a convenient alternative for people who don't have a lot of space. It is a portable and efficient piece of equipment that can be stashed away in a closet or even under your desk at work.

It can be hard to find a mini exercise bike that is both durable and convenient to carry with you when you're on the move. To make things easier for you, we've put together comprehensive reviews of the best mini exercise bikes out there.

<https://garagegymplanner.com/best-mini-exercise-bike-reviews/>

How often should you use it?

If weight loss is your goal then spending more time pedaling will decrease your body fat, increase your calorie burn, and help you lose more weight. Opt for at least 10 minutes of intense pedaling per day.

Muscles worked

Using this device regularly can help improve your cardiovascular health and help you attain your weight loss goals. It strengthens your leg muscles and reduces total fat mass and body fat percentage.

Tips for optimal performance

Use this device when working at your desk or cycle away whilst you're watching TV, and you'll soon start seeing positive results.

Best Places to Buy: E-tailers like Walmart and eBay

What we Recommend: Sunny Health & Fitness SF-B0418 Magnetic Mini Exercise Bike

48. Metal plates/bumper plates



These are the typical weight plates you find in your gym. They are either plain metal plates or they come coated with a rubber layer. The coated plates are called bumper plates.

While you're shopping for plates, you need to make sure that you also get good bar clamps to go with it. Bar clamps or muscle clamps as they are known stop your plates from falling off and are invaluable. For more info about bumper plates and accessories, take a look at our [detailed review](#) of all bumper plates in the market.

How often should you use it?

They get used frequently if you are into heavy lifting or free weights.

Muscles worked: Depends on the machine/equipment used.

Tips for optimal performance

From a home gym perspective, if you've got good rubber floors (or even if you don't) we'd recommend you getting the bumper variety as it's a lot easier on your general infrastructure.

Best places to buy: You can get them at any place, online or offline, which is even remotely associated with fitness! Just make sure you don't skimp here because they are going to have to last you a really long time.

What we recommend: [Rogue HG 2.0 Bumper Plates](#)

49. Medicine Balls



Also called a med ball, exercise ball or a fitness ball, it is a weighted ball which is used in a number of exercise movements to improve fitness, strength and coordination. It is also used for helping people recover from various types of injury.

When choosing medicine balls, there are a variety of factors that you need to keep in mind when buying a medicine ball. If you need more info, we've got you covered with our [2019 medicine ball review](#).

How often should you use it?

You can use this every day as it improves coordination and fitness all at once.

Muscles worked: Core muscles, chest, arms

Tips for optimal performance

There are several ways to use a medicine ball effectively and it can improve strength, agility, fitness and coordination depending on what you choose. Basic exercises like burpees, planks, pushups, and lunges can be combined with medicine balls to give you new types of movement.

Best places to buy: Etailers like Walmart or Tesco, specialist fitness brands like Rogue

What we recommend: [Dynamax Medicine Balls](#)

50. Pedometers



A small electronic device mostly worn like a watch that helps you calculate the number of steps that you take in a day. Also called a step counter, some models have increased functionalities like step count per hour, calories burned, total distance, and move reminder.

You can get pedometers for all price ranges but the cheaper ones will have problems with varying accuracy. Even the best pedometers have a small margin of error when it comes to step counts.

How often should you use it?

You can leave it on your wrist all the time.

Tips for optimal performance

Since this is an electronic device, it usually performs well day in and day out. You need to make sure that you charge/replace batteries when due, clean the watch adequately especially the heart rate sensor and you should be good.

Best places to buy: Etailers like Walmart or Tesco

What we recommend: Fitbit Zip

51. Pull up Bands



Considering pull-ups are one of the best exercises for your upper body and these bands can help making it a lot easier for you if you are just getting into it.

These pull up bands come in various resistances so people of all weights and abilities can use them effectively. Read our take on what we think is the best pull up band for you.

How often should you use it?

You should use them as often as you do your pull-ups. Once you start to accomplish unassisted pull-ups more often, then you can give these up.

Muscles worked: Trunk, arms, shoulders, scapular (traps and pectorals), abs, arms.

Tips for optimal performance

You can vary your assistance with these bands. You can either wrap it around your knee for lower amount of assistance, and through your step for increased assistance.

Best places to buy: E-tailers like Walmart and Tesco or specialist fitness brands like Rogue fitness

What we recommend: Rogue Monster Bands

52. Blood Glucose Meter



This is an electronic device that is designed to test the concentration of glucose in blood. A small drop of blood is placed on a test strip that is connected to a digital meter. In a few seconds, you will be able to know your blood glucose levels.

You can choose from basic blood glucose meters to advanced kits that have everything you need to test your blood sugar levels.

How often should you use it?

It is recommended to consult your doctor before deciding on the frequency of use.

Best places to buy: E-tailers or specialist brand websites.

What we recommend: Bayer Contour NEXT EZ Diabetic Testing Starter Kit

53. GPS Watch



This is an advanced watch with an integrated GPS receiver. A smart GPS enabled watch also comes with other capabilities and features depending on its purpose.

GPS Technology has revolutionized the way we move around the planet. Take a look at this [in-depth review of GPS watches](#) to decide which one is right for your needs.

How often should you use it?

It is a good idea to strap on this watch whenever you exercise to get useful data such as heart rate, calories burned, etc.

54. Gymnastics Grips & CrossFit Gloves



Gymnastics Grips & CrossFit Gloves are meant to protect your hands during a workout while offering a firm grip. They can also help you avoid painful rips and blisters.

The right crossfit gloves can take your workouts up a notch.

How often should you use it?

It is recommended to protect your hands with gloves every time you perform a CrossFit Workout.

Tips for optimal performance

Ensure that you use gloves that fit well and have adequate grip to avoid any accidents.

Best Places to Buy: E-tailers or specialist brand websites.

What we Recommend: 5.10 Original Gymnastics Bar Lifting Grips

55. Hoverboards



Hoverboards are self-balancing personal transporters. They consist of two motorized wheels, articulated pads and offer good control over stability and balance.

Hoverboards are a cost-effective and eco-friendly way to get around. For more information of the best hoverboards available, check out our [list](#):

How often should you use it?

Hoverboards are a great form of transportation and recreation, but make sure you use it in a safe environment to avoid accidents.

Tips for optimal performance

Wear Safety equipment and take precautions when using a hoverboard, especially at night.

Best Places to Buy: E-tailers like Walmart and eBay

What we Recommend: TOMOLOO Hoverboard

56. Jump Rope



A jump rope is a fitness tool that provides a total body conditioning and strengthening workout. It is a simple device that helps you burn more calories than many other forms of exercise.

Ensure that you select a jump rope with suitable rope length. For tips on choosing a jump rope, check out our comprehensive article.

<https://garagegymplanner.com/best-jump-rope/>

How often should you use it?

An average person trying to get fit needs to jump 10 minutes a day, whereas a serious athlete would need to spend more time to improve strength and endurance.

Muscles worked:

Your quadriceps, deltoids hamstrings, glutes, abdominals, forearms, and gastrocnemius muscles get a workout from jump rope exercises.

Tips for optimal performance

Jump in short bursts that last only 30 seconds to a minute. Rest for one to two minutes in between intervals.

Best Places to Buy: E-tailers like Walmart and specialist fitness brands like Rogue Fitness

What we Recommend: [Jump Rope Sold By FMS International](#)

57. Resistance Band



These are elastic bands that allow you to perform strength training exercises without weights. They allow you to add variety and intensity to your training and can deliver a full body workout.

You can incorporate resistance bands into numerous workouts. For tips on how to get started check out our [detailed post](#).

How often should you use it?

Resistance bands can be used 2 to 3 times a week if you're a beginner and more often if you're an advanced fitness seeker.

Muscles worked: Resistance bands can be used to target almost any muscle group.

Tips for optimal performance

Since they are convenient and portable, you can also carry them with you when traveling to work out on the go.

Best Places to Buy: E-tailers like Walmart or specialist fitness brands like Rogue Fitness

What we Recommend: [Rogue Monster Bands](#)

58. Yoga Mat



These mats are used in the practice of yoga to keep your feet and hands in place when performing an asana. Today yoga mats come in many different materials and finishes such as cotton, jute and rubber.

Always opt for a good quality yoga mat to avoid distractions and enjoy your yoga sessions. The right yoga mat can make all the difference between a bad and good workout.

How often should you use it?

The best way to start seeing results from yoga is by performing it every day.

Muscles worked

Different yoga poses are meant to work specific parts of your body, such as your lower back muscles, your hamstrings and more.

Tips for optimal performance

Grip is the most important factor when selecting a yoga mat. Look for a mat that will give you adequate grip. A good yoga mat must be thick enough to protect your knees and joints while still being light.

Best Places to Buy: E-tailers like Walmart and eBay

What we Recommend: Manduka PROlite Yoga Mat

59. Fitness Tracker



A fitness tracker also known as an activity tracker is a device that is used to monitor fitness-related metrics. Today fitness trackers can do much more than just count your steps. It can help you stay active, achieve your fitness goals, and even get better sleep.

With so many good fitness trackers available right now, and promising ones on the horizon, it can be hard to decide which one to go for. To make things easier, we put together comprehensive reviews of the best fitness trackers out there.

<https://garagegymplanner.com/best-fitness-tracker/>

How often should you use it?

Use a fitness tracker whenever you're performing a workout of any kind to get useful data that can help you maximize your workouts.

Best Places to Buy: E-tailers like Walmart, Tesco and eBay

What we Recommend: Fitbit Charge 2

60. Barbell



Barbell is a piece of weight training equipment with a long bar and weights mounted at each end. It is used in weightlifting, bodybuilding, powerlifting and more. It's an essential piece of equipment for power racks and smith machine.

They have been around for decades but have now gotten extremely complex considering the amount of technology available today. Find out how barbells are even used on CrossFit these days!

<https://garagegymplanner.com/crossfit-arms-workout/>

How often should you use it?

Barbells get used all the time as they are the primary pieces of strength building equipment in a gym.

Muscles worked: Depends on the exercise.

Tips for optimal performance

Get barbells that have at least 175,000 PSI. The best ones are >200,000 PSI

Best places to buy: Specialist fitness brands like RogueFitness.com and Eleikoshop.com

What we recommend: [US Weight Duracast Barbell Weight Set](#)

61. Wall Ball



You can build your core and lower body strength using a wall ball. All you have to do is fling the ball against a wall and catch it. You can combine this with sit-ups for better results.

These can be tedious but are great for working the whole body at once and increasing agility and balance. They also help with improved footwork and multiple core strengthening. This exercise requires medium or intermediate level of workout experience and is definitely not for beginners. Boosting your endurance to high-intensity workouts is essential, so take a look at this:

<https://garagegymplanner.com/crossfit-workout-of-day/>

How often should you use it?

Twice a week

Muscles worked: Works the quads, glutes, calves, hamstrings, abs, lower back, chest, front deltoids, back deltoid, biceps and tricep

Tips for optimal performance

Trying doing it in intervals between other exercises

Best Places to Buy: E-tailers like Walmart and specialist fitness brands like Rogue Fitness

What we Recommend: [Rep Soft Medicine Ball / Wall Ball](#)

62. Foam Roller



Foam rollers are useful to help you with your workout routines and also to treat muscle soreness. They also promote flexibility and relax overactive muscles.

Foam rollers open up contracted muscles, expand the tissues and then soothe them from the soreness one feels after a workout. The foam roller also increases blood circulation in the body and is a great piece of equipment to recover from injuries and catches.

How often should you use it?

Work as a great warm-up for the muscles, especially before and after a rigorous workout. Can be used every day.

Muscles worked: Can be used for the glutes, hamstrings and calf muscles

Tips for optimal performance

It's good to foam roll both before and after a workout

Best Places to Buy: E-tailers like Walmart and specialist fitness brands like Rogue Fitness

What we Recommend: LuxFit Premium High Density Foam Roller

63. Stability ball



These giant inflatable rubber balls can help you perform balance training exercises such as planks and also help in stretching. It is a versatile tool which is excellent for insulating your core as well.

Stability balls are also called Swiss balls and can be used creatively to make a lot of workouts more interesting and effective. It is a full-body workout tool and increases stability and balance. Can be used for drills like planks, hip thrusts, lying hamstring curls and so on. Want to know which stability ball would be the best choice for you? Take a look:

How often should you use it?

2-3 times a week

Muscles worked: Works on the thighs, core and also back muscles. Varies according to the exercise.

Tips for optimal performance

Women love using them or pairing them up with other workouts. For example, frog crunches, the Russian twist, back extension and so on are great to be done using a stability ball.

Best Places to Buy: E-tailers like Walmart and specialist fitness brands like Rogue Fitness

What we Recommend: Black Mountain 2000lbs Static Strength Exercise Stability Ball with Pump

64. Agility Ladder



A fitness aid which you can roll out on a flat surface and perform various movements on such as sideways shuffles, skipping through rungs, etc. Improves agility of the individual.

Great for speed and agility drills. Preferred equipment by athletic trainers and sports people. Forward, lateral and backward movements can be improvised and achieved using this equipment. It is also widely used as a strength and conditioning tool. [Check out](#) the best agility ladders for 2019:

How often should you use it?

Twice a week

Muscles worked: Leg and core muscles

Tips for optimal performance

Perform later, front and back movements. Skipping and shuffles are also effective

Best Places to Buy: E-tailers like Walmart and specialist fitness brands like Rogue Fitness

What we Recommend: [GHB Pro Agility Ladder](#) [Agility Training Ladder](#) [Speed Flat Rung](#)

65. Balance Boards



An essential piece of equipment in balance training, active exercise, and other athletic training. It can also be used for recreation and brain development.

Great for increasing stability and balance in the sagittal and axial plane of the body. Can also be used by older people to generate muscle strength and also increase flexibility in movement. Balance boards are also useful for improvement in hand-eye coordination and brain development through better focusing skills.

How often should you use it?

Can be used twice a week

Muscles worked: Works on core muscles

Tips for optimal performance

Pair it with core exercises and athletic training to see maximum results

Best Places to Buy: E-tailers like Walmart and specialist fitness brands like Rogue Fitness

What we Recommend: Sportneer Wooden Balance Board Platform

Conclusion

We've tried to give you an exhaustive amount of equipment and notes on how it can help develop muscularity and fitness.

Remember:

You need to understand what each machine does in order for you to get the most out of it.

We at SPORTiVA Academy believe that you need knowledge about fitness before you can become the best you are able to.

